CONTINUOUS-TIME MOTION PLANNING AND STEERING WITH GLOBAL CONVERGENCE

Mohsen Amiri[†], **Sima Didari**[†], **Davoud Ataee Tarzanagh**[†], **Hankyu Moon, Yeongdae Kwon**
Samsung SDS Research America, Mountain View, CA, USA

Project Page

ABSTRACT

Motion planning is central to robotics, and it is increasingly used to correct the motions proposed by learned planners and policies. In both uses, constraints must hold at every instant of the motion, not only at sampled times. To our knowledge, no algorithm provably converges to a globally optimal motion once a smoothness cost is combined with such constraints. We cast planning and steering as a single continuous-time program: robots avoid one another and static or moving obstacles, and a learned proposal is steered by keeping the motion close to it. The program is elementary to state, yet nonconvex. We propose GlobMP (Global Motion Planning), a gradient projection method over piecewise-polynomial curves, on which each constraint is one polynomial per piece that must stay nonnegative over the whole motion; it alternates cost reduction with feasibility restoration. Under mild assumptions, we show that, from a fine grid start, its cost approaches the global optimum as the number of polynomial pieces grows, with every constraint met to any chosen tolerance, or exactly on stable instances. We demonstrate its superior performance over strong baselines in real and synthetic experiments, including teams of up to 200 robots, multi-robot planning on the Robotarium testbed and an extended benchmark, maze navigation, and steering tactile-reactive policies on a bimanual humanoid.

1 INTRODUCTION

Motion planning is a fundamental problem in robotics: it asks for a trajectory that connects a given initial state to a goal while satisfying motion constraints, such as avoiding collisions (LaValle, 1998). It underlies navigation, manipulation (Sundaralingam & Hermans, 2019; Gao et al., 2024; Amice et al., 2024), multi-robot coordination (Li et al., 2021; Tang et al., 2025; Shaoul et al., 2025), and, increasingly, the *steering* of learned planners and policies (Janner et al., 2022; Carvalho et al., 2023; Chi et al., 2023; Giaretta et al., 2026).

Motion planners approach this problem in several ways, each with a gap. Sampling-based planners explore many routes (Karaman & Frazzoli, 2011), yet smoothing their waypoints can introduce collisions; optimization-based planners improve a motion locally (Zucker et al., 2013; Tedrake & the Drake Development Team, 2019), with TrajOpt (Schulman et al., 2014) also supporting continuous collision checks. Constraints enforced at sampled times do not ensure feasibility between them (Hargraves & Paris, 1987; Figure 1). Convex-region planners search a decomposition of free space (Deits & Tedrake, 2015a; Marcucci et al., 2023), and semidefinite relaxations certify a polynomial motion as globally optimal only when the relaxation is tight (Graesdal et al., 2026). Even shortest-path complexity depends on the geometry: polynomial-time in the real-RAM model among planar discs (Chang et al., 2005), NP-hard among polyhedra in $\mathbb{R}^3$ (Canny & Reif, 1987), and still open in $\mathbb{R}^3$ even among disjoint unit discs or axis-aligned unit cubes (Mitchell, 2017).

Our setting is a point robot in $\mathbb{R}^n$ that travels between prescribed boundary states around obstacles while minimizing a smoothness cost; steering a learned planner or policy is the same problem with one added constraint that keeps the robot near its proposal. Although elementary to state, this setting already contains the core difficulty of avoiding collisions; to our knowledge, no algorithm converges to its global optimum once a derivative cost such as acceleration or snap is combined with collision

[†]Equal contribution.

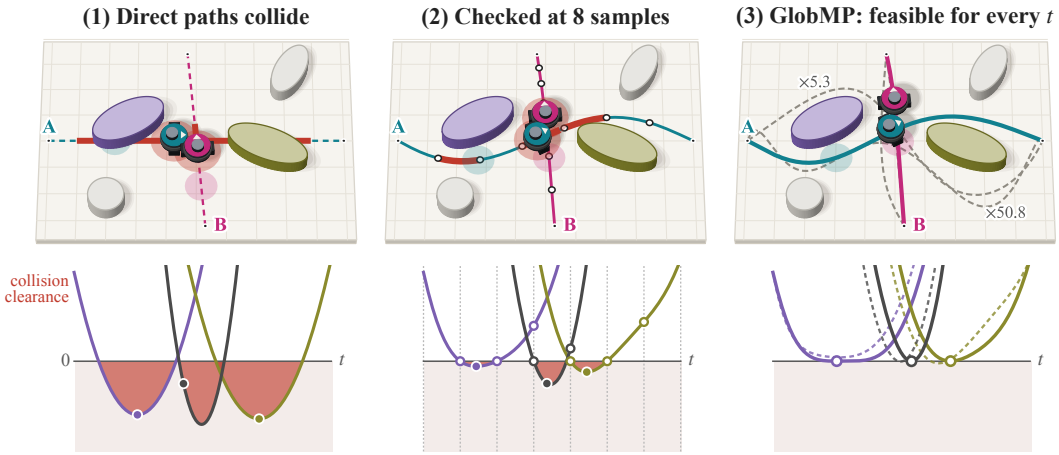

Figure 1: Two robots, A and B, planned jointly among ellipsoidal obstacles; each obstacle and the other robot enter the program as a collision constraint (ellipses through the weighted norm of Appendix A2.1). Top: each robot's path in its colour (A teal, B magenta), with both robots drawn at the time of the dark-grey A−B mark and a faded disc where each was earlier; red marks a collision. Bottom: robot A's clearance to each coloured obstacle and the A−B clearance (dark grey) over time; below zero is a collision. (1) The direct (dashed) paths collide. (2) Checking the constraints at 8 sample times (open dots) misses collisions between them. (3) The final solid curves keep every clearance nonnegative on $[0,1]$; coloured dashed: an earlier descent step; grey dashed: numerically refined alternatives, with cost relative to GlobMP's.

constraints on the whole time interval. We introduce GlobMP and show that, from a fine grid start, its cost approaches the global optimum as the number of pieces grows, within any constraint tolerance or exactly on stable instances.

GlobMP combines exact whole-interval constraints, gradient-based refinement and a systematic search over initial curves. We consider

$$
\begin{aligned}
&\min_{\boldsymbol{\gamma}\in\mathscr{H}^k} && \int_0^1 \left\|\boldsymbol{\gamma}^{(k)}(t)\right\|^2 dt && && \text{smoothness} && \\
&\text{subject to} && \left\|\boldsymbol{\gamma}(t)-\mathbf{c}_j(t)\right\| \geq r_j, && \forall t\in[0,1],\ j=1,\dots,p, && \text{collision} && (\mathrm{P}_\infty)\\
&&& \left\|\boldsymbol{\gamma}(t)-\boldsymbol{\rho}(t)\right\| \leq \bar{r}, && \forall t\in[0,1], && \text{steering} && \\
&&& \boldsymbol{\gamma}^{(s)}(0)=\boldsymbol{\gamma}_0^{(s)},\ \boldsymbol{\gamma}^{(s)}(1)=\boldsymbol{\gamma}_1^{(s)}, && s=0,\dots,\ell, && \text{boundary} &&
\end{aligned}
$$

Here $\boldsymbol{\gamma}\colon[0,1]\to\mathbb{R}^n$ is the robot's trajectory, with time normalized to $[0,1]$, and $\mathscr{H}^k=\mathscr{H}^k([0,1];\mathbb{R}^n)$ denotes the Sobolev space of curves whose derivatives through order k are square-integrable: the objective is finite exactly on $\mathscr{H}^k$, and its curves have $k-1$ continuous derivatives, so the constraints and boundary values are defined at every instant. The objective measures smoothness through the kth derivative: $k=2$ penalizes acceleration and $k=4$ snap (Mellinger & Kumar, 2011). The collision constraints keep the robot at distance at least $r_j>0$ from each of p obstacle centres $\mathbf{c}_j(t)$, which may move, and the steering constraint keeps it within $\bar{r}>0$ of a reference $\boldsymbol{\rho}(t)$, both at every instant. The boundary conditions fix the position and its derivatives through order ℓ at both ends. Unlike Graesdal et al. (2026), whose spheres are static, we let the centres move in time; this single extension covers several robots, each a moving centre for the others, and steering, where the robot must stay near the moving centre $\boldsymbol{\rho}(t)$.

Several robots are planned by stacking their curves into one $\boldsymbol{\gamma}$ and keeping robots a and b (radii r_a, r_b) apart: $\|\boldsymbol{\gamma}_a(t)-\boldsymbol{\gamma}_b(t)\|\geq r_a+r_b$ (Shaoul et al., 2025). The reference $\boldsymbol{\rho}$ can be the waypoints of a learned planner (Janner et al., 2022) or the action chunk of a policy (Chi et al., 2023), which the steering constraint corrects rather than replaces; with $\bar{r}=\infty$ it is dropped and (P_∞) is plain motion planning. Boundary orders $s\geq k$ bind only our polynomial curves (Assumption A). Ellipsoids, walls, corridors and other variants are in Appendix A2.1.

Our contributions address two difficulties of (P_∞): constraints must hold at every instant, and improving one feasible curve need not find the best route.

C1. We enforce collision constraints *exactly in continuous time*. At a fixed polynomial degree, each constraint becomes a univariate polynomial that must be nonnegative on $[0,1]$ (Powers & Reznick, 2000), so it holds between sample times too; collision, robot separation and steering share one block formulation (§2).

C2. We propose *GlobMP*, a gradient projection algorithm that alternates a cost step with feasibility restoration (§2). Cone projections give coefficient corrections for violated blocks, which Jacobians turn into control-point updates, block by block and with the boundary states kept.

C3. We prove *global convergence* (Theorem 1): GlobMP keeps every curve feasible without increasing its cost and, from a fine grid start with N polynomial pieces, approaches the best continuous-time motion as N grows, within any constraint tolerance and exactly on stable instances.

C4. We *deploy* GlobMP on real robots and show superior performance over strong baselines: multi-robot teams on the Robotarium testbed (§4.2), and steering state-of-the-art expert policies on a bimanual humanoid with dexterous hands in tactile manipulation tasks (§4.3). The appendix adds an extended multi-robot benchmark, maze navigation and end-effector obstacle avoidance on D3IL (Appendix A4).

2 Polynomial formulation and gradient projection

We optimize the control points of a polynomial curve, alternating a gradient step on the cost with restoration of feasibility; because the curve is a polynomial, each constraint is checked over the whole motion. We represent the curve by a polynomial of degree d (piecewise when the horizon is split) in the *Bernstein basis* $[\mathbf{b}_d(t)]_i = \binom{d}{i} t^i (1-t)^{d-i}$, $i = 0,\dots,d$, whose weights are nonnegative and sum to one on $[0,1]$:

$$\boldsymbol{\gamma}(t) = \boldsymbol{\Gamma}\mathbf{b}_d(t), \qquad \boldsymbol{\Gamma} \in \mathbb{R}^{n\times(d+1)}. \tag{1a}$$

The columns of $\boldsymbol{\Gamma}$ are the *control points*, whose convex hull contains the curve. Substituting this representation into the objective gives

$$\mathrm{cost}(\boldsymbol{\Gamma}) := \int_0^1 \left\|\boldsymbol{\Gamma}\mathbf{b}_d^{(k)}(t)\right\|^2 dt = \mathrm{tr}(\mathbf{Q}\boldsymbol{\Gamma}^\top\boldsymbol{\Gamma}), \tag{1b}$$

where tr is the trace and $\mathbf{Q} := \int_0^1 \mathbf{b}_d^{(k)}\mathbf{b}_d^{(k)\top} dt$ is precomputed exactly, so each cost step uses a matrix gradient without reevaluating the integral. Since $\mathbf{Q} \succeq 0$ (positive semidefinite), the objective is convex; the constraints carry the nonconvexity.

Let constraint j be $g_j(\boldsymbol{\gamma}(t),t) \geq 0$, where $g_j(\mathbf{x},t) := \|\mathbf{x} - \mathbf{c}_j(t)\|^2 - r_j^2$ for the collision constraints $j \leq p$, and $g_{p+1}(\mathbf{x},t) := \bar{r}^2 - \|\mathbf{x} - \boldsymbol{\rho}(t)\|^2$ for the steering constraint when $\bar{r} < \infty$. With several robots, each robot's own position $\mathbf{x}_a$ replaces $\mathbf{x}$, and each pair of robots forms an additional collision block, with $\mathbf{x}_a - \mathbf{x}_b$ and $r_a + r_b$ in place of $\mathbf{x} - \mathbf{c}_j$ and r_j; from here on, p counts both the obstacle blocks and the pair blocks. For polynomial obstacle centres and a polynomial reference, the value along the curve is a polynomial in time,

$$g_j^{\boldsymbol{\Gamma}}(t) := g_j\big(\boldsymbol{\Gamma}\mathbf{b}_d(t),t\big) = \boldsymbol{\pi}_j(\boldsymbol{\Gamma})^\top \mathbf{b}_{m_j}(t), \tag{1c}$$

with degree bound $m_j = 2d$, or $2\max(d,d_j)$ for a centre moving on a polynomial path of degree d_j (the reference plays the role of a centre for $j = p+1$); piecewise-polynomial data are handled interval by interval (Appendix A3.3). Its coefficient vector $\boldsymbol{\pi}_j(\boldsymbol{\Gamma})$ is quadratic in $\boldsymbol{\Gamma}$. The constraint holds at every instant exactly when this vector belongs to the closed convex cone

$$\mathscr{K}_j := \{\boldsymbol{\zeta} \in \mathbb{R}^{m_j+1} : \boldsymbol{\zeta}^\top \mathbf{b}_{m_j}(t) \geq 0 \text{ for every } t \in [0,1]\}. \tag{1d}$$

This tests the entire polynomial: $(2t-1)^2$ has Bernstein coefficients $(1,-1,1)$ yet is nonnegative, so a coefficient-sign test would wrongly reject it. The coefficients are quadratic in $\boldsymbol{\Gamma}$, so the set of control points that satisfy a block need not be convex. Integral budgets, an optional extension beyond (P_∞), fit the same notation as scalar blocks,

$$\pi_j(\boldsymbol{\Gamma}) := \int_0^1 \psi_j\big(\boldsymbol{\Gamma}\mathbf{b}_d(t),t\big)\, dt \in \mathscr{K}_j := \mathbb{R}_+, \tag{1e}$$

where ψ_j encodes the signed budget margin; for uniform notation we also write $\boldsymbol{\pi}_j := \pi_j$ for budgets. Let $\bar{p}$ denote the number of pointwise blocks ($\bar{p} = p+1$ with the steering block, $\bar{p} = p$ without); the q budgets are blocks $j = \bar{p}+1, \dots, \bar{p}+q$. We collect the constraints as

$$\boldsymbol{\pi}(\boldsymbol{\Gamma}) := \big(\boldsymbol{\pi}_1(\boldsymbol{\Gamma}), \dots, \boldsymbol{\pi}_{\bar{p}}(\boldsymbol{\Gamma}), \pi_{\bar{p}+1}(\boldsymbol{\Gamma}), \dots, \pi_{\bar{p}+q}(\boldsymbol{\Gamma})\big), \qquad \mathscr{K} := \mathscr{K}_1 \times \cdots \times \mathscr{K}_{\bar{p}} \times \mathbb{R}^q_+. \tag{1f}$$

Now, substituting (1a) into the objective and the constraints of (P_∞) gives (1b) and (1c)–(1f); with the boundary conditions written as linear equations $\boldsymbol{\Gamma}\mathbf{A} = \mathbf{B}$ in the control points, we obtain the degree-d reformulation of (P_∞) as

$$\inf_{\boldsymbol{\Gamma} \in \mathbb{R}^{n\times(d+1)}} \ \operatorname{cost}(\boldsymbol{\Gamma}) \quad \text{s.t.} \quad \boldsymbol{\pi}(\boldsymbol{\Gamma}) \in \mathscr{K}, \qquad \boldsymbol{\Gamma}\mathbf{A} = \mathbf{B}. \tag{1g}$$

Let $\mathscr{A}_{\mathrm{bc}} := \{\boldsymbol{\Gamma} : \boldsymbol{\Gamma}\mathbf{A} = \mathbf{B}\}$ denote the control points that satisfy them. We fix one such point $\boldsymbol{\Gamma}_{\mathrm{aff}}$ and allow only changes that keep the equations. When $2(\ell+1) \le d$, every $\boldsymbol{\Gamma} \in \mathscr{A}_{\mathrm{bc}}$ is $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}_{\mathrm{aff}} + \boldsymbol{\Gamma}_{\mathrm{free}}\mathbf{N}^\top$, where the columns of $\mathbf{N}$ are an orthonormal basis of the null space of $\mathbf{A}^\top$ and $\boldsymbol{\Gamma}_{\mathrm{free}} \in \mathbb{R}^{n\times d_f}$, $d_f = d+1-2(\ell+1)$, collects the free coordinates (Lemma 3, Appendix A3.1); distances between free and full control points agree. In these free coordinates

$$\begin{aligned} \inf_{\boldsymbol{\Gamma}_{\mathrm{free}}} \quad & \operatorname{cost}\big(\boldsymbol{\Gamma}_{\mathrm{aff}} + \boldsymbol{\Gamma}_{\mathrm{free}}\mathbf{N}^\top\big) \\ \text{subject to} \quad & \boldsymbol{\Gamma}_{\mathrm{free}} \in \mathscr{F} := \Big\{\boldsymbol{\Gamma}_{\mathrm{free}} : \ \boldsymbol{\pi}\big(\boldsymbol{\Gamma}_{\mathrm{aff}} + \boldsymbol{\Gamma}_{\mathrm{free}}\mathbf{N}^\top\big) \in \mathscr{K}\Big\}. \end{aligned} \tag{P}$$

Every reconstructed curve meets the boundary conditions; for piecewise curves the junction conditions join the linear equations. Fixing the degree restricts the curves but neither discretizes time nor weakens the whole-interval constraints.

Write $\boldsymbol{\pi}_j(\boldsymbol{\Gamma}_{\mathrm{free}})$ for $\boldsymbol{\pi}_j(\boldsymbol{\Gamma}_{\mathrm{aff}} + \boldsymbol{\Gamma}_{\mathrm{free}}\mathbf{N}^\top)$. Restoration, Algorithm 2, takes gradient steps on the summed squared distances of the blocks to their cones, where $P_{\mathscr{K}_j}$ maps a vector to its nearest point in $\mathscr{K}_j$,

$$V(\boldsymbol{\Gamma}_{\mathrm{free}}) := \sum_{j=1}^{\bar{p}+q} \big\| \boldsymbol{\pi}_j(\boldsymbol{\Gamma}_{\mathrm{free}}) - P_{\mathscr{K}_j}\big(\boldsymbol{\pi}_j(\boldsymbol{\Gamma}_{\mathrm{free}})\big) \big\|^2. \tag{2}$$

The squared distance to a closed convex cone is differentiable, with gradient twice the residual $\mathbf{e}_j = \boldsymbol{\pi}_j - P_{\mathscr{K}_j}(\boldsymbol{\pi}_j)$, which vanishes exactly when block j is satisfied. The chain rule gives $2D\boldsymbol{\pi}_j^\top \mathbf{e}_j$, where $D\boldsymbol{\pi}_j$ is the Jacobian (matrix of first derivatives) in free coordinates and the gradient is reshaped as a control-point matrix. Only violated blocks contribute (lines 4–5). GlobMP (Algorithm 1) alternates a cost step along $-2\boldsymbol{\Gamma}\mathbf{Q}\mathbf{N}$ with restoration (lines 4–6), and rebuilding $\boldsymbol{\Gamma}$ from the free coordinates preserves the boundary states. $P_{\mathscr{F}}$ denotes Algorithm 2, which need not return an exactly feasible or nearest curve in finitely many steps; the theory analyses the repair below.

Each cone projection is a small convex problem. Evaluating a polynomial at time t is an inner product of its coefficients with $\mathbf{b}_{m_j}(t)$, so the vectors $\mathbf{b}_{m_j}(t)$, $t \in [0,1]$, generate the dual cone $\mathscr{K}_j^*$, and by Moreau's decomposition (Moreau, 1962) the projection adds to $\boldsymbol{\pi}_j$ the point of $\mathscr{K}_j^*$ closest to $-\boldsymbol{\pi}_j$. We build $\boldsymbol{\omega}^\star$ greedily as a nonnegative combination $\boldsymbol{\omega}$ of $\mathbf{b}_{m_j}(t)$ over a growing set of selected times, with $\boldsymbol{\xi}$ the part of $-\boldsymbol{\pi}_j$ still to be corrected:

$$\begin{aligned} P_{\mathscr{K}_j}(\boldsymbol{\pi}_j) = \boldsymbol{\pi}_j + \boldsymbol{\omega}^\star, \qquad \boldsymbol{\omega}^\star = \operatorname*{arg\,min}_{\boldsymbol{\omega} \in \mathscr{K}_j^*} \|\boldsymbol{\omega} + \boldsymbol{\pi}_j\|, \\ t_* \in \operatorname*{arg\,max}_{t\in[0,1]} \frac{\langle \boldsymbol{\xi}, \mathbf{b}_{m_j}(t)\rangle}{\|\mathbf{b}_{m_j}(t)\|}, \qquad \boldsymbol{\xi} := -\boldsymbol{\pi}_j - \boldsymbol{\omega}. \end{aligned} \tag{3}$$

Each step adds the time t_*, where the corrected polynomial $\boldsymbol{\pi}_j + \boldsymbol{\omega}$ is most negative relative to its scale, and refits nonnegative weights on all selected times (Algorithm 3; Locatello et al., 2017), with error bounds in Appendix A3.1.4; scalar budgets use $u \mapsto \max\{u, 0\}$. The selected times locate the violation rather than fix a list of checks, and blocks are projected independently, so the work per iteration grows linearly with the number of blocks at fixed degree and cone-solve iterations.

For the analysis we use a feasible version of the repair step. Let $\mathbf{z}$ denote the free coordinates $\boldsymbol{\Gamma}_{\mathrm{free}}$ stacked into one vector and $f(\mathbf{z})$ the cost; we keep the notation $\boldsymbol{\pi}_j(\mathbf{z})$, $D\boldsymbol{\pi}_j(\mathbf{z})$ and $\mathscr{F}$ for the blocks, their Jacobians and the feasible set. Each collision block is convex in the control points,

Algorithm 1 GlobMP

Require: anchor $\mathbf{\Gamma}_{\text{aff}}$ and basis $\mathbf{N}$, start $\mathbf{\Gamma}_{\text{free}}$, step η, iterations T

1: $\mathbf{\Gamma}_{\text{free}} \leftarrow P_{\mathscr{F}}(\mathbf{\Gamma}_{\text{free}})$ *restore start*
2: **for** $\nu = 1, \dots, T$ **do**
3: $\quad \mathbf{\Gamma} \leftarrow \mathbf{\Gamma}_{\text{aff}} + \mathbf{\Gamma}_{\text{free}}\mathbf{N}^{\top}$ *reconstruct curve*
4: $\quad \mathbf{\Delta} \leftarrow 2\mathbf{\Gamma}\mathbf{Q}\mathbf{N}$ *cost gradient*
5: $\quad \mathbf{Y} \leftarrow \mathbf{\Gamma}_{\text{free}} - \eta\,\mathbf{\Delta}$ *cost step*
6: $\quad \mathbf{\Gamma}_{\text{free}} \leftarrow P_{\mathscr{F}}(\mathbf{Y})$ *restore, Alg. 2*
7: **end for**
8: **return** $\mathbf{\Gamma}_{\text{free}}$ *candidate*

Algorithm 2 Feasibility restoration for (P)

Require: free coordinates $\mathbf{\Gamma}_{\text{free}}$, blocks $\boldsymbol{\pi}_j$, tolerance tol, step η_{proj}, iterations T_{proj}

1: **for** $\nu = 1, \dots, T_{\text{proj}}$ **do**
2: $\quad \mathbf{e}_j \leftarrow (\mathbf{I} - P_{\mathscr{K}_j})\,\boldsymbol{\pi}_j(\mathbf{\Gamma}_{\text{free}})$ *all blocks*
3: $\quad$ **if** $\max_j \|\mathbf{e}_j\|^2 \le \text{tol}$ **then break**
4: $\quad \mathscr{V} \leftarrow \{j : \mathbf{e}_j \neq \mathbf{0}\}$ *violated blocks*
5: $\quad \mathbf{\Delta} \leftarrow 2\sum_{j\in\mathscr{V}} D\boldsymbol{\pi}_j^{\top}\mathbf{e}_j$ *merit gradient*
6: $\quad \mathbf{\Gamma}_{\text{free}} \leftarrow \mathbf{\Gamma}_{\text{free}} - \eta_{\text{proj}}\,\mathbf{\Delta}$ *restoration step*
7: **end for**
8: **return** $\mathbf{\Gamma}_{\text{free}}$ *candidate*

so its linearization at any $\mathbf{z}$ lies below it at every instant. For a sphere the linearized constraint is a half-space that excludes the whole obstacle, so each step keeps every robot behind a moving plane that separates it from each obstacle and from each other robot. Let $\mathscr{D}$ denote the points that satisfy the steering block exactly, a speed limit (Appendix A2.1) when present, and the optional budgets; steering and speed sets are convex, since balls are convex and the curve depends affinely on the free control points; the budgets are assumed convex ($\mathscr{D}$ is all points when none is present). Let

$$\mathscr{C}(\mathbf{z}) := \big\{\mathbf{w} \in \mathscr{D} :\ \boldsymbol{\pi}_j(\mathbf{z}) + D\boldsymbol{\pi}_j(\mathbf{z})(\mathbf{w}-\mathbf{z}) \in \mathscr{K}_j, \quad j = 1, \dots, p\big\}. \tag{4}$$

Then $\mathscr{C}(\mathbf{z})$ is convex and contained in $\mathscr{F}$, and it contains $\mathbf{z}$ when $\mathbf{z}$ is feasible. Line 6 of Algorithm 1 becomes the convex solve

$$\mathbf{z}^{+} = \operatorname*{arg\,min}_{\mathbf{w}\in\mathscr{C}(\mathbf{z})} \tfrac{1}{2}\|\mathbf{w} - (\mathbf{z} - \eta\nabla f(\mathbf{z}))\|^2. \tag{5}$$

Every iterate is feasible, and $\mathscr{C}(\mathbf{z})$ is rebuilt after each step, so this is not a convex reformulation of (P); it needs a feasible start and an exact solve. In the analysis, GlobMP starts from the cheapest feasible point of a grid of free control points, each tested exactly over the whole interval. For (P_∞) it adds pieces (Theorem 1) or raises the degree (Algorithm 4, under a condition on configurations).

3 Theory

We state one theorem, for (P_∞); its fixed-degree counterpart and all proofs are in Appendix A3. Let $E_\infty^\star$ denote the optimal value of (P_∞).

Assumption A (problem data)**.** *The cost order is $k \ge 1$ and the endpoints fix derivatives of orders $0, \dots, \ell$ (if $\ell \ge k$, (P_∞) keeps those of order below k); the obstacle centres and the reference are continuous and piecewise polynomial in t; the endpoints are strictly feasible; and (P_∞) is feasible.*

Assumption B (enough boundary conditions)**.** *The two endpoints fix at least as many values per coordinate as the order of the cost: $2(\ell+1) \ge k$.*

A change of the curve that keeps the boundary values and has zero kth-derivative cost is a polynomial of degree below k with a root of order $\ell+1$ at each endpoint, which is zero under Assumption B. Hence f is strongly convex, with curvature at most $\beta = 2\lambda_{\max}(\mathbf{H})$, $\mathbf{H} = \mathbf{N}^{\top}\mathbf{Q}\mathbf{N}$. For $|\tilde{r}| \le \tilde{r}_{\max}$, $\tilde{r}_{\max} > 0$ below every radius and half the endpoint clearance (Appendix A3.2.2), let ($\text{P}_\infty^{\tilde{r}}$) denote ($\text{P}_\infty$) with every obstacle radius raised by $\tilde{r}$, every pair separation by $2\tilde{r}$ and $\bar{r}$ lowered by $\tilde{r}$ ($\tilde{r} < 0$, a tolerance, loosens them), and $E_\infty^\star(\tilde{r})$ its optimal value.

Let $\mathscr{S}_N$ denote the curves of N polynomial pieces of a fixed degree $d \ge \max\{2k-1, \ell\}$ on equal intervals, joined with $k-1$ continuous derivatives. Let $\mathbf{z}_h$ denote a cheapest point of the grid of spacing h in their free control points that is feasible for the problem being solved; grid points can be checked exactly by root isolation for rational data.

Theorem 1 (global convergence for (P_∞))**.** *Under Assumptions A and B, run GlobMP (Algorithm 1, line 6 solved exactly by (5)) over $\mathscr{S}_N$ on ($\text{P}_\infty^{\tilde{r}}$) with a step $0 < \eta < 2/\beta$ (β computed for $\mathscr{S}_N$).*

(i) *From a feasible start, every iterate is feasible for ($\text{P}_\infty^{\tilde{r}}$) at every instant, with nonincreasing cost.*

(ii) *If a curve of cost at most U is feasible for $(\mathrm{P}_\infty^{\tilde{r}+\mu})$, $\mu > 0$, $\tilde{r}+\mu \le \tilde{r}_{\max}$, and $\delta_N := CN^{-r} \le \mu$, $r = \min\{k-1/2, 1\}$, with C depending on U, $\tilde{r}_{\max}$ and the data, then for small h, $\mathbf{z}_h$ exists and every cost from it lies in $[E^\star_\infty(\tilde{r}), E^\star_\infty(\tilde{r}+\delta_N)+\delta_N]$.*

(iii) *For $\tilde{r} = -\mu \ge -\tilde{r}_{\max}$ and $U \ge E^\star_\infty$, these curves violate no constraint of (P_∞) by more than μ (2μ for a separation) and cost at most $E^\star_\infty + \delta_N$; as $\mu \to 0$ (so $N \to \infty$), the costs tend to $E^\star_\infty$ and the curves have subsequences converging uniformly to global minimizers of (P_∞).*

At $\tilde{r} = 0$ the costs converge to $E^\star_\infty$ when the optimal value is stable, $E^\star_\infty(\tilde{r}) \to E^\star_\infty$ as $\tilde{r} \downarrow 0$, that is, when the robust-optimum analogue of sampling-based planners (Solovey et al., 2020) is the optimum; this holds at all but countably many $\tilde{r}$ in place of 0; if $E^\star_\infty(\tilde{r}) \le E^\star_\infty + c\tilde{r}$, the rate is $(c+1)\delta_N$. Methods that keep a positive clearance never reach $E^\star_\infty$ on unstable instances, such as two touching discs (Appendix A3.3).

The proof (Appendix A3.2) has three steps. First, collision constraints are convex in position, so $\mathscr{C}(\mathbf{z})$ in (4) is convex and feasible, and the repair (5) lowers f. Second, interpolating a curve of $(\mathrm{P}_\infty^{\tilde{r}+\delta_N})$ with N pieces never raises its cost and moves it by less than δ_N, so a ball around it, which the grid hits, is feasible for $(\mathrm{P}_\infty^{\tilde{r}})$; if $\ell \ge k$, an end correction costs $O(N^{-1})$. Third, curves of bounded cost have uniform limits, which are optimal as $\mu \to 0$.

Corollary 2. *Under Assumptions A and B,* **(i)** *if $\ell < k$, Theorem 1(ii) holds with $r = k - 1/2$ and $C = 1 + 2\sqrt{U}$;* **(ii)** *for either (r, C), $0 < \mu \le \tilde{r}_{\max}$, $\varepsilon > 0$ and $N \ge (C/\min\{\varepsilon, \mu\})^{1/r}$, the curves of Theorem 1(iii), within tolerance μ, cost at most $E^\star_\infty + \varepsilon$.*

In minimum-snap planning ($k = 4$) with position through jerk fixed at both ends ($\ell = 3$), accuracy $\varepsilon = \mu$ takes order $\varepsilon^{-2/7}$ pieces (ε^{-2} and $\varepsilon^{-2/3}$ for $k = 1, 2$, $\ell < k$). Fixing the snap too ($\ell = k$) adds cost of order $1/N$ at the end pieces, hence order ε^{-1} pieces.

4 Experiments

We evaluate GlobMP on synthetic teams of 2 to 200 robots among spheres (§4.1), a multi-robot benchmark on the Robotarium testbed (§4.2) and tactile-reactive manipulation (§4.3). Algorithm 2 restores from heuristic seeds (route classes, prioritized plans, the executing curve).

4.1 Synthetic experiments

The synthetic study (Problem 1, Appendix A4.2) has robots exchange places in pairs past static spheres, under the minimum-snap cost with piecewise degree-9 polynomials; each pair or sphere can be passed on either side, so an instance has many route classes. A curve is *feasible* if it makes no contact at any t; GlobMP keeps a 0.5 cm margin. GlobMP runs Algorithms 1–2 from one waypoint path per route class and keeps the cheapest feasible result. The baselines are RRT-Connect (Kuffner & LaValle, 2000), RRT* (Karaman & Frazzoli, 2011) and GCS (Marcucci et al., 2023) on IRIS regions (Deits & Tedrake, 2015a) in the joint configuration space, whose paths are fitted by the cheapest collision-free (else least-colliding) curve in GlobMP's class (Richter et al., 2016), and SDP-MP of Graesdal et al. (2026) on the same program (Appendix A4.2).

Each baseline imposes the constraints off the curve it returns (along a sampled path, inside convex regions, or on a relaxed variable), keeps centimetres to decimetres of slack, and fails more often as robots are added; SDP-MP collides or times out on every instance (Figure 2a,b). GlobMP is feasible at every t on every instance and ends at the margin. Feasible baseline curves cost from about twice to hundreds of times more, with a gap that mostly grows with team size (Figure 2c); where SDP-MP returns a lower bound, GlobMP is within 6% of it at the median. Theorem 1 covers this curve class ($\ell = k$, rate N^{-1}) but not its seeds, so the lowest cost found is not a proven optimum. From 2 to 200 robots GlobMP is feasible at every t on every seed, at 200 robots (82,000 blocks) in about two minutes on CPU and 28 s on one GPU, setup included (Figure 3a,b), with time per iteration at most linear in the blocks (Figure 3c); the baselines are feasible on at most 2 of 5 seeds at eight robots and on none from 16.

To check the rates of Theorem 1 and Corollary 2, we solve one two-robot instance again and again with more pieces N and track how far GlobMP's best cost stays above the limit (Figure 4). When

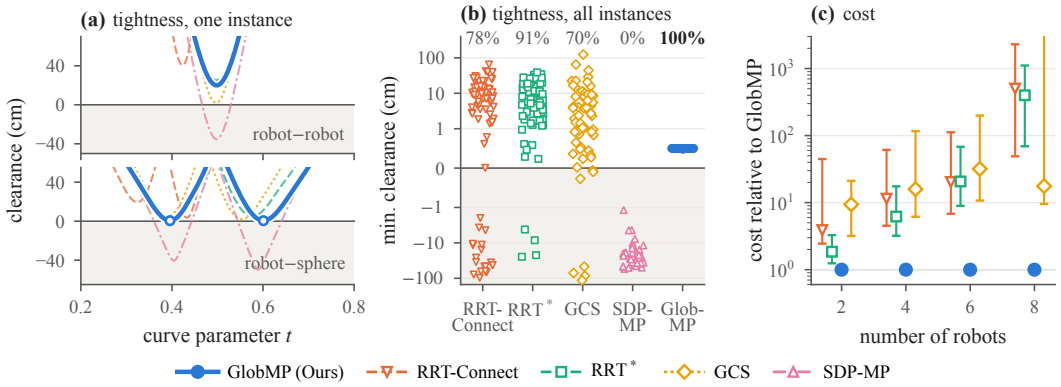

Figure 2: Tightness and cost, 20 instances each with 2, 4, 6 and 8 robots. (a) Clearance over time on one 2-robot instance, between the robots (top) and to the spheres (bottom). (b) Minimum clearance of each returned curve, with the share of runs feasible at every t. (c) Cost of feasible curves relative to GlobMP's (median and interquartile range).

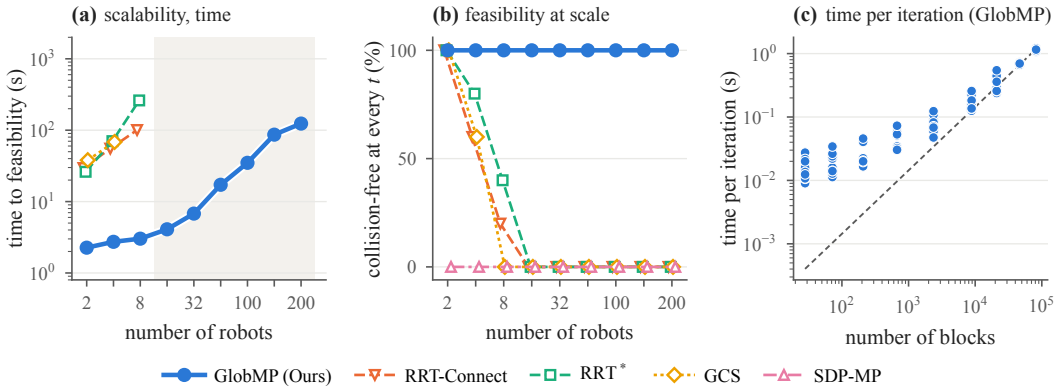

Figure 3: Scalability from 2 to 200 robots (10 runs per size for GlobMP, 5 per baseline). (a) Median CPU time to the first collision-free trajectory, setup included; where shaded, no baseline finds one. (b) Share of runs collision-free at every instant. (c) GlobMP's time per iteration against the number of blocks; dashed: linear growth.

the ends fix every derivative up to the one in the cost, the extra cost halves each time N doubles, for velocity, acceleration, jerk and snap: this is the cost of the end pieces in the discussion after Corollary 2, and the bound is tight for $k \geq 2$. With one derivative fewer fixed and $k \geq 2$, the extra cost falls much faster than the corollary's bound and is below the numerical precision by $N = 8$ for jerk, $N = 10$ for snap and $N = 20$ for acceleration, so the bound holds but is loose here. The cost also grows linearly with the safety margin around the obstacles, so the rate $(c+1)\delta_N$ after Theorem 1 fits.

4.2 Multi-robot motion planning

Multi-robot motion planning coordinates the simultaneous motion of several robots toward assigned goals while avoiding collisions with obstacles and with one another. We evaluate GlobMP on five multi-robot environments, `Empty`, `Conveyor`, `Cross-Room`, `Drop-Region` and `Corridor-Bay`, from Shaoul et al. (2025) (`Cross-Room` and `Corridor-Bay` reconstructed from Liang et al. 2025), following the test-time velocity constraint and evaluation protocol of Giaretta et al. (2026) on the Georgia Tech Robotarium (Wilson et al., 2020), with teams of $N_a \in \{3, 6, 9, 12\}$ robots. The environments are fitted to the 3.2×2.0 m Robotarium arena, with the common velocity limit selected for the Robotarium setting and inflated robot radius $r = 0.13$ m. For each (map, team-size) pair, we use ten start–goal instances: nine sampled from the benchmark

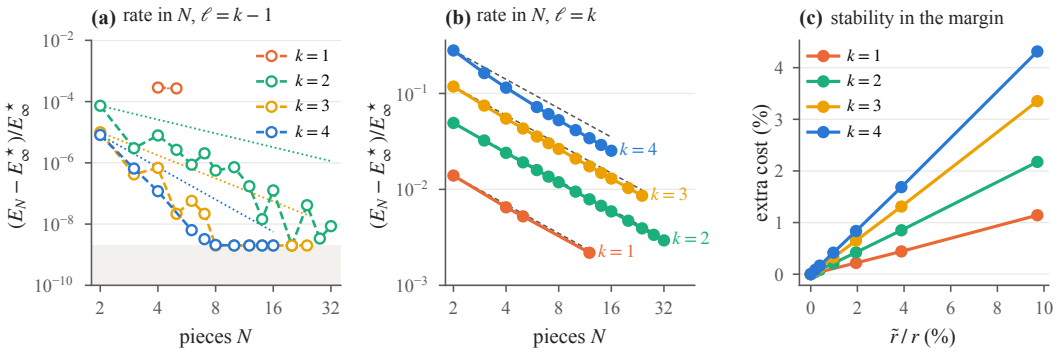

Figure 4: Relative extra cost over the limit as the number of pieces grows (one two-robot instance), for velocity to snap ($k = 1,\dots,4$). (a) One derivative fewer fixed at the ends; dotted: rates of Corollary 2; shaded: numerical precision. (b) The cost's own derivative also fixed; dashed: halving per doubling. (c) Extra cost against the margin $\tilde{r}/r$, with r the smaller sphere radius.

distribution and one "ring" instance across the workspace (`Conveyor` below $N_a = 12$: ten sampled). Hardware details of the Robotarium setup are provided in Appendix A4.1.

GlobMP needs no training: at test time it solves each instance directly as one (P), with the robots' curves stacked, a collision constraint at radius $2r$ for every pair and affine blocks for boxes (Appendix A2.1.1), so its plan is collision-free at every t. The baselines are CD, PCD-LB and PCD-SHD (Luan et al., 2026), Final Proj. and DiRecT (Giaretta et al., 2026) and MMD-CBS (Shaoul et al., 2025). Each method is evaluated on the same start–goal instances. The sampling baselines draw 64 candidates, whereas GlobMP returns one trajectory.

Figure 5 reports the success of a single returned plan across the five environments and team sizes. On `Conveyor` and `Cross-Room`, baseline performance drops rapidly as the number of robots increases: every baseline reaches zero on `Conveyor` at $N_a = 12$ and stays at or below 0.50 on `Cross-Room` from $N_a = 9$, whereas GlobMP's single plan succeeds on every instance except two of the ten on `Conveyor` at $N_a = 12$. The stronger projection baselines remain more reliable on `Empty`, `Drop-Region`, and `Corridor-Bay`, but most also degrade with team size. Figure 6 presents representative 12-robot ring trajectories on `Cross-Room` and `Conveyor`. Appendix A4.5 extends the evaluation to eight environments, teams of up to 21 robots, three velocity caps, and a larger set of start–goal instances.

4.3 Steering tactile-reactive dexterous policies

A Dexmate Vega-1 with two XHAND1 hands with fingertip force sensing (Appendix A4.1) performs four bimanual tasks learned from teleoperation (Figure 7a). The expert policies are two state-of-the-art vision-language-action models, GR00T N1.7 (NVIDIA, 2026), built on the GR00T N1 architecture (NVIDIA et al., 2025), and $\pi_{0.5}$ (Physical Intelligence et al., 2025), each with and without tactile input. Both command absolute joint targets for both arms and hands ($n = 38$) at 30 Hz in chunks of 16 executed steps.

Deployed, the policies fail geometrically (they replay memorized targets), temporally (consecutive chunks disagree by up to 0.5 rad) and at contact, mostly because of observations outside the training distribution, which no motion constraint can observe; so we separate *deciding* which constraints apply from *enforcing* them (Figure 7b). A *meta-reasoner*, the vision-language model Claude Opus 5.5 (Anthropic, 2026), defines the constraints once, at reset, from the images, the task and the measured state, for example the keep-out, keep-in and grip of Figure 7b. A *certifier* then solves (P) per chunk, with the policy's chunk as the reference $\boldsymbol{\rho}$, so the executed curve meets every active constraint at every t. Keep-out and keep-in blocks act on hand positions, which depend on the joints through forward kinematics; the certifier enforces them by linearizing the kinematics at the current curve and re-solving, so for these blocks feasibility at every t holds up to that linearization, while blocks on the joints themselves, such as the grip, are exact. The meta-reasoner only chooses constraints and never solves them, so a wrong choice changes *which* feasible curve is returned, never

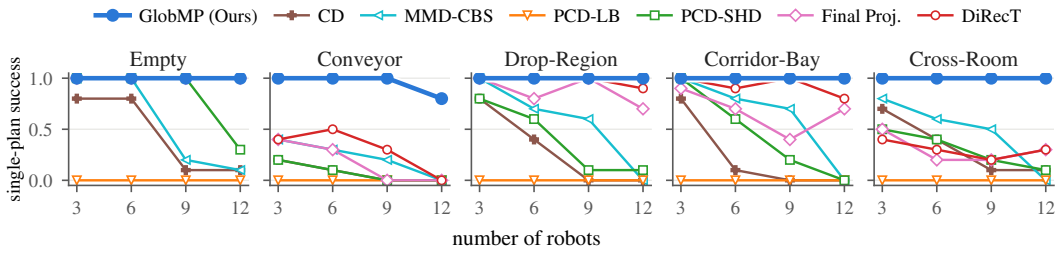

Figure 5: Success of a *single* returned plan on the Robotarium arena for $N_a \in \{3, 6, 9, 12\}$ robots, one panel per map; for GlobMP, the one plan it returns, and for a sampling baseline, the first of its 64 candidates. PCD-LB is at zero throughout: its log-barrier pushes agents into the arena boundary. MMD-CBS returns a single plan, with 60 s per instance.

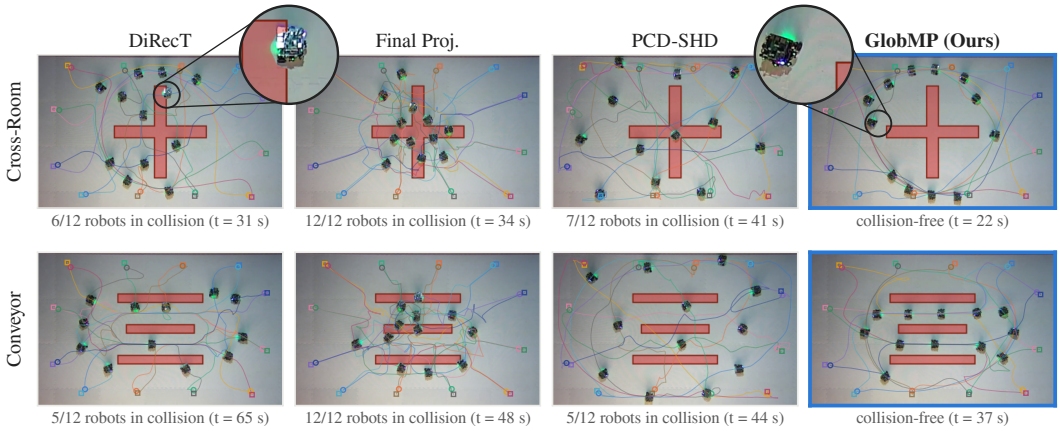

Figure 6: Executed Robotarium trajectories for the 12-robot ring instance on `Cross-Room` and `Conveyor`. Obstacle-hit counts span the whole run; circles magnify highlighted regions.

whether it is feasible, unlike model-written programs (Huang et al., 2024). If the policy slows, a demonstrated time law rescales the reference (Kant & Zucker, 1986).

The baselines are the expert policies alone, with and without touch (none with touch on the tissue), and every condition run has 9 or 10 real trials per task. Steering matches or raises every task (Figure 7c): it turns the tissue, which the policies alone rarely complete, into a mostly successful task, and completes the handover in every trial. Touch pays off where the hand must regulate contact, as in placing the toy, pulling the tissue and stowing the mug, and adds little where the task is guided by vision, as in the handover, where it even hurts the policies alone. In a closed-loop simulator replaying failed real tissue deployments, the pinch lands where and when demonstrated only with steering.

5 Conclusion

We developed GlobMP, a gradient projection method with convex feasibility restoration for continuous-time motion planning over polynomial curves; it enforces every collision constraint at every instant, with planning and steering in one program. Under mild assumptions and from a fine grid start, its cost approaches the continuous-time optimum as the number of pieces grows, within any constraint tolerance and exactly on stable instances. In real and synthetic experiments, from teams of up to 200 robots to the Robotarium testbed and tactile manipulation on a bimanual humanoid, it outperforms strong baselines. Extending the guarantee to speed limits and corridor constraints is a natural next step.

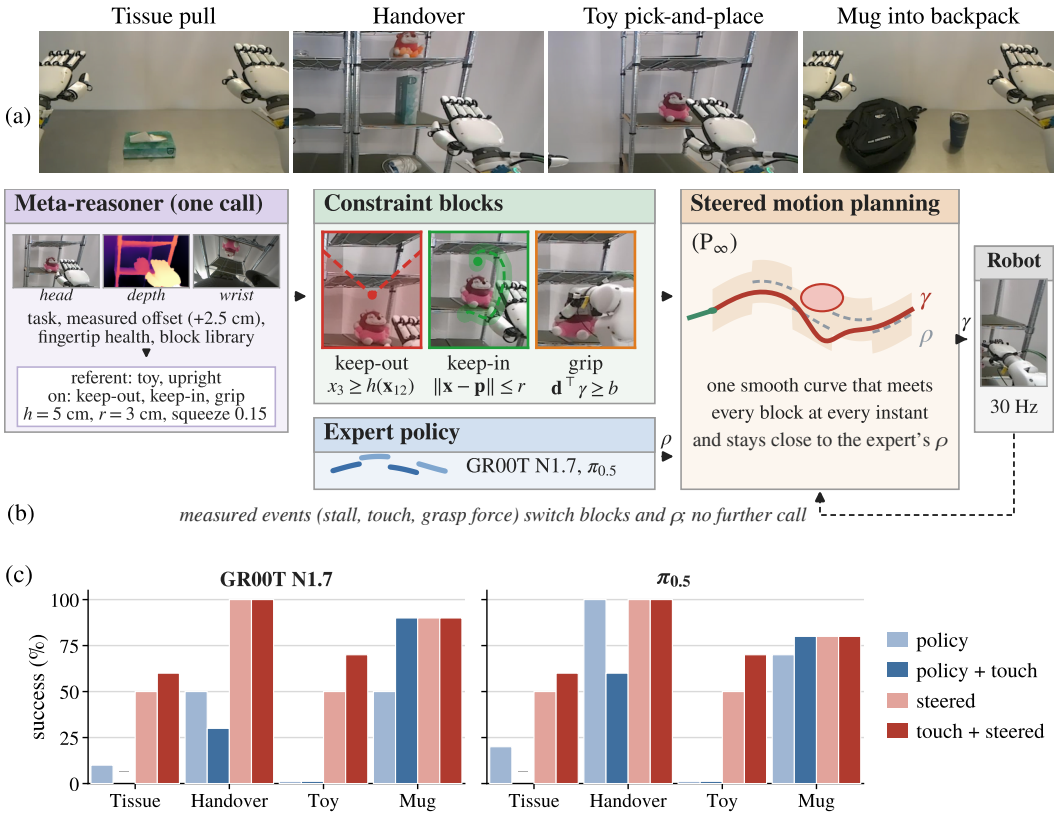

Figure 7: (a) The initial head-camera frame of each task. (b) Steered motion planning on one real toy trial: one meta-reasoner call at reset sets the active blocks and their parameters; (P) takes the policy's chunk as $\boldsymbol{\rho}$ and returns a curve that is smooth, collision-free and within $\bar{r}$ of $\boldsymbol{\rho}$ at every t; measured events then switch blocks. (c) Real-robot success of GR00T and $\pi_{0.5}$ per task (9 or 10 trials per bar): the policy without and with touch input, and steered without and with touch; dash: not run.

AI USE STATEMENT

We disclose the use of large language models (LLMs) to aid and polish the writing, editing the text for clarity and length; for retrieval and discovery, helping find related work and check citations; and to generate the synthetic datasets, whose instances are produced by LLM-assisted scripts from fixed seeds (Appendix A4.2). The authors take full responsibility for all content.

REPRODUCIBILITY STATEMENT

We will release all code: the GlobMP solver (restoration and projected gradient steps), the scripts and configurations for every experiment and figure, and the steering and deployment code for learned policies on the Vega-1 with XHAND1 hands (certifier, grounding, time law, policy servers), together with the teleoperation demonstrations and fine-tuned policy checkpoints of §4.3. The project page, https://simadidari.github.io/globmp/, collects the videos and results.

REFERENCES

Alexandre Amice, Peter Werner, and Russ Tedrake. Certifying bimanual RRT motion plans in a second. In *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 9293–9299, 2024. doi: 10.1109/ICRA57147.2024.10611296. arXiv:2310.16603.

Anthropic. Claude Opus 5.5. https://www.anthropic.com/claude-opus-5-5, 2026. Released September 22, 2026.

Jacek Bochnak, Michel Coste, and Marie-Françoise Roy. *Real Algebraic Geometry*, volume 36 of *Ergebnisse der Mathematik und ihrer Grenzgebiete*. Springer, 1998.

Jérôme Bolte, Aris Daniilidis, Adrian Lewis, and Masahiro Shiota. Clarke subgradients of stratifiable functions. *SIAM Journal on Optimization*, 18(2):556–572, 2007.

John Canny and John Reif. New lower bound techniques for robot motion planning problems. In *28th Annual Symposium on Foundations of Computer Science (FOCS)*, pp. 49–60, 1987.

João Carvalho, An T. Le, Mark Baierl, Dorothea Koert, and Jan Peters. Motion planning diffusion: Learning and planning of robot motions with diffusion models. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 1916–1923, 2023.

Ee-Chien Chang, Sung Woo Choi, DoYong Kwon, Hyungju Park, and Chee-Keng Yap. Shortest path amidst disc obstacles is computable. In *Proceedings of the 21st Annual Symposium on Computational Geometry (SoCG)*, pp. 116–125, 2005. doi: 10.1145/1064092.1064112.

Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.026. URL https://www.roboticsproceedings.org/rss19/p026.html.

Jacob K. Christopher, Stephen Baek, and Ferdinando Fioretto. Constrained synthesis with projected diffusion models. *Advances in Neural Information Processing Systems*, 37:89307–89333, 2024.

Hyungjin Chung, Jeongsol Kim, Michael T. McCann, Marc L. Klasky, and Jong Chul Ye. Diffusion posterior sampling for general noisy inverse problems. In *International Conference on Learning Representations*, 2023.

Robin Deits and Russ Tedrake. Computing large convex regions of obstacle-free space through semidefinite programming. In *Algorithmic Foundations of Robotics XI*, volume 107 of *Springer Tracts in Advanced Robotics*, pp. 109–124, 2015a. doi: 10.1007/978-3-319-16595-0_7.

Robin Deits and Russ Tedrake. Efficient mixed-integer planning for UAVs in cluttered environments. In *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 42–49, 2015b. doi: 10.1109/ICRA.2015.7138978.

Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems*, volume 34, pp. 8780–8794, 2021.

Shiying Dong, Zhipeng Shen, Rudolf Reiter, Hailong Huang, Bingzhao Gao, Hong Chen, and Wen-Hua Chen. A fast semidefinite convex relaxation for optimal control problems with spatio-temporal constraints, 2026. arXiv:2601.03055.

Justin Fu, Aviral Kumar, Ofir Nachum, George Tucker, and Sergey Levine. D4RL: Datasets for deep data-driven reinforcement learning, 2020. arXiv:2004.07219.

Fei Gao, William Wu, Yi Lin, and Shaojie Shen. Online safe trajectory generation for quadrotors using fast marching method and Bernstein basis polynomial. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 344–351, 2018.

Xiao Gao, Kunpeng Yao, Farshad Khadivar, and Aude Billard. Enhancing dexterity in confined spaces: Real-time motion planning for multifingered in-hand manipulation. *IEEE Robotics & Automation Magazine*, 31(4):100–112, 2024. doi: 10.1109/MRA.2024.3433110. arXiv:2309.06955.

Paolo Giaretta, Zeyang Li, and Navid Azizan. DiRecT: Safe diffusion-based planning via receding-horizon denoising, 2026. arXiv:2606.15359.

Bernhard Paus Graesdal, Shao Yuan Chew Chia, Tobia Marcucci, Savva Morozov, Alexandre Amice, Pablo A. Parrilo, and Russ Tedrake. Towards tight convex relaxations for contact-rich manipulation. In *Robotics: Science and Systems (RSS)*, 2024.

Bernhard Paus Graesdal, Alexandre Amice, Pablo A. Parrilo, and Russ Tedrake. Semidefinite relaxations for collision-free motion planning, 2026. arXiv:2606.14063.

C. R. Hargraves and S. W. Paris. Direct trajectory optimization using nonlinear programming and collocation. *Journal of Guidance, Control, and Dynamics*, 10(4):338–342, 1987. doi: 10.2514/3.20223.

Hannah Häring, Dennis Gramlich, Christian Ebenbauer, and Carsten W. Scherer. Trajectory generation for the unicycle model using semidefinite relaxations. *European Journal of Control*, 80:101063, 2024. doi: 10.1016/j.ejcon.2024.101063.

Zhilin He, Yorai Shaoul, and Jiaoyang Li. Model-based diffusion optimal control for multi-robot motion planning, 2026. arXiv:2607.12423.

Wenlong Huang, Chen Wang, Yunzhu Li, Ruohan Zhang, and Li Fei-Fei. ReKep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation. In *Proceedings of the 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pp. 4573–4602, 2024.

Michael Janner, Yilun Du, Joshua B. Tenenbaum, and Sergey Levine. Planning with diffusion for flexible behavior synthesis. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 9902–9915, 2022.

Xiaogang Jia, Denis Blessing, Xinkai Jiang, Moritz Reuss, Atalay Donat, Rudolf Lioutikov, and Gerhard Neumann. Towards diverse behaviors: A benchmark for imitation learning with human demonstrations. In *International Conference on Learning Representations (ICLR)*, 2024.

Shucheng Kang, Guorui Liu, and Heng Yang. Global contact-rich planning with sparsity-rich semidefinite relaxations. In *Robotics: Science and Systems (RSS)*, 2025. doi: 10.15607/RSS.2025.XXI.046. arXiv:2502.02829.

Kamal Kant and Steven W. Zucker. Toward efficient trajectory planning: The path-velocity decomposition. *The International Journal of Robotics Research*, 5(3):72–89, 1986.

Sertac Karaman and Emilio Frazzoli. Sampling-based algorithms for optimal motion planning. *The International Journal of Robotics Research*, 30(7):846–894, 2011.

James J. Kuffner and Steven M. LaValle. RRT-Connect: An efficient approach to single-query path planning. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, volume 2, pp. 995–1001, 2000. doi: 10.1109/ROBOT.2000.844730.

Jean-Bernard Lasserre. Global optimization with polynomials and the problem of moments. *SIAM Journal on Optimization*, 11(3):796–817, 2001.

Steven M. LaValle. Rapidly-exploring random trees: A new tool for path planning. Technical Report TR 98-11, Computer Science Dept., Iowa State University, 1998.

Jiaoyang Li, Wheeler Ruml, and Sven Koenig. EECBS: A bounded-suboptimal search for multi-agent path finding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pp. 12353–12362, 2021.

Jinhao Liang, Jacob K. Christopher, Sven Koenig, and Ferdinando Fioretto. Simultaneous multi-robot motion planning with projected diffusion models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 37162–37180, 2025.

Francesco Locatello, Michael Tschannen, Gunnar Rätsch, and Martin Jaggi. Greedy algorithms for cone constrained optimization with convergence guarantees. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pp. 773–784, 2017.

George G. Lorentz. *Bernstein Polynomials*. Chelsea Publishing, New York, 2nd edition, 1986.

Hao Luan, Yi Xian Goh, See-Kiong Ng, and Chun Kai Ling. Projected coupled diffusion for test-time constrained joint generation. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=1FEm5JLpvg.

Ishaan Mahajan, Jon Arrizabalaga, Andrea Grillo, Fausto Vega, James Anderson, Zachary Manchester, and Brian Plancher. TinySDP: Real time semidefinite optimization for certifiable and agile edge robotics, 2026. arXiv:2605.13748.

Tobia Marcucci, Mark Petersen, David von Wrangel, and Russ Tedrake. Motion planning around obstacles with convex optimization. *Science Robotics*, 8(84):eadf7843, 2023.

Tobia Marcucci, Jack Umenberger, Pablo A. Parrilo, and Russ Tedrake. Shortest paths in graphs of convex sets. *SIAM Journal on Optimization*, 34(1):507–532, 2024. doi: 10.1137/22M1523790.

Tobia Marcucci, Mathew Halm, Will Yang, Dongchan Lee, and Andrew D. Marchese. A biconvex method for minimum-time motion planning through sequences of convex sets. In *Robotics: Science and Systems (RSS)*, 2025. arXiv:2504.18978.

Daniel Mellinger and Vijay Kumar. Minimum snap trajectory generation and control for quadrotors. In *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2520–2525, 2011.

Joseph S. B. Mitchell. Shortest paths and networks. In Jacob E. Goodman, Joseph O'Rourke, and Csaba D. Tóth (eds.), *Handbook of Discrete and Computational Geometry*, chapter 31. CRC Press, 3rd edition, 2017. doi: 10.1201/9781315119601.

Jean-Jacques Moreau. Décomposition orthogonale d'un espace hilbertien selon deux cônes mutuellement polaires. *Comptes Rendus de l'Académie des Sciences de Paris*, 255:238–240, 1962.

NVIDIA. Isaac GR00T N1.7. https://github.com/NVIDIA/Isaac-GR00T, 2026. Model release.

NVIDIA, Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, et al. GR00T N1: An open foundation model for generalist humanoid robots, 2025. URL https://arxiv.org/abs/2503.14734. arXiv:2503.14734.

Alan Papalia, Andrew Fishberg, Brendan W. O'Neill, Jonathan P. How, David M. Rosen, and John J. Leonard. Certifiably correct range-aided SLAM. *IEEE Transactions on Robotics*, 40:4265–4283, 2024.

Pablo A. Parrilo. Semidefinite programming relaxations for semialgebraic problems. *Mathematical Programming, Series B*, 96:293–320, 2003. doi: 10.1007/s10107-003-0387-5.

Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. In *Conference on Robot Learning (CoRL)*, volume 305 of *Proceedings of Machine Learning Research*, pp. 17–40, 2025. arXiv:2504.16054.

Victoria Powers and Bruce Reznick. Polynomials that are positive on an interval. *Transactions of the American Mathematical Society*, 352(10):4677–4692, 2000.

Charles Richter, Adam Bry, and Nicholas Roy. Polynomial trajectory planning for aggressive quadrotor flight in dense indoor environments. In *Robotics Research: The 16th International Symposium ISRR*, volume 114 of *Springer Tracts in Advanced Robotics*, pp. 649–666, 2016.

Stephen M. Robinson. Stability theory for systems of inequalities, part II: Differentiable nonlinear systems. *SIAM Journal on Numerical Analysis*, 13(4):497–513, 1976. doi: 10.1137/0713043.

R. Tyrrell Rockafellar and Roger J.-B. Wets. *Variational Analysis*. Springer, 1998.

David M. Rosen, Luca Carlone, Afonso S. Bandeira, and John J. Leonard. SE-Sync: A certifiably correct algorithm for synchronization over the special Euclidean group. *The International Journal of Robotics Research*, 38(2-3):95–125, 2019.

John Schulman, Yan Duan, Jonathan Ho, Alex Lee, Ibrahim Awwal, Henry Bradlow, Jia Pan, Sachin Patil, Ken Goldberg, and Pieter Abbeel. Motion planning with sequential convex optimization and convex collision checking. *The International Journal of Robotics Research*, 33(9):1251–1270, 2014.

Yorai Shaoul, Itamar Mishani, Shivam Vats, Jiaoyang Li, and Maxim Likhachev. Multi-robot motion planning with diffusion models. In *International Conference on Learning Representations (ICLR)*, 2025.

Naum Z. Shor. Quadratic optimization problems. *Soviet Journal of Computer and Systems Sciences*, 25:1–11, 1987.

Kiril Solovey, Lucas Janson, Edward Schmerling, Emilio Frazzoli, and Marco Pavone. Revisiting the asymptotic optimality of RRT*. In *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2189–2195, 2020. arXiv:1909.09688.

Josef Stoer and Roland Bulirsch. *Introduction to Numerical Analysis*, volume 12 of *Texts in Applied Mathematics*. Springer, 3rd edition, 2002.

Balakumar Sundaralingam and Tucker Hermans. Relaxed-rigidity constraints: kinematic trajectory optimization and collision avoidance for in-grasp manipulation. *Autonomous Robots*, 43(2):469–483, 2019. doi: 10.1007/s10514-018-9772-z. arXiv:1806.00942.

Jingtao Tang, Zining Mao, Lufan Yang, and Hang Ma. Space-time graphs of convex sets for multi-robot motion planning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 8683–8690, 2025. doi: 10.1109/IROS60139.2025.11246429.

Russ Tedrake and the Drake Development Team. Drake: Model-based design and verification for robotics, 2019. URL https://drake.mit.edu.

Sangli Teng, Ashkan Jasour, Ram Vasudevan, and Maani Ghaffari. Convex geometric motion planning on Lie groups via moment relaxation. In *Robotics: Science and Systems XIX*, 2023. doi: 10.15607/RSS.2023.XIX.058.

Kevin Tracy, Taylor A. Howell, and Zachary Manchester. Differentiable collision detection for a set of convex primitives. In *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3663–3670, 2023. doi: 10.1109/ICRA48891.2023.10160716.

Fausto Vega, Jon Arrizabalaga, Ryan Watson, and Zachary Manchester. Convex maneuver planning for spacecraft collision avoidance. In *International Conference on Space Robotics (iSpaRo)*, pp. 626–633, 2025. doi: 10.1109/iSpaRo66239.2025.11437385.

David von Wrangel and Russ Tedrake. Using graphs of convex sets to guide nonconvex trajectory optimization. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 9863–9870, 2024. doi: 10.1109/IROS58592.2024.10802426.

Andreas Wächter and Lorenz T. Biegler. On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming. *Mathematical Programming*, 106(1): 25–57, 2006.

Zhongqi Wei and Frederike Dümbgen. Global sampling-based trajectory optimization for contact-rich manipulation via KernelSOS, 2026. arXiv:2604.27175.

Peter Werner, Richard Cheng, Tom Stewart, Russ Tedrake, and Daniela Rus. Superfast configuration-space convex set computation on GPUs for online motion planning. In *Robotics: Science and Systems (RSS)*, 2025. doi: 10.15607/RSS.2025.XXI.045.

Peter Werner, Thomas Cohn, Rebecca H. Jiang, Tim Seyde, Max Simchowitz, Russ Tedrake, and Daniela Rus. Faster algorithms for growing collision-free convex polytopes in robot configuration space. *The International Journal of Robotics Research*, 2026. doi: 10.1177/02783649261436917.

Sean Wilson, Paul Glotfelter, Li Wang, Siddharth Mayya, Gennaro Notomista, Mark Mote, and Magnus Egerstedt. The Robotarium: Globally impactful opportunities, challenges, and lessons learned in remote-access, distributed control of multirobot systems. *IEEE Control Systems Magazine*, 40(1):26–44, 2020.

Wei Xiao, Tsun-Hsuan Wang, Chuang Gan, Ramin Hasani, Mathias Lechner, and Daniela Rus. SafeDiffuser: Safe planning with diffusion probabilistic models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=ig2wk7kK9J.

Heng Yang, Jingnan Shi, and Luca Carlone. TEASER: Fast and certifiable point cloud registration. *IEEE Transactions on Robotics*, 37(2):314–333, 2021. doi: 10.1109/TRO.2020.3033695.

Lujie Yang, Tobia Marcucci, Pablo A. Parrilo, and Russ Tedrake. A new semidefinite relaxation for linear and piecewise-affine optimal control with time scaling. In *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 9228–9235, 2025. doi: 10.1109/ICRA55743.2025.11128850.

Jichen Zhang, Liqun Zhao, Antonis Papachristodoulou, and Jack Umenberger. Constrained diffusers for safe planning and control. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, pp. 39283–39316, 2025.

Matt Zucker, Nathan Ratliff, Anca D. Dragan, Mihail Pivtoraiko, Matthew Klingensmith, Christopher M. Dellin, J. Andrew Bagnell, and Siddhartha S. Srinivasa. CHOMP: Covariant Hamiltonian optimization for motion planning. *The International Journal of Robotics Research*, 32(9-10):1164–1193, 2013. doi: 10.1177/0278364913488805.

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A1 Addendum to Section 1: related work and Figure 1

A1.1 Extended related work

Relation to existing formulations. Collocation replaces each block by inequalities at sampled times (Hargraves & Paris, 1987); the control-polygon rule asks every coefficient to be nonnegative (Gao et al., 2018), which is sufficient for $\mathscr{K}_j$ but conservative; convex decompositions (Deits &

Tedrake, 2015a; Marcucci et al., 2023) are exact for the decomposition rather than for the free space; semidefinite relaxations (Lasserre, 2001; Parrilo, 2003; Graesdal et al., 2026) certify a trajectory only when tight. We keep the original variables and free space, test the exact condition on each block, and approach the optimum of (P_∞) as the class is refined, within any constraint tolerance and exactly on stable instances (Theorem 1); under a condition on configurations (Assumption C) we also reach global minimizers over each fine enough spline class (Proposition 13).

Sampling-based planning and trajectory optimization. The two classical families both act on the nonconvex problem itself, and neither certifies global optimality. Sampling-based planners (LaValle, 1998; Karaman & Frazzoli, 2011) are probabilistically complete and, for RRT*, asymptotically optimal, but in probability and only to the robust optimum (Solovey et al., 2020), the least cost approached with a positive clearance, which can exceed the optimum (Appendix A3.3). Trajectory optimizers instead discretize and hand the result to local nonlinear programming: direct collocation (Hargraves & Paris, 1987), covariant gradient descent (Zucker et al., 2013), sequential convexification (Schulman et al., 2014), and differentiable collision geometry (Tracy et al., 2023); this pipeline is fast and widely deployed but converges to a stationary point with no test for whether it is global. We keep this setting and add a proof of global convergence.

Convex decomposition and graphs of convex sets. A second line makes the problem tractable by convexifying the *domain*. IRIS inflates large obstacle-free convex regions (Deits & Tedrake, 2015a), first used for mixed-integer planning (Deits & Tedrake, 2015b); shortest-paths-in-graphs-of-convex-sets then plans over that decomposition as a mixed-integer convex program, solved through its convex relaxation and rounding (Marcucci et al., 2024; 2023), with subsequent work on scalability (Marcucci et al., 2025; Werner et al., 2026; 2025), on space-time variants for multiple robots (Tang et al., 2025), and on using the convex solution to warm-start nonconvex trajectory optimization (von Wrangel & Tedrake, 2024). These methods bound their optimality gap *for the decomposition*: the regions are a modelling choice made before optimizing, and a curve is admissible only if it respects them. Our guarantee is stated on the original free space, with no decomposition.

Learning-based planners. Diffusion models have recently been trained to sample trajectories directly, for single (Carvalho et al., 2023) and multiple (Shaoul et al., 2025) robots, with constraints imposed during sampling by projection (Luan et al., 2026) or by receding-horizon denoising (Giaretta et al., 2026). These amortize planning into fast learned sampling and represent several routes, but they enforce feasibility only approximately at sampling time and produce no optimality certificate. We compare against them in §4; our contribution is a guarantee rather than a speedup, and the two can be combined (Appendix A2.2).

Convex relaxations of planning and control. Beyond Graesdal et al. (2026), several works apply semidefinite or moment-hierarchy relaxations (Shor, 1987; Lasserre, 2001; Parrilo, 2003) to planning and control problems and report tight solutions empirically, without analysing when or why tightness holds. Teng et al. (2023) formulate kinodynamic motion planning on Lie groups as polynomial optimization and apply the moment hierarchy; Häring et al. (2024) study minimum-energy unicycle control with a semidefinite relaxation; Vega et al. (2025) apply the Shor relaxation to spacecraft manoeuvre planning with an ellipsoidal, probability-of-collision keep-out constraint; Mahajan et al. (2026) use it for real-time predictive control around ball-shaped obstacles on embedded hardware. Control problems with free timing (Yang et al., 2025; Dong et al., 2026) and contact-rich planning (Graesdal et al., 2024; Kang et al., 2025) have been relaxed in the same way, and a sum-of-squares surrogate guides contact-rich search (Wei & Dümbgen, 2026). Outside planning, such relaxations underlie certifiable estimation, for example registration and SLAM (Yang et al., 2021; Rosen et al., 2019; Papalia et al., 2024), which we do not review. In contrast to the planning and control work above, which observes tightness empirically, we prove that a feasible descent method on the original nonconvex program, started from a fine grid, approaches the optimal cost of the original program as the class is refined, within any constraint tolerance (Theorem 1).

A1.2 Inexact sampled checks and the data for Figure 1

Sampling and control-polygon tests are not exact. *Sampling* t at a finite set is a strict *relaxation*. For any finite $\mathscr{T} = \{t_1 < \cdots < t_M\} \subset [0,1]$, choosing $a < b$ inside any gap between consecutive samples makes $g(t) = (t-a)(t-b)$ (of degree $2 \le 2d$) nonnegative at every sample and negative in between, with $\min_{[0,1]} g = -\left(\frac{b-a}{2}\right)^2$. So the sampled feasible set is in general a proper superset

of $\mathscr{F}$ at every M and every degree; no finite sample is ever exact. Structurally, each $\mathbf{b}_{2d}(t_0)$ is an extreme ray of the dual cone $\mathscr{K}_{2d}^*$ (12) of the cone (1d) of degree-$2d$ nonnegative polynomials (indexed by degree, Appendix A3.1.2): if $\mathbf{b}_{2d}(t_0) = \mathbf{x} + \mathbf{y}$ with $\mathbf{x}, \mathbf{y} \in \mathscr{K}_{2d}^*$, pairing with $(t - t_0)^2$ forces $\mathbf{x}$'s representing measure onto $\{t_0\}$, so a continuum of extreme rays would be replaced by finitely many. The direction of the error is the unsafe one for a planner: the sampled problem gives a valid *lower* bound on the optimal cost of (P) when solved globally, but the curve it returns may penetrate the obstacle, and the margin violation is only $\Theta(1/M^2)$: on a uniform grid a gap of width $\tau = 1/(M-1)$ admits violations approaching $\min_{[0,1]} g = -\tau^2/4$ above.

Requiring all Bernstein coefficients to be nonnegative is a strict *restriction*, and degree elevation does not close it. If $g \geq 0$ vanishes at some $t^\star \in (0,1)$, then since every entry of $\mathbf{b}_D(t^\star)$ is strictly positive for every degree D, nonnegative coefficients $\boldsymbol{\zeta}$ of g would force $\boldsymbol{\zeta}^\top \mathbf{b}_D(t^\star) = 0$ with all terms nonnegative, hence $\boldsymbol{\zeta} = \mathbf{0}$ and $g \equiv 0$: no elevation degree ever works for a polynomial with an interior zero. This case arises here, because whenever a minimizer of (P) touches an obstacle at an interior time, its avoidance polynomial has an interior double root. The smallest Bernstein-D coefficient of such a polynomial stays negative and decays only like $-t^\star(1-t^\star)g''(t^\star)/(2D)$: for $g(t) = (t - \frac{1}{2})^2$ it is exactly $-1/\big(4(D-1)\big)$ for even D and $-1/(4D)$ for odd D, so no elevation is ever enough. Approaching tangency the required degree likewise diverges: for $g_\varepsilon(t) = (t - \frac{1}{2})^2 + \varepsilon$, so that $\min_{[0,1]} g_\varepsilon = \varepsilon$, the least D with all Bernstein-D coefficients nonnegative satisfies $D_{\min} = 1/(4\varepsilon) + O(1)$; in workspace terms $D_{\min} \approx \|\boldsymbol{\gamma}'\|^2 t^\star(1-t^\star)/(2r_j\delta)$ for clearance δ. The avoidance condition is therefore carried exactly, as written in (1d).

Data for Figure 1. The figure shows two robots, A (left to right) and B (bottom to top), planned jointly with one degree-6 Bernstein segment each; each robot is a moving centre for the other (§1). The obstacles are three ellipses and a circle, each enlarged by the robot radius 0.055 so that the constraint on the robot's centre keeps the whole robot clear, and the robots keep their centres at least 0.11 apart. The cost is the smoothness objective of (P_∞) with $k = 2$ (minimum acceleration), positions fixed at both ends ($\ell = 0$), and no steering constraint. Panel (2) imposes the constraints only at 8 sample times; panel (3) runs GlobMP with every constraint on all of $[0,1]$. The instance has eight route classes (A passes each of the two coloured ellipses on either side, and either robot crosses first); one opening curve per class, restored and then refined by Algorithm 1, gives feasible KKT curves in five of them, with costs 1, 1.2, 5.3, 5.6 and 51 times the cheapest; panel (3) shows the cheapest and, in grey, the routes that take a different path in space.

A2 Addendum to Section 2: Formulation and Algorithm

A2.1 Variants of the formulation

The experiments use four variants of the blocks of (P_∞). Each is convex in the position or has a convex feasible set imposed exactly, so Proposition 10 covers all four. Theorem 1 covers the position blocks (ellipsoids, robot pairs, boxes and walls); the speed limit constrains $\boldsymbol{\gamma}'$ and lies outside it.

Ellipsoidal obstacles. A weighted norm $\|\mathbf{x}\|_{\boldsymbol{\Sigma}} = (\mathbf{x}^\top \boldsymbol{\Sigma} \mathbf{x})^{1/2}$, $\boldsymbol{\Sigma} \succeq 0$, in the collision constraint, $\|\boldsymbol{\gamma}(t) - \mathbf{c}_j(t)\|_{\boldsymbol{\Sigma}_j} \geq r_j$, keeps every block quadratic; with $\boldsymbol{\Sigma}_j \succ 0$ it describes an ellipsoid.

Several robots. Several robots are planned jointly by stacking their curves into one $\boldsymbol{\gamma}$; let $\mathbf{E}_a$ select robot a's rows, so $\boldsymbol{\gamma}_a = \mathbf{E}_a \boldsymbol{\gamma}$. For a robot of radius r_a, a sphere of radius r and centre $\mathbf{m}_j$ is $\boldsymbol{\Sigma}_j = \mathbf{E}_a^\top \mathbf{E}_a$, $\mathbf{E}_a \mathbf{c}_j = \mathbf{m}_j$, $r_j = r + r_a$; two robots a, b are kept apart by $\boldsymbol{\Sigma}_j = \mathbf{E}_{ab}^\top \mathbf{E}_{ab}$ with $\mathbf{E}_{ab} = \mathbf{E}_a - \mathbf{E}_b$ and $r_j = r_a + r_b$, so that $\|\boldsymbol{\gamma}\|_{\boldsymbol{\Sigma}_j} = \|\boldsymbol{\gamma}_a - \boldsymbol{\gamma}_b\|$. The multi-robot experiments of §A4.5 use this separation.

Workspace box and speed limit. Box constraints use $x_i - \underline{x}_i \geq 0$ and $\bar{x}_i - x_i \geq 0$ (Appendix A4.6); a speed limit uses $(T_{\mathrm{dur}} v_{\max})^2 - \|\boldsymbol{\gamma}'(t)\|^2 \geq 0$, where T_{dur} is the trajectory duration, so the physical velocity is $\boldsymbol{\gamma}'/T_{\mathrm{dur}}$. Both are affine or concave quadratic, so their feasible sets are convex.

A2.1.1 HALF-SPACE AND CORRIDOR CONSTRAINTS

Confining the curve to one side of a wall (a workspace bound, or a face of a corridor) is the constraint $\mathbf{a}_w^\top \boldsymbol{\gamma}(t) - \xi_w \geq 0$ for all t. The polynomial $\mathbf{a}_w^\top \boldsymbol{\gamma}(t) - \xi_w$ has degree d, so its Bernstein coefficient vector $\boldsymbol{\Gamma}^\top \mathbf{a}_w - \xi_w \mathbf{u}$ ($\mathbf{u}$ the vector of ones) is raised to degree $2d$ by the elevation map $\mathbf{W} \in \mathbb{R}^{(2d+1)\times(d+1)}$, $W_{si} = \binom{d}{i}\binom{d}{s-i}/\binom{2d}{s}$:

$$\boldsymbol{\pi}_w(\boldsymbol{\Gamma}) \;:=\; \mathbf{W}\big(\boldsymbol{\Gamma}^\top \mathbf{a}_w - \xi_w \mathbf{u}\big) \;\in\; \mathscr{K}_{2d}. \tag{6}$$

Elevation rewrites one polynomial in a higher basis and changes nothing else, so (6) holds exactly when the wall is respected at every t, and the block enters the same cone $\mathscr{K}_{2d}$ as the collision blocks. It is affine in $\boldsymbol{\Gamma}$, so its linearization is exact. An obstacle block is a squared length and a wall block a length, so the implementation divides each obstacle block by r_j^2 and each wall block by the workspace scale; since $\mathscr{K}$ is a cone, this changes the numerics but not the feasible set.

A2.1.2 REFERENCE MOTIONS FROM LEARNED POLICIES

The reference $\boldsymbol{\rho}$ of (P_∞) is a motion supplied by a learned planner or policy (one fixed proposal per solve), a demonstration or a classical planner. For positions $\mathbf{y}_0, \ldots, \mathbf{y}_L \in \mathbb{R}^n$ at normalized times $0 = t_0 < \cdots < t_L = 1$, we use the piecewise-linear interpolant

$$\boldsymbol{\rho}(t) = \frac{t_{i+1} - t}{t_{i+1} - t_i}\,\mathbf{y}_i + \frac{t - t_i}{t_{i+1} - t_i}\,\mathbf{y}_{i+1}, \qquad t \in [t_i, t_{i+1}], \quad i = 0, \ldots, L-1. \tag{7}$$

It need not have the derivatives required of $\boldsymbol{\gamma}$, and it enters only the steering constraint, so the cost is unchanged. Relative action commands are first composed from the current configuration into absolute waypoints.

A2.2 STEERING A GENERATIVE PLANNER

Learned planners propose motion as waypoints. Diffusion and flow-matching planners (Janner et al., 2022; Carvalho et al., 2023; Shaoul et al., 2025; Giaretta et al., 2026) denoise an array $\mathbf{y} = (\mathbf{y}_0, \ldots, \mathbf{y}_L)$ of positions at fixed times, one model per map, with start and goal written into the array at every step. Constraints enter either as gradient guidance, which does not enforce them, or as a projection of the waypoints, which enforces them at the $L+1$ sampled times only; Section 4.1 measures what happens between samples. We use (P) as that projection, so the object the sampler carries is a polynomial trajectory that satisfies every block at every t, and the generator is left untouched.

Given waypoints $\mathbf{y}$, the reference $\boldsymbol{\rho}$ is their piecewise-linear interpolant (7), and the steering constraint of (P_∞), $\|\boldsymbol{\gamma}(t) - \boldsymbol{\rho}(t)\| \leq \bar{r}$ for every t, keeps the curve near the proposal. Nothing else changes: Algorithms 1–2 run as in §2.

A3 ADDENDUM TO SECTION 3: THEORY AND PROOFS

Appendix A3.1 sets up the chart, the matrices and the compactness facts used throughout. Appendix A3.2 proves Theorem 1 and, under Assumption C stated there, its counterpart at a fixed number of pieces (Proposition 13); Appendix A3.3 discusses the assumptions and the stability of the optimal value.

A3.1 PRELIMINARIES

A3.1.1 BOUNDARY CHART

Lemma 3. *If $\mathbf{A}$ has full column rank, then $\boldsymbol{\Gamma}_{\mathrm{aff}} = \mathbf{B}(\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$ belongs to $\mathscr{A}_{\mathrm{bc}}$. If $\mathbf{N}$ is an orthonormal basis for $\ker \mathbf{A}^\top$, then $\mathbf{Z} \longmapsto \boldsymbol{\Gamma}_{\mathrm{aff}} + \mathbf{Z}\mathbf{N}^\top$ ($\mathbf{Z} = \boldsymbol{\Gamma}_{\mathrm{free}}$) is an affine isometry from $\mathbb{R}^{n\times d_f}$ onto $\mathscr{A}_{\mathrm{bc}}$, with inverse $\boldsymbol{\Gamma} \longmapsto (\boldsymbol{\Gamma} - \boldsymbol{\Gamma}_{\mathrm{aff}})\mathbf{N}$.*

Proof of Lemma 3. Full column rank makes $\mathbf{A}^\top\mathbf{A}$ invertible, and multiplying the displayed anchor by $\mathbf{A}$ gives $\mathbf{B}$. Any difference of two solutions has rows in $\ker \mathbf{A}^\top$ and thus uniquely equals

$\mathbf{Z}\mathbf{N}^\top$. The inverse formula follows from $\mathbf{N}^\top\mathbf{N} = \mathbf{I}$. Finally, with $\|\cdot\|_F$ the Frobenius norm, $\|\mathbf{Z}\mathbf{N}^\top\|_F^2 = \mathrm{tr}(\mathbf{Z}^\top\mathbf{Z}) = \|\mathbf{Z}\|_F^2$. The map and its inverse are continuous, so relative local minima and set membership transfer between the two coordinates. □

A3.1.2 Matrices, cone and bounds

Polynomial cost, boundary, and constraint matrices. The costs we use are all of the quadratic form

$$\mathrm{cost}(\mathbf{\Gamma}) = \mathrm{tr}(\mathbf{Q}\mathbf{\Gamma}^\top\mathbf{\Gamma}) - 2\,\mathrm{tr}(\mathbf{L}^\top\mathbf{\Gamma}) + c_0, \tag{8}$$

with $\mathbf{Q}$ symmetric. The main-text cost is the case $\mathbf{L} = 0$, $c_0 = 0$; the linear and constant terms cover tracking costs, and the gradient is $2(\mathbf{\Gamma}\mathbf{Q} - \mathbf{L})\mathbf{N}$; the constraint blocks do not depend on the cost. For reference fitting, define $\mathbf{G}_d = \int_0^1 \mathbf{b}_d(t)\mathbf{b}_d(t)^\top dt$ and $\mathbf{R}_d = \int_0^1 \boldsymbol{\rho}(t)\mathbf{b}_d(t)^\top dt$. Then (8) uses $\mathbf{Q} = \frac{1}{2}\mathbf{G}_d$, $\mathbf{L} = \frac{1}{2}\mathbf{R}_d$, and $c_0 = \frac{1}{2}\int_0^1 \|\boldsymbol{\rho}(t)\|^2 dt$. The reference need only be square-integrable. For the integrated squared-derivative objective in (P_∞), let $\mathbf{D}_k$ be the matrix with $\mathbf{b}_d^{(k)} = \mathbf{D}_k\mathbf{b}_{d-k}$ and use $\mathbf{Q} = \mathbf{G}^{(k)} := \mathbf{D}_k\mathbf{G}_{d-k}\mathbf{D}_k^\top$, $\mathbf{L} = 0$, $c_0 = 0$. The endpoint evaluation matrix is

$$\mathbf{\Gamma}\mathbf{A} = \mathbf{B}, \qquad \mathbf{A} = [\,\mathbf{b}_d(0)\ \mathbf{b}_d(1)\ \cdots\ \mathbf{b}_d^{(\ell)}(0)\ \mathbf{b}_d^{(\ell)}(1)\,] \in \mathbb{R}^{(d+1)\times 2(\ell+1)}. \tag{9}$$

The columns of $\mathbf{B}$ give the prescribed values in the same order.

For the signed quadratic derivative blocks used in the examples, let $0 \le i_j \le k$, $\mathbf{\Sigma}_j \succeq 0$, $\bar{d}_j = d - i_j$, and $\mathbf{D}_0 = \mathbf{I}$. Then

$$\mathbf{M}_j(\mathbf{\Gamma}) = \sigma_j\big(\mathbf{\Gamma}_j^\top\mathbf{\Sigma}_j\mathbf{\Gamma}_j - r_j^2\mathbf{u}\mathbf{u}^\top\big), \qquad \mathbf{\Gamma}_j = \mathbf{\Gamma}\mathbf{D}_{i_j} - \mathbf{C}_j, \quad \sigma_j \in \{-1,+1\}, \tag{10}$$

where $\mathbf{u}$ is the vector of ones and the columns of $\mathbf{C}_j$ are the Bernstein coefficients of the centre $\mathbf{c}_j$. Let S_e denote the linear map that sends a symmetric $(e+1)\times(e+1)$ matrix $\mathbf{M}$ to the degree-$2e$ Bernstein coefficients of $\mathbf{b}_e(t)^\top\mathbf{M}\mathbf{b}_e(t)$. Then

$$g_j^{\mathbf{\Gamma}}(t) = \mathbf{b}_{\bar{d}_j}(t)^\top\mathbf{M}_j(\mathbf{\Gamma})\mathbf{b}_{\bar{d}_j}(t), \qquad \boldsymbol{\pi}_j(\mathbf{\Gamma}) = S_{\bar{d}_j}[\mathbf{M}_j(\mathbf{\Gamma})]. \tag{11}$$

Here $m_j = 2\bar{d}_j$. If the centre has higher degree, first elevate both coefficient representations to a common degree and use the corresponding linear derivative/elevation map in place of $\mathbf{D}_{i_j}$. Affine box blocks have degree d, are raised to degree $2d$ (Appendix A4.6), and have affine coefficient maps.

Properties of the cone. Write $\mathscr{K}_m$ for the cone (1d) of degree m, so $\mathscr{K}_j = \mathscr{K}_{m_j}$, and $p_{\boldsymbol{\zeta}}(t) := \boldsymbol{\zeta}^\top\mathbf{b}_m(t)$ for the polynomial with coefficients $\boldsymbol{\zeta}$. All four properties of $\mathscr{K}_m$ are immediate from (1d). It is an intersection of closed homogeneous half-spaces, hence a *closed convex cone*. It is *pointed*, since $\boldsymbol{\zeta}, -\boldsymbol{\zeta} \in \mathscr{K}_m$ forces the polynomial to vanish, hence $\boldsymbol{\zeta} = \mathbf{0}$ as $\mathbf{b}_m$ is a basis. It has *nonempty interior*, since $\mathbf{b}_m$ is a nonnegative partition of unity, so $|p_{\boldsymbol{\zeta}}(t)| \le \|\boldsymbol{\zeta}\|_\infty \le \|\boldsymbol{\zeta}\|$ and any $\boldsymbol{\zeta}$ with $\min_{[0,1]} p_{\boldsymbol{\zeta}} = \mu > 0$ carries the open μ-ball. And it is *semialgebraic*, by Tarski–Seidenberg applied to $\forall t\,(0 \le t \le 1 \to p_{\boldsymbol{\zeta}}(t) \ge 0)$.

The *dual cone*, the set of vectors with nonnegative inner product with every element of $\mathscr{K}_m$, is

$$\mathscr{K}_m^* = \mathrm{cone}\{\mathbf{b}_m(t) : t \in [0,1]\}, \tag{12}$$

the nonnegative combinations of the evaluation vectors used in §2. This holds because $\mathscr{K}_m$ is by definition the dual of that conic hull, the hull is closed (each element needs at most $m+1$ generators by Carathéodory, and $\mathbf{u}^\top\mathbf{b}_m(t) \equiv 1$ bounds the total weight, so the generating data lie in a compact set), and the bipolar theorem (a closed convex cone is the dual of its own dual) applies.

Lipschitz bound for the restoration merit. For general C^2 coefficient maps on a compact convex region $\mathscr{R}$, set $\mathbf{e}_j(\mathbf{z}) = \boldsymbol{\pi}_j(\mathbf{z}) - P_{\mathscr{K}_j}(\boldsymbol{\pi}_j(\mathbf{z}))$. Projection residuals are nonexpansive and $0 \in \mathscr{K}_j$, hence, with $B_{\pi,j}$ and $B_{J,j}$ bounds on $\|\boldsymbol{\pi}_j\|$ and $\|D\boldsymbol{\pi}_j\|_{\mathrm{op}}$ over $\mathscr{R}$ and $L_{J,j}$ a Lipschitz constant of $D\boldsymbol{\pi}_j$ (computed below for spherical exclusions), $\|\mathbf{e}_j(\mathbf{z})\| \le B_{\pi,j}$ and $\|\mathbf{e}_j(\mathbf{z}) - \mathbf{e}_j(\mathbf{y})\| \le B_{J,j}\|\mathbf{z} - \mathbf{y}\|$. The gradient $2D\boldsymbol{\pi}_j(\mathbf{z})^\top\mathbf{e}_j(\mathbf{z})$ therefore has Lipschitz constant at most $2B_{J,j}^2 + 2L_{J,j}B_{\pi,j}$, and summing over blocks bounds the full merit; the step scale of Appendix A3.1.4 uses the larger, also valid, constant with $4B_{J,j}^2$. The remaining calculation specializes to spherical position exclusions ($\mathbf{\Sigma}_j = \mathbf{I}$).

$L_{J,j}$, the Lipschitz constant of the Jacobian $D\boldsymbol{\pi}_j$ of §2 as a function of the base point, is fixed and $\mathscr{R}$-independent. By the exact quadratic expansion of the coefficient map, the Jacobian of $\boldsymbol{\pi}_j$ at base point $\boldsymbol{\Gamma}$, applied to a direction $\boldsymbol{\Delta}$, is $D\boldsymbol{\pi}_j(\boldsymbol{\Gamma})[\boldsymbol{\Delta}] = S_d(\boldsymbol{\Delta}^\top \boldsymbol{\Gamma}_j + \boldsymbol{\Gamma}_j^\top \boldsymbol{\Delta})$. For two base points $\boldsymbol{\Gamma}^{(1)}, \boldsymbol{\Gamma}^{(2)} \in \mathscr{A}_{\mathrm{bc}}$, writing $\mathbf{F} := \boldsymbol{\Gamma}^{(1)} - \boldsymbol{\Gamma}^{(2)}$ (so $\boldsymbol{\Gamma}_j^{(1)} - \boldsymbol{\Gamma}_j^{(2)} = \mathbf{F}$ as well, the obstacle-centring shift being a constant offset; for a robot pair both $\mathbf{F}$ and $\boldsymbol{\Delta}$ enter through the difference of the two robots' rows, each at most $\sqrt{2}$ times longer, so $L_{J,j}$ below gains a factor 2),

$$\big(D\boldsymbol{\pi}_j(\boldsymbol{\Gamma}^{(1)}) - D\boldsymbol{\pi}_j(\boldsymbol{\Gamma}^{(2)})\big)[\boldsymbol{\Delta}] \;=\; S_d\big(\boldsymbol{\Delta}^\top \mathbf{F} + \mathbf{F}^\top \boldsymbol{\Delta}\big),$$

linear in $\boldsymbol{\Delta}$ and in $\mathbf{F}$. Since S_d is a fixed linear map, this gives $\|D\boldsymbol{\pi}_j(\boldsymbol{\Gamma}^{(1)}) - D\boldsymbol{\pi}_j(\boldsymbol{\Gamma}^{(2)})\|_{\mathrm{op}} \le 2\|S_d\|_{\mathrm{op}}\|\mathbf{F}\|_F$: $D\boldsymbol{\pi}_j$ is Lipschitz on all of $\mathscr{A}_{\mathrm{bc}}$ with constant $2\|S_d\|_{\mathrm{op}}$, a fixed number depending only on d. Restricting to the free coordinates via Lemma 3's chart $\mathbf{Z} \mapsto \boldsymbol{\Gamma}_{\mathrm{aff}} + \mathbf{Z}\mathbf{N}^\top$ composes this bound with the fixed linear map $\mathbf{N}$, giving $L_{J,j} := 2\|S_d\|_{\mathrm{op}}$ (as $\|\mathbf{N}\|_{\mathrm{op}} = 1$): an explicit constant depending only on d, and independent of any compact set $\mathscr{R}$.

Quadratic structure and curvature. For matrices $\mathbf{X}, \mathbf{Y}$ of the same shape, $\langle \mathbf{X}, \mathbf{Y} \rangle := \mathrm{tr}(\mathbf{X}^\top \mathbf{Y})$ is the Frobenius inner product.

Lemma 4. *Fix the chart anchor $\boldsymbol{\Gamma}_{\mathrm{aff}} \in \mathscr{A}_{\mathrm{bc}}$ and write $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}_{\mathrm{aff}} + \mathbf{Z}\mathbf{N}^\top$ (Lemma 3). Then, exactly, with no remainder,*

$$\mathrm{cost}(\boldsymbol{\Gamma}_{\mathrm{aff}} + \mathbf{Z}\mathbf{N}^\top) \;=\; \mathrm{cost}(\boldsymbol{\Gamma}_{\mathrm{aff}}) \;+\; 2\big\langle (\boldsymbol{\Gamma}_{\mathrm{aff}}\mathbf{Q} - \mathbf{L})\mathbf{N}, \mathbf{Z} \big\rangle \;+\; \mathrm{tr}\big(\mathbf{H}\mathbf{Z}^\top \mathbf{Z}\big), \qquad \mathbf{H} \;:=\; \mathbf{N}^\top \mathbf{Q} \mathbf{N}.$$

Proof of Lemma 4. The cost (8) is a polynomial of degree two in $\boldsymbol{\Gamma}$ and the chart is affine in $\mathbf{Z}$, so the expansion stops at degree two. To see the terms, write $\mathbf{W} := \mathbf{Z}\mathbf{N}^\top$, so $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}_{\mathrm{aff}} + \mathbf{W}$, and recall that $\mathbf{Q}$ is symmetric. Expanding the product,

$$\boldsymbol{\Gamma}^\top \boldsymbol{\Gamma} = \boldsymbol{\Gamma}_{\mathrm{aff}}^\top \boldsymbol{\Gamma}_{\mathrm{aff}} + \boldsymbol{\Gamma}_{\mathrm{aff}}^\top \mathbf{W} + \mathbf{W}^\top \boldsymbol{\Gamma}_{\mathrm{aff}} + \mathbf{W}^\top \mathbf{W}.$$

We multiply by $\mathbf{Q}$ and take the trace, term by term, using only $\mathrm{tr}(\mathbf{X}) = \mathrm{tr}(\mathbf{X}^\top)$ and $\mathrm{tr}(\mathbf{X}\mathbf{Y}) = \mathrm{tr}(\mathbf{Y}\mathbf{X})$. The two middle terms are equal, since $\mathrm{tr}(\mathbf{Q}\mathbf{W}^\top \boldsymbol{\Gamma}_{\mathrm{aff}}) = \mathrm{tr}(\boldsymbol{\Gamma}_{\mathrm{aff}}^\top \mathbf{W}\mathbf{Q}) = \mathrm{tr}(\mathbf{Q}\boldsymbol{\Gamma}_{\mathrm{aff}}^\top \mathbf{W})$, and together they give

$$2\,\mathrm{tr}(\mathbf{Q}\boldsymbol{\Gamma}_{\mathrm{aff}}^\top \mathbf{Z}\mathbf{N}^\top) = 2\,\mathrm{tr}(\mathbf{N}^\top \mathbf{Q}\boldsymbol{\Gamma}_{\mathrm{aff}}^\top \mathbf{Z}) = 2\langle \boldsymbol{\Gamma}_{\mathrm{aff}}\mathbf{Q}\mathbf{N}, \mathbf{Z} \rangle.$$

The last term gives $\mathrm{tr}(\mathbf{Q}\mathbf{N}\mathbf{Z}^\top \mathbf{Z}\mathbf{N}^\top) = \mathrm{tr}(\mathbf{N}^\top \mathbf{Q}\mathbf{N}\mathbf{Z}^\top \mathbf{Z}) = \mathrm{tr}(\mathbf{H}\mathbf{Z}^\top \mathbf{Z})$. The linear part of the cost splits in the same way: $-2\,\mathrm{tr}(\mathbf{L}^\top \boldsymbol{\Gamma}) = -2\,\mathrm{tr}(\mathbf{L}^\top \boldsymbol{\Gamma}_{\mathrm{aff}}) - 2\,\mathrm{tr}(\mathbf{N}^\top \mathbf{L}^\top \mathbf{Z}) = -2\,\mathrm{tr}(\mathbf{L}^\top \boldsymbol{\Gamma}_{\mathrm{aff}}) - 2\langle \mathbf{L}\mathbf{N}, \mathbf{Z} \rangle$. The terms without $\mathbf{Z}$, namely $\mathrm{tr}(\mathbf{Q}\boldsymbol{\Gamma}_{\mathrm{aff}}^\top \boldsymbol{\Gamma}_{\mathrm{aff}}) - 2\,\mathrm{tr}(\mathbf{L}^\top \boldsymbol{\Gamma}_{\mathrm{aff}}) + c_0$, add up to $\mathrm{cost}(\boldsymbol{\Gamma}_{\mathrm{aff}})$, and the two terms linear in $\mathbf{Z}$ add up to $2\langle (\boldsymbol{\Gamma}_{\mathrm{aff}}\mathbf{Q} - \mathbf{L})\mathbf{N}, \mathbf{Z} \rangle$. This is the claimed identity, for tracking and for planning alike. Finally, writing $\mathbf{z}_1, \ldots, \mathbf{z}_n$ for the rows of $\mathbf{Z}$, $\mathrm{tr}(\mathbf{H}\mathbf{Z}^\top \mathbf{Z}) = \sum_i \mathbf{z}_i \mathbf{H} \mathbf{z}_i^\top$, which lies between $\lambda_{\min}(\mathbf{H})\|\mathbf{Z}\|_F^2$ and $\lambda_{\max}(\mathbf{H})\|\mathbf{Z}\|_F^2$; this is how the eigenvalues of $\mathbf{H}$ control the curvature below. □

Lemma 5. *For $\mathbf{Q} \succeq 0$ and a full-column-rank $\mathbf{N}$, $\mathbf{H} = \mathbf{N}^\top \mathbf{Q}\mathbf{N} \succ 0$ if and only if $\ker \mathbf{Q} \cap \mathrm{range}(\mathbf{N}) = \{0\}$. For the derivative objective in (P_∞) with the endpoint data (9) and $\mathbf{N}$ the chart basis of Lemma 3, $\mathbf{H} \succ 0$ for $d \ge \max\{k, 2(\ell+1)\}$ under Assumption B.*

Proof of Lemma 5. For any vector $\mathbf{v}$, put $\mathbf{w} := \mathbf{N}\mathbf{v}$. Then $\mathbf{v}^\top \mathbf{H}\mathbf{v} = \mathbf{w}^\top \mathbf{Q}\mathbf{w} \ge 0$ because $\mathbf{Q} \succeq 0$, so $\mathbf{H} \succeq 0$, and $\mathbf{H} \succ 0$ means that $\mathbf{v}^\top \mathbf{H}\mathbf{v} = 0$ only for $\mathbf{v} = \mathbf{0}$. For a positive semidefinite $\mathbf{Q}$, $\mathbf{w}^\top \mathbf{Q}\mathbf{w} = \|\mathbf{Q}^{1/2}\mathbf{w}\|^2$ ($\mathbf{Q}^{1/2}$ the symmetric square root of $\mathbf{Q}$), which is zero exactly when $\mathbf{Q}\mathbf{w} = \mathbf{0}$. Hence $\mathbf{v}^\top \mathbf{H}\mathbf{v} = 0$ exactly when $\mathbf{N}\mathbf{v} \in \ker \mathbf{Q}$. If $\ker \mathbf{Q} \cap \mathrm{range}(\mathbf{N}) = \{0\}$, this forces $\mathbf{N}\mathbf{v} = \mathbf{0}$, and then $\mathbf{v} = \mathbf{0}$ because $\mathbf{N}$ has full column rank; so $\mathbf{H} \succ 0$. Conversely, a nonzero $\mathbf{w} = \mathbf{N}\mathbf{v}$ in $\ker \mathbf{Q}$ has $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{v}^\top \mathbf{H}\mathbf{v} = 0$, so $\mathbf{H}$ is not positive definite.

For tracking, $\mathbf{Q} = \frac{1}{2}\mathbf{G}_d$ and $\mathbf{w}^\top \mathbf{G}_d \mathbf{w} = \int_0^1 p_{\mathbf{w}}(t)^2\,dt$, where $p_{\mathbf{w}}$ is the polynomial with coefficient vector $\mathbf{w}$. This is positive unless $p_{\mathbf{w}} = 0$, so $\ker \mathbf{Q} = \{0\}$ and the criterion holds.

For minimum-derivative planning, $\mathbf{Q} = \mathbf{G}^{(k)}$; the condition $d \ge k$ makes the kth-derivative cost meaningful, and $d \ge 2(\ell+1)$ leaves free coefficients ($d_f \ge 1$). Take $\mathbf{w}$ in both sets and read it as a polynomial $p_{\mathbf{w}}$ of degree at most d, a change of the curve that keeps the endpoint data. Being in $\ker \mathbf{G}^{(k)}$ means $\int_0^1 |p_{\mathbf{w}}^{(k)}(t)|^2 dt = 0$, so $p_{\mathbf{w}}^{(k)} \equiv 0$ and $p_{\mathbf{w}}$ has degree below k. Being in $\mathrm{range}(\mathbf{N}) =$

$\ker \mathbf{A}^\top$ means that $p_{\mathbf{w}}$ and its first ℓ derivatives vanish at $t=0$ and at $t=1$, that is, $p_{\mathbf{w}}$ has a root of order $\ell+1$ at each endpoint, so $p_{\mathbf{w}} = [t(1-t)]^{\ell+1} q$ for a polynomial q (Lemma 9). If $q \neq 0$, then $p_{\mathbf{w}}$ has degree at least $2(\ell+1) \geq k$ by Assumption B, contradicting degree below k. So $q=0$, $\mathbf{w}=\mathbf{0}$, and the criterion holds. For example, for acceleration ($k=2$) with only positions fixed ($\ell=0$), such a change is a straight line ($p''_{\mathbf{w}}=0$) that is zero at $t=0$ and $t=1$, hence zero everywhere. □

Hence the constants

$$\alpha \;:=\; 2\lambda_{\min}(\mathbf{H}) \;>\; 0, \qquad \beta \;:=\; 2\lambda_{\max}(\mathbf{H}) \;<\; \infty \tag{13}$$

are well defined, and by Lemma 4 they bound the curvature on all of $\mathscr{A}_{\mathrm{bc}}$, with no remainder term. For tracking with orthonormal $\mathbf{N}$, $\lambda_{\min}(\mathbf{H}) \geq \frac{1}{2}\lambda_{\min}(\mathbf{G}_d) > 0$.

Compactness and existence on each component. Lemma 6 gives the compact sublevel sets used in the proof of Proposition 10. Lemmas 7 and 8 are not needed for the main results; they record that the feasible set splits into finitely many connected components, each with its own cheapest curve, which is why GlobMP is run from several seeds in §4.

Lemma 6. *For any $\bar{\mathbf{\Gamma}} \in \mathscr{A}_{\mathrm{bc}}$, writing $\mathbf{\Gamma} = \mathbf{\Gamma}_{\mathrm{aff}} + \mathbf{Z}\mathbf{N}^\top$ in the chart of Lemma 3, $\mathscr{E}(\bar{\mathbf{\Gamma}}) := \{\mathbf{\Gamma} \in \mathscr{A}_{\mathrm{bc}} : \mathrm{cost}(\mathbf{\Gamma}) \leq \mathrm{cost}(\bar{\mathbf{\Gamma}})\}$ is, under $\mathbf{H} \succ 0$ (Lemma 5), an ellipsoid in $\mathbf{Z}$-coordinates centred at the unconstrained minimizer $\mathbf{Z}^\star = -(\mathbf{\Gamma}_{\mathrm{aff}}\mathbf{Q} - \mathbf{L})\mathbf{N}\mathbf{H}^{-1}$ of* cost *on $\mathscr{A}_{\mathrm{bc}}$, hence a compact convex subset of $\mathscr{A}_{\mathrm{bc}}$.*

Proof of Lemma 6. The proof completes the square, as in the scalar identity $ax^2 + 2bx = a(x + b/a)^2 - b^2/a$ for $a > 0$, which writes a quadratic as a constant plus a nonnegative square centred at its minimizer.

Let $c_{\mathrm{a}} := \mathrm{cost}(\mathbf{\Gamma}_{\mathrm{aff}})$ and $\mathbf{V} := (\mathbf{\Gamma}_{\mathrm{aff}}\mathbf{Q} - \mathbf{L})\mathbf{N}$, so Lemma 4 reads $\mathrm{cost}(\mathbf{\Gamma}_{\mathrm{aff}} + \mathbf{Z}\mathbf{N}^\top) = c_{\mathrm{a}} + 2\langle \mathbf{V}, \mathbf{Z}\rangle + \mathrm{tr}(\mathbf{H}\mathbf{Z}^\top\mathbf{Z})$. Since $\mathbf{H} \succ 0$ it is invertible, and $\mathbf{Z}^\star := -\mathbf{V}\mathbf{H}^{-1}$ satisfies $\mathbf{Z}^\star\mathbf{H} = -\mathbf{V}$; as $\mathbf{H}$ is symmetric, also $\mathbf{H}\mathbf{Z}^{\star\top} = -\mathbf{V}^\top$. Expanding the square around $\mathbf{Z}^\star$,

$$\mathrm{tr}\left(\mathbf{H}(\mathbf{Z}-\mathbf{Z}^\star)^\top(\mathbf{Z}-\mathbf{Z}^\star)\right) = \mathrm{tr}(\mathbf{H}\mathbf{Z}^\top\mathbf{Z}) - \mathrm{tr}(\mathbf{H}\mathbf{Z}^\top\mathbf{Z}^\star) - \mathrm{tr}(\mathbf{H}\mathbf{Z}^{\star\top}\mathbf{Z}) + \mathrm{tr}(\mathbf{H}\mathbf{Z}^{\star\top}\mathbf{Z}^\star).$$

The two middle traces are equal (transpose one and use the symmetry of $\mathbf{H}$), and $\mathrm{tr}(\mathbf{H}\mathbf{Z}^{\star\top}\mathbf{Z}) = -\mathrm{tr}(\mathbf{V}^\top\mathbf{Z}) = -\langle \mathbf{V}, \mathbf{Z}\rangle$. So the right side is $\mathrm{tr}(\mathbf{H}\mathbf{Z}^\top\mathbf{Z}) + 2\langle \mathbf{V}, \mathbf{Z}\rangle + \mathrm{tr}(\mathbf{H}\mathbf{Z}^{\star\top}\mathbf{Z}^\star)$, and substituting into the cost gives

$$\mathrm{cost}(\mathbf{\Gamma}_{\mathrm{aff}} + \mathbf{Z}\mathbf{N}^\top) = c_{\mathrm{a}} - \mathrm{tr}(\mathbf{H}\mathbf{Z}^{\star\top}\mathbf{Z}^\star) + \mathrm{tr}\left(\mathbf{H}(\mathbf{Z}-\mathbf{Z}^\star)^\top(\mathbf{Z}-\mathbf{Z}^\star)\right).$$

The last term is nonnegative and vanishes only at $\mathbf{Z} = \mathbf{Z}^\star$ (as $\mathbf{H} \succ 0$), so $\mathbf{Z}^\star$ is the unconstrained minimizer of the cost on $\mathscr{A}_{\mathrm{bc}}$.

Write $\bar{\mathbf{Z}}$ for the coordinates of $\bar{\mathbf{\Gamma}}$. By the display, $\mathrm{cost}(\mathbf{\Gamma}) \leq \mathrm{cost}(\bar{\mathbf{\Gamma}})$ holds exactly when

$$\mathrm{tr}\left(\mathbf{H}(\mathbf{Z}-\mathbf{Z}^\star)^\top(\mathbf{Z}-\mathbf{Z}^\star)\right) \leq \varsigma, \qquad \varsigma := \mathrm{cost}(\bar{\mathbf{\Gamma}}) - c_{\mathrm{a}} + \mathrm{tr}(\mathbf{H}\mathbf{Z}^{\star\top}\mathbf{Z}^\star) = \mathrm{tr}\left(\mathbf{H}(\bar{\mathbf{Z}}-\mathbf{Z}^\star)^\top(\bar{\mathbf{Z}}-\mathbf{Z}^\star)\right) \geq 0.$$

With $\mathbf{Y} := \mathbf{Z} - \mathbf{Z}^\star$ and rows $\mathbf{y}_1, \ldots, \mathbf{y}_n$, the left side is $\sum_i \mathbf{y}_i\mathbf{H}\mathbf{y}_i^\top$, a positive definite quadratic form in the entries of $\mathbf{Y}$; so $\{\mathbf{Y} : \sum_i \mathbf{y}_i\mathbf{H}\mathbf{y}_i^\top \leq \varsigma\}$ is an ellipsoid centred at $\mathbf{Y} = \mathbf{0}$. It is closed (the set where a continuous function is at most a constant), bounded (by the last line of the proof of Lemma 4, each point has $\|\mathbf{Y}\|_F^2 \leq \varsigma/\lambda_{\min}(\mathbf{H})$), hence compact, and convex (a sublevel set of a convex quadratic). Shifting by $\mathbf{Z}^\star$ and mapping through the chart $\mathbf{Z} \mapsto \mathbf{\Gamma}_{\mathrm{aff}} + \mathbf{Z}\mathbf{N}^\top$, an affine isometry (Lemma 3), keeps all three properties, so $\mathscr{E}(\bar{\mathbf{\Gamma}})$ is a compact convex ellipsoid in $\mathscr{A}_{\mathrm{bc}}$; $\bar{\mathbf{\Gamma}}$ enters only through the size ς. □

Lemma 7. *Suppose every coefficient map is continuous and semialgebraic, that is, described by finitely many polynomial equations and inequalities (polynomial maps qualify). Consider the set of feasible curves, $\mathscr{A}_{\mathrm{bc}} \cap \bigcap_j \boldsymbol{\pi}_j^{-1}(\mathscr{K}_j)$, identified with $\mathscr{F}$ through the chart of Lemma 3. Then:*

- *(i)* Finitely many pieces. *$\mathscr{F}$ splits into finitely many separate pieces $\mathscr{F}_1, \ldots, \mathscr{F}_{\bar{c}}$, its connected components: two feasible curves lie in the same piece exactly when one can be deformed continuously into the other without leaving $\mathscr{F}$.*

(ii) Each piece is closed. *Each $\mathscr{F}_c$ is closed in $\mathbb{R}^{n\times(d+1)}$, and also open relative to $\mathscr{F}$ (clopen in $\mathscr{F}$).*

(iii) The pieces are a positive distance apart. *If $\mathscr{U}$ is a nonempty compact subset of one piece $\mathscr{F}_c$, then every feasible curve outside that piece stays at distance at least $d_0(\mathscr{U}) := \operatorname{dist}(\mathscr{U}, \mathscr{F} \setminus \mathscr{F}_c) > 0$ from $\mathscr{U}$; we set $d_0(\mathscr{U}) = +\infty$ when $\mathscr{F}$ has only the one piece.*

Proof of Lemma 7. We first show that $\mathscr{F}$ is closed and semialgebraic, and then prove the three items in order. A curve is feasible when, for every block j, its coefficients $\boldsymbol{\pi}_j$ lie in the cone $\mathscr{K}_j$. Each $\mathscr{K}_j$ is closed and each $\boldsymbol{\pi}_j$ is continuous, so the set of curves meeting block j is closed (the preimage of a closed set under a continuous map), and $\mathscr{F}$, the intersection of these finitely many closed sets with the closed set $\mathscr{A}_{\mathrm{bc}}$, is closed.

The set $\mathscr{F}$ is semialgebraic: each cone $\mathscr{K}_j$ (for example, the polynomials that are nonnegative on $[0,1]$) and each $\boldsymbol{\pi}_j$ is described by polynomial equations and inequalities (the condition 'for every $t \in [0,1]$' can be written without t by the Tarski–Seidenberg theorem), and preimages and finite intersections of such sets are again of this kind (Bochnak et al., 1998).

(i) A semialgebraic set has only finitely many connected components, and each of them is path connected (Bochnak et al., 1998). Without this property a closed set can have infinitely many components, as $\{x \in (0,1] : \sin(1/x) \geq 0\}$ shows.

(ii) A connected component of any set is closed relative to that set (the closure of a connected set is connected, so a component already contains its limit points in the set). Since $\mathscr{F}$ is closed in $\mathbb{R}^{n\times(d+1)}$, each $\mathscr{F}_c$ is closed there too. Its complement in $\mathscr{F}$ is the union of the other components, a finite union of closed sets, hence closed; so $\mathscr{F}_c$ is also open relative to $\mathscr{F}$.

(iii) Let $\mathscr{U} \subset \mathscr{F}_c$ be nonempty and compact and let $\mathscr{W} := \mathscr{F} \setminus \mathscr{F}_c$, which is closed by (ii) and disjoint from $\mathscr{U}$. If $\mathscr{W}$ is empty, $d_0(\mathscr{U}) = +\infty$ by definition. Otherwise the function $x \mapsto \operatorname{dist}(x, \mathscr{W})$ is continuous, and it is positive at every $x \in \mathscr{U}$ because $x \notin \mathscr{W}$ and $\mathscr{W}$ is closed. A continuous positive function on a compact set attains its minimum, which is therefore positive; that minimum is $d_0(\mathscr{U})$. Compactness is needed here: the graph of $1/x$ and the x-axis are disjoint closed sets at distance 0. □

Lemma 8. *Assume $\mathbf{H} \succ 0$ and continuous coefficient maps. Then every connected component $\mathscr{F}_c$ of the feasible set contains a cheapest curve: there is a $\boldsymbol{\Gamma}_c^\star \in \mathscr{F}_c$ with $\operatorname{cost}(\boldsymbol{\Gamma}_c^\star) \leq \operatorname{cost}(\boldsymbol{\Gamma})$ for every $\boldsymbol{\Gamma} \in \mathscr{F}_c$ (it need not be unique; we fix one).*

Proof of Lemma 8. A continuous function on a nonempty compact set attains its minimum (Weierstrass). Since $\mathscr{F}_c$ may be unbounded, we first restrict to a compact subset that contains every minimizer.

Pick any curve $\bar{\boldsymbol{\Gamma}} \in \mathscr{F}_c$. A curve that costs more than $\bar{\boldsymbol{\Gamma}}$ is not a minimizer, so it suffices to minimize over the curves of $\mathscr{F}_c$ that cost at most $\operatorname{cost}(\bar{\boldsymbol{\Gamma}})$, that is, over $\mathscr{S} := \mathscr{F}_c \cap \mathscr{E}(\bar{\boldsymbol{\Gamma}})$, where $\mathscr{E}(\bar{\boldsymbol{\Gamma}})$ is the sublevel set of Lemma 6. This set $\mathscr{S}$ is nonempty, since it contains $\bar{\boldsymbol{\Gamma}}$. It is closed, since $\mathscr{E}(\bar{\boldsymbol{\Gamma}})$ is closed and so is $\mathscr{F}_c$: continuous coefficient maps make $\mathscr{F}$ closed, and a connected component of a closed set is closed. It is bounded, since it lies in the bounded ellipsoid $\mathscr{E}(\bar{\boldsymbol{\Gamma}})$; this is where $\mathbf{H} \succ 0$ is used. So $\mathscr{S}$ is compact, and the cost, a quadratic and hence continuous function, attains its minimum on $\mathscr{S}$ at some curve $\boldsymbol{\Gamma}_c^\star$. This curve is cheapest on all of $\mathscr{F}_c$: a curve of $\mathscr{F}_c$ inside $\mathscr{S}$ costs at least $\operatorname{cost}(\boldsymbol{\Gamma}_c^\star)$ by the choice of $\boldsymbol{\Gamma}_c^\star$, and a curve of $\mathscr{F}_c$ outside $\mathscr{S}$ costs more than $\operatorname{cost}(\bar{\boldsymbol{\Gamma}}) \geq \operatorname{cost}(\boldsymbol{\Gamma}_c^\star)$. □

Cumulative constraints and derivative-cost curvature. The finite-dimensional reduction of (P_∞) also admits scalar budget blocks. Write $\mathbf{z} = \operatorname{vec}(\boldsymbol{\Gamma}_{\mathrm{free}})$ and let $\pi_j(\mathbf{z})$ denote a budget block (1e). Its cone is $\mathbb{R}_+$, so its contribution to the restoration merit and gradient is

$$V_j(\mathbf{z}) = \min\{\pi_j(\mathbf{z}), 0\}^2, \qquad \nabla V_j(\mathbf{z}) = 2\min\{\pi_j(\mathbf{z}), 0\}\nabla\pi_j(\mathbf{z}).$$

These formulas do not require the integrand to be polynomial in time. For a general integrand, differentiation under the integral is justified, for example, if it is measurable in time and, for almost every t, C^2 in the trajectory arguments, with its value and first two composed $\mathbf{z}$-derivatives dominated by integrable functions on each compact set of free coordinates.

For a single segment of degree $d \geq k$ with only start and goal fixed, the objective of (P_∞) has $\mathbf{H} \succ 0$ if and only if $k \leq 2$ (the case $\ell = 0$ of Assumption B): a free perturbation has zero derivative cost exactly when its degree is below k, and among those vanishing at $t = 0$ and $t = 1$ the first nonzero one is $t(1-t)$, of degree 2. When the cost has free zero-curvature directions, compactness must be checked separately. Additional endpoint derivatives can remove these directions as in Lemma 9; adding them changes the boundary data of (P_∞).

A3.1.3 The structural assumption

Assumption D (finite-dimensional representation)**.** *$d \geq k$ and $2(\ell+1) \leq d$, and every pointwise coefficient map and scalar cumulative map is C^2 on an open neighbourhood of the regions used in the analysis.*

The quadratic obstacle and separation blocks, and budget blocks whose integrand meets the condition of Appendix A3.1.2, satisfy the smoothness part. The degree condition makes $\mathbf{A}$ of full column rank, since its columns are independent Hermite interpolation conditions (Stoer & Bulirsch, 2002); $\mathbf{N}$ is an orthonormal basis of $\ker \mathbf{A}^\top$, and $\mathbf{Q} \succeq 0$ always. Under Assumption D, Assumption B is equivalent to $\mathbf{H} \succ 0$ for the derivative objective, with $\mathbf{H}$ as in Lemma 4 (Lemma 9). Write $\mathscr{P}_r$ for scalar polynomials of degree at most r, with $\mathscr{P}_r = \{0\}$ for $r < 0$, and coeff$(\cdot)$ for the Bernstein-d coefficient vector of a polynomial.

Lemma 9 (kernels)**.** *Let $0 \leq k \leq d$, let $\mathbf{A}$ be the endpoint matrix of (9), and read each coefficient vector as the polynomial of degree at most d it represents. The vectors with zero kth-derivative cost are exactly the polynomials of degree below k, and the vectors with zero endpoint data are exactly the multiples of $[t(1-t)]^{\ell+1}$:*

$$\ker \mathbf{G}^{(k)} = \mathrm{coeff}(\mathscr{P}_{k-1}), \qquad \ker \mathbf{A}^\top = \mathrm{coeff}\big([t(1-t)]^{\ell+1}\,\mathscr{P}_{d-2(\ell+1)}\big).$$

Under Assumption D, the second space has dimension $d_f = d + 1 - 2(\ell+1)$.

Proof of Lemma 9. Write $p_{\boldsymbol{\zeta}}$ for the polynomial with coefficient vector $\boldsymbol{\zeta}$.

By the definition of $\mathbf{G}^{(k)}$,

$$\boldsymbol{\zeta}^\top \mathbf{G}^{(k)} \boldsymbol{\zeta} = \int_0^1 \big|p_{\boldsymbol{\zeta}}^{(k)}(t)\big|^2 dt. \tag{14}$$

As $\mathbf{G}^{(k)} \succeq 0$, a vector lies in its kernel exactly when (14) is zero (the argument in the proof of Lemma 5). The integral of the square of a continuous function is zero only if the function is zero, so this happens exactly when $p_{\boldsymbol{\zeta}}^{(k)} \equiv 0$, that is, when $p_{\boldsymbol{\zeta}}$ has degree below k.

The columns of $\mathbf{A}$ evaluate a polynomial and its first ℓ derivatives at $t = 0$ and at $t = 1$, so $\mathbf{A}^\top \boldsymbol{\zeta} = \mathbf{0}$ says $p_{\boldsymbol{\zeta}}^{(s)}(0) = p_{\boldsymbol{\zeta}}^{(s)}(1) = 0$ for $s = 0, \ldots, \ell$. By Taylor's formula at 0,

$$p_{\boldsymbol{\zeta}}(t) = \sum_{s=0}^{d} \frac{p_{\boldsymbol{\zeta}}^{(s)}(0)}{s!}\, t^s, \tag{15}$$

and the first $\ell+1$ terms of (15) vanish, so $t^{\ell+1}$ divides $p_{\boldsymbol{\zeta}}$. In the same way $(1-t)^{\ell+1}$ divides it, and since the two factors have no common root, their product does:

$$p_{\boldsymbol{\zeta}}(t) = \big[t(1-t)\big]^{\ell+1} q(t), \qquad \deg q \leq d - 2(\ell+1). \tag{16}$$

Conversely, every such product has zero values and zero derivatives of orders $0, \ldots, \ell$ at both ends, by the product rule.

The polynomials q in (16) form a space of dimension $d - 2(\ell+1) + 1$, and $q \mapsto [t(1-t)]^{\ell+1} q$ is linear and one-to-one, since a product of nonzero polynomials is nonzero. So the second kernel has dimension $d + 1 - 2(\ell+1) = d_f$, which is at least 1 under Assumption D. □

Algorithm 3 Approximate projection onto $\mathscr{K}_m$ by fully corrective nonnegative matching pursuit

Require: $\boldsymbol{\pi} \in \mathbb{R}^{m+1}$, degree $m \geq 0$, threshold $\varepsilon \geq 0$, iteration budget T_{cone}

if $\boldsymbol{\pi} = \mathbf{0}$ **then return** $\mathbf{0}$
$\mathbf{y} \leftarrow -\boldsymbol{\pi}/\|\boldsymbol{\pi}\|$, $\mathscr{T} \leftarrow \emptyset$, $\boldsymbol{\omega} \leftarrow \mathbf{0}$, $\mathbf{v} \leftarrow \mathbf{y}$
for $\nu = 1, \ldots, T_{\text{cone}}$ **do**
  $t_\nu \leftarrow \arg\max_{t \in [0,1]} \langle \mathbf{v}, \mathbf{b}_m(t) \rangle / \|\mathbf{b}_m(t)\|$
  if $\langle \mathbf{v}, \mathbf{b}_m(t_\nu) \rangle / \|\mathbf{b}_m(t_\nu)\| \leq \varepsilon$ **then break**
  $\mathscr{T} \leftarrow \mathscr{T} \cup \{t_\nu\}$
  $\mathbf{w} \leftarrow \arg\min_{w_t \geq 0, t \in \mathscr{T}} \| \sum_{t \in \mathscr{T}} w_t \mathbf{b}_m(t) - \mathbf{y} \|^2$
  $\boldsymbol{\omega} \leftarrow \sum_{t \in \mathscr{T}} w_t \mathbf{b}_m(t)$, $\mathbf{v} \leftarrow \mathbf{y} - \boldsymbol{\omega}$
end for
return $\boldsymbol{\pi} + \|\boldsymbol{\pi}\| \boldsymbol{\omega}$

A3.1.4 Step size and the cone solve

Step size. A valid conservative step scale, provided the step stays in a compact convex region $\mathscr{R}$, is

$$L_{\text{proj}} = \sum_j (4B_{J,j}^2 + 2L_{J,j} B_{\pi,j}), \qquad \eta_{\text{proj}} = 1/L_{\text{proj}},$$

where $B_{J,j} = \max_{\mathscr{R}} \|D\boldsymbol{\pi}_j\|_{\text{op}}$, $B_{\pi,j} = \max_{\mathscr{R}} \|\boldsymbol{\pi}_j\|$, and $L_{J,j}$ is the Lipschitz constant of the Jacobian $D\boldsymbol{\pi}_j$ of §2.

On a compact convex region containing both endpoints of a restoration step, the Lipschitz bound of Appendix A3.1.2 gives

$$V(\mathbf{z} - \eta_{\text{proj}} \nabla V(\mathbf{z})) \leq V(\mathbf{z}) - \frac{\eta_{\text{proj}}}{2} \|\nabla V(\mathbf{z})\|^2,$$

for $0 < \eta_{\text{proj}} \leq 1/L_{\text{proj}}$. The step must be kept in the region, which compactness does not do by itself, and constants taken over a ball around the unknown minimizer are not computable. Algorithm 2 keeps every nonzero residual, so omitted blocks have zero gradient and every update is a step on the full merit $V(\mathbf{z}) = \sum_j \text{dist}^2(\boldsymbol{\pi}_j(\mathbf{z}), \mathscr{K}_j)$, the merit (2) in the coordinates $\mathbf{z}$. With exact cone projections and the region condition above, telescoping T_{proj} executed steps gives

$$\min_{0 \leq \nu < T_{\text{proj}}} \|\nabla V(\mathbf{z}^\nu)\|^2 \leq \frac{2V(\mathbf{z}^0)}{\eta_{\text{proj}} T_{\text{proj}}}.$$

The bound covers scalar cumulative blocks and does not need $\mathbf{H} \succ 0$. It gives a small merit gradient, not feasibility: for the scalar block $\pi(z) = z^2 - 1 \in \mathbb{R}_+$ of one coordinate z, $V(z) = (1 - z^2)^2$ near zero and $V'(0) = 0$ although $V(0) = 1$.

The cone solve and its accuracy. Algorithm 3 approximates the projection onto $\mathscr{K}_m$ for any degree $m \geq 0$, with $m = m_j$ for block j. By Moreau's identity it projects $-\boldsymbol{\pi}/\|\boldsymbol{\pi}\|$ onto the dual cone $\mathscr{K}_m^*$ with the atoms $\mathbf{b}_m(t)$ and adds $\|\boldsymbol{\pi}\|$ times the result to $\boldsymbol{\pi}$; T_{cone} is a numerical budget.

The normalized correlation gives the best decrease along a single nonnegative ray. It is not a polynomial in t: if $\mu(t) = \langle \mathbf{v}, \mathbf{b}_m(t) \rangle$ and $\|\mathbf{b}_m(t)\|^2 > 0$, its interior stationary points satisfy $2\mu' \|\mathbf{b}_m\|^2 - \mu(\|\mathbf{b}_m\|^2)' = 0$. Evaluate the correlation at these roots and at the endpoints; an identically zero derivative means the correlation is constant.

A direct rate argument specializes the conic pursuit framework of Locatello et al. (2017). Let $\boldsymbol{\omega}_\nu$ be an exact fully corrective iterate, $\boldsymbol{\omega}_* = P_{\mathscr{K}_m^*}(\mathbf{y})$, and $\theta_\nu = \frac{1}{2}\|\mathbf{y} - \boldsymbol{\omega}_\nu\|^2 - \frac{1}{2}\|\mathbf{y} - \boldsymbol{\omega}_*\|^2$, which is nonnegative because $\boldsymbol{\omega}_*$ is the nearest point of the cone. For the normalized atoms $\mathbf{a}_t = \mathbf{b}_m(t)/\|\mathbf{b}_m(t)\|$, nonnegative entries summing to one give $\|\mathbf{b}_m(t)\| \leq 1$, so $\mathbf{u}^\top \mathbf{a}_t \geq 1$. Since $\|\boldsymbol{\omega}_*\| \leq \|\mathbf{y}\| = 1$, it has an atomic representation of total weight at most $\sqrt{m+1}$. Exact nonnegative least squares gives $\langle \mathbf{y} - \boldsymbol{\omega}_\nu, \boldsymbol{\omega}_\nu \rangle = 0$. Convexity then bounds θ_ν by $\sqrt{m+1}$ times the largest positive atom correlation. Adding that unit atom with its optimal nonnegative weight decreases the objective by half the squared correlation; full correction does at least as well. Hence

$$\theta_{\nu+1} \leq \theta_\nu - \frac{\theta_\nu^2}{2(m+1)}, \qquad \theta_\nu \leq \frac{2(m+1)}{\nu + 4(m+1)},$$

using $\theta_0 \le 1/2$, for exact maximization and weight solves without a stopping threshold; it gives neither a linear rate nor finite termination.

If the correlation test terminates at threshold ε, then $\theta_v \le \sqrt{m+1}\,\varepsilon$. Strong convexity of the squared-distance objective gives projection error at most $\|\boldsymbol{\pi}\|\sqrt{2\sqrt{m+1}\,\varepsilon}$ for the returned vector. Its polynomial values are at least $-\|\boldsymbol{\pi}\|\varepsilon$, since that vector is $-\|\boldsymbol{\pi}\|\mathbf{v}$. For $\varepsilon > 0$ these bound the error rather than give exact membership, and stopping on the budget gives neither bound. Proposition 10 and Theorem 1 assume an exact solve of the convex problem (5), which is not this cone projection; Algorithm 3 plays no role there.

A3.2 Proof of Theorem 1

Throughout this subsection, the blocks are polynomial in $\boldsymbol{\Gamma}$, as the sphere, ellipsoid, pair and affine blocks are, and $\mathbf{z}$ stands for $\boldsymbol{\Gamma}_{\text{free}}$. It uses the boundary chart (Lemma 3) and the compact cost sublevel sets (Lemma 6) of Appendix A3.1; the push maps that move a curve off the obstacles (Lemma 14) serve only Appendix A3.2.3 and the stability results of Appendix A3.3.

A3.2.1 Fixed degree

Let $E^\star$ denote the optimum of (P), whose collision blocks ($j \le p$) are polynomial and convex in the position, and whose steering block, speed limit and (assumed convex) budgets have convex feasible sets, all imposed through $\mathscr{D}$. Let $\mathbf{z}_h$ be a cheapest point of the feasible grid $\mathscr{F} \cap h\mathbb{Z}^{nd_f} \cap \mathscr{B}$ (in the rational chart of free control points described after the grid part below), for a box $\mathscr{B}$ with a global minimizer of (P) in its interior; $\mathscr{B}$ may be the whole chart, since f has compact sublevel sets and a cheapest feasible grid point then exists whenever one grid point is feasible. The start of Theorem 1 and Proposition 13 is this point for the class $\mathscr{S}_N$, with $\mathscr{B}$ the whole chart and the blocks of the problem being solved, where d_f now counts the free control points per coordinate left by the boundary and junction equations. Proposition 13 rests on the following result at fixed degree, which uses a constraint qualification on the control points; Theorem 1 uses only its inner-set and descent steps, which do not need Assumption E.

Assumption E (constraint qualification)**.** *For some $U > E^\star$, every $\mathbf{z} \in \mathscr{F}$ with $f(\mathbf{z}) \le U$ has a $\mathbf{v}$ with $\boldsymbol{\pi}(\mathbf{z}) + D\boldsymbol{\pi}(\mathbf{z})\,\mathbf{v} \in \operatorname{int}\mathscr{K}$.*

Assumption E is Robinson's condition (Robinson, 1976): some change of the control points makes every block strictly feasible to first order. It is needed only on this sublevel set, and on fine splines it follows from Assumption C (Appendix A3.2.3, Lemma 16).

Proposition 10 (GlobMP at fixed degree)**.** *Assume Assumptions D and B (so that $\mathbf{H} \succ 0$) and Assumption E, and use a step $0 < \eta < 2/\beta$. Run the repair (5) from a feasible start of cost at most U. Then every iterate is feasible, each step lowers the cost by at least $(1/\eta - \beta/2)\|\mathbf{z}^{\nu+1} - \mathbf{z}^{\nu}\|^2$, and every limit point of the iterates is a KKT point of (P) (defined in the grid part of the proof). If the start is the cheapest feasible grid point $\mathbf{z}_h$ and the grid spacing h is small enough, then moreover the costs decrease to the optimum, $f(\mathbf{z}^{\nu}) \downarrow E^\star$, and every limit point is a global minimizer of (P).*

The proof has a descent part and a grid part. With uniform feasible random starts, a run with globally minimizing limit points is obtained with probability one after finitely many starts (Corollary 11).

Proof of Proposition 10, descent part. Each step solves a convex problem over a set $\mathscr{C}(\mathbf{z})$ contained in the feasible set, so feasibility is preserved; the cost decreases as in gradient descent; and limit points satisfy the first-order optimality conditions.

Fix a feasible $\mathbf{z}$ and a point $\mathbf{w} \in \mathscr{C}(\mathbf{z})$, and write $\boldsymbol{\gamma}_z$, $\boldsymbol{\gamma}_w$ for their curves. Take a collision block $j \le p$. Its function $g_j(\cdot,t)$ is convex in the position, and a convex function lies above each of its tangent planes:

$$g_j(\boldsymbol{\gamma}_w(t),t) \;\ge\; g_j(\boldsymbol{\gamma}_z(t),t) + \nabla_{\mathbf{x}} g_j(\boldsymbol{\gamma}_z(t),t)^{\top}\big(\boldsymbol{\gamma}_w(t) - \boldsymbol{\gamma}_z(t)\big) \qquad \text{for every } t \in [0,1]. \tag{17}$$

The right side of (17) is the first-order part of the block, the piecewise polynomial with coefficients $\boldsymbol{\pi}_j(\mathbf{z}) + D\boldsymbol{\pi}_j(\mathbf{z})(\mathbf{w} - \mathbf{z})$. For a sphere, for example, expanding the square in $g_j = \|\mathbf{x} - \mathbf{c}_j\|^2 - r_j^2$ shows that the two sides differ by exactly $\|\boldsymbol{\gamma}_w - \boldsymbol{\gamma}_z\|^2 \ge 0$; for a robot pair the difference is the

squared norm of the difference between the two robots' changes. If the coefficients of the right side of (17) lie in $\mathscr{K}_j$, the right side is nonnegative for every t, and then so is the left side. The steering block and the budgets are imposed exactly through $\mathscr{D}$, a closed convex set. Hence every point of $\mathscr{C}(\mathbf{z})$ is feasible. Also $\mathscr{C}(\mathbf{z})$ is closed and convex (its constraints say that affine functions of $\mathbf{w}$ lie in closed convex cones, intersected with $\mathscr{D}$), and it contains $\mathbf{z}$ itself, whose linearized blocks are $\boldsymbol{\pi}_j(\mathbf{z}) \in \mathscr{K}_j$.

By (5), $\mathbf{z}^{\nu+1}$ is the nearest point of $\mathscr{C}(\mathbf{z}^\nu)$ to $\mathbf{y} := \mathbf{z}^\nu - \eta\nabla f(\mathbf{z}^\nu)$. The nearest point $\mathbf{p}$ of a closed convex set to $\mathbf{y}$ is characterized by

$$\langle \mathbf{y}-\mathbf{p}, \mathbf{w}-\mathbf{p}\rangle \le 0 \qquad \text{for every } \mathbf{w} \text{ in the set.} \tag{18}$$

We apply (18) with $\mathbf{w} = \mathbf{z}^\nu$, which lies in the set. Writing $\boldsymbol{\Delta} := \mathbf{z}^{\nu+1} - \mathbf{z}^\nu$, it reads $\langle -\eta\nabla f(\mathbf{z}^\nu) - \boldsymbol{\Delta}, -\boldsymbol{\Delta}\rangle \le 0$, that is,

$$\langle \nabla f(\mathbf{z}^\nu), \boldsymbol{\Delta}\rangle \le -\frac{1}{\eta}\|\boldsymbol{\Delta}\|^2. \tag{19}$$

The cost is quadratic with curvature at most β, so, by (19),

$$f(\mathbf{z}^{\nu+1}) \le f(\mathbf{z}^\nu) + \langle \nabla f(\mathbf{z}^\nu), \boldsymbol{\Delta}\rangle + \frac{\beta}{2}\|\boldsymbol{\Delta}\|^2 \le f(\mathbf{z}^\nu) - c_\eta\|\boldsymbol{\Delta}\|^2, \qquad c_\eta := \frac{1}{\eta} - \frac{\beta}{2}, \tag{20}$$

and $c_\eta > 0$ because $\eta < 2/\beta$. Adding (20) for $\nu = 0, \dots, T-1$, the costs telescope:

$$c_\eta \sum_{\nu=0}^{T-1} \|\mathbf{z}^{\nu+1} - \mathbf{z}^\nu\|^2 \le f(\mathbf{z}^0) - f(\mathbf{z}^T) \le f(\mathbf{z}^0) - E^\star, \tag{21}$$

since every iterate is feasible. By (21) the steps are square summable and tend to zero, and, as the cost never rises, all iterates stay in the compact sublevel set of Lemma 6.

Let a subsequence $\mathbf{z}^{\nu_i}$ converge to $\bar{\mathbf{z}}$. The steps tend to zero, so $\mathbf{z}^{\nu_i+1} \to \bar{\mathbf{z}}$ as well, and $\bar{\mathbf{z}}$ is feasible because $\mathscr{F}$ is closed. We show in three steps that $\bar{\mathbf{z}}$ cannot be improved to first order.

First, a strictly feasible point. Assumption E gives a $\mathbf{v}$ with $\boldsymbol{\pi}(\bar{\mathbf{z}}) + D\boldsymbol{\pi}(\bar{\mathbf{z}})\mathbf{v} \in \operatorname{int}\mathscr{K}$. For small $\tau > 0$ put $\mathbf{w}_0 = \bar{\mathbf{z}} + \tau\mathbf{v}$. Its linearized blocks at $\bar{\mathbf{z}}$ are

$$\boldsymbol{\pi}(\bar{\mathbf{z}}) + D\boldsymbol{\pi}(\bar{\mathbf{z}})(\mathbf{w}_0 - \bar{\mathbf{z}}) = (1-\tau)\,\boldsymbol{\pi}(\bar{\mathbf{z}}) + \tau\big(\boldsymbol{\pi}(\bar{\mathbf{z}}) + D\boldsymbol{\pi}(\bar{\mathbf{z}})\mathbf{v}\big), \tag{22}$$

a combination of a point of the convex cone $\mathscr{K}$ and a point of its interior, hence in the interior. The steering block and the budgets, imposed exactly, have the same expansion up to $O(\tau^2)$, so they too hold strictly for small τ; the set $\mathscr{D}$ does not depend on $\mathbf{z}$. So $\mathbf{w}_0$ satisfies every constraint of $\mathscr{C}(\bar{\mathbf{z}})$ strictly.

Second, the limit of the nearest-point inequality. Take any $\mathbf{w} \in \mathscr{C}(\bar{\mathbf{z}})$ and $\theta \in (0,1]$. The point $\mathbf{w}_\theta = (1-\theta)\mathbf{w} + \theta\mathbf{w}_0$ satisfies the constraints of $\mathscr{C}(\bar{\mathbf{z}})$ strictly, and these constraints change continuously with the base point, so $\mathbf{w}_\theta \in \mathscr{C}(\mathbf{z}^{\nu_i})$ for all large i. Inequality (18) at step ν_i with this test point is

$$\big\langle \mathbf{z}^{\nu_i} - \eta\nabla f(\mathbf{z}^{\nu_i}) - \mathbf{z}^{\nu_i+1},\ \mathbf{w}_\theta - \mathbf{z}^{\nu_i+1}\big\rangle \le 0.$$

Letting $i \to \infty$ gives $\langle -\eta\nabla f(\bar{\mathbf{z}}), \mathbf{w}_\theta - \bar{\mathbf{z}}\rangle \le 0$, and letting $\theta \to 0$ gives

$$\langle \nabla f(\bar{\mathbf{z}}), \mathbf{w} - \bar{\mathbf{z}}\rangle \ge 0 \qquad \text{for every } \mathbf{w} \in \mathscr{C}(\bar{\mathbf{z}}). \tag{23}$$

For the convex function f on the convex set $\mathscr{C}(\bar{\mathbf{z}})$, (23) says that $\bar{\mathbf{z}}$ is a minimizer of f over $\mathscr{C}(\bar{\mathbf{z}})$.

Third, multipliers. The convex problem of minimizing f over $\mathscr{C}(\bar{\mathbf{z}})$ has the strictly feasible point $\mathbf{w}_0$, so its minimizer $\bar{\mathbf{z}}$ satisfies the usual optimality conditions with multipliers. The steering block and the budgets enter through $\mathscr{D}$, whose normal cone at $\bar{\mathbf{z}}$ is $D\boldsymbol{\pi}_{\mathscr{D}}(\bar{\mathbf{z}})^\top \mathscr{N}_{\mathscr{K}_{\mathscr{D}}}(\boldsymbol{\pi}_{\mathscr{D}}(\bar{\mathbf{z}}))$ by Assumption E and Rockafellar & Wets (1998, Thm. 6.14); here $\boldsymbol{\pi}_{\mathscr{D}}$ stacks their blocks, $\mathscr{K}_{\mathscr{D}}$ is the product of their cones, and $\mathscr{N}_{\mathscr{K}}(\mathbf{y})$, $\mathscr{N}_{\mathscr{K}_{\mathscr{D}}}(\mathbf{y})$ are the normal cones of $\mathscr{K}$, $\mathscr{K}_{\mathscr{D}}$ at $\mathbf{y}$ (the directions pointing out of them). These conditions involve only the values $\boldsymbol{\pi}(\bar{\mathbf{z}})$ and the derivative $D\boldsymbol{\pi}(\bar{\mathbf{z}})$ of the blocks of (P), and they are exactly the KKT conditions (26) of (P) at $\bar{\mathbf{z}}$ (stated in the grid part below).

Proof of Proposition 10, grid part. Given the descent part, it remains to show (i) $f(\mathbf{z}_h) \to E^\star$ as $h \to 0$, with every limit point of $\mathbf{z}_h$ a global minimizer, and (ii) for small h, GlobMP from $\mathbf{z}_h$ has $f(\mathbf{z}^\nu) \downarrow E^\star$ and only globally minimizing limit points.

(i) A global minimizer exists, because $\mathscr{F}$ is closed and the cost has compact sublevel sets (Lemma 6), and by the choice of $\mathscr{B}$ one, $\mathbf{z}^\star$, lies in its interior. The minimizer itself may lie on the boundary of $\mathscr{F}$, so we first move into the interior. Let $\mathbf{v}$ be the vector that Assumption E gives at $\mathbf{z}^\star$. For $\tau \in (0,1)$, Taylor's formula gives

$$\boldsymbol{\pi}(\mathbf{z}^\star + \tau\mathbf{v}) = (1-\tau)\,\boldsymbol{\pi}(\mathbf{z}^\star) + \tau\big(\boldsymbol{\pi}(\mathbf{z}^\star) + D\boldsymbol{\pi}(\mathbf{z}^\star)\mathbf{v}\big) + o(\tau). \tag{24}$$

In (24), the first term lies in the cone $\mathscr{K}$. The second is τ times a point of $\operatorname{int}\mathscr{K}$, so for some $r_0 > 0$ a ball of radius τr_0 about it lies in $\mathscr{K}$; adding a point of the convex cone $\mathscr{K}$ keeps this ball inside $\mathscr{K}$. The remainder $o(\tau)$ is smaller than τr_0 for small τ, so $\boldsymbol{\pi}(\mathbf{z}^\star + \tau\mathbf{v})$ lies in $\operatorname{int}\mathscr{K}$. As $\boldsymbol{\pi}$ is continuous, a closed ball $\overline{\mathbb{B}}(\mathbf{z}^\star + \tau\mathbf{v}, \delta)$ with $\delta > 0$ lies in $\mathscr{F} \cap \mathscr{B}$. Given $\varepsilon > 0$, choosing τ and then δ small makes the cost below $E^\star + \varepsilon$ on this ball, by continuity of f.

Every point lies within distance $h\sqrt{nd_f}/2$ of the grid in the rational chart, and the change to $\mathbf{z}$ coordinates is an affine bijection, so for small h the ball contains a grid point. That point is feasible and costs less than $E^\star + \varepsilon$, so the cheapest feasible grid point does too, while no feasible point costs less than $E^\star$:

$$E^\star \le f(\mathbf{z}_h) < E^\star + \varepsilon. \tag{25}$$

As ε is arbitrary, (25) gives $f(\mathbf{z}_h) \to E^\star$. A limit point of $\mathbf{z}_h$ lies in the closed set $\mathscr{F} \cap \mathscr{B}$ and, by continuity, costs $E^\star$, so it is a global minimizer.

(ii) Let $\varphi(\mathbf{z})$ equal $f(\mathbf{z})$ for feasible $\mathbf{z}$ and $+\infty$ otherwise. The set $\mathscr{F}$ is closed and *semialgebraic*, that is, described by finitely many polynomial equations and inequalities: $\boldsymbol{\pi}$ is polynomial, and the condition "for every $t \in [0,1]$" in each $\mathscr{K}_j$ can be rewritten without the quantifier by the Tarski–Seidenberg theorem. So φ is lower semicontinuous and semialgebraic, and continuous on its closed domain $\mathscr{F}$. For such a function the critical points, roughly the points where the negative cost gradient points out of $\mathscr{F}$ (Clarke critical points), take only finitely many cost values, the *critical values* (Bolte et al., 2007, Cor. 9(ii)). With the normal cone $\mathscr{N}_{\mathscr{K}}$ as above, a *KKT point* of (P) is a feasible $\mathbf{z}$ with

$$-\nabla f(\mathbf{z}) \in D\boldsymbol{\pi}(\mathbf{z})^\top \mathscr{N}_{\mathscr{K}}\big(\boldsymbol{\pi}(\mathbf{z})\big). \tag{26}$$

Every KKT point is such a critical point, since $D\boldsymbol{\pi}(\mathbf{z})^\top \mathscr{N}_{\mathscr{K}}(\boldsymbol{\pi}(\mathbf{z}))$ lies in the normal cone of $\mathscr{F}$ at $\mathbf{z}$ (Rockafellar & Wets, 1998, Thm. 6.14).

Since there are only finitely many critical values, there is a gap above $E^\star$ that contains none. Let $\varepsilon_0 > 0$ be the smaller of $U - E^\star$ and this gap (the distance from $E^\star$ to the next critical value, or $+\infty$ if there is none). By (i), that is, by (25) with $\varepsilon = \varepsilon_0$, there is h_0 with $f(\mathbf{z}_h) < E^\star + \varepsilon_0$ for every $h \le h_0$. By the descent part ((20) and the three steps after it), the costs of GlobMP from $\mathbf{z}_h$ decrease to some limit $\bar{E} \le f(\mathbf{z}_h) < E^\star + \varepsilon_0$, every limit point has cost $\bar{E}$ and is a KKT point, and $\bar{E} \ge E^\star$ because every iterate is feasible. So $\bar{E}$ is a critical value in $[E^\star, E^\star + \varepsilon_0)$, and the only one there is $E^\star$. Hence $\bar{E} = E^\star$ and every limit point is a global minimizer. The iterates are bounded, so their distance to the compact set of global minimizers tends to zero.

The chart of Lemma 3 has irrational entries in general. The control points that the boundary conditions leave free form a rational affine chart of $\mathscr{A}_{\mathrm{bc}}$: the conditions fix the others in terms of these and the data (for one segment, the data alone). For a single segment, a move of the free coordinates moves $\boldsymbol{\Gamma}$ by the same amount in the Frobenius norm and the free coordinates have norm at most $\|\boldsymbol{\Gamma}\|_F$, which is all Proposition 17 uses of Lemma 3; for a spline class such as $\mathscr{S}_N$ the chart is still an affine bijection, which is all the grid part of Proposition 10 needs. In both cases the membership test is exact in rational arithmetic for rational data and breakpoints.

Corollary 11 (random starts)**.** *Assume the hypotheses of Proposition 10, with $\mathscr{B}$ bounded. Draw starting points independently and uniformly at random from $\mathscr{F} \cap \mathscr{B}$, run the exact repair (5) from each (each run understood in its infinite-iteration limit), and keep the cheapest limiting cost. There is a number $\vartheta > 0$, depending on the instance, such that after n_s starts the kept cost is above $E^\star$ with probability at most $(1-\vartheta)^{n_s}$. So, with probability one, some run reaches globally minimizing limit points after finitely many starts.*

Proof of Corollary 11. Part (i) of the grid part of the proof of Proposition 10 gives a ball inside $\mathscr{F} \cap \mathscr{B}$ on which $f < E^\star + \varepsilon_0$. The ball has positive volume, and so has $\mathscr{F} \cap \mathscr{B}$, which contains it; let $\vartheta \in (0,1]$ be the ratio of the two volumes. A uniform start then lands in the ball with probability ϑ. Such a start is feasible and costs less than $E^\star + \varepsilon_0$, and part (ii) of the grid part uses nothing else about its start, so the run from it reaches only global minimizers. The starts are independent, so the probability that none of the first n_s lands in the ball is $(1-\vartheta)^{n_s}$, which tends to zero; hence the probability that every start fails is zero. Membership in $\mathscr{F}$ is decided exactly, so uniform starts can be drawn by rejection from $\mathscr{B}$. By contrast, the grid of Proposition 10 has $O((\operatorname{diam}\mathscr{B}/h)^{nd_f})$ points, so that proposition establishes global convergence, not an efficient search.

A3.2.2 Theorem 1: the continuous-time problem

Let $E(\boldsymbol{\gamma}) = \int_0^1 \|\boldsymbol{\gamma}^{(k)}\|^2 dt$ denote the cost of (P_∞) on $\mathscr{H}^k$. For a margin $\tilde{r}$, the problem ($\mathrm{P}_\infty^{\tilde{r}}$) replaces each r_j by $r_j + \tilde{r}$, each pair separation $r_a + r_b$ by $r_a + r_b + 2\tilde{r}$ and $\bar{r}$ by $\bar{r} - \tilde{r}$, and keeps the boundary data of (P_∞); $E_\infty^\star(\tilde{r})$ denotes its optimal value, $+\infty$ when it is infeasible. For the position blocks of Appendix A2.1, an ellipsoid's radius is raised by $\tilde{r}$ in its own norm (a robot pair keeps the pair rule even when written as an ellipsoid block), and a wall or box face, written with a unit normal, is moved inward by $\tilde{r}$. The *clearance* σ of the endpoints is the largest $\tilde{r}$ for which the start and goal positions are feasible for ($\mathrm{P}_\infty^{\tilde{r}}$); it is positive by Assumption A, and $\tilde{r}_{\max}$ is below $\sigma/2$ and below every obstacle and robot radius. For $|\tilde{r}| \le \tilde{r}_{\max}$ the radii $r_j + \tilde{r}$, the separations and $\bar{r} - \tilde{r}$ stay positive, so every block keeps the form of §2: the collision and pair blocks are convex in the position and the steering set is a ball. The endpoints keep a clearance of at least $\sigma/2$ in ($\mathrm{P}_\infty^{\tilde{r}}$). The feasible sets shrink as $\tilde{r}$ grows, so $E_\infty^\star$ is nondecreasing, and ($\mathrm{P}_\infty^{\tilde{r}}$) is feasible for $\tilde{r} \le 0$. Whenever ($\mathrm{P}_\infty^{\tilde{r}}$) is feasible it has a minimizer, by the existence step of the proof of Proposition 17, which uses only a feasible curve and Assumption B. The chart, the coercivity step and the end correction borrowed from Appendix A3.2.3 do not use Assumption C either.

Lemma 12 (interpolation keeps a margin)**.** *Let $-\tilde{r}_{\max} \le \tilde{r} < \tilde{r}' \le \tilde{r}_{\max}$ and $U \ge 0$, and let $\boldsymbol{\gamma}$ be a curve that is feasible for the stricter problem ($\mathrm{P}_\infty^{\tilde{r}'}$) and costs $E(\boldsymbol{\gamma}) \le U$. Let $\mathbf{q}$ be its piecewise Hermite interpolant of degree $2k-1$ at the nodes i/N, $i = 0, \dots, N$, the piecewise polynomial that matches $\boldsymbol{\gamma}$ and its first $k-1$ derivatives at every node. Then:*

- *(a) $\mathbf{q}$ costs no more than $\boldsymbol{\gamma}$, $E(\mathbf{q}) \le E(\boldsymbol{\gamma})$, and stays uniformly close to it, $\|\mathbf{q} - \boldsymbol{\gamma}\|_\infty \le N^{-(k-1/2)}\sqrt{E(\boldsymbol{\gamma})}$;*
- *(b) if the two margins are far enough apart, $(1 + 2\sqrt{U})N^{-(k-1/2)} \le \tilde{r}' - \tilde{r}$, then $\mathbf{q}$ satisfies every constraint of the looser problem ($\mathrm{P}_\infty^{\tilde{r}}$) strictly at every instant, and if moreover $\ell < k$, then $\mathbf{q} \in \mathscr{S}_N$;*
- *(c) if $\ell \ge k$ and $\boldsymbol{\gamma}$ minimizes ($\mathrm{P}_\infty^{\tilde{r}'}$), there are N_d and C_d, depending only on U and the data, such that for $N \ge N_d$ a correction $\mathbf{p}$ supported on the first and last two pieces makes $\mathbf{q} + \mathbf{p}$ a curve of $\mathscr{S}_N$ with all the boundary data and $E(\mathbf{q}+\mathbf{p}) \le E(\boldsymbol{\gamma}) + C_d/N$, along which, under the condition of (b), every constraint of ($\mathrm{P}_\infty^{\tilde{r}}$) holds strictly at every instant.*

Proof. Interpolation removes part of the kth derivative, so it does not raise the cost, and it moves the curve by an amount that the gap $\tilde{r}' - \tilde{r}$ between the two margins absorbs. When $\ell \ge k$ the interpolant lacks some end data, which a correction at the two ends supplies.

(a) Fix one piece $[a, a + 1/N]$. The error $\mathbf{e} = \boldsymbol{\gamma} - \mathbf{q}$ and its first $k-1$ derivatives vanish at both ends of the piece, because $\mathbf{q}$ matches them there. Let $\boldsymbol{\varepsilon} = \mathbf{e}^{(k)}$ and let $\mathbf{w}$ be any polynomial of degree below k. Integrating $\int \boldsymbol{\varepsilon} \cdot \mathbf{w}\, dt$ by parts moves one derivative from $\mathbf{e}$ onto $\mathbf{w}$; the boundary terms vanish because the derivatives of $\mathbf{e}$ of order below k vanish at both ends. After k such steps we reach $\pm \int \mathbf{e} \cdot \mathbf{w}^{(k)} dt = 0$, since $\mathbf{w}^{(k)} = 0$. So $\boldsymbol{\varepsilon}$ is orthogonal, on the piece, to every polynomial of degree below k, and in particular to $\mathbf{q}^{(k)}$, which is one. From $\boldsymbol{\gamma}^{(k)} = \mathbf{q}^{(k)} + \boldsymbol{\varepsilon}$ and Pythagoras, $\int |\boldsymbol{\gamma}^{(k)}|^2 = \int |\mathbf{q}^{(k)}|^2 + \int |\boldsymbol{\varepsilon}|^2$ on every piece. Adding over the pieces,

$$E(\mathbf{q}) = E(\boldsymbol{\gamma}) - \|\boldsymbol{\varepsilon}\|_{L^2}^2 \le E(\boldsymbol{\gamma}), \qquad \|\boldsymbol{\varepsilon}\|_{L^2} \le \sqrt{E(\boldsymbol{\gamma})}. \tag{27}$$

On the same piece, $\mathbf{e}$ and its first $k-1$ derivatives vanish at the left node a, so Taylor's formula with integral remainder reads

$$\mathbf{e}(t) = \int_a^t \frac{(t-s)^{k-1}}{(k-1)!}\, \boldsymbol{\varepsilon}(s)\, ds. \tag{28}$$

By the Cauchy–Schwarz inequality applied to (28), with $\int_a^t (t-s)^{2k-2}ds = (t-a)^{2k-1}/(2k-1)$, and by (27),

$$\|\mathbf{e}(t)\| \le \frac{(t-a)^{k-1/2}}{(k-1)!\sqrt{2k-1}}\,\|\boldsymbol{\varepsilon}\|_{L^2} \le N^{-(k-1/2)}\sqrt{E(\boldsymbol{\gamma})}, \tag{29}$$

since $t-a \le 1/N$ and $(k-1)!\sqrt{2k-1} \ge 1$.

(b) As $\sqrt{E(\boldsymbol{\gamma})} \le \sqrt{U} < 1+2\sqrt{U}$, the condition of (b) and part (a) give $\|\mathbf{q}-\boldsymbol{\gamma}\|_\infty < \tilde r' - \tilde r$: at every instant each robot moves by less than the gap between the margins. The two problems differ exactly by such a gap. An obstacle distance of at least $r_j + \tilde r'$ along $\boldsymbol{\gamma}$ stays above $r_j + \tilde r$ along $\mathbf{q}$. Two robots each move by less than $\tilde r' - \tilde r$, so their distance changes by less than $2(\tilde r' - \tilde r)$, and a separation of at least $r_a + r_b + 2\tilde r'$ stays above $r_a + r_b + 2\tilde r$. A distance to the reference of at most $\bar r - \tilde r'$ stays below $\bar r - \tilde r$, and a wall offset of at least $\tilde r'$ stays above $\tilde r$. For an ellipsoid with matrix $\boldsymbol{\Sigma}_j$, the motion measured in its own norm is at most $\lambda_{\max}(\boldsymbol{\Sigma}_j)^{1/2}$ times larger, so for that block the condition carries this factor. Finally, $\mathbf{q}$ has $k-1$ continuous derivatives (they match at the nodes), degree $2k-1 \le d$ (raised to d by degree elevation, which does not change the curve), and the derivatives of $\boldsymbol{\gamma}$ of order below k at both endpoints. When $\ell < k$ these are all the boundary data, so $\mathbf{q} \in \mathscr{S}_N$.

(c) Let $\ell \ge k$. Curves with the boundary data of order below k and cost at most U are bounded in $\mathscr{H}^k$ (the coercivity step of the proof of Proposition 17), hence equicontinuous. The endpoints have clearance at least $\sigma/2$ in $(\mathrm{P}_\infty^{\tilde r'})$ and the data are uniformly continuous, so there is $\tau_0 \in (0, 1/4]$, depending only on U and the data, such that along every such curve that is feasible for $(\mathrm{P}_\infty^{\tilde r'})$, every constraint of $(\mathrm{P}_\infty^{\tilde r'})$ holds with clearance at least $\sigma/4$ on $[0,\tau_0]$ and on $[1-\tau_0, 1]$.

By the previous step, adding to $\boldsymbol{\gamma}$ a small multiple $s\boldsymbol{\varphi}$ of a smooth $\boldsymbol{\varphi}$ supported in $(0,\tau_0)$ keeps it feasible, and it keeps the boundary data. Since $\boldsymbol{\gamma}$ is a minimizer, the derivative of $E(\boldsymbol{\gamma}+s\boldsymbol{\varphi})$ at $s=0$ is zero: $\int_0^1 \boldsymbol{\gamma}^{(k)}\cdot\boldsymbol{\varphi}^{(k)}dt = 0$ for every such $\boldsymbol{\varphi}$. This says that $\boldsymbol{\gamma}^{(2k)} = 0$ on $(0,\tau_0)$ in the weak sense, so there $\boldsymbol{\gamma}^{(k)}$ is a polynomial of degree below k and $\boldsymbol{\gamma}$ one of degree at most $2k-1$. The L^2 norm of $\boldsymbol{\gamma}^{(k)}$ on $[0,\tau_0]$ is at most $\sqrt{U}$, and on polynomials of bounded degree all norms are equivalent, so $\boldsymbol{\gamma}^{(k)}$ and all derivatives of $\boldsymbol{\gamma}$ at 0 are bounded by a constant depending only on U, τ_0 and the boundary data. The same holds at $t=1$.

For $N \ge 2/\tau_0$ the first two pieces lie in $[0,\tau_0]$, where $\boldsymbol{\gamma}$ is a polynomial of degree at most $2k-1$, and Hermite interpolation of degree $2k-1$ reproduces such polynomials; so $\mathbf{q} = \boldsymbol{\gamma}$ on the first and last two pieces. There $\mathbf{q}$ has the right derivatives of order below k, but its derivatives of orders $k,\dots,\ell$ at $t=0$ are those of $\boldsymbol{\gamma}$, not the prescribed ones. The missing amounts $\mathbf{a}_s = \boldsymbol{\gamma}_0^{(s)} - \boldsymbol{\gamma}^{(s)}(0)$, $k \le s \le \ell$, are bounded by a number $\bar a$ that does not depend on $\tilde r$, $\tilde r'$ or N.

Let $\mathbf{p}(t) = N^{-k}\mathbf{P}(Nt)$ on $[0, 2/N]$ be the correction of the proof of Lemma 15 (a construction that does not use Assumption C), built from these $\mathbf{a}_s$, together with its mirror image at $t=1$. It has zero derivatives of order below k at both ends of $[0,2/N]$ and the derivatives $\mathbf{p}^{(s)}(0) = \mathbf{a}_s$ for $k \le s \le \ell$, so $\mathbf{q}+\mathbf{p}$ lies in $\mathscr{S}_N$ and has all the boundary data. Moreover $\mathbf{P}$ is bounded by a constant K, depending on k, ℓ and d, times $\sum_s \|\mathbf{a}_s\| \le \bar a$. So the correction moves the curve by at most $N^{-k}K\bar a$, and only on $[0,2/N]\cup[1-2/N,1]$, where the clearance is at least $\sigma/4$. For N large this move is below $\sigma/4$, and every constraint of $(\mathrm{P}_\infty^{\tilde r})$ still holds strictly.

At each end, the cost rises by $\int|\mathbf{p}^{(k)}|^2 + 2\int \mathbf{q}^{(k)}\cdot\mathbf{p}^{(k)}$. Since $\mathbf{p}^{(k)}(t) = \mathbf{P}^{(k)}(Nt)$, substituting $u = Nt$ and applying the Cauchy–Schwarz inequality to the cross term give

$$\int_0^{2/N} |\mathbf{p}^{(k)}|^2 dt = \frac{1}{N}\int_0^2 \|\mathbf{P}^{(k)}\|^2 du, \qquad 2\left|\int_0^{2/N} \mathbf{q}^{(k)}\cdot\mathbf{p}^{(k)}dt\right| \le \frac{2\sqrt{2}}{N}\,\|\boldsymbol{\gamma}^{(k)}\|_{L^\infty(0,\tau_0)}\|\mathbf{P}^{(k)}\|_{L^2(0,2)}. \tag{30}$$

Both terms in (30) are a constant times $1/N$, which gives C_d; N_d is the least $N \ge \max\{4, 2/\tau_0\}$ for which $N^{-k}K\bar a < \sigma/4$. □

Proof of Theorem 1. (i) For $|\tilde r| \le \tilde r_{\max}$ the blocks of $(\mathrm{P}_\infty^{\tilde r})$ have the form of §2: changing a radius by $\tilde r$ does not change the shape of a block. So the descent part of the proof of Proposition 10 applies to $(\mathrm{P}_\infty^{\tilde r})$ over $\mathscr{S}_N$. It needs $\mathbf{H} \succ 0$ and an affine chart; both are shown, without Assumption C, in the proof of Proposition 13, where β is computed in the orthonormal chart of $\mathscr{S}_N$. Hence, from a feasible start, every iterate is feasible for $(\mathrm{P}_\infty^{\tilde r})$ at every instant, and each step lowers f by at least

$(1/\eta - \beta/2)\|\mathbf{z}^{\nu+1} - \mathbf{z}^{\nu}\|^2$. The start $\mathbf{z}_h$ exists as soon as one grid point is feasible, since f has compact sublevel sets in the chart.

(ii) The optimal curve of the stricter problem with margin $\tilde{r} + \delta_N$, interpolated with N pieces, is feasible for margin $\tilde{r}$ and costs at most δ_N more; so do the grid points near it. Hence the cheapest grid point, and every later iterate, costs at most $E^\star_\infty(\tilde{r} + \delta_N) + \delta_N$.

Let $r = \min\{k - \frac{1}{2}, 1\}$, take N_d and C_d from Lemma 12 ($C_d = 0$ if $\ell < k$), and let

$$C = 1 + 2\sqrt{U} \text{ if } \ell < k, \qquad C = \max\left\{1 + 2\sqrt{U},\ 2C_d,\ 2\tilde{r}_{\max} N_d^r\right\} \text{ if } \ell \geq k. \tag{31}$$

If $\ell \geq k$, the hypothesis $\delta_N = CN^{-r} \leq \mu \leq 2\tilde{r}_{\max}$ and $C \geq 2\tilde{r}_{\max} N_d^r$ give $N \geq N_d$, and $C \geq 2C_d$ with $r \leq 1$ gives $C_d/N \leq \frac{1}{2} CN^{-r} < \delta_N$.

Put $\tilde{r}' = \tilde{r} + \delta_N$, which lies in $(\tilde{r}, \tilde{r} + \mu]$. The given curve of cost at most U is feasible for $(\mathrm{P}_\infty^{\tilde{r}+\mu})$, hence for the looser $(\mathrm{P}_\infty^{\tilde{r}'})$, so this problem has a minimizer $\boldsymbol{\gamma}$ with $E(\boldsymbol{\gamma}) = E^\star_\infty(\tilde{r}') \leq U$. Since $(1 + 2\sqrt{U})N^{-(k-1/2)} \leq CN^{-r} = \tilde{r}' - \tilde{r}$, Lemma 12 applies and gives a curve $\tilde{\mathbf{q}}$ of $\mathscr{S}_N$ with all the boundary data ($\mathbf{q}$ if $\ell < k$, $\mathbf{q} + \mathbf{p}$ otherwise), along which every constraint of $(\mathrm{P}_\infty^{\tilde{r}})$ holds strictly at every instant, and with $E(\tilde{\mathbf{q}}) \leq E^\star_\infty(\tilde{r}') + C_d/N < E^\star_\infty(\tilde{r}') + \delta_N$.

The blocks are continuous in the free control points and in t, and positive along $\tilde{\mathbf{q}}$ on the compact interval $[0, 1]$, so they stay positive for all curves whose free control points are close to those of $\tilde{\mathbf{q}}$; the cost is continuous too, and the chart is an affine bijection. So a closed ball about the free coordinates of $\tilde{\mathbf{q}}$ consists of points feasible for $(\mathrm{P}_\infty^{\tilde{r}})$ that cost less than $E^\star_\infty(\tilde{r}') + \delta_N$. For h small compared with the radius of this ball the grid has a point in it, so $\mathbf{z}_h$ exists and

$$f(\mathbf{z}_h) < E^\star_\infty(\tilde{r} + \delta_N) + \delta_N. \tag{32}$$

By (i), no later cost is larger. The threshold on h depends on N and on the unknown minimizer, so Corollary 2 bounds the number of pieces but not the grid spacing.

Every iterate is a curve of $\mathscr{S}_N$ that is feasible for $(\mathrm{P}_\infty^{\tilde{r}})$ and has its boundary data, so it costs at least the optimum $E^\star_\infty(\tilde{r})$ of that problem.

(iii) Take $\tilde{r} = -\mu$. Then $(\mathrm{P}_\infty^{\tilde{r}+\mu})$ is (P_∞) itself, which has a minimizer of cost $E^\star_\infty \leq U$, so (ii) applies. Since $-\mu + \delta_N \leq 0$ and $E^\star_\infty(\cdot)$ is nondecreasing, (32) bounds every cost by

$$E^\star_\infty(-\mu + \delta_N) + \delta_N \leq E^\star_\infty + \delta_N. \tag{33}$$

A curve feasible for $(\mathrm{P}_\infty^{-\mu})$ keeps a distance of at least $r_j - \mu$ from obstacle j, separations of at least $r_a + r_b - 2\mu$ and a distance of at most $\bar{r} + \mu$ from the reference, so it violates no constraint of (P_∞) by more than μ (2μ for a separation).

Let $\mu_i \to 0$, let N_i and h_i be as in (ii), and let $\boldsymbol{\gamma}_i$ be either a curve that GlobMP produces from $\mathbf{z}_{h_i}$ on $(\mathrm{P}_\infty^{-\mu_i})$ or a minimizer of $(\mathrm{P}_\infty^{-\mu_i})$. These curves cost at most $E^\star_\infty + \mu_i$, so they are bounded in $\mathscr{H}^k$. A bounded sequence in $\mathscr{H}^k$ has a subsequence that converges weakly in $\mathscr{H}^k$, and uniformly together with its derivatives below k, to a curve $\boldsymbol{\gamma}$ with the boundary data. At every t the limit keeps a distance of at least r_j from obstacle j, separations of at least $r_a + r_b$ and a distance of at most $\bar{r}$ from the reference, as limits of the loosened inequalities, so it is feasible for (P_∞). By weak lower semicontinuity of E, $E(\boldsymbol{\gamma}) \leq \liminf E(\boldsymbol{\gamma}_i) \leq E^\star_\infty$, and $\boldsymbol{\gamma}$ is a global minimizer. Applied to the minimizers, this gives $E^\star_\infty(-\mu_i) \to E^\star_\infty$. So the costs from $\mathbf{z}_{h_i}$, which lie between $E^\star_\infty(-\mu_i)$ and $E^\star_\infty + \delta_{N_i}$, tend to $E^\star_\infty$.

For the position blocks of Appendix A2.1 the proof is the same, with the bound $K\bar{a}$ that defines N_d multiplied by $\lambda := \max\{1, \max_j \lambda_{\max}(\boldsymbol{\Sigma}_j)^{1/2}\}$ over the ellipsoids, and C, computed with this N_d, multiplied by λ (Lemma 12).

Proof of Corollary 2. Item (i) is the case $\ell < k$ of the proof of Theorem 1(ii), which needs no end correction: the margin condition and the cost bound hold with $\delta_N = (1 + 2\sqrt{U})N^{-(k-1/2)}$. For item (ii), the choice of N gives $\delta_N = CN^{-r} \leq \min\{\varepsilon, \mu\}$, so Theorem 1(iii) applies: the curves violate no constraint of (P_∞) by more than μ (2μ for a separation) and cost at most $E^\star_\infty + \delta_N \leq E^\star_\infty + \varepsilon$.

A3.2.3 GLOBAL MINIMIZERS OVER $\mathscr{S}_N$ UNDER A CONDITION ON CONFIGURATIONS

Theorem 1 does not say that GlobMP reaches a global minimizer over a fixed class $\mathscr{S}_N$. It does for (P_∞) and large N under the following condition.

Assumption C (free configurations)**.** *With g_j as in §2, at every time and feasible position, some* $\mathbf{v}$ *increases every active constraint: $\nabla_{\mathbf{x}} g_j(\mathbf{x},t)^\top \mathbf{v} > 0$ whenever $g_j(\mathbf{x},t) = 0$.*

Assumption C is the Mangasarian–Fromovitz condition on configurations; it rules out jamming and is used only at feasible positions within distance 1 of the points visited by curves of cost at most $E^\star_\infty + 1$ (Appendix A3.3). Under it, GlobMP from $\mathbf{z}_h$ reaches global minimizers over every fine enough class $\mathscr{S}_N$; the proof chains four results and is at the end of this subsection.

Proposition 13 (global minimizers over $\mathscr{S}_N$)**.** *Under Assumptions A, B and C:*

(i) *There is N_0 such that, for every $N \geq N_0$, GlobMP over $\mathscr{S}_N$ on (P_∞) with the repair (5), a step $0 < \eta < 2/\beta$ (β computed for $\mathscr{S}_N$) and the start $\mathbf{z}_h$ with h small enough produces curves feasible at every instant, with nonincreasing cost, and all its limit points are global minimizers over $\mathscr{S}_N$.*

(ii) *For $N \geq N_0$, the optimal cost $E^\star_N$ over $\mathscr{S}_N$ satisfies $E^\star_\infty \leq E^\star_N \leq E^\star_\infty + CN^{-\min\{k-1/2,\,1\}}$ with C independent of N, and as $N \to \infty$ subsequences of these minimizers converge uniformly to global minimizers of (P_∞).*

Lemma 14 (pushing into the interior)**.** *Fix a time and let the g_j be C^1 functions of the position, so that $\mathscr{X} := \{\mathbf{x} \in \mathbb{R}^n : g_j(\mathbf{x}) \geq 0 \ \forall j\}$ is the set of feasible positions. Assume Assumption C at this time: at every $\mathbf{x} \in \mathscr{X}$ some direction $\mathbf{v}$ increases all active constraints, $\nabla g_j(\mathbf{x}) \cdot \mathbf{v} > 0$ whenever $g_j(\mathbf{x}) = 0$. Assume also that the start and goal positions are strictly feasible. Then for every compact $\mathscr{Q} \subset \mathbb{R}^n$ there are numbers $\mu_0, G > 0$ and, for each $0 < \mu \leq \mu_0$, a smooth map $\mathbf{\Phi}_\mu : \mathbb{R}^n \to \mathbb{R}^n$, a* push*, such that:*

(a) $\mathbf{\Phi}_\mu$ is the identity near the start and goal positions;
(b) it pushes feasible points at least μ inside: $g_j(\mathbf{\Phi}_\mu(\mathbf{x})) \geq \mu$ for all j and all $\mathbf{x} \in \mathscr{X} \cap \mathscr{Q}$;
(c) it moves points only by order μ: $\|\mathbf{\Phi}_\mu - \mathrm{id}\|_{C^i(\mathscr{Q})} \leq a_i \mu$ for every i, with a_i independent of μ;
(d) a point $\mathbf{y}$ within distance 1 of $\mathscr{Q}$ with $g_j(\mathbf{y}) \geq \mu$ for all j has the whole open ball $\mathbb{B}(\mathbf{y}, \mu/G)$ inside $\mathscr{X}$.

Proof. Assumption C gives, near each point, a direction that increases every nearly active constraint. These directions are combined into one smooth vector field $\boldsymbol{\phi}$, and $\mathbf{\Phi}_\mu$ is its flow for time μ; small clearances grow by about 2μ, and points move by $O(\mu)$.

Let $\mathscr{Q}'$ be the closed 1-neighbourhood of $\mathscr{Q}$. For $\mathbf{x} \in \mathscr{X} \cap \mathscr{Q}'$, let $\mathscr{J}(\mathbf{x}) = \{j : g_j(\mathbf{x}) = 0\}$ be its active constraints, and choose $\mathbf{v}_\mathbf{x}$ with $\nabla g_j(\mathbf{x}) \cdot \mathbf{v}_\mathbf{x} > 0$ for $j \in \mathscr{J}(\mathbf{x})$. Scaling $\mathbf{v}_\mathbf{x}$ and using continuity gives an open neighbourhood $\mathscr{O}_\mathbf{x}$ of $\mathbf{x}$ on which $\nabla g_j \cdot \mathbf{v}_\mathbf{x} \geq 2$ for the active j and $g_j \geq \kappa_\mathbf{x}$ for the inactive j, with some $\kappa_\mathbf{x} > 0$.

The set $\mathscr{X} \cap \mathscr{Q}'$ is compact, so finitely many of these neighbourhoods, $\mathscr{O}_{\mathbf{x}_1}, \ldots, \mathscr{O}_{\mathbf{x}_L}$, cover it. Let $\kappa > 0$ be below every $\kappa_{\mathbf{x}_l}$ and below $\frac{1}{2}\min_j g_j$ on small balls about the start and goal positions. Take a smooth partition of unity $\sum_l \varpi_l = 1$ on $\mathscr{X} \cap \mathscr{Q}'$, with each $\varpi_l \geq 0$ zero outside $\mathscr{O}_{\mathbf{x}_l}$, and a smooth cutoff χ that vanishes on those balls and equals 1 wherever some $g_j < \kappa$. Put

$$\boldsymbol{\phi} = \chi \sum_{l=1}^{L} \varpi_l \, \mathbf{v}_{\mathbf{x}_l}. \tag{34}$$

Now take $\mathbf{y} \in \mathscr{X} \cap \mathscr{Q}'$ with $g_j(\mathbf{y}) < \kappa$. If $\varpi_l(\mathbf{y}) > 0$, then $\mathbf{y} \in \mathscr{O}_{\mathbf{x}_l}$, and j must be active at $\mathbf{x}_l$ (an inactive j would have $g_j(\mathbf{y}) \geq \kappa_{\mathbf{x}_l} > \kappa$), so $\nabla g_j(\mathbf{y}) \cdot \mathbf{v}_{\mathbf{x}_l} \geq 2$. Averaging with the weights ϖ_l in (34) gives

$$\nabla g_j(\mathbf{y}) \cdot \boldsymbol{\phi}(\mathbf{y}) \geq 2 \qquad \text{whenever } \mathbf{y} \in \mathscr{X} \cap \mathscr{Q}' \text{ and } g_j(\mathbf{y}) < \kappa. \tag{35}$$

Let $\mathbf{\Phi}_\mu$ be the time-μ flow of $\boldsymbol{\phi}$: the point reached at time μ by moving with velocity $\boldsymbol{\phi}$. Let $M_\phi = \sup\|\boldsymbol{\phi}\|$ and let G bound $\|\nabla g_j\|$ on the closed 2-neighbourhood of $\mathscr{Q}$. Along a trajectory started at $\mathbf{x} \in \mathscr{X} \cap \mathscr{Q}$, the value of g_j changes at rate $\nabla g_j \cdot \boldsymbol{\phi}$: by (35) it increases at rate at least 2

while it is below κ, and it decreases at rate at most GM_ϕ otherwise. Take

$$\mu \le \mu_0 := \min\Big\{\frac{\kappa}{4},\ \frac{\kappa}{2GM_\phi},\ \frac{1}{M_\phi},\ G\Big\}. \tag{36}$$

The trajectory moves by at most $M_\phi\mu \le 1$, so it stays in $\mathscr{Q}'$. A value that starts below κ grows, and a value that starts at or above κ can lose at most $GM_\phi\mu \le \kappa/2$; so the trajectory stays in $\mathscr{X}$, and at time μ every g_j is at least $\min\{2\mu, \kappa/2\} \ge \mu$. This is (b), and (a) holds because $\boldsymbol{\phi}$ vanishes near the start and goal.

For (c), write

$$\boldsymbol{\Phi}_\mu - \mathrm{id} = \int_0^\mu \boldsymbol{\phi}\circ\boldsymbol{\Phi}_s\, ds. \tag{37}$$

The derivatives of $\boldsymbol{\Phi}_s$ up to order i are bounded on $\mathscr{Q}$ uniformly in $s \le \mu_0$, since they solve the flow equation differentiated in $\mathbf{x}$. So the integrand in (37) is bounded in $C^i(\mathscr{Q})$ by a constant a_i, and the integral over an interval of length μ is at most $a_i\mu$.

For (d), the gradient bound $\|\nabla g_j\| \le G$ holds on the ball $\mathbb{B}(\mathbf{y},1)$, which lies in the 2-neighbourhood of $\mathscr{Q}$. So g_j can drop by at most $G\rho$ over a distance ρ, and $g_j(\mathbf{y}) \ge \mu$ gives $g_j > 0$ on $\mathbb{B}(\mathbf{y},\mu/G)$, as $\mu/G \le 1$. Only points of $\mathscr{X}\cap\mathscr{Q}'$ were used, so the condition is needed only there. □

Lemma 15 (rate in the number of pieces)**.** *Write $E_N^\star$ for the optimum over $\mathscr{S}_N$, whose curves carry the boundary data. Under the hypotheses of Proposition 13, the best spline with N pieces is almost as cheap as the best curve: there are C and N_1, depending on the instance, such that for all $N \ge N_1$*

$$E_\infty^\star \le E_N^\star \le E_\infty^\star + CN^{-(k-1/2)} \ \text{ if } \ell < k, \qquad E_\infty^\star \le E_N^\star \le E_\infty^\star + CN^{-\min\{k-1/2,\,1\}} \ \text{ if } \ell \ge k.$$

Proof of Lemma 15. An optimal curve may touch the obstacles, so its interpolant may violate the constraints. The optimal curve is therefore first pushed inward by μ, at cost $O(\mu)$, and then interpolated with N pieces, which does not raise the cost and moves the curve by $O(N^{-(k-1/2)})$; choosing μ of the same order makes the push absorb this displacement.

Every spline of $\mathscr{S}_N$ is a curve of $\mathscr{H}^k$ with the boundary data, so $E_N^\star \ge E_\infty^\star$.

Let $\boldsymbol{\gamma}^\star$ minimize (P_∞); it exists by the existence step of the proof of Proposition 17. Apply the pushes $\boldsymbol{\Phi}_\mu$ of Lemma 14 to every block, including the steering block. When the centres and the reference move, we use the same construction in space and time together: on the compact set $\{(\mathbf{x},t) \in \mathscr{Q}' \times [0,1] : g_j(\mathbf{x},t) \ge 0\ \forall j\}$, with $\mathscr{Q}'$ the closed 1-neighbourhood of a compact set containing $\boldsymbol{\gamma}^\star([0,1])$, with the cutoff χ vanishing near $(\boldsymbol{\gamma}_0,0)$ and $(\boldsymbol{\gamma}_1,1)$, and with a partition of unity in $(\mathbf{x},t)$. This gives a field $\boldsymbol{\phi}(\mathbf{x},t)$ that is smooth in $(\mathbf{x},t)$ (the partition of unity is smooth even where the centres are only continuous) and maps $\boldsymbol{\Phi}_\mu(\cdot,t)$ that push at each instant. They are the identity near the endpoints, give every clearance at least μ, and satisfy $\|\boldsymbol{\Phi}_\mu - \mathrm{id}\|_{C^k} \le a_k\mu$ on a compact set containing the curve; their derivatives in t are also $O(\mu)$. Put $\boldsymbol{\gamma}_\mu(t) = \boldsymbol{\Phi}_\mu(\boldsymbol{\gamma}^\star(t),t)$.

By the chain rule, $\boldsymbol{\gamma}_\mu^{(k)} - \boldsymbol{\gamma}^{\star(k)}$ is a sum of terms, each containing a derivative of $\boldsymbol{\Phi}_\mu - \mathrm{id}$ (of size $O(\mu)$), bounded derivatives of $\boldsymbol{\gamma}^\star$ of order below k, and at most one factor $\boldsymbol{\gamma}^{\star(k)} \in L^2$. So, expanding the square in the cost,

$$\|\boldsymbol{\gamma}_\mu^{(k)} - \boldsymbol{\gamma}^{\star(k)}\|_{L^2} \le c_1\mu, \qquad E(\boldsymbol{\gamma}_\mu) \le E_\infty^\star + c_2\mu. \tag{38}$$

By Lemma 14(d), every point of $\boldsymbol{\gamma}_\mu$ is at distance at least μ/G from the boundary of the feasible region.

Let $\mathbf{q}$ be the piecewise Hermite interpolant of $\boldsymbol{\gamma}_\mu$ of degree $2k-1$ at the $N+1$ nodes. By (27) and (29), applied to $\boldsymbol{\gamma}_\mu$,

$$E(\mathbf{q}) \le E(\boldsymbol{\gamma}_\mu), \qquad \|\boldsymbol{\gamma}_\mu - \mathbf{q}\|_\infty \le N^{-(k-1/2)}\|\boldsymbol{\gamma}_\mu^{(k)}\|_{L^2} \le c_3N^{-(k-1/2)}, \tag{39}$$

where $c_3 = \sqrt{E_\infty^\star + c_2\mu_0}$ bounds $\|\boldsymbol{\gamma}_\mu^{(k)}\|_{L^2}$ for every $\mu \le \mu_0$. The interpolant matches the derivatives of order below k at every node, so when $\ell < k$ it has the boundary data and lies in $\mathscr{S}_N$ (after degree elevation if $d > 2k-1$).

Take

$$\mu = 2Gc_3N^{-(k-1/2)}, \tag{40}$$

which is at most μ_0 for large N. Then $\mathbf{q}$ stays within $c_3N^{-(k-1/2)} = \mu/(2G) < \mu/G$ of $\boldsymbol{\gamma}_\mu$, so it stays inside the feasible region at every instant, and, by (38), (39) and (40), it costs at most $E^\star_\infty + c_2\mu = E^\star_\infty + 2c_2Gc_3N^{-(k-1/2)}$. So for $\ell < k$ the lemma holds with $C = 2c_2Gc_3$ and N_1 the least N with $2Gc_3N^{-(k-1/2)} \le \mu_0$.

If $\ell \ge k$, then $\mathbf{q}$ meets only the end conditions of order below k. No constraint is active near an endpoint, so there $\boldsymbol{\gamma}^\star$, being optimal, solves $\boldsymbol{\gamma}^{\star(2k)} = 0$: it is a polynomial of degree at most $2k-1$. The push $\boldsymbol{\Phi}_\mu$ is the identity there, so for large N the interpolant equals $\boldsymbol{\gamma}^\star$ on the end pieces. The missing amounts $\mathbf{a}_s = \boldsymbol{\gamma}_0^{(s)} - \boldsymbol{\gamma}^{\star(s)}(0)$, $k \le s \le \ell$, do not depend on N.

Let $\mathbf{P}$ be a spline on $[0,2]$ with two unit pieces of degree d and $k-1$ continuous derivatives, with zero derivatives of order below k at 0 and at 2, and with $\mathbf{P}^{(s)}(0) = N^{k-s}\mathbf{a}_s$ for $k \le s \le \ell$. It exists: its first piece is the Taylor polynomial with these values ($d \ge \ell$), and the second joins it to zero by Hermite interpolation ($d \ge 2k-1$). Since $N^{k-s} \le 1$ for $s \ge k$, $\mathbf{P}$ is bounded uniformly in N. The rescaled $\mathbf{p}(t) = N^{-k}\mathbf{P}(Nt)$ on $[0,2/N]$ has $\mathbf{p}^{(s)}(0) = \mathbf{a}_s$, so adding it, and its mirror image at $t = 1$, gives a curve of $\mathscr{S}_N$ with all the boundary data. It moves the curve by $O(N^{-k})$, inside the clearance near the endpoints. It raises the cost by $N^{-1}\int_0^2 \|\mathbf{P}^{(k)}\|^2$ (substitute $u = Nt$) plus a cross term at most $2\|\boldsymbol{\gamma}^{\star(k)}\|_{L^\infty(0,2/N)}(2/N)^{1/2}\|\mathbf{p}^{(k)}\|_{L^2}$ (by the Cauchy–Schwarz inequality), and both are $O(N^{-1})$. Together with the $\ell < k$ bound this gives $E^\star_N \le E^\star_\infty + CN^{-\min\{k-1/2,1\}}$.

Lemma 16 (Assumption C gives Assumption E)**.** *Under the hypotheses of Lemma 15, fix any U. For all large enough N, every feasible curve of $\mathscr{S}_N$ with cost at most U satisfies Robinson's condition: some change $\mathbf{v}$ of its control points makes every block strictly feasible to first order, as in Assumption E.*

Proof of Lemma 16. The push field of Lemma 14, evaluated along the curve, increases every small clearance. It is not a change of control points of the spline, but its interpolant is, and on short pieces the interpolant is close enough to retain this property.

Let $\boldsymbol{\phi}(\mathbf{x},t)$ be the space-time field built from Lemma 14 in the proof of Lemma 15, on a compact set that contains every curve of cost at most U; such curves are bounded by the coercivity step of the proof of Proposition 17, which uses Assumption B. The field is of class C^k, vanishes near the endpoints, and has $\nabla_{\mathbf{x}} g_j \cdot \boldsymbol{\phi} \ge 2$ wherever $g_j < \kappa$, for some $\kappa > 0$. Let $\boldsymbol{\gamma}$ be a feasible curve of the class with cost at most U.

Let $\mathbf{v}$ be the piecewise Hermite interpolant of $t \mapsto \boldsymbol{\phi}(\boldsymbol{\gamma}(t),t)$ at the nodes. Curves of cost at most U are equicontinuous, so for large N their first and last pieces lie where $\boldsymbol{\phi}$ vanishes. Then $\mathbf{v}$ vanishes on those pieces, so it has zero boundary data of every order, and it is an admissible change of the control points.

By the chain rule, $\|(\boldsymbol{\phi}\circ\boldsymbol{\gamma})^{(k)}\|_{L^2}$ is bounded by a constant c_4 that depends only on U. So, by (29),

$$\|\mathbf{v} - \boldsymbol{\phi}\circ\boldsymbol{\gamma}\|_\infty \le c_4N^{-(k-1/2)}, \tag{41}$$

which is below $1/G$ (G as in Lemma 14) for large N. Since $\|\nabla_{\mathbf{x}} g_j\| \le G$, this changes $\nabla_{\mathbf{x}} g_j \cdot \mathbf{v}$ by less than 1 from $\nabla_{\mathbf{x}} g_j \cdot \boldsymbol{\phi}$, so $\nabla_{\mathbf{x}} g_j \cdot \mathbf{v} \ge 1$ wherever $g_j < \kappa$ along $\boldsymbol{\gamma}$.

For small $s > 0$ consider the first-order change $g_j + s\nabla_{\mathbf{x}} g_j \cdot \mathbf{v}$, a piecewise polynomial in t with coefficients $\boldsymbol{\pi}_j + D\boldsymbol{\pi}_j(s\mathbf{v})$. Where $g_j < \kappa$ it is at least $g_j + s > 0$, since $g_j \ge 0$ along the feasible curve; where $g_j \ge \kappa$ it is at least $\kappa - sG\sup\|\mathbf{v}\| > 0$ for small s. So it is positive on all of $[0,1]$, that is, its coefficients lie in $\operatorname{int}\mathscr{K}_j$. This is Assumption E at $\boldsymbol{\gamma}$, with the vector $s\mathbf{v}$.

The degree-refinement procedure. At level r, take degrees $d_r \uparrow \infty$ satisfying Assumption D, grid spacing $h_r = (2d_r)^{-(k+1)}/r$, and a box $\mathscr{B}_r$ in the free control points (the rational chart after the proof of Proposition 10), centred at zero with half-width $r\sqrt{d_r+1}$. Write $d_f(r) = d_r + 1 - 2(\ell+1)$. Degree elevation keeps the previous output without changing its curve or cost. Algorithm 1 may run for finitely many iterations, since a trial replaces its seed only when it is feasible and no more expensive. Membership in the grid candidate set is tested exactly over $[0,1]$, piece by piece, for

Algorithm 4 Degree and seed refinement for (P_∞)

Require: degrees d_r with charts $(\mathbf{\Gamma}_{\text{aff}}^{(r)}, \mathbf{N}_r)$, spacings h_r, boxes $\mathscr{B}_r$; levels R; finite iteration caps for Algorithm 1

$\mathbf{\Gamma}_{\text{best}} \leftarrow$ none
for $r = 1, \dots, R$ **do**
  $\mathscr{G}_r \leftarrow$ feasible degree-d_r curves from $h_r \mathbb{Z}^{n d_f(r)} \cap \mathscr{B}_r$
  if $\mathbf{\Gamma}_{\text{best}} \neq$ none **then** add its degree-d_r elevation to $\mathscr{G}_r$
  if $\mathscr{G}_r = \emptyset$ **then continue**
  $\mathbf{\Gamma}_{\text{seed}} \leftarrow \arg\min_{\mathbf{\Gamma} \in \mathscr{G}_r} \text{cost}(\mathbf{\Gamma})$
  $\mathbf{\Gamma}_{\text{trial,free}} \leftarrow$ Algorithm 1 with chart $(\mathbf{\Gamma}_{\text{aff}}^{(r)}, \mathbf{N}_r)$, initialized at $(\mathbf{\Gamma}_{\text{seed}} - \mathbf{\Gamma}_{\text{aff}}^{(r)})\mathbf{N}_r$
  $\mathbf{\Gamma}_{\text{trial}} \leftarrow \mathbf{\Gamma}_{\text{aff}}^{(r)} + \mathbf{\Gamma}_{\text{trial,free}} \mathbf{N}_r^\top$
  $\mathbf{\Gamma}_{\text{best}} \leftarrow$ cheaper feasible member of $\{\mathbf{\Gamma}_{\text{seed}}, \mathbf{\Gamma}_{\text{trial}}\}$
end for
return $\mathbf{\Gamma}_{\text{best}}$ *or no feasible curve found within the R levels*

rational piecewise-polynomial data and breakpoints by Sturm sequences (an exact count of the real roots of a polynomial in an interval).

Proposition 17 (degree refinement)**.** *Assume Assumptions A, B and C. Once Algorithm 4 has found a first feasible curve, its output costs never increase and converge to the optimum $E_\infty^\star$; the optimum $E^\star(d)$ of (P) at degree d also decreases to $E_\infty^\star$; and $E_\infty^\star$ is attained by some curve. Every limit point of the output curves (weakly in $\mathscr{H}^k$, and uniformly together with the derivatives below k) is a global minimizer of (P_∞).*

Proof of Proposition 17. A smooth near-optimal curve with a positive distance from every constraint boundary is approximated by a polynomial, and the polynomial by a grid point; both remain feasible and near-optimal, and the refinement does not increase the cost of the grid point. We prove the conclusions for constraints g_j and ψ_j that depend continuously on t and on derivatives of order below k, assuming two properties: a *coercive* cost (curves of bounded cost are bounded) and the *approach from inside* (for every $\varepsilon > 0$ some C^∞ curve with the boundary data costs at most $E_\infty^\star + \varepsilon$ and has every constraint bounded below by a positive number). The last two steps derive these two properties from the hypotheses. As in Appendix A3.2.2, E denotes the cost.

Take a sequence of feasible curves whose costs tend to $E_\infty^\star$. By coercivity it is bounded in $\mathscr{H}^k$, so a subsequence converges weakly in $\mathscr{H}^k$ and, with its derivatives below k, uniformly. The limit keeps the boundary data and the constraints, which depend only on those derivatives. The cost E, a continuous convex quadratic, is weakly lower semicontinuous, so the limit costs at most $E_\infty^\star$ and is a minimizer. The approach from inside shows that the infimum over the smooth curves C^k is the same number.

Fix $\varepsilon > 0$ and a C^∞ curve $\boldsymbol{\gamma}_\varepsilon$ from the approach from inside, with $E(\boldsymbol{\gamma}_\varepsilon) \le E_\infty^\star + \varepsilon$ and every constraint bounded below by a positive number. By continuity there is $\delta > 0$ such that every curve with the boundary data within 2δ of $\boldsymbol{\gamma}_\varepsilon$ in C^k is feasible and costs at most $E_\infty^\star + 2\varepsilon$. The Bernstein polynomials of $\boldsymbol{\gamma}_\varepsilon$ converge to it in $C^{\max\{k,\ell\}}$ (Lorentz, 1986). Their boundary data are slightly off; adding the Hermite polynomial of degree $2\ell + 1$ that corrects them, whose size tends to zero, gives curves $\boldsymbol{\gamma}_d$ of degree d with the boundary data and $\|\boldsymbol{\gamma}_d - \boldsymbol{\gamma}_\varepsilon\|_{C^k} \le \delta$ for all $d \ge d(\varepsilon)$.

For a degree-d curve with control points $\mathbf{\Gamma}$, the ith derivative is

$$\boldsymbol{\gamma}^{(i)} = \frac{d!}{(d-i)!} \sum_l (\Delta^i \mathbf{\Gamma})_l \, [\mathbf{b}_{d-i}]_l, \tag{42}$$

where Δ takes differences of consecutive control points. Each $(\Delta^i \mathbf{\Gamma})_l$ has norm at most $2^i \|\mathbf{\Gamma}\|_F$, the weights $[\mathbf{b}_{d-i}]_l$ are nonnegative and sum to one, and $d!/(d-i)! \le d^i$, so (42) gives

$$\|\boldsymbol{\gamma}^{(i)}\|_\infty \le (2d)^i \|\mathbf{\Gamma}\|_F. \tag{43}$$

This applies to the difference of two curves, since the curve depends linearly on its control points.

For one segment the rational chart (after the proof of Proposition 10) is an isometry. So moving the free coordinates of $\boldsymbol{\gamma}_{d_r}$ to a nearest point of $h_r\mathbb{Z}^{nd_f(r)}$, a move of at most $h_r\sqrt{n(d_r+1)}/2$, moves the curve, by (43) and the choice of h_r, by at most

$$(2d_r)^k h_r\sqrt{n(d_r+1)}/2 \le \sqrt{n}/(2r) \qquad \text{in } C^k,$$

which is below δ for large r. The grid point must also lie in the box $\mathscr{B}_r$. The free coordinates of $\boldsymbol{\gamma}_d$ are entries of its control points, which are bounded by some C_ε independent of d, so they have norm at most $\sqrt{n(d+1)}C_\varepsilon$, and for large r the box, of half-width $r\sqrt{d_r+1}$, contains the grid point.

Hence for $r \ge r(\varepsilon)$ the seed costs at most $E^\star_\infty + 2\varepsilon$, and so does the output, which is never more expensive than the seed. Every output is feasible at degree d_r, so it costs at least $E^\star(d_r) \ge E^\star_\infty$. The previous output, written at the higher degree, is a candidate at each level, so the costs never increase. Also $E^\star(d)$ is nonincreasing in d (degree elevation keeps every curve) and lies between $E^\star_\infty$ and the output costs. As ε was arbitrary, both tend to $E^\star_\infty$.

The outputs have bounded cost, hence are bounded in $\mathscr{H}^k$. A weak limit is feasible, as in the existence step, and costs at most $\lim E = E^\star_\infty$, so it is a global minimizer.

To prove coercivity, suppose to the contrary that some curves $\boldsymbol{\gamma}_i$ with the boundary data have bounded cost but $\|\boldsymbol{\gamma}_i\|_{\mathscr{H}^k} \to \infty$. Divide by the norm. The normalized curves have norm one, kth derivatives tending to zero, and boundary data tending to zero. By the compact embedding of $\mathscr{H}^k$ in $\mathscr{H}^{k-1}$, a subsequence converges in $\mathscr{H}^k$ to a curve of norm one with zero kth derivative, a polynomial of degree below k, whose derivatives of orders $0,\dots,\min\{\ell, k-1\}$ vanish at both ends. Under Assumption B, $2(\ell+1) \ge k$, the argument in the proof of Lemma 5 shows that such a polynomial is zero, which contradicts norm one.

For the approach from inside, the position constraints satisfy the Mangasarian–Fromovitz condition at every t and the endpoints are strictly feasible. Let $\boldsymbol{\gamma}^\star$ minimize over $\mathscr{H}^k$ and apply the pushes $\boldsymbol{\Phi}_\mu$ of Lemma 14. The curve $t \mapsto \boldsymbol{\Phi}_\mu(\boldsymbol{\gamma}^\star(t), t)$ keeps the boundary data (the push is the identity near the endpoints), has every $g_j \ge \mu$, and costs at most $E^\star_\infty + c_2\mu$ by the chain-rule estimate in the proof of Lemma 15. Smoothing it in $\mathscr{H}^k$ and restoring the boundary data by a small Hermite correction keeps the cost within ε and every $g_j \ge \mu/2$. If $\ell \ge k$, the end correction of Lemma 15, with a smooth $\mathbf{P}$ on a layer of width w, adds the end data of orders $k,\dots,\ell$ at cost $O(w)$ and moves the curve by $O(w^k)$, so for small w every $g_j \ge \mu/4$.

Proof of Proposition 13. The proof chains four results. On fine splines, Assumption C gives Robinson's condition (Lemma 16). The fixed-degree result then makes GlobMP find the cheapest spline (Proposition 10). The rate lemma bounds how far that cost is from $E^\star_\infty$ (Lemma 15). And compactness passes to the limit $N \to \infty$. A global minimizer of (P_∞) exists by the existence step of the proof of Proposition 17, which needs only a feasible curve and Assumption B.

Let $U = E^\star_\infty + 1$. Let N_0 exceed the thresholds of Lemmas 15 and 16 for this U, and satisfy $CN_0^{-\min\{k-1/2,1\}} < 1$; then $E^\star_N < U$ for $N \ge N_0$ by Lemma 15.

By Lemma 16, for $N \ge N_0$ Assumption E holds on the feasible curves of $\mathscr{S}_N$ that cost at most U. Since $U > E^\star_N$, this set contains the global minimizers over $\mathscr{S}_N$ and every limit point of GlobMP from a start of cost at most U, which is all that Proposition 10 uses.

Assumption D enters the proofs of Proposition 10 and Corollary 11 only through $\mathbf{H} \succ 0$ and an affine chart of the boundary equations. First, $\mathbf{H} \succ 0$ on $\mathscr{S}_N$: a spline with $k-1$ continuous derivatives and zero kth derivative is a single polynomial of degree below k, and under Assumption B the zero boundary data force it to be zero (Lemma 5). Second, the chart: on $\mathscr{S}_N$ the boundary and junction equations are linear, and they are consistent for $N \ge 3$, and for every N when $\ell < k$ (a middle piece of degree at least $2k-1$ joins the end pieces); so the chart exists. In it, $\beta = 2\lambda_{\max}(\mathbf{N}^\top\mathbf{Q}\mathbf{N})$, with $\mathbf{Q}$ the block-diagonal cost matrix of the pieces (each block N^{2k-1} times the cost matrix of one piece on $[0,1]$) and $\mathbf{N}$ an orthonormal basis of the directions that the boundary and junction equations leave free; $\mathbf{z}_h$ is mapped to these coordinates before GlobMP runs.

(i) Proposition 10, applied over $\mathscr{S}_N$ with $\mathscr{B}$ the whole chart, which contains a global minimizer in its interior, shows that GlobMP from $\mathbf{z}_h$ has only global minimizers of $\mathscr{S}_N$ as limit points. Corollary 11 gives the same, with probability one after finitely many starts, for independent uniform starts in a

bounded box that contains a global minimizer in its interior. These limit points are feasible at every instant and cost $E_N^\star$.

(ii) The bounds on $E_N^\star$ are Lemma 15. As $N \to \infty$ the minimizers over $\mathscr{S}_N$ have bounded cost, so by Assumption B they are bounded in $\mathscr{H}^k$ (the coercivity step of the proof of Proposition 17). A subsequence converges weakly in $\mathscr{H}^k$ and uniformly with the derivatives below k. The limit is feasible and has the boundary data of order below k, and, since the cost cannot jump up in a weak limit, it costs at most $\lim E_N^\star = E_\infty^\star$. So it is a global minimizer of (P_∞).

A3.3 Scope of the assumptions and stability

This subsection records which parts of Assumptions A and B the proofs need and which can be weakened, when the optimal value is stable, and when Assumption C, used only in Propositions 13 and 17, holds and why Proposition 13 cannot drop it. Theorem 1 needs only Assumptions A and B.

Piecewise-polynomial data. Assumption A asks for continuous, piecewise-polynomial obstacle centres and reference, which covers the piecewise-linear reference (7) of a learned policy. The exact test and the finite-critical-values step use polynomial data. The cone test stays exact for rational data and breakpoints: on each interval of a common partition of $[0,1]$ into the pieces of the curve and of the data, the block is a polynomial in t, and $\mathscr{K}_j$ becomes the product of the cones of these intervals, still closed and convex. The problem over $\mathscr{S}_N$ stays semialgebraic, which is all the finite-critical-values step needs. The push, the rate and Lemma 16 use only continuity of the centres and the reference: the field $\boldsymbol{\phi}$ of Lemma 14 is a smooth partition of unity times fixed vectors, and continuity of g_j and $\nabla_{\mathbf{x}} g_j$ in $(\mathbf{x},t)$ is all its construction needs. Beyond the exact test, the proof of Theorem 1 uses only continuity of the data, while the finiteness sketch for unstable margins below also uses polynomial data.

Bounded curves instead of Assumption B. Assumption B enters only through boundedness: curves of bounded cost are bounded. The descent step uses the upper curvature β and not α. When the steering constraint is present ($\bar{r} < \infty$), every feasible curve has $\|\boldsymbol{\gamma}\|_\infty \le \|\boldsymbol{\rho}\|_\infty + \bar{r}$, and the coercivity argument of Proposition 17 applies unchanged: a normalized sequence with vanishing kth derivatives and vanishing sup norm cannot converge to a polynomial of norm one. So the conclusions of Theorem 1 and Proposition 13 hold with $\bar{r} < \infty$ in place of Assumption B, with f convex rather than strongly convex over $\mathscr{S}_N$ and compactness supplied by the bounded feasible set.

Stability of the optimal value. The value $E_\infty^\star(\tilde{r})$ of Appendix A3.2.2 does not decrease as $\tilde{r}$ grows, and it is left-continuous on $(-\tilde{r}_{\max}, \tilde{r}_{\max}]$: minimizers for margins $\tilde{r}_i \uparrow \tilde{r}$ cost at most $E_\infty^\star(\tilde{r})$, so, as in the proof of Theorem 1(iii), a subsequence converges weakly in $\mathscr{H}^k$ and uniformly to a curve feasible for ($\mathrm{P}_\infty^{\tilde{r}}$) of cost at most $\lim_i E_\infty^\star(\tilde{r}_i)$ (if ($\mathrm{P}_\infty^{\tilde{r}}$) is infeasible, the same argument gives $E_\infty^\star(\tilde{r}_i) \to \infty$). Its discontinuities are therefore upward jumps on the right. Let $E_{\mathrm{rob}} := \lim_{\tilde{r}\downarrow 0} E_\infty^\star(\tilde{r})$. A curve along which every constraint holds strictly keeps, by compactness, a positive margin, so E_{rob} is the least cost approached by feasible curves with a positive clearance, the analogue for this cost of the robust optimum of Solovey et al. (2020), and every method whose curves keep a positive clearance has costs at least E_{rob}. We call (P_∞) *stable* when $E_{\mathrm{rob}} = E_\infty^\star$. Then ($\mathrm{P}_\infty^{\tilde{r}}$) is feasible for small $\tilde{r} > 0$, and Theorem 1(ii) at $\tilde{r} = 0$ gives curves feasible for (P_∞) with costs in $[E_\infty^\star, E_\infty^\star(\delta_N) + \delta_N]$, which tend to $E_\infty^\star$; if $E_\infty^\star(\tilde{r}) \le E_\infty^\star + c\,\tilde{r}$ for small $\tilde{r} > 0$, they lie within $(c+1)\delta_N$ of it. This linear bound holds when Assumption C holds at the positions of one optimal curve: the construction of Lemma 14 on a neighbourhood of its graph, used as in the proof of Lemma 15, gives a curve with every block at least ρ and cost at most $E_\infty^\star + c_2\rho$, and a block value of at least ρ is a margin of at least $\rho/(4G')$ for a constant G' of the data. So Theorem 1 recovers the rate of Proposition 13(ii) with exactly feasible curves when the condition holds along one optimal curve.

Unstable margins are countable, and finitely many for polynomial data (sketch). A monotone function has at most countably many jumps, so ($\mathrm{P}_\infty^{\tilde{r}_0}$) is stable, $E_\infty^\star(\tilde{r}) \to E_\infty^\star(\tilde{r}_0)$ as $\tilde{r} \downarrow \tilde{r}_0$, for all but countably many $\tilde{r}_0 \in [0, \tilde{r}_{\max})$. With piecewise-polynomial data the exceptional margins are finitely many; we sketch the argument. Call a position $\mathbf{x}$ *jammed* at time t in ($\mathrm{P}_\infty^{\tilde{r}}$) if it is feasible and Assumption C fails there, that is, by Gordan's theorem, some weights $\lambda_j \ge 0$, not all zero, on the active blocks give $\sum_j \lambda_j \nabla_{\mathbf{x}} g_j = 0$. Take such weights with the fewest nonzero entries, on

a set S; every linear dependency of these gradients is then a multiple of $\boldsymbol{\lambda}$. Where $g_j = 0$, the derivative of block j in $\tilde{r}$ is $-2(r_j + \tilde{r})$ for an obstacle, $-4(r_a + r_b + 2\tilde{r})$ for a pair and $-2(\bar{r} - \tilde{r})$ for steering, all negative, so $\sum_{j \in S} \lambda_j \partial_{\tilde{r}} g_j < 0$, and the blocks of S have a derivative of full rank in $(\mathbf{x}, \tilde{r})$. On each interval where the data are polynomial, the points of their common zero set in $(\mathbf{x}, t, \tilde{r})$ where this derivative has full rank form a smooth semialgebraic manifold, and a jam with $a := \sum_{j \in S} \lambda_j \partial_t g_j = 0$ is a critical point of its projection to $\tilde{r}$; at a fixed time (static data or a breakpoint of the data) every jam is such a critical point. By the semialgebraic Sard theorem (Bochnak et al., 1998, Thm. 9.6.2), these critical values, over the finitely many sets S, intervals and breakpoints, form a finite set $\mathscr{M}$. For $\tilde{r} \notin \mathscr{M}$, a feasible curve passing a jammed position $\mathbf{x}_0$ at a time $t_0 \in (0,1)$ would give $\psi(\tau) = \sum_{j \in S} \lambda_j g_j(\boldsymbol{\gamma}(t_0 + \tau), t_0 + \tau) \geq 0$ with $\psi(0) = 0$ and $\psi(\tau) = a\tau + o(\tau)$, $a \neq 0$, since $\|\boldsymbol{\gamma}(t_0 + \tau) - \mathbf{x}_0\|^2 = o(|\tau|)$; this changes sign. So for $\tilde{r}_0 \notin \mathscr{M}$ every optimal curve of $(\mathrm{P}_\infty^{\tilde{r}_0})$ avoids jammed positions, and the push of the previous paragraph, applied to $(\mathrm{P}_\infty^{\tilde{r}_0})$, makes its value Lipschitz from the right at $\tilde{r}_0$.

Two unstable examples. Two obstacles that touch can make an instance unstable. Take a point robot in $\mathbb{R}^2$ with $k = 1$, endpoints $(0, \mp 1)$ and two discs of radius $1/4$ centred at $(\pm 1/4, 0)$, which touch at the origin. The straight line through the origin is optimal, $E_\infty^\star = 4$, but for every $\tilde{r} > 0$ the enlarged discs overlap, every feasible curve goes around their union, hence around one disc, and $E_\infty^\star(\tilde{r}) \geq E_{\mathrm{rob}} \approx 5.04$, the squared length of the shortest path around one disc. Every method whose curves keep a positive clearance thus costs at least E_{rob}, while Theorem 1(iii) gives costs at most $4 + \delta_N$ within any tolerance μ. GlobMP from $\mathbf{z}_h$ at $\tilde{r} = 0$ can still reach 4, when the grid contains the straight line, so stability is not necessary for its convergence from that start. Some instances need a tolerance even to start. Take a point robot in $\mathbb{R}^2$ with $k = 1$, start and goal at the origin, the steering constraint $\|\boldsymbol{\gamma}(t)\| \leq 1$ ($\boldsymbol{\rho} = \mathbf{0}$, $\bar{r} = 1$), a static disc of radius 0.9 centred at $(-0.7, 0.7)$, and a moving disc of radius 1 whose centre is $(0, -2(1 - 3t))$ on $[0, 1/3]$, the origin on $[1/3, 2/3]$ and $(2(3t - 2), 0)$ on $[2/3, 1]$. The curve that runs to the unit circle along the ray at $45°$, turns clockwise along the circle to $225°$ while the moving disc rests at the origin, and returns along that ray is feasible, with cost $6 + 3\pi^2$, so Assumptions A and B hold. On $[1/3, 2/3]$ every feasible curve has $\|\boldsymbol{\gamma}\| = 1$, and a polynomial curve of constant norm on an interval is constant (the leading coefficient of $\|\boldsymbol{\gamma}\|^2$ is the squared norm of the leading coefficient of $\boldsymbol{\gamma}$), so a piecewise-polynomial curve rests at one point $\mathbf{u}$ there. Just before $t = 1/3$ the two constraints give $\gamma_2 \geq -(1 - 3t)$, and just after $t = 2/3$ they give $\gamma_1 \leq 3t - 2$, so $u_1 \leq 0 \leq u_2$; that quarter of the unit circle lies inside the static disc. So no piecewise-polynomial curve, of any degree and with any knots, is feasible, $E_\infty^\star(\tilde{r}) = \infty$ for every $\tilde{r} > 0$, and GlobMP has no feasible start at $\tilde{r} = 0$, while $(\mathrm{P}_\infty^{-\mu})$ widens the circle to an annulus and Theorem 1(iii) applies. At $\tilde{r} = 0$, a tolerance or a condition such as stability is therefore needed.

When Assumption C holds. It is needed only at feasible positions within distance 1 of curves of cost at most $E_\infty^\star + 1$. It always holds for robots without obstacles or steering: the direction $\mathbf{v} = \mathbf{x}$, which spreads every robot away from the origin, gives $\nabla g_j \cdot \mathbf{v} = 2\|\mathbf{x}_a - \mathbf{x}_b\|^2 > 0$ for every touching pair; spreading about the centre of a single ball obstacle does the same for that obstacle and any number of robots. For static constraints it holds for almost every choice of margins: give block j the radius $r_j + \delta_j$. For each set $\mathscr{J}$ of blocks, the map $\mathbf{x} \mapsto (g_j(\mathbf{x}))_{j \in \mathscr{J}}$ is polynomial, so by Sard's theorem almost every value is regular, and at every configuration where exactly the blocks of $\mathscr{J}$ are active their gradients are then linearly independent, which implies Assumption C. There are finitely many sets $\mathscr{J}$, so the exceptional margins $\boldsymbol{\delta}$ form a null set. For moving obstacles or a moving reference the same count suggests that, generically, the condition fails only at isolated points of space-time, and a cheapest curve may pass through one.

Proposition 13 cannot drop Assumption C. The condition is needed by the algorithm, not only by the proof. Take a point robot in $\mathbb{R}^2$ and two discs of radius r centred at $(\pm r, 0)$, which touch at the origin, where Assumption C fails, and the cost $c\,x_2$ with $c > 0$ over $x_2 \geq -1$ (adding $(\lambda/2)\|\mathbf{x}\|^2$, $\lambda > 0$, changes nothing below); the vertical line $x_1 = 0$ is feasible throughout. From $(0, y)$ with $0 < y \leq \eta c$, the linearized set is $\{\mathbf{w} : w_2 \geq y/2 + (r/y)|w_1|\}$, and the repair returns its apex $(0, y/2)$. The iterates halve their distance to the origin at every step and converge to it, a point that is not a KKT point, since the cost still decreases along the feasible line. A cheapest curve through a zero-width gap between two touching obstacles can stall GlobMP in the same way.

A4 Addendum to Section 4: Experiments

A4.1 Experimental setup

This subsection summarizes the hardware, robot platforms and software shared across all experiments. GlobMP's runs below are instances of (P), each built from blocks of (P_∞), Appendix A2.1 or (44) and solved by Algorithms 1–2 under the protocol of its source. All reported baseline results are obtained from our own reproductions.

Hardware. The D3IL and maze baselines ran on a workstation with a 24-core Intel Core Ultra 9 285K at 3.7 GHz and one NVIDIA GeForce RTX 5090 (32 GB); the synthetic study of §4.1 and the multi-robot suite of §A4.5 ran on a shared node with two 32-core AMD EPYC 7502 CPUs and eight NVIDIA A100-PCIE GPUs (40 GB each), as did GlobMP's maze runs, so the maze times compare different machines. Training, sampling and model evaluation were performed on a single GPU. The RRT, GCS and SDP-MP baselines ran on CPU with each run pinned to fixed cores, and Algorithms 1–2 ran on CPU or GPU as stated with each experiment (for the synthetic study, in Appendix A4.2).

Robot platforms. The bimanual demonstrations are collected on the setup of Figure 8, using a Dexmate Vega-1 with two XHAND1 dexterous hands. Two ZED X One S cameras are mounted at the wrists and one ZED X Mini camera is mounted on the head. Teleoperation uses Manus gloves for hand target gestures and HTC VIVE trackers for wrist target poses. Only the resulting trajectories enter this paper; none of the teleoperation hardware is used at inference. Figure 9 shows the modalities, the task string, the camera images and the fingertip tactile arrays, at the instant the box changes hands.

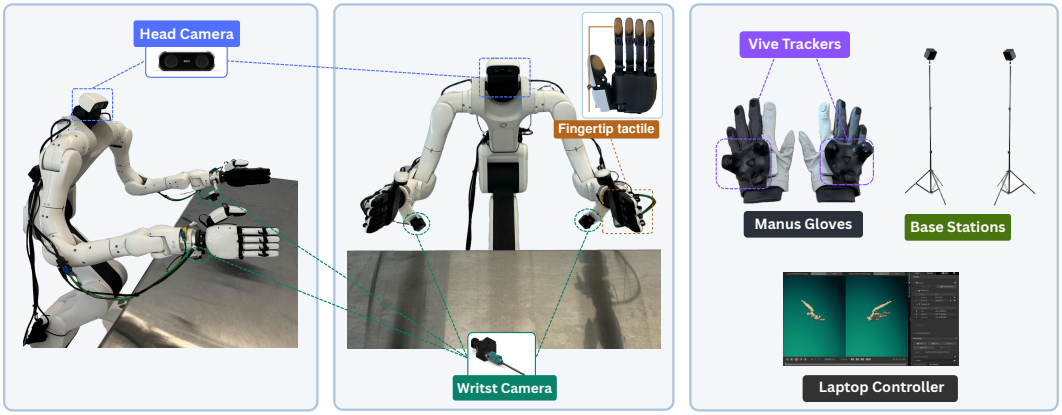
The multi-robot experiments use the Georgia Tech Robotarium (Wilson et al., 2020), with its 3.2×2.0 m arena, GRITSBot geometry, and differential-drive dynamics. The benchmark environments of §4.2 are adapted to the Robotarium workspace, with their obstacle geometry represented virtually in the arena. The GRITSBots have physical radius 0.055 m, while planning uses an inflated radius $r = 0.13$ m to account for tracking error, the testbed's barrier-certificate safety layer and the nonholonomic conversion of the plan. The planned trajectories are tracked through the Robotarium's single-integrator-to-unicycle controller, and an overhead camera records the executed motion. The virtual obstacles are overlaid from the corresponding map definitions, and clearance is evaluated from the robot trajectories tracked in the video after correcting for lens distortion. Across the four obstacle maps, no GlobMP robot comes closer than 0.06 m to an obstacle; the closest pass occurs on `Conveyor` and is shown in Figure 10; Figures 17–21 show all methods on each map.

Software. All experiments are based on PyTorch (2.13; 2.6 for Maze2D) for automatic differentiation, JAX 0.6.2 for DiRecT's SCP–ADMM projector, and IPOPT through Pyomo for the augmented-Lagrangian projection, under an external wall-clock cap. The RRT variants run through OMPL; IRIS and GCS through Drake; and the curve fits and SDP-MP through CVXPY with Clarabel. The multi-robot baselines build on MMD (Shaoul et al., 2025) (MIT), MDOC (He et al., 2026) (MIT) and DiRecT (Giaretta et al., 2026) (MIT), itself an extension of Projected Coupled Diffusion (Luan et al., 2026) (MIT). Every learned baseline is trained from scratch under its source's protocol rather than scored from a released checkpoint.

A4.2 Synthetic experiments

The synthetic study of §4.1 tests continuous-time feasibility, the cost of the returned curve, and scalability on one family. Every instance is a minimum-derivative problem with snap cost $k = 4$, endpoint data $\ell = 4$, piecewise degree-9 polynomials, C^3 junctions and margin 0.005 m (added once to pair separations), so the piecewise conditions of Appendix A4.6 apply and the planning curve lies in the $\mathscr{H}^4$ class of (P_∞).

Problem 1 (robots exchanging places among spheres)**.** N_a robots, N_a even, are stacked into one curve $\boldsymbol{\gamma}$ with values in $\mathbb{R}^{2N_a}$ and form $N_a/2$ exchanging pairs: in pair i one robot moves from (x_0, y_i) to $(x_0 + D_i, y_i)$ and the other the reverse way along $y_i + o_i$, with $D_i \sim U(6,8)$ m and $o_i \sim U(0.3, 0.9)\, R_S$ with $R_S = 1.2$ m. Every pair of robots carries a separation block ($\boldsymbol{\Sigma}_j = \mathbf{E}_{ab}^\top \mathbf{E}_{ab}$, $\mathbf{c}_j = \mathbf{0}$,



(a) Robot Platform **(b)** operator Station

Figure 8: Robot platform and operator station. The operator's finger joints come from Manus gloves and the wrist poses from VIVE trackers against two base stations; retargeting turns both into absolute joint targets for the 38 actuated degrees of freedom of a Dexmate Vega-1 carrying two XHAND1 hands, sent at 30 Hz. Head and wrist cameras and the fingertip force sensors record alongside. The close-up is the same hand photographed on the bench and cut out of its background; the five fingertip pads, 120 taxels each, are tinted amber and the leader points at one of them; they are the only tactile surfaces the hand has, the thumb's included. The palm and the finger links sense nothing. Only the resulting trajectories are used in this paper; none of the operator hardware is present at inference.

$r_j = R_S$; Appendix A2.1), and every robot a ball block for each static sphere, with $\mathbf{E}_a\mathbf{c}_j$ its centre, $\mathbf{\Sigma}_j = \mathbf{E}_a^\top \mathbf{E}_a$ and r_j its radius. Two spheres of radius $U(0.45, 0.65)$ m sit 0.3 to 0.9 radii below the first row, so the first robot's straight path enters both, and every start and goal clears them by 0.2 m. There is no workspace bound, so every inequality block has a nonconvex feasible set. Each exchanging pair, and each sphere a straight path enters, is passed on one of two sides, which gives 8 to 128 route classes per instance.

The main comparison uses $N = 7$ pieces and 20 instances at each of $N_a = 2, 4, 6, 8$. The scaling study uses $N = 4$ pieces, rows spaced $3R_S$ apart with $o_i = 0.6R_S$ and one D_i shared by all pairs, three spheres, and, from 2 to 200 robots, 10 seeds per size for GlobMP and 5 for the baselines.

Methods. GlobMP opens each route class with a waypoint path that passes every pair and every crossed sphere on the chosen side, runs Algorithms 1–2 (in the main comparison also an SQP polish and the finishing step of Appendix A4.6), and returns the cheapest feasible result over the classes. RRT-Connect and RRT* are the OMPL implementations (Kuffner & LaValle, 2000; Karaman & Frazzoli, 2011) in $\mathbb{R}^{2N_a}$ with the exact validity test (every pair at least R_S apart, every robot outside every sphere); they sample in the rectangle spanned by the instance plus 3 m, since a sampling planner needs a bounded region, and get a 60 s budget (600 s in the scaling study). Their path is collision-free as a polyline but is not a curve of the class, so the returned curve is the cheapest collision-free fit among several (else the least colliding), each scored on the true geometry: the minimum-snap interpolant through the path at the segment breakpoints, which cuts corners; fits that trade snap against the squared distance to the path; and minimum-snap fits kept within a corridor around the path wherever the current fit collides, with the corridor width set by the path's own clearance (Mellinger & Kumar, 2011; Richter et al., 2016), each on the path and on a shortcut version of it. GCS (Marcucci et al., 2023) works in the same joint space, where both collision sets are convex: robot a inside a sphere is $\{\mathbf{x} : \|\mathbf{x}_a - \mathbf{E}_a\mathbf{c}_j\| \le r_j\}$ and a colliding pair is $\{\mathbf{x} : \|\mathbf{x}_a - \mathbf{x}_b\| \le R_S\}$. IRIS (Deits & Tedrake, 2015a) grows regions from seeds every 0.5 m along an RRT-Connect path, Drake's GCS trajectory optimization finds the minimum path-energy path through them (plus a 10^{-3} time cost) with Bézier order 5, falling back to 3 and 2 where its convex solve fails numerically, and its curve is fitted in the same way. SDP-MP, the semidefinite relaxation of Graesdal et al. (2026), is written on the same program in its own affine chart, so that the endpoint and junction conditions hold by construction; the per-segment form with lifted junctions stops with a numerical error in Clarabel

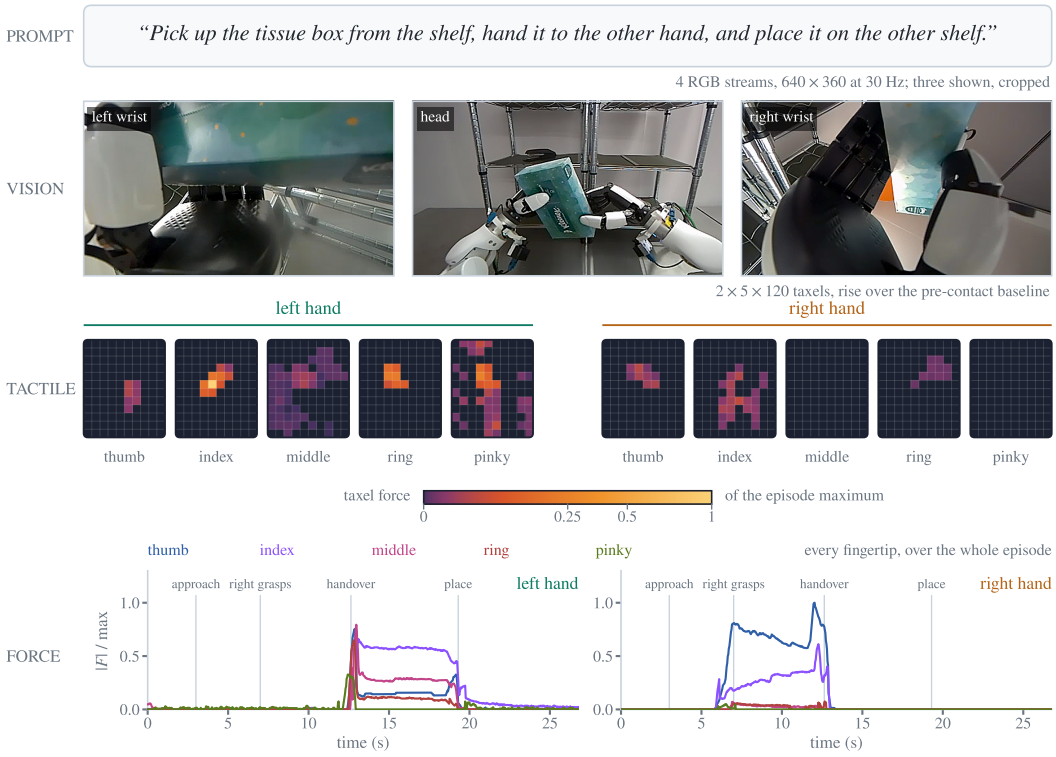


Figure 9: Given modalities, at the instant the tissue box changes hands (one episode of the handover task, 27 s at 30 Hz). *Prompt*: the task string the policy is conditioned on, one per episode. *Vision*: three of the four RGB streams. *Tactile*: both hands at that same instant, five fingertip pads of 120 taxels each, every pad unrolled into its 10×12 measurement grid, one square per taxel, shown as its rise over the pre-contact baseline on a stretched colour scale. *Force*: the same fingertip sensors over the whole episode, every finger on its own, one panel per hand. Both scales are divided by the largest value the episode contains, the maps by the largest taxel rise and the traces by the largest fingertip force: the hand reports integers and the vendor documents no conversion to newtons, so the figure normalizes rather than claiming a force unit. The load passes from the right hand to the left at the handover and falls when the box is placed.

from two segments on. The chart form is joint over all robots and segments, and each block must lie on each segment in the cone of polynomials nonnegative on $[0,1]$ of §2, written as sums of squares. On an obstacle-free instance it reproduces the exact minimum-snap optimum. Its rounded curve is the first-order part of the solution. The baselines run on CPU, one run per four pinned cores, and a run that returns nothing within one hour counts as a failure. GlobMP's route classes run on CPU in the same way at $N_a = 2, 4$ and on the first 10 instances at $N_a = 6$, and on A100 GPUs, four to ten runs per GPU, for the rest, so its times are totals of work rather than a like-for-like speed comparison.

Rate in the number of pieces. Figure 4 uses instance 0 of the main comparison with $N_a = 2$, degree 9, C^{k-1} joins and the 0.5 cm margin, for $k \in \{1,2,3,4\}$ and $\ell \in \{k-1, k\}$, and N up to 12 ($k = 1$), 16 ($k = 4$), 24 ($k = 3$) and 32 ($k = 2$). All route-class runs converge to the same class, so $E_N^\star$ is estimated by the cost GlobMP reaches from it: Algorithms 1–2, the SQP polish and the finishing step, from the waypoint path for $N \le 8$ and from the $N = 8$ curve, refitted by least squares, for larger N; a run counts only at a KKT point that keeps the margin. The cost is evaluated in global time from the kth-derivative coefficients, with the cost quadratic expanded around the obstacle-free minimizer (a change of chart): the expansion around the default chart point loses all digits beyond about 16 pieces at $k = 4$, and two starts agree to $5 \cdot 10^{-10}$ relative; Figure 4a floors at $2 \cdot 10^{-9}$. Since (P_∞) keeps only the orders below k, both values of ℓ share $E_\infty^\star$, estimated by the cheapest $\ell = k-1$ value at $N \ge 8$. Panel (c) takes $N = 8$, $\ell = k-1$ and margins (added once to the pair separation) from

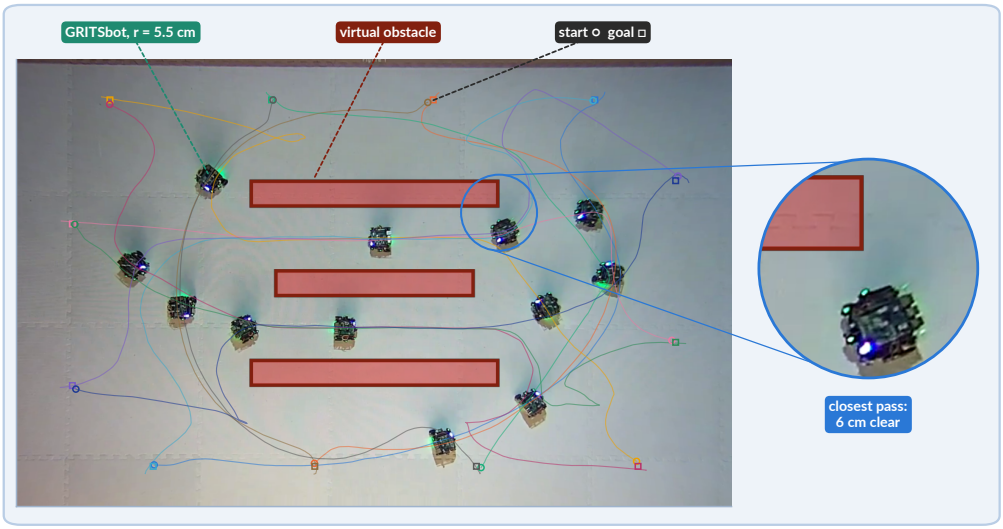


Figure 10: GlobMP on the Georgia Tech Robotarium: twelve GRITSBots on `Conveyor` in the 3.2×2.0 m arena, at $t = 25$ s. The paths the robots drove, starts (circles), goals (squares), and the virtual obstacles drawn from the map. The circle magnifies the run's closest pass: 6 cm between the robot's body and a belt.

0 to 5 cm, up to 10% of the smaller sphere radius (0.51 m; the other is 0.58 m); for $k \geq 2$ the pair block stays inactive, at least 19 cm clear, so these are the margins $\tilde{r}$ of Theorem 1 (for $k = 1$ it is active, so the $k = 1$ curve in (c) raises the pair separation by $\tilde{r}$, not $2\tilde{r}$, and is not $E^{\star}_{\infty}(\tilde{r})$). Velocity ($k = 1$) wraps the spheres along arcs, and its finishing step reaches a KKT point only on some runs, each started from the waypoint path or the curve at $N/2$: the panels show the runs that do, against $E^{\star}_{\infty}$ and, in (c), the zero-margin cost, each estimated by the cheapest $\ell = 0$ curve feasible on the true geometry ($N = 10$ and $N = 8$).

A4.3 Maze navigation

We examine whether the proposed formulation remains effective in a nonconvex maze. We consider the safe maze navigation benchmark of Xiao et al. (2025), following the setting reported by Giaretta et al. (2026). A point robot traverses the D4RL `maze2d-large` environment (Fu et al., 2020) between fixed endpoints while avoiding keep-out regions placed within the maze corridors. The state contains planar position and velocity, and the action is a planar force. A rollout is considered safe if the executed trajectory remains outside every keep-out region. We evaluate two settings, `Broad`, containing two keep-outs, and `Narrow`, containing six.

This benchmark combines both constraint constructions used in this paper. The maze walls form a union of cells and therefore define a disjunctive free space that cannot be represented by a single membership in $\mathscr{K}$. A breadth-first search on the cell graph selects a corridor in $\approx$1 ms, after which the continuous trajectory is optimized and verified within the selected corridor. Each of the 12 corridor boxes along the route contributes four affine blocks (6) and one Bézier segment, with adjacent segments joined C^{ℓ}, resulting in 48 affine blocks. Each keep-out contributes one block (44), quadratic for $e = 2$ and quartic for $e = 4$, beyond the quadratic blocks of (P_{∞}). We use $k = \ell = 2$ with pieces of degree 9, so Corollary 2(i) ($\ell < k$) does not apply here. The maze geometry is not learned. The baselines rely on the diffusion prior to remain within the maze and explicitly constrain only the keep-outs, whereas our formulation enforces the corridor boundaries through affine blocks and does not use a learned prior.

Baselines. We compare against ten diffusion planners, nine of them constrained: *Diffuser* (Janner et al., 2022), *Classifier Guidance* (Dhariwal & Nichol, 2021), *Gradient Guidance* (Chung et al., 2023), *Projected Diffusion* (Christopher et al., 2024), *Primal–Dual* and *Augmented Lagrangian* (Zhang et al., 2025), the *SafeDiffuser* variants RoS, ReS, and TVS (Xiao et al., 2025), and *DiRecT* (Giaretta et al., 2026), on the task setup of Giaretta et al. (2026). Methods requiring a nonlinear

Table 1: `Maze2D Broad`, in the format of Giaretta et al. (2026)'s Table 5. Mean ± standard deviation. The four *Gen.* columns score the generated trajectory. Best value(s) in bold, Time excepted. All baselines are run here over 10 evaluation episodes, sampling from the same diffusion planner we train, so the values are measured rather than quoted; our rows are deterministic and report a single trajectory.

Method	SR ↑	Viol. ↓	Score ↑	Time (s) ↓	Gen. SR ↑	Gen. Viol. ↓	Gen. AS ↓	Gen. CS ↓
Diffuser	0.000	44.600 ± 11.298	1.612 ± 0.004	0.256 ± 0.099	0.000	39.700 ± 18.815	0.0396 ± 0.0022	0.4281 ± 0.0427
Classifier Guidance	0.100	27.800 ± 13.934	1.612 ± 0.004	0.548 ± 0.091	0.300	9.000 ± 11.340	0.0400 ± 0.0022	0.4352 ± 0.0552
Gradient Guidance	0.200	13.000 ± 6.885	1.613 ± 0.005	2.663 ± 0.107	**1.000**	**0.000** ± 0.000	0.0404 ± 0.0023	0.4471 ± 0.0460
Projected Diffusion	0.400	38.500 ± 45.247	1.314 ± 0.463	103.729 ± 26.086	**1.000**	**0.000** ± 0.000	0.0026 ± 0.0002	0.0092 ± 0.0024
Primal–Dual	0.000	30.600 ± 12.492	0.111 ± 0.320	4.426 ± 0.118	0.800	0.300 ± 0.640	20.1566 ± 0.3115	1.8145 ± 0.0146
Augmented Lagrangian	0.500	9.600 ± 10.394	−0.025 ± 0.000	4.229 ± 0.212	0.000	9.900 ± 7.382	1.8568 ± 0.4291	1.1980 ± 0.0542
SafeDiffuser-RoS	0.000	59.900 ± 41.293	1.070 ± 0.562	145.924 ± 21.730	**1.000**	**0.000** ± 0.000	0.0082 ± 0.0066	0.0115 ± 0.0043
SafeDiffuser-ReS	0.400	9.800 ± 8.600	0.655 ± 0.833	30.553 ± 2.242	**1.000**	**0.000** ± 0.000	0.0039 ± 0.0048	0.0164 ± 0.0059
SafeDiffuser-TVS	0.400	42.400 ± 38.733	1.180 ± 0.625	100.873 ± 18.575	**1.000**	**0.000** ± 0.000	0.0048 ± 0.0029	0.0117 ± 0.0056
DiRecT	**1.000**	**0.000** ± 0.000	1.610 ± 0.004	7.746 ± 0.485	**1.000**	**0.000** ± 0.000	0.0024 ± 0.0000	0.0080 ± 0.0003
GlobMP (Ours), disc cover	**1.000**	**0.000**	1.591	270.3	**1.000**	**0.000**	0.0009	0.0010
GlobMP (Ours), exact e=4	**1.000**	**0.000**	**1.629**	32.8	**1.000**	**0.000**	**0.0005**	**0.0001**

solver use IPOPT (Wächter & Biegler, 2006). The nine constrained methods impose the keep-out regions but leave satisfaction of the maze walls to the learned prior, and none guarantees feasibility in continuous time. Their constraints are enforced pointwise during sampling or projection, so feasibility holds at the sampled states rather than along the whole curve.

Training and environment details. Each keep-out region is represented by an ellipse or super-ellipse centred at (c_x, c_y) with semi-axes (r_x, r_y) and order e,

$$\left|\frac{x-c_x}{r_x}\right|^e + \left|\frac{y-c_y}{r_y}\right|^e \geq 1. \tag{44}$$

The `Broad` setting uses one region with $e = 2$ and one with $e = 4$, while all six keep-outs in `Narrow` use $e = 4$. Each region straddles a wall stub and extends into the corridor, and GlobMP's route touches the 0.4 margin of both keep-outs in `Broad` and of four of the six in `Narrow`, as illustrated in Figures 11 and 12. A single continuous-time diffusion model with a temporal UNet is used by all baselines. It is trained for 500,000 steps with Adam, batch size 256, on the D4RL `maze2d-large-v1` dataset using min–max normalization. The learning rate is 10^{-3} with cosine annealing over the first 10,000 steps, the exponential moving average decay is 0.995, and the planning horizon is 384 steps. The model outputs a state–action trajectory. Dynamics are identified from the same dataset by ordinary least squares over 3,993,503 transitions and are imposed on the baselines as equality constraints. Because open-loop execution of the fitted linear model diverges over this horizon, all methods execute their planned trajectories using a proportional–derivative controller with proportional gain 5 and derivative gain 1.

Evaluation. We follow the protocol of Giaretta et al. (2026). Each baseline generates 10 i.i.d. trajectories conditioned on the same fixed endpoints, with each trajectory executed for at most 800 environment steps. We report *SR*, the fraction of collision-free rollouts; *Viol.*, the mean number of steps spent inside a keep-out; *Score*, the D4RL normalized return; and *Time*, the planning time. The *Gen.* columns evaluate the generated trajectory before execution and additionally report acceleration smoothness, *AS*, and curvature smoothness, *CS* (not the constraint satisfaction of Appendix A4.5). Baseline values are means over 10 samples, whereas our method is deterministic and reports a single trajectory.

Tables 1 and 2 reveal an important difference between planning and execution. In `Broad`, several baselines generate trajectories with *Gen. SR* = 1.00 but lose safety during execution: Gradient Guidance falls to 0.20, SafeDiffuser-RoS to 0.00, and Projected Diffusion, SafeDiffuser-ReS and SafeDiffuser-TVS to 0.40. DiRecT is the only baseline that carries *Gen. SR* = 1.00 through execution. Both of our formulations maintain *SR* = 1.00 with zero violations before and after execution. The disc-cover formulation attains a score of 1.591, while the exact $e = 4$ formulation improves the score to 1.629 and achieves the lowest *AS* and *CS*. Both satisfy the keep-out constraints for every t.

The distinction becomes more pronounced in `Narrow`, where feasibility is difficult already at the planning stage. Gradient Guidance, SafeDiffuser-ReS and DiRecT are the only baselines reaching *Gen. SR* = 1.00, and of these only DiRecT keeps it through execution; the other two fall to 0.10. Our trajectory maintains *SR* = 1.00 with zero violations, achieves a score of 1.625 compared with 1.615 for DiRecT, and obtains the lowest *AS* and *CS*, at a planning time of 90.0 s compared with 16.0 s

Table 2: `Maze2D Narrow`, the six-keep-out setting. Mean ± standard deviation. The four *Gen.* columns score the generated trajectory. Best value(s) in bold, Time excepted. All baselines are run here over 10 evaluation episodes, sampling from the same diffusion planner we train, so the values are measured rather than quoted; our row is deterministic and reports a single trajectory.

Method	SR ↑	Viol. ↓	Score ↑	Time (s) ↓	Gen. SR ↑	Gen. Viol. ↓	Gen. AS ↓	Gen. CS ↓
Diffuser	0.000	94.100 ± 7.327	1.612 ± 0.004	0.260 ± 0.106	0.000	82.800 ± 16.726	0.0396 ± 0.0022	0.4281 ± 0.0427
Classifier Guidance	0.000	64.400 ± 14.908	1.619 ± 0.005	0.549 ± 0.104	0.000	27.700 ± 23.367	0.0402 ± 0.0023	0.4385 ± 0.0559
Gradient Guidance	0.100	21.700 ± 17.211	1.622 ± 0.006	2.607 ± 0.086	**1.000**	**0.000** ± 0.000	0.0404 ± 0.0028	0.4475 ± 0.0535
Projected Diffusion	0.100	83.200 ± 158.026	1.057 ± 0.721	756.264 ± 166.998	0.800	4.800 ± 9.765	0.0145 ± 0.0225	0.0221 ± 0.0258
Primal–Dual	0.000	125.900 ± 114.102	0.103 ± 0.294	4.379 ± 0.138	0.000	11.400 ± 4.409	20.1512 ± 0.3103	1.8129 ± 0.0146
Augmented Lagrangian	0.000	222.200 ± 181.693	−0.025 ± 0.000	4.394 ± 0.204	0.000	159.300 ± 40.020	2.5444 ± 0.6313	1.2043 ± 0.0519
SafeDiffuser-RoS	0.000	258.200 ± 206.817	0.927 ± 0.628	1215.129 ± 163.413	0.000	20.800 ± 7.859	0.5856 ± 0.0694	0.2983 ± 0.0335
SafeDiffuser-ReS	0.100	223.800 ± 230.466	0.808 ± 0.834	179.773 ± 90.523	**1.000**	**0.000** ± 0.000	0.0350 ± 0.0569	0.0283 ± 0.0121
SafeDiffuser-TVS	0.000	266.300 ± 200.686	0.908 ± 0.622	559.372 ± 114.038	0.000	31.700 ± 19.990	0.0458 ± 0.0083	0.0306 ± 0.0076
DiRecT	**1.000**	**0.000** ± 0.000	1.615 ± 0.003	15.957 ± 1.406	**1.000**	**0.000** ± 0.000	0.0024 ± 0.0000	0.0084 ± 0.0008
GlobMP (Ours), exact e=4	**1.000**	**0.000**	**1.625**	90.0	**1.000**	**0.000**	**0.0006**	**0.0002**

for DiRecT (different machines, Appendix A4.1). That time is not a single descent: Algorithm 1 is within 1% of its own final cost after 19 s and stops at 57 s (60 s cap), and an SQP polish then supplies the final 6% of the cost (30 s, plus 3 s for the finishing step); letting Algorithm 1 run to its own stopping rule instead (532 s) changes neither the cost nor the score. A 20 s cap gives 32.8 s on `Broad` against 7.7 s for DiRecT. On both maps the returned trajectory satisfies the keep-out constraints at every $t \in [0, 1]$, not only at the sampled states that the benchmark checks.

In Tables 1 and 2, several baselines lose safety between the generated plan and the executed rollout, whereas every GlobMP row is at 1.00 on both. Figures 11 and 12 show the executed rollouts of every method on the two maps. GlobMP keeps the left side of (44) at least 1.4, a margin of 0.4; the plan touches this margin exactly at its active contacts (minimum clearance 3.1×10^{-7} on `Broad` and 3.7×10^{-10} on `Narrow`). Tracking error takes the executed rollout up to 0.24 into the margin band, which is what the margin is for: the rollout stays outside every keep-out, and the benchmark counts zero violations.

A4.4 Collision-free obstacle avoidance in end-effector space

We evaluate GlobMP against the baselines on the constrained manipulation benchmark of Jia et al. (2024), with GlobMP as the projection step of DiRecT's own diffusion sampler. A planar end-effector must reach a goal region while avoiding a set of pillars (Figure 13). The state lies in $\mathbb{R}^4$ and contains the current and desired positions, while the action lies in $\mathbb{R}^2$ and specifies velocity. The initial position is fixed, the goal lies on $y = 0.36$, and safety requires the entire trajectory to remain outside every keep-out region. Circular keep-outs are represented by the quadratic block (10) at degree $2d$, while planar constraints use the affine block of §A2.1.1. The formulation would use one Bézier segment of degree $d = 12$, with position and velocity fixed at both endpoints ($\ell = 1$), leaving $d_f = 9$ free parameters per coordinate, and minimize acceleration with $k = 2$.

Baselines. We run the ten diffusion planners of the maze benchmark (§A4.3), which cover soft guidance, projection, dual optimization, barrier-function guidance, and receding-horizon projection. Methods requiring a nonlinear solver use IPOPT (Wächter & Biegler, 2006).

Training and environment details. A continuous-time diffusion model with a temporal UNet is trained for 500,000 steps with Adam on the D3IL `avoiding` dataset using batch size 32, learning rate 2×10^{-4}, cosine annealing over the first 100,000 steps, and an exponential moving average with decay 0.995, at horizon H=16. The same model is shared by all baselines, and constraints are introduced only during sampling, so no retraining is required when the constraint set changes. The original task, learned from 96 human demonstrations, contains six circular pillars, 5–6 cm across. Following Giaretta et al. (2026), we introduce unseen test-time constraints by enlarging the first pillar from radius 0.03 to 0.08 and restricting lateral motion with two planar regions. We further activate three keep-outs of radius 0.02 centred at $(0.5, 0.16)$, $(0.4, 0.175)$, and $(0.6, 0.175)$, which are included in their release but disabled by default. The resulting problem contains ten circular keep-outs and two planar constraints. These constraints deliberately differ from the training distribution. Of the 96 demonstrations, 77 violate the enlarged pillar, 62 violate the planar constraints, and 53

Figure 11: Executed rollouts on `Maze2D Broad`, one panel per method, under the benchmark's PD controller. Baseline panels show all ten evaluation runs behind Table 1, unfiltered; GlobMP is deterministic and returns one. Stretches drawn dark and marked $\times$ are where the robot is inside a keep-out, and the badge counts the runs that entered one, which is $10(1-SR)$ of the table. Every baseline of the table appears, in the table's order, with GlobMP's exact e=4 row. The last cell magnifies every stretch where GlobMP runs inside the enforced margin, the region marked by the dashed box on its own panel: the continuous plan (dashed) is held off the keep-out by the margin, and the executed rollout (solid) dips into the margin band but stays outside the keep-out, which is what the margin is for.

violate the three additional keep-outs. The resulting free space contains four route classes, with a minimum passage width of 0.063 m.

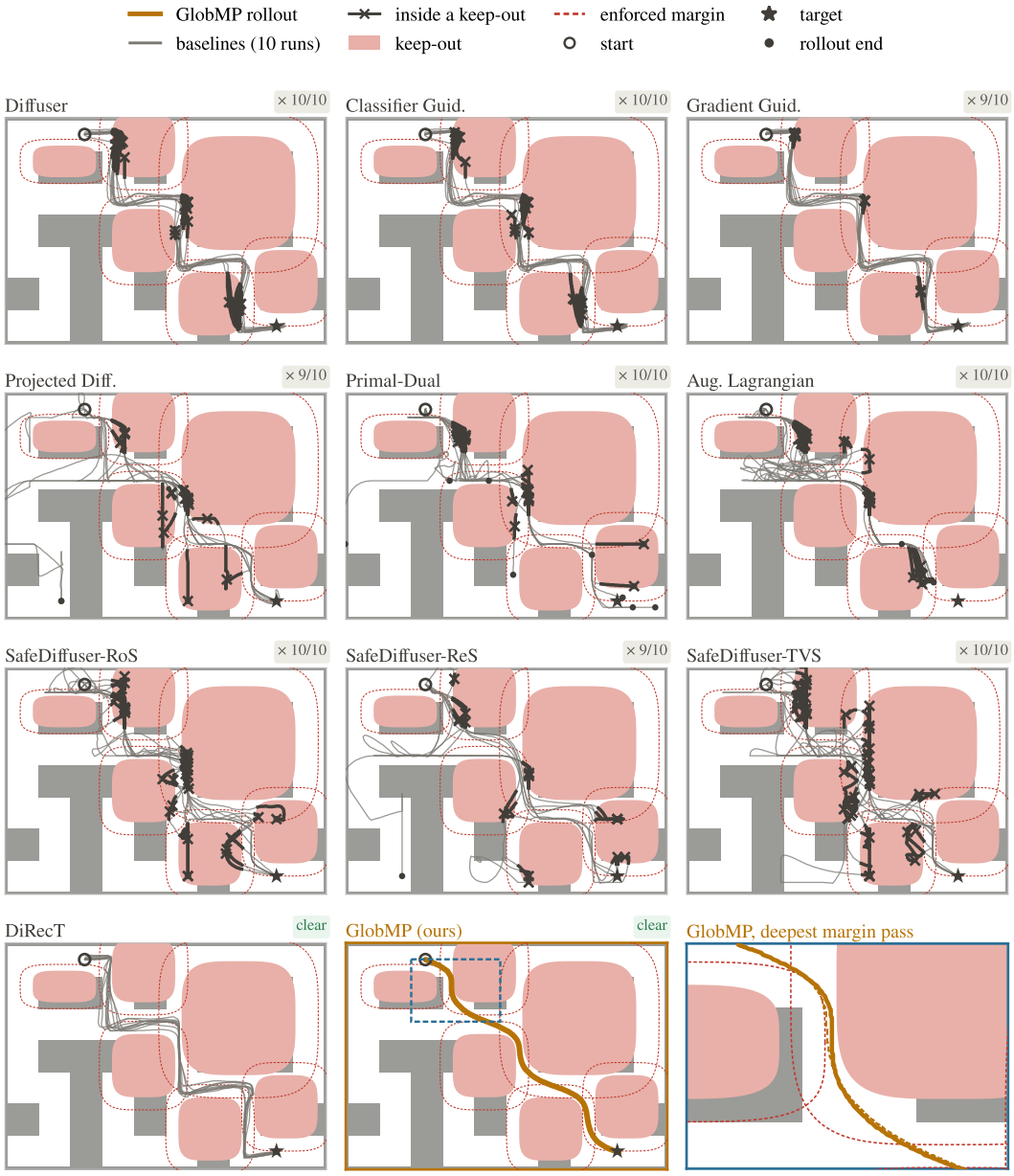


Figure 12: Executed rollouts on `Maze2D Narrow`, drawn as in Figure 11, over the ten evaluation runs behind Table 2. Only DiRecT and GlobMP keep every run outside the keep-outs; the keep-outs leave little of the corridor, and most baselines enter one on every run. So little is left that GlobMP's trajectory is inside the enforced margin along most of the route, so the last cell magnifies its deepest pass rather than every stretch.

Evaluation. We evaluate each method over 100 i.i.d. rollouts from the same initial position at seed 0, with at most 100 environment steps, planning horizon H=16, and execution horizon of 4 steps. We report *SR*, the fraction of rollouts satisfying all constraints; *TS*, the fraction reaching the goal within the episode budget; *Steps(Safe)*, the mean steps among safe rollouts; and *Time*, the mean computation time per replanning step. All values are obtained from our executions of the released implementations.

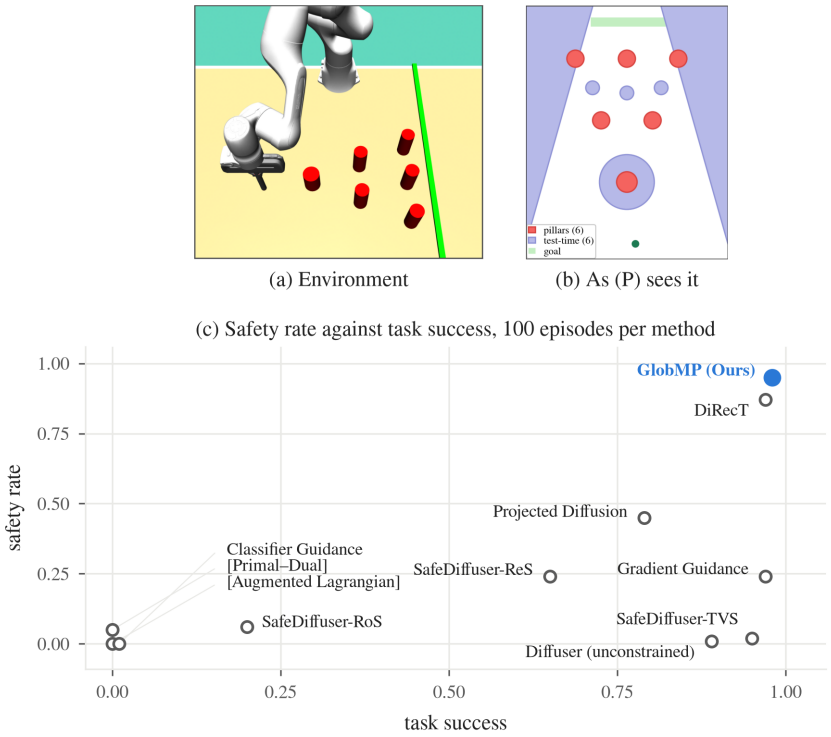


Figure 13: D3IL `avoiding-v0`. *(a)* The environment's own view; the test-time constraints have no physical counterpart, so they appear only in *(b)*. *(b)* The scene as (P) sees it, with all ten discs and both half-planes. *(c)* Safety rate against task success over 100 episodes, one point per method: a point at the lower right reaches the goal without avoiding, and only the upper right does both. Brackets as in Table 3.

GlobMP. We keep DiRecT's checkpoint, sampler, dynamics fit and evaluation, and replace only its projection: at every denoising step GlobMP projects the sampler's plan, minimizing DiRecT's own cost, onto plans whose straight-line motion between the 16 states clears every keep-out by the margin at every instant, not only at the states. The actions follow from the planned positions through the dynamics fit, so their limits (the environment clips each step to at most 1.25 cm, and never backwards) and their rate of change enter as linear blocks, and every plan can be executed as planned. A plan that starts closer to a keep-out than the margin must win the margin back, by at most 1 cm per step. Within one plan the start state is fixed, so each projection starts from the previous one's solution, which is feasible; Algorithm 1 then lowers the cost from a feasible start.

Table 3 shows that safety and task completion do not necessarily coincide, and Figure 14 shows the rollouts behind it. On the same 100 episodes, GlobMP is safe in 0.95 of them and reaches the goal in 0.98, ahead of DiRecT (0.87 and 0.97), and every plan it returns is feasible at every instant. Over four seeds (400 episodes) GlobMP averages 0.86 and 0.97, on par with DiRecT's single seed, and it is slower: safe rollouts take 78 steps against 61, and each plan takes longer to compute. Projected Diffusion attains the next-highest safety rate at 0.45. In contrast, Gradient Guidance and SafeDiffuser-TVS reach the goal in 0.97 and 0.95 of episodes but achieve safety rates of only 0.24 and 0.02, respectively. The two metrics should therefore be interpreted jointly, since high task success may still correspond to frequent constraint violations.

For DiRecT, the remaining safety failures occur primarily during execution rather than planning. IPOPT converges on every solve and 815 of 819 generated plans are feasible; moreover, in 10 of the 13 unsafe episodes, every generated plan is feasible before execution. Thus, even when the planner satisfies the test-time constraints, tracking can still drive the executed trajectory outside the feasible

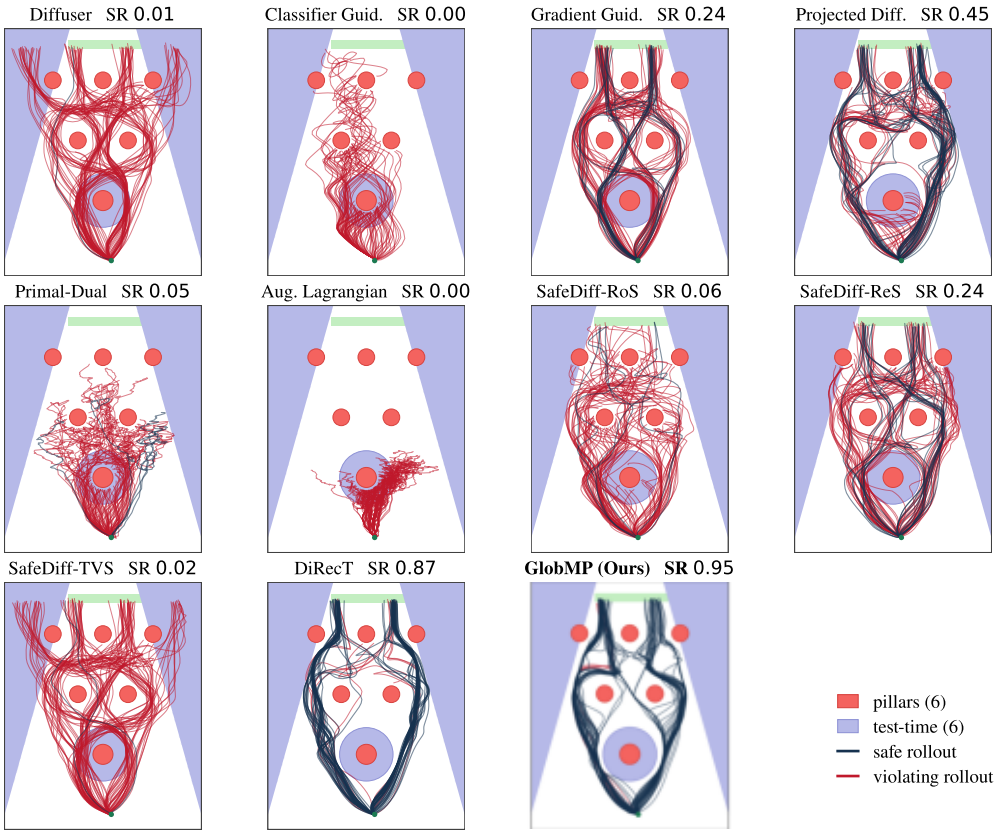


Figure 14: All 100 rollouts of every method on D3IL `avoiding-v0`, from the common start (green), GlobMP last. A rollout is drawn red if it violates any of the twelve constraints under the benchmark's own predicate and navy otherwise, so the fraction of navy is the safety rate in the panel title. The six pillars belong to the environment; the six lavender regions are imposed only at test time.

set. As in Giaretta et al. (2026), these experiments evaluate inference-time constraint handling: the additional constraints are introduced only at test time and no method is retrained.

Table 3: D3IL `avoiding-v0`, in the format of Giaretta et al. (2026)'s Table 6, 100 episodes per method. Every value is our own execution of the released code from a single shared checkpoint. Bracketed rows required a non-default `--guidance-last-steps 50` to run at all: at the default both crash the simulator. Boldface marks the best safety rate and task success among the methods that enforce the constraints; the cost columns carry none. GlobMP is the projection step of DiRecT's sampler (replanning every 4 steps, 28 CPU workers for 50 episodes at a time), on the same seed; over four seeds it reaches 0.86 and 0.97.

Method	SR ↑	TS ↑	Steps(Safe) ↓	Time (s) ↓
Diffuser (unconstrained)	0.01	0.89	61.00	0.09 ± 0.00
Classifier Guidance	0.00	0.01	—	0.17 ± 0.02
Gradient Guidance	0.24	0.97	62.96 ± 7.27	1.16 ± 0.07
Projected Diffusion	0.45	0.79	65.18 ± 8.09	7.16 ± 3.11
[Primal–Dual]	0.05	0.00	100.00 ± 0.00	0.74 ± 0.02
[Augmented Lagrangian]	0.00	0.00	—	0.79 ± 0.01
SafeDiffuser-RoS	0.06	0.20	67.50 ± 6.55	7.34 ± 0.18
SafeDiffuser-ReS	0.24	0.65	62.17 ± 5.10	5.44 ± 0.27
SafeDiffuser-TVS	0.02	0.95	58.50 ± 1.50	5.26 ± 0.09
DiRecT	0.87	0.97	61.38 ± 4.84	3.72 ± 0.33
GlobMP (Ours)	**0.95**	**0.98**	78.20 ± 9.22	15.12 ± 9.82

A4.5 Multi-robot motion planning

We extend the multi-robot evaluation of §4.2 to a broader benchmark with eight environments, team sizes $N_a \in \{3, 6, \ldots, 21\}$, and three test-time velocity caps (results in Figures 15–16). The benchmark follows Shaoul et al. (2025), with the velocity constraint of Luan et al. (2026) and the evaluation protocol of Giaretta et al. (2026). Six environments are from Shaoul et al. (2025); `Corridor-Bay` and `Cross-Room` are reconstructed from Liang et al. (2025). Start–goal pairs follow the sampling procedure of Shaoul et al. (2025), which places every endpoint at least $2.1r + 0.01$ from an obstacle of the source map (before the enlargement below) and $\max(0.15, 2.2r) + 0.01$ from every other endpoint.

Problem setup. Each agent follows a planar Bézier trajectory, with one block per agent and obstacle and one per pair of agents at separation $2r$ (§A4.6). Box obstacles use the affine blocks of §A2.1.1, one half-space per agent, segment and box, with the side selected by an independent grid seed.

The eight environments cover different forms of geometric and coordinated motion:

- `Empty` contains no obstacles and isolates inter-agent coordination.
- `Basic` and `Dense` contain ten and twenty scattered circular obstacles, respectively.
- `Conveyor` contains two narrow corridors with opposing preferred traffic directions, requiring coordinated single-file motion.
- `Highways` contains a central block and eight boundary blocks that leave a ring road with a preferred counterclockwise circulation.
- `Drop-Region` contains four square obstacles and rewards pausing at drop-off points beside them.
- `Corridor-Bay` contains three parallel channels joined by a narrow crossing, requiring teams to coordinate their passage.
- `Cross-Room` contains a plus-shaped divider that forms four quadrants connected only through gaps near the workspace boundary.

The demonstrations associated with each environment encode its characteristic motion pattern, which is measured by the data-adherence metric *DA*; on `Corridor-Bay` and `Cross-Room`, which have no preferred pattern, *DA* is the start-to-goal distance divided by the path length.

In the normalized workspace $[-1, 1]^2$, robots have radius $r = 0.05$ on every map except `Empty`, where $r = 0.10$, with minimum pairwise separation $2r$. Obstacles are those of the source maps, enlarged by 2.20 (`Basic`), 1.50 (`Conveyor`), 1.45 (`Highways`), 1.60 (`Drop-Region`),

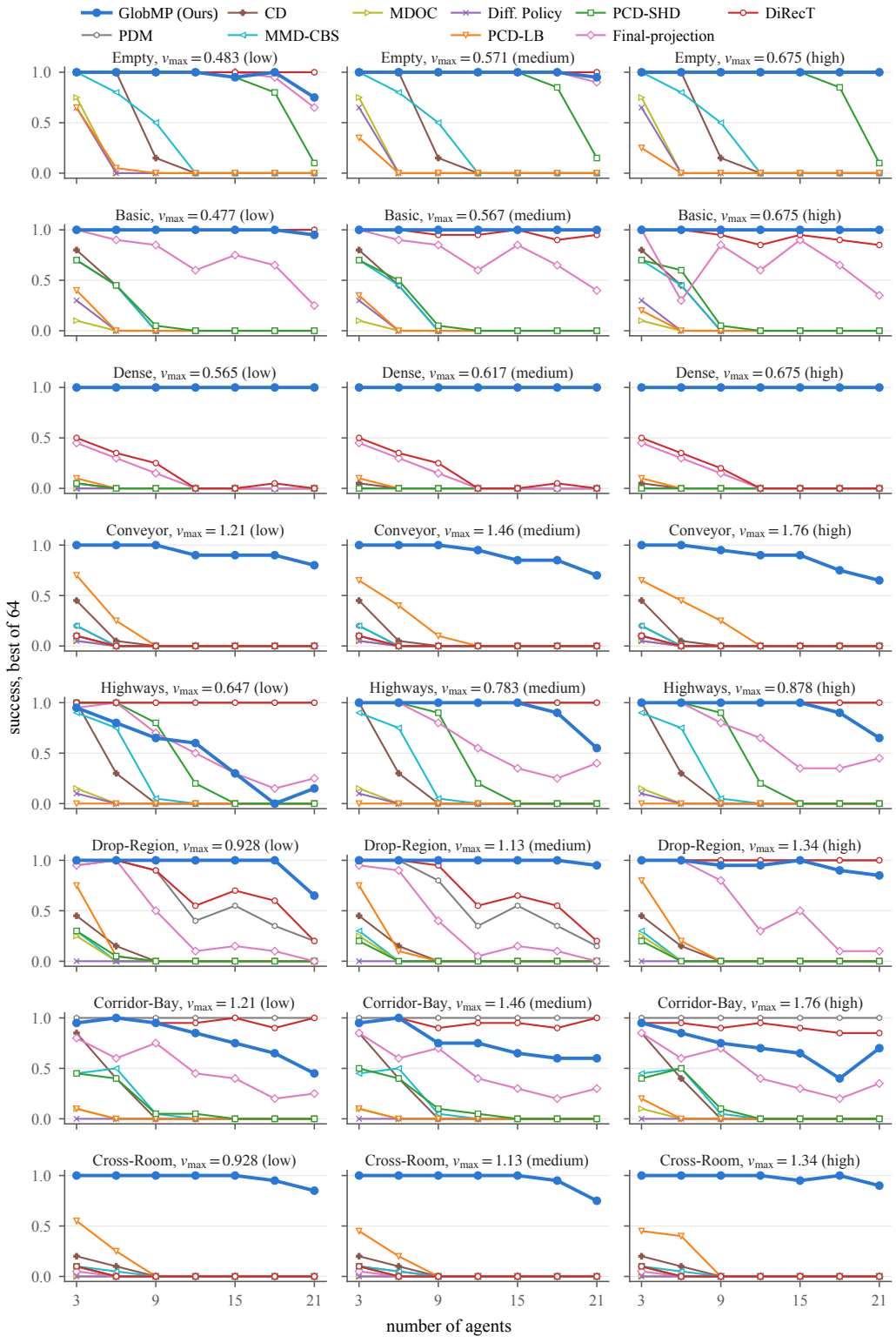


Figure 15: Best-of-64 success rate for $N_a \in \{3, 6, \ldots, 21\}$ agents on every cell of the suite, one row per map and one column per velocity cap: an instance counts as solved when any of the 64 candidates a baseline samples is collision-free. This is the SR reported in Tables 4–11; our method still returns one trajectory.

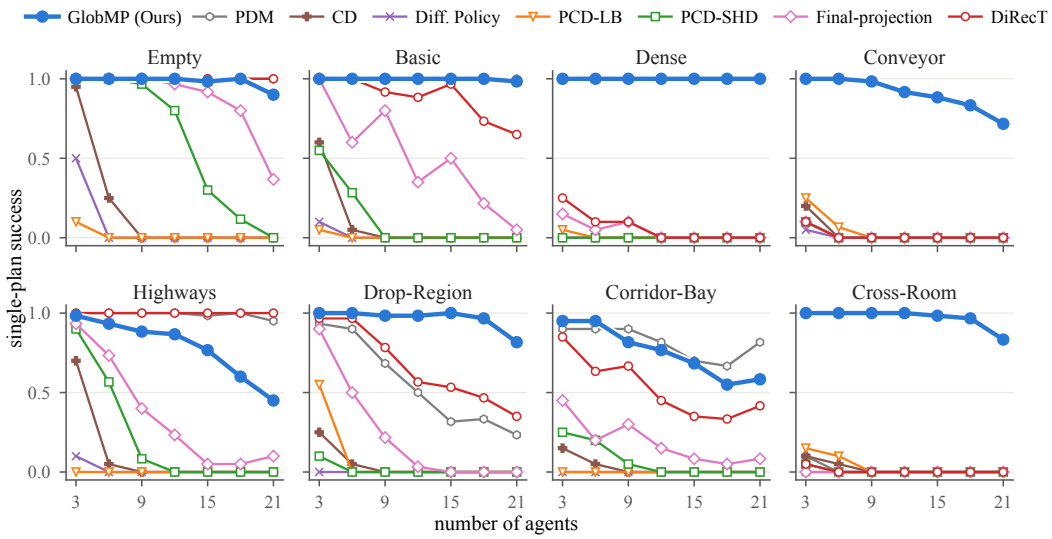


Figure 16: Success of a *single* returned plan for $N_a \in \{3, 6, \ldots, 21\}$ agents, one panel per map, averaged over the three velocity caps: the first candidate each baseline samples, and the one trajectory our method returns. MDOC and MMD-CBS store no per-candidate data and are omitted. Figure 15 gives the best-of-64 counterpart.

1.15 (`Corridor-Bay`) and 1.10 (`Cross-Room`); `Dense` keeps its published sizes. Each environment is evaluated under Low, Medium, and High test-time velocity caps. In the order Low, Medium, High, these are $(0.483, 0.571, 0.675)$ for `Empty`, $(0.477, 0.567, 0.675)$ for `Basic`, $(0.565, 0.617, 0.675)$ for `Dense`, $(1.21, 1.46, 1.76)$ for `Conveyor`, $(0.647, 0.783, 0.878)$ for `Highways`, $(0.928, 1.13, 1.34)$ for `Drop-Region`, $(1.21, 1.46, 1.76)$ for `Corridor-Bay`, and $(0.928, 1.13, 1.34)$ for `Cross-Room`.

Baselines. We compare GlobMP against Diffusion Policy (Chi et al., 2023), CD, PCD-LB, and PCD-SHD (Luan et al., 2026), Final Proj. and DiRecT (Giaretta et al., 2026), MMD-CBS (Shaoul et al., 2025), and MDOC (He et al., 2026). Diffusion Policy provides the unguided reference; CD uses constraint guidance; PCD-LB and PCD-SHD augment guidance with projection; Final Proj. applies projection after sampling; MMD-CBS combines sampled single-agent motions through conflict-based search; DiRecT applies projection throughout denoising; and MDOC uses model-based diffusion optimal control. Each method is evaluated on the same start–goal instances.

Evaluation. We evaluate each method for $N_a \in \{3, 6, \ldots, 21\}$ on all eight maps and under the three velocity caps. For each (map, team size) pair, the same 20 initial configurations are shared across methods, and each sampling baseline generates 64 candidate trajectories per configuration; MMD-CBS and MDOC return one plan, and GlobMP one trajectory. Following Giaretta et al. (2026), we report *SR*, the fraction of configurations with at least one collision-free trajectory; *CS*, the fraction of all trajectories satisfying the collision and kinematic constraints; *Coll*, the mean number of colliding agent pairs; *DA*, environment-specific motion-pattern adherence; and *Time*, wall-clock time per configuration in seconds. Tables 4–11 give SR and CS in percent; the text quotes them as fractions. A dash marks a quantity that was not measured or is not comparable, rather than a zero; the conventions of each table are restated in its caption.

Figures 15–16 and Tables 4–11 summarize the results. We interpret *DA* together with feasibility, since adherence to the demonstrated motion pattern is not meaningful if the generated trajectory violates the constraints, consistent with the evaluation of Giaretta et al. (2026). At the Medium velocity cap, averaged across all maps and team sizes, *GlobMP (Ours)* attains a mean *SR* of 0.941, ahead of DiRecT at 0.601, Final Proj. at 0.419, PCD-SHD at 0.211, CD at 0.132, MMD-CBS at 0.121, PCD-LB at 0.063, MDOC at 0.025 and Diffusion Policy at 0.020.

Table 4: Multi-robot planning performance on `Empty` for $N_a \in \{3, 6, \ldots, 21\}$ under the Low, Medium and High velocity caps. *SR* and *CS* measure trajectory feasibility, *Coll* reports inter-robot collisions, *DA* adherence to the environment-specific motion pattern, and *Time* the planning time per configuration: for *GlobMP (Ours)* the median over the configurations with the standard deviation as subscript, for the baselines the mean, so the two are not directly comparable. † marks a method that applies no velocity limit, so its three blocks report the same run; MMD-CBS also returns one plan per configuration rather than 64 candidates. A dash under *Coll* marks a count that is not commensurable with the others, as MDOC and MMD-CBS count colliding time steps where every other row counts agent pairs with any collision; MDOC's *CS* is omitted because its plans that do not cover every agent were counted as feasible. MMD-CBS plans under 60 s per configuration and *GlobMP (Ours)* under 300 s per team.

		Low ($v_{\max} = 0.483$)					Medium ($v_{\max} = 0.571$)					High ($v_{\max} = 0.675$)				
N_a	Method	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓
3	Diff. Policy†	65.0	48.2$_{\pm 46.5}$	0.67$_{\pm 0.77}$	0.999$_{\pm 0.001}$	0.5	65.0	48.2$_{\pm 46.5}$	0.67$_{\pm 0.77}$	0.999$_{\pm 0.001}$	0.3	65.0	48.2$_{\pm 46.5}$	0.67$_{\pm 0.77}$	0.999$_{\pm 0.001}$	0.5
	CD†	100.0	1.1$_{\pm 4.4}$	0.14$_{\pm 0.28}$	0.517$_{\pm 0.149}$	1.0$_{\pm 0.1}$	100.0	7.1$_{\pm 20.2}$	0.14$_{\pm 0.28}$	0.517$_{\pm 0.149}$	1.1$_{\pm 0.1}$	100.0	20.5$_{\pm 26.1}$	0.14$_{\pm 0.28}$	0.517$_{\pm 0.149}$	0.5$_{\pm 0.0}$
	MMD-CBS†	100.0	70.0$_{\pm 45.8}$	—	0.946$_{\pm 0.071}$	3.1$_{\pm 2.5}$	100.0	70.0$_{\pm 45.8}$	—	0.946$_{\pm 0.071}$	3.1$_{\pm 2.5}$	100.0	70.0$_{\pm 45.8}$	—	0.946$_{\pm 0.071}$	3.1$_{\pm 2.5}$
	MDOC†	75.0	—	—	0.572$_{\pm 0.400}$	39.0$_{\pm 6.1}$	75.0	—	—	0.572$_{\pm 0.400}$	34.8$_{\pm 4.3}$	75.0	—	—	0.572$_{\pm 0.400}$	39.6$_{\pm 4.8}$
	PCD-LB	65.0	12.3$_{\pm 20.2}$	0.02$_{\pm 0.05}$	0.325$_{\pm 0.135}$	2.8$_{\pm 0.8}$	35.0	4.6$_{\pm 11.9}$	0.01$_{\pm 0.02}$	0.316$_{\pm 0.133}$	4.2$_{\pm 0.9}$	25.0	3.3$_{\pm 9.1}$	0.01$_{\pm 0.01}$	0.307$_{\pm 0.130}$	3.2$_{\pm 0.6}$
	PCD-SHD	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.828$_{\pm 0.113}$	3.0$_{\pm 0.4}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.831$_{\pm 0.109}$	1.4$_{\pm 0.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.833$_{\pm 0.107}$	1.4$_{\pm 0.2}$
	Final Proj.	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.998**$_{\pm 0.003}$	3.7$_{\pm 0.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.998**$_{\pm 0.003}$	2.0$_{\pm 0.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.998**$_{\pm 0.003}$	3.7$_{\pm 0.5}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.993$_{\pm 0.018}$	31.5$_{\pm 0.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.993$_{\pm 0.018}$	31.4$_{\pm 0.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.993$_{\pm 0.017}$	31.0$_{\pm 1.3}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.933$_{\pm 0.149}$	4.0$_{\pm 17.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.934$_{\pm 0.149}$	4.0$_{\pm 6.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.984$_{\pm 0.043}$	0.2$_{\pm 0.1}$
6	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	4.77$_{\pm 2.73}$	0.999$_{\pm 0.001}$	0.5	0.0	0.0$_{\pm 0.0}$	4.77$_{\pm 2.73}$	0.999$_{\pm 0.001}$	0.6	0.0	0.0$_{\pm 0.0}$	4.77$_{\pm 2.73}$	0.999$_{\pm 0.001}$	0.3
	CD†	100.0	0.0$_{\pm 0.0}$	0.96$_{\pm 0.85}$	0.582$_{\pm 0.079}$	1.1$_{\pm 0.1}$	100.0	0.0$_{\pm 0.0}$	0.96$_{\pm 0.85}$	0.582$_{\pm 0.079}$	0.9$_{\pm 0.3}$	100.0	0.2$_{\pm 0.7}$	0.96$_{\pm 0.85}$	0.582$_{\pm 0.079}$	1.1$_{\pm 0.1}$
	MMD-CBS†	80.0	15.0$_{\pm 35.7}$	—	0.879$_{\pm 0.065}$	30.3$_{\pm 21.2}$	80.0	15.0$_{\pm 35.7}$	—	0.879$_{\pm 0.065}$	30.3$_{\pm 21.2}$	80.0	15.0$_{\pm 35.7}$	—	0.879$_{\pm 0.065}$	30.3$_{\pm 21.2}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	57.6$_{\pm 4.8}$	0.0	—	—	0.000$_{\pm 0.000}$	58.2$_{\pm 4.8}$	0.0	—	—	0.000$_{\pm 0.000}$	67.8$_{\pm 5.2}$
	PCD-LB	5.0	0.1$_{\pm 0.3}$	1.01$_{\pm 0.73}$	0.299$_{\pm 0.050}$	3.8$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	1.04$_{\pm 0.72}$	0.300$_{\pm 0.056}$	5.8$_{\pm 1.1}$	0.0	0.0$_{\pm 0.0}$	1.08$_{\pm 0.69}$	0.302$_{\pm 0.060}$	2.3$_{\pm 0.3}$
	PCD-SHD	**100.0**	99.8$_{\pm 0.5}$	0.00$_{\pm 0.00}$	0.608$_{\pm 0.073}$	4.3$_{\pm 1.4}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.612$_{\pm 0.071}$	2.0$_{\pm 0.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.613$_{\pm 0.068}$	1.6$_{\pm 0.1}$
	Final Proj.	**100.0**	98.3$_{\pm 7.5}$	0.04$_{\pm 0.15}$	**0.995**$_{\pm 0.006}$	3.2$_{\pm 0.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.996**$_{\pm 0.005}$	1.8$_{\pm 0.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.996**$_{\pm 0.004}$	1.8$_{\pm 0.8}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.985$_{\pm 0.017}$	27.6$_{\pm 3.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.985$_{\pm 0.016}$	24.6$_{\pm 5.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.985$_{\pm 0.016}$	24.0$_{\pm 6.4}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.975$_{\pm 0.090}$	4.4$_{\pm 22.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.976$_{\pm 0.090}$	4.3$_{\pm 4.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.960$_{\pm 0.089}$	0.2$_{\pm 0.1}$
9	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	9.22$_{\pm 3.01}$	0.999$_{\pm 0.001}$	0.6	0.0	0.0$_{\pm 0.0}$	9.22$_{\pm 3.01}$	0.999$_{\pm 0.001}$	0.6	0.0	0.0$_{\pm 0.0}$	9.22$_{\pm 3.01}$	0.999$_{\pm 0.001}$	0.6
	CD†	15.0	0.0$_{\pm 0.0}$	3.80$_{\pm 1.33}$	0.569$_{\pm 0.075}$	1.1$_{\pm 0.1}$	15.0	0.0$_{\pm 0.0}$	3.80$_{\pm 1.33}$	0.569$_{\pm 0.075}$	0.5$_{\pm 0.1}$	15.0	0.0$_{\pm 0.0}$	3.80$_{\pm 1.33}$	0.569$_{\pm 0.075}$	1.1$_{\pm 0.1}$
	MMD-CBS†	50.0	0.0$_{\pm 0.0}$	—	0.877$_{\pm 0.062}$	57.5$_{\pm 8.9}$	50.0	0.0$_{\pm 0.0}$	—	0.877$_{\pm 0.062}$	57.5$_{\pm 8.9}$	50.0	0.0$_{\pm 0.0}$	—	0.877$_{\pm 0.062}$	57.5$_{\pm 8.9}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	83.3$_{\pm 5.9}$	0.0	—	—	0.000$_{\pm 0.000}$	84.1$_{\pm 5.7}$	0.0	—	—	0.000$_{\pm 0.000}$	97.4$_{\pm 7.2}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	4.78$_{\pm 1.47}$	0.253$_{\pm 0.038}$	4.6$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	5.26$_{\pm 1.65}$	0.260$_{\pm 0.037}$	6.4$_{\pm 0.8}$	0.0	0.0$_{\pm 0.0}$	5.78$_{\pm 1.64}$	0.271$_{\pm 0.037}$	4.6$_{\pm 0.5}$
	PCD-SHD	**100.0**	94.8$_{\pm 8.7}$	0.05$_{\pm 0.09}$	0.547$_{\pm 0.052}$	5.4$_{\pm 0.9}$	**100.0**	96.2$_{\pm 5.0}$	0.04$_{\pm 0.05}$	0.557$_{\pm 0.052}$	2.2$_{\pm 0.2}$	**100.0**	97.5$_{\pm 3.8}$	0.02$_{\pm 0.04}$	0.562$_{\pm 0.052}$	1.9$_{\pm 0.1}$
	Final Proj.	**100.0**	99.4$_{\pm 2.1}$	0.01$_{\pm 0.04}$	**0.993**$_{\pm 0.006}$	4.1$_{\pm 0.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.995**$_{\pm 0.005}$	2.2$_{\pm 0.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.995**$_{\pm 0.004}$	2.1$_{\pm 0.8}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.972$_{\pm 0.020}$	33.5$_{\pm 7.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.973$_{\pm 0.020}$	33.3$_{\pm 7.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.974$_{\pm 0.019}$	33.2$_{\pm 8.0}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.989$_{\pm 0.062}$	6.1$_{\pm 19.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.990$_{\pm 0.062}$	4.7$_{\pm 1.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.932$_{\pm 0.116}$	0.4$_{\pm 0.2}$
12	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	17.68$_{\pm 4.12}$	0.999$_{\pm 0.001}$	0.6	0.0	0.0$_{\pm 0.0}$	17.68$_{\pm 4.12}$	0.999$_{\pm 0.001}$	0.6	0.0	0.0$_{\pm 0.0}$	17.68$_{\pm 4.12}$	0.999$_{\pm 0.001}$	0.6
	CD†	0.0	0.0$_{\pm 0.0}$	8.90$_{\pm 2.05}$	0.555$_{\pm 0.045}$	0.5$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	8.90$_{\pm 2.05}$	0.555$_{\pm 0.045}$	1.1$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	8.90$_{\pm 2.05}$	0.555$_{\pm 0.045}$	0.5$_{\pm 0.0}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.961$_{\pm 0.022}$	67.5$_{\pm 4.9}$	0.0	0.0$_{\pm 0.0}$	—	0.961$_{\pm 0.022}$	67.5$_{\pm 4.9}$	0.0	0.0$_{\pm 0.0}$	—	0.961$_{\pm 0.022}$	67.5$_{\pm 4.9}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	110.0$_{\pm 8.6}$	0.0	—	—	0.000$_{\pm 0.000}$	111.2$_{\pm 8.5}$	0.0	—	—	0.000$_{\pm 0.000}$	128.9$_{\pm 10.2}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	11.68$_{\pm 1.53}$	0.259$_{\pm 0.038}$	5.8$_{\pm 0.8}$	0.0	0.0$_{\pm 0.0}$	13.11$_{\pm 1.52}$	0.265$_{\pm 0.038}$	8.0$_{\pm 1.0}$	0.0	0.0$_{\pm 0.0}$	14.36$_{\pm 1.63}$	0.279$_{\pm 0.040}$	3.1$_{\pm 0.4}$
	PCD-SHD	**100.0**	74.0$_{\pm 15.1}$	0.27$_{\pm 0.19}$	0.515$_{\pm 0.047}$	7.1$_{\pm 1.4}$	**100.0**	80.6$_{\pm 13.5}$	0.18$_{\pm 0.13}$	0.521$_{\pm 0.043}$	2.9$_{\pm 0.5}$	**100.0**	81.6$_{\pm 12.9}$	0.17$_{\pm 0.13}$	0.524$_{\pm 0.040}$	2.3$_{\pm 0.1}$
	Final Proj.	**100.0**	97.1$_{\pm 6.5}$	0.05$_{\pm 0.11}$	0.988$_{\pm 0.009}$	6.4$_{\pm 0.6}$	**100.0**	99.9$_{\pm 0.3}$	0.00$_{\pm 0.01}$	0.990$_{\pm 0.007}$	6.5$_{\pm 0.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.992**$_{\pm 0.006}$	3.0$_{\pm 0.8}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.955$_{\pm 0.023}$	45.2$_{\pm 9.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.957$_{\pm 0.022}$	45.8$_{\pm 9.8}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.958$_{\pm 0.021}$	50.0$_{\pm 5.4}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.994**$_{\pm 0.024}$	9.7$_{\pm 31.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.996**$_{\pm 0.016}$	7.0$_{\pm 5.7}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.904$_{\pm 0.133}$	0.6$_{\pm 0.3}$
15	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	26.96$_{\pm 5.94}$	0.999$_{\pm 0.001}$	0.6	0.0	0.0$_{\pm 0.0}$	26.96$_{\pm 5.94}$	0.999$_{\pm 0.001}$	0.6	0.0	0.0$_{\pm 0.0}$	26.96$_{\pm 5.94}$	0.999$_{\pm 0.001}$	0.3
	CD†	0.0	0.0$_{\pm 0.0}$	15.53$_{\pm 2.54}$	0.587$_{\pm 0.059}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	15.53$_{\pm 2.54}$	0.587$_{\pm 0.059}$	0.9$_{\pm 0.3}$	0.0	0.0$_{\pm 0.0}$	15.53$_{\pm 2.54}$	0.587$_{\pm 0.059}$	1.2$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.989$_{\pm 0.013}$	74.4$_{\pm 9.3}$	0.0	0.0$_{\pm 0.0}$	—	0.989$_{\pm 0.013}$	74.4$_{\pm 9.3}$	0.0	0.0$_{\pm 0.0}$	—	0.989$_{\pm 0.013}$	74.4$_{\pm 9.3}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	139.2$_{\pm 10.6}$	0.0	—	—	0.000$_{\pm 0.000}$	139.0$_{\pm 10.6}$	0.0	—	—	0.000$_{\pm 0.000}$	153.1$_{\pm 12.1}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	20.89$_{\pm 2.48}$	0.280$_{\pm 0.038}$	6.8$_{\pm 0.9}$	0.0	0.0$_{\pm 0.0}$	23.55$_{\pm 2.73}$	0.284$_{\pm 0.037}$	9.2$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	26.34$_{\pm 2.90}$	0.300$_{\pm 0.037}$	6.8$_{\pm 0.4}$
	PCD-SHD	95.0	33.7$_{\pm 20.2}$	0.99$_{\pm 0.52}$	0.506$_{\pm 0.028}$	9.5$_{\pm 2.0}$	**100.0**	37.2$_{\pm 16.9}$	0.79$_{\pm 0.44}$	0.517$_{\pm 0.026}$	2.7$_{\pm 0.3}$	**100.0**	37.5$_{\pm 18.1}$	0.73$_{\pm 0.43}$	0.523$_{\pm 0.027}$	2.7$_{\pm 0.1}$
	Final Proj.	**100.0**	82.8$_{\pm 26.1}$	0.60$_{\pm 1.61}$	0.975$_{\pm 0.013}$	5.2$_{\pm 0.4}$	**100.0**	94.7$_{\pm 13.7}$	0.13$_{\pm 0.38}$	0.979$_{\pm 0.011}$	8.5$_{\pm 2.0}$	**100.0**	98.9$_{\pm 3.4}$	0.02$_{\pm 0.07}$	**0.982**$_{\pm 0.010}$	4.2$_{\pm 0.8}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.926$_{\pm 0.028}$	61.9$_{\pm 11.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.930$_{\pm 0.025}$	68.1$_{\pm 0.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.932$_{\pm 0.024}$	46.4$_{\pm 18.9}$
	GlobMP (Ours)	95.0	95.0$_{\pm 21.8}$	1.65$_{\pm 7.19}$	**0.983**$_{\pm 0.043}$	24.5$_{\pm 61.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.987**$_{\pm 0.033}$	14.6$_{\pm 40.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.905$_{\pm 0.134}$	1.1$_{\pm 1.6}$
18	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	39.06$_{\pm 6.29}$	0.999$_{\pm 0.001}$	0.7	0.0	0.0$_{\pm 0.0}$	39.06$_{\pm 6.29}$	0.999$_{\pm 0.001}$	0.7	0.0	0.0$_{\pm 0.0}$	39.06$_{\pm 6.29}$	0.999$_{\pm 0.001}$	0.7
	CD†	0.0	0.0$_{\pm 0.0}$	22.94$_{\pm 1.70}$	0.565$_{\pm 0.049}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	22.94$_{\pm 1.70}$	0.565$_{\pm 0.049}$	0.6$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	22.94$_{\pm 1.70}$	0.565$_{\pm 0.049}$	1.2$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.996$_{\pm 0.009}$	81.6$_{\pm 17.1}$	0.0	0.0$_{\pm 0.0}$	—	0.996$_{\pm 0.009}$	81.6$_{\pm 17.1}$	0.0	0.0$_{\pm 0.0}$	—	0.996$_{\pm 0.009}$	81.6$_{\pm 17.1}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	171.7$_{\pm 13.7}$	0.0	—	—	0.000$_{\pm 0.000}$	172.6$_{\pm 14.8}$	0.0	—	—	0.000$_{\pm 0.000}$	185.6$_{\pm 17.7}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	29.66$_{\pm 2.84}$	0.271$_{\pm 0.038}$	8.1$_{\pm 0.9}$	0.0	0.0$_{\pm 0.0}$	33.68$_{\pm 2.86}$	0.276$_{\pm 0.037}$	11.0$_{\pm 1.2}$	0.0	0.0$_{\pm 0.0}$	38.10$_{\pm 2.86}$	0.289$_{\pm 0.037}$	4.1$_{\pm 0.5}$
	PCD-SHD	80.0	7.1$_{\pm 7.0}$	2.21$_{\pm 0.82}$	0.485$_{\pm 0.034}$	5.2$_{\pm 1.0}$	85.0	11.2$_{\pm 9.5}$	1.83$_{\pm 0.78}$	0.494$_{\pm 0.035}$	3.3$_{\pm 0.5}$	85.0	9.9$_{\pm 8.8}$	1.69$_{\pm 0.73}$	0.502$_{\pm 0.036}$	3.0$_{\pm 0.2}$
	Final Proj.	95.0	67.3$_{\pm 36.5}$	1.60$_{\pm 2.70}$	0.966$_{\pm 0.015}$	13.6$_{\pm 0.6}$	**100.0**	85.0$_{\pm 28.5}$	0.45$_{\pm 0.95}$	0.971$_{\pm 0.012}$	10.2$_{\pm 3.0}$	**100.0**	95.4$_{\pm 11.3}$	0.08$_{\pm 0.24}$	**0.975**$_{\pm 0.011}$	5.8$_{\pm 0.8}$
	DiRecT	**100.0**	99.1$_{\pm 4.1}$	0.00$_{\pm 0.00}$	0.901$_{\pm 0.025}$	38.0$_{\pm 0.6}$	**100.0**	99.1$_{\pm 3.7}$	0.00$_{\pm 0.00}$	0.907$_{\pm 0.024}$	38.2$_{\pm 0.6}$	**100.0**	99.2$_{\pm 3.4}$	**0.00**$_{\pm 0.00}$	0.911$_{\pm 0.023}$	38.2$_{\pm 0.6}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.977**$_{\pm 0.047}$	27.5$_{\pm 51.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.984**$_{\pm 0.034}$	15.0$_{\pm 25.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.876$_{\pm 0.161}$	1.6$_{\pm 1.9}$
21	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	60.02$_{\pm 7.50}$	0.999$_{\pm 0.001}$	0.7	0.0	0.0$_{\pm 0.0}$	60.02$_{\pm 7.50}$	0.999$_{\pm 0.001}$	0.3	0.0	0.0$_{\pm 0.0}$	60.02$_{\pm 7.50}$	0.999$_{\pm 0.001}$	0.7
	CD†	0.0	0.0$_{\pm 0.0}$	35.83$_{\pm 3.71}$	0.587$_{\pm 0.041}$	1.3$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	35.83$_{\pm 3.71}$	0.587$_{\pm 0.041}$	1.2$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	35.83$_{\pm 3.71}$	0.587$_{\pm 0.041}$	1.3$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	1.000$_{\pm 0.000}$	82.4$_{\pm 9.9}$	0.0	0.0$_{\pm 0.0}$	—	1.000$_{\pm 0.000}$	82.4$_{\pm 9.9}$	0.0	0.0$_{\pm 0.0}$	—	1.000$_{\pm 0.000}$	82.4$_{\pm 9.9}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	193.5$_{\pm 25.2}$	0.0	—	—	0.000$_{\pm 0.000}$	192.8$_{\pm 25.2}$	0.0	—	—	0.000$_{\pm 0.000}$	217.6$_{\pm 36.3}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	45.06$_{\pm 2.97}$	0.255$_{\pm 0.026}$	9.9$_{\pm 1.5}$	0.0	0.0$_{\pm 0.0}$	51.43$_{\pm 2.86}$	0.260$_{\pm 0.022}$	13.2$_{\pm 1.8}$	0.0	0.0$_{\pm 0.0}$	57.66$_{\pm 3.16}$	0.273$_{\pm 0.020}$	9.3$_{\pm 0.8}$
	PCD-SHD	10.0	0.2$_{\pm 0.7}$	5.13$_{\pm 1.31}$	0.467$_{\pm 0.028}$	6.8$_{\pm 1.5}$	15.0	0.2$_{\pm 0.6}$	4.10$_{\pm 1.05}$	0.480$_{\pm 0.024}$	4.3$_{\pm 0.6}$	10.0	0.4$_{\pm 1.2}$	3.64$_{\pm 0.92}$	0.491$_{\pm 0.023}$	3.6$_{\pm 0.2}$
	Final Proj.	65.0	19.2$_{\pm 27.0}$	7.83$_{\pm 6.74}$	0.940$_{\pm 0.020}$	10.8$_{\pm 2.2}$	90.0	45.0$_{\pm 35.2}$	3.88$_{\pm 4.47}$	0.948$_{\pm 0.018}$	14.5$_{\pm 4.3}$	**100.0**	62.4$_{\pm 34.9}$	1.70$_{\pm 2.11}$	**0.954**$_{\pm 0.016}$	7.9$_{\pm 0.8}$
	DiRecT	**100.0**	**99.8**$_{\pm 0.7}$	**0.00**$_{\pm 0.00}$	0.857$_{\pm 0.033}$	99.1$_{\pm 31.9}$	**100.0**	**99.9**$_{\pm 0.3}$	**0.00**$_{\pm 0.00}$	0.867$_{\pm 0.031}$	91.9$_{\pm 33.8}$	**100.0**	99.9$_{\pm 0.3}$	**0.00**$_{\pm 0.00}$	0.873$_{\pm 0.028}$	85.6$_{\pm 34.6}$
	GlobMP (Ours)	75.0	75.0$_{\pm 43.3}$	17.15$_{\pm 30.27}$	**0.972**$_{\pm 0.055}$	72.7$_{\pm 81.0}$	95.0	95.0$_{\pm 21.8}$	4.50$_{\pm 19.62}$	**0.972**$_{\pm 0.050}$	31.2$_{\pm 46.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.839$_{\pm 0.185}$	3.1$_{\pm 3.8}$

Table 5: Multi-robot planning performance on `Basic` for $N_a \in \{3, 6, \ldots, 21\}$ under the Low, Medium and High velocity caps. *SR* and *CS* measure trajectory feasibility, *Coll* reports inter-robot collisions, *DA* adherence to the environment-specific motion pattern, and *Time* the planning time per configuration: for *GlobMP (Ours)* the median over the configurations with the standard deviation as subscript, for the baselines the mean, so the two are not directly comparable. † marks a method that applies no velocity limit, so its three blocks report the same run; MMD-CBS also returns one plan per configuration rather than 64 candidates. A dash under *Coll* marks a count that is not commensurable with the others, as MDOC and MMD-CBS count colliding time steps where every other row counts agent pairs with any collision; MDOC's *CS* is omitted because its plans that do not cover every agent were counted as feasible. MMD-CBS plans under 60 s per configuration and *GlobMP (Ours)* under 300 s per team.

		Low ($v_{\max} = 0.477$)					Medium ($v_{\max} = 0.567$)					High ($v_{\max} = 0.675$)				
N_a	Method	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓
3	Diff. Policy†	30.0	13.0$_{\pm 29.6}$	0.35$_{\pm 0.64}$	0.998$_{\pm 0.001}$	0.3	30.0	13.0$_{\pm 29.6}$	0.35$_{\pm 0.64}$	0.998$_{\pm 0.001}$	0.3	30.0	13.0$_{\pm 29.6}$	0.35$_{\pm 0.64}$	0.998$_{\pm 0.001}$	0.3
	CD†	80.0	0.0$_{\pm 0.0}$	0.05$_{\pm 0.08}$	0.611$_{\pm 0.138}$	1.1$_{\pm 0.0}$	80.0	0.0$_{\pm 0.0}$	0.05$_{\pm 0.08}$	0.611$_{\pm 0.138}$	0.4$_{\pm 0.0}$	80.0	0.4$_{\pm 1.0}$	0.05$_{\pm 0.08}$	0.611$_{\pm 0.138}$	1.1$_{\pm 0.1}$
	MMD-CBS†	70.0	35.0$_{\pm 47.7}$	—	0.895$_{\pm 0.113}$	2.5$_{\pm 1.7}$	70.0	50.0$_{\pm 50.0}$	—	0.895$_{\pm 0.113}$	2.5$_{\pm 1.7}$	70.0	55.0$_{\pm 49.7}$	—	0.895$_{\pm 0.113}$	2.5$_{\pm 1.7}$
	MDOC†	10.0	—	—	0.090$_{\pm 0.269}$	33.4$_{\pm 4.5}$	10.0	—	—	0.090$_{\pm 0.269}$	33.6$_{\pm 4.4}$	10.0	—	—	0.090$_{\pm 0.269}$	33.5$_{\pm 4.6}$
	PCD-LB	40.0	5.2$_{\pm 10.1}$	0.01$_{\pm 0.03}$	0.361$_{\pm 0.114}$	5.2$_{\pm 1.6}$	35.0	2.2$_{\pm 3.6}$	0.01$_{\pm 0.03}$	0.349$_{\pm 0.112}$	4.4$_{\pm 0.6}$	20.0	2.1$_{\pm 6.1}$	0.01$_{\pm 0.03}$	0.336$_{\pm 0.106}$	3.6$_{\pm 0.4}$
	PCD-SHD	70.0	61.2$_{\pm 43.9}$	0.06$_{\pm 0.14}$	0.847$_{\pm 0.120}$	3.1$_{\pm 0.2}$	70.0	61.1$_{\pm 43.9}$	0.06$_{\pm 0.14}$	0.848$_{\pm 0.118}$	1.5$_{\pm 0.1}$	70.0	60.7$_{\pm 43.8}$	0.05$_{\pm 0.12}$	0.849$_{\pm 0.118}$	3.0$_{\pm 0.1}$
	Final Proj.	**100.0**	95.1$_{\pm 21.5}$	**0.00**$_{\pm 0.00}$	**0.964**$_{\pm 0.050}$	4.0$_{\pm 0.7}$	**100.0**	95.1$_{\pm 21.5}$	**0.00**$_{\pm 0.00}$	**0.969**$_{\pm 0.043}$	4.0$_{\pm 0.8}$	**100.0**	95.1$_{\pm 21.5}$	**0.00**$_{\pm 0.00}$	0.973$_{\pm 0.037}$	2.2$_{\pm 0.8}$
	DiRecT	**100.0**	99.8$_{\pm 1.0}$	**0.00**$_{\pm 0.00}$	0.925$_{\pm 0.098}$	26.3$_{\pm 6.4}$	**100.0**	95.6$_{\pm 19.1}$	**0.00**$_{\pm 0.00}$	0.930$_{\pm 0.093}$	26.8$_{\pm 6.5}$	**100.0**	95.5$_{\pm 19.4}$	**0.00**$_{\pm 0.00}$	0.931$_{\pm 0.092}$	27.7$_{\pm 7.3}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.878$_{\pm 0.169}$	2.1$_{\pm 2.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.851$_{\pm 0.184}$	0.2$_{\pm 2.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.987**$_{\pm 0.026}$	0.0$_{\pm 0.1}$
6	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	3.63$_{\pm 1.90}$	0.998$_{\pm 0.001}$	0.3	0.0	0.0$_{\pm 0.0}$	3.63$_{\pm 1.90}$	0.998$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	3.63$_{\pm 1.90}$	0.998$_{\pm 0.001}$	0.5
	CD†	45.0	0.0$_{\pm 0.0}$	1.14$_{\pm 0.58}$	0.633$_{\pm 0.079}$	1.1$_{\pm 0.1}$	45.0	0.0$_{\pm 0.0}$	1.14$_{\pm 0.58}$	0.633$_{\pm 0.079}$	1.2$_{\pm 0.0}$	45.0	0.0$_{\pm 0.0}$	1.14$_{\pm 0.58}$	0.633$_{\pm 0.079}$	1.1$_{\pm 0.1}$
	MMD-CBS†	45.0	0.0$_{\pm 0.0}$	—	0.797$_{\pm 0.102}$	9.3$_{\pm 9.6}$	45.0	5.0$_{\pm 21.8}$	—	0.797$_{\pm 0.102}$	9.3$_{\pm 9.6}$	45.0	10.0$_{\pm 30.0}$	—	0.797$_{\pm 0.102}$	9.3$_{\pm 9.6}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	90.6$_{\pm 21.6}$	0.0	—	—	0.000$_{\pm 0.000}$	69.7$_{\pm 5.3}$	0.0	—	—	0.000$_{\pm 0.000}$	77.3$_{\pm 14.4}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	0.69$_{\pm 0.50}$	0.360$_{\pm 0.064}$	6.4$_{\pm 0.9}$	0.0	0.0$_{\pm 0.0}$	0.72$_{\pm 0.44}$	0.347$_{\pm 0.056}$	4.9$_{\pm 0.5}$	0.0	0.0$_{\pm 0.0}$	0.78$_{\pm 0.43}$	0.336$_{\pm 0.051}$	1.8$_{\pm 0.1}$
	PCD-SHD	45.0	22.2$_{\pm 33.5}$	0.46$_{\pm 0.35}$	0.773$_{\pm 0.084}$	3.3$_{\pm 0.2}$	50.0	21.4$_{\pm 33.1}$	0.45$_{\pm 0.33}$	0.777$_{\pm 0.082}$	3.2$_{\pm 0.1}$	60.0	20.8$_{\pm 31.9}$	0.44$_{\pm 0.31}$	0.780$_{\pm 0.079}$	3.1$_{\pm 0.1}$
	Final Proj.	90.0	85.6$_{\pm 32.3}$	0.09$_{\pm 0.33}$	**0.959**$_{\pm 0.031}$	4.3$_{\pm 0.8}$	90.0	88.6$_{\pm 29.9}$	0.02$_{\pm 0.06}$	**0.965**$_{\pm 0.026}$	4.4$_{\pm 0.7}$	30.0	9.1$_{\pm 22.6}$	0.08$_{\pm 0.11}$	**0.983**$_{\pm 0.011}$	4.2$_{\pm 0.6}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.919$_{\pm 0.059}$	30.6$_{\pm 7.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.921$_{\pm 0.058}$	29.4$_{\pm 7.8}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.922$_{\pm 0.057}$	30.9$_{\pm 8.8}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.932$_{\pm 0.141}$	4.2$_{\pm 18.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.934$_{\pm 0.137}$	4.2$_{\pm 5.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.980$_{\pm 0.041}$	0.2$_{\pm 0.1}$
9	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	6.58$_{\pm 2.27}$	0.999$_{\pm 0.001}$	0.3	0.0	0.0$_{\pm 0.0}$	6.58$_{\pm 2.27}$	0.999$_{\pm 0.001}$	0.3	0.0	0.0$_{\pm 0.0}$	6.58$_{\pm 2.27}$	0.999$_{\pm 0.001}$	0.3
	CD†	0.0	0.0$_{\pm 0.0}$	2.86$_{\pm 1.16}$	0.630$_{\pm 0.065}$	0.5$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	2.86$_{\pm 1.16}$	0.630$_{\pm 0.065}$	1.2$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	2.86$_{\pm 1.16}$	0.630$_{\pm 0.065}$	1.1$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.756$_{\pm 0.082}$	15.1$_{\pm 23.5}$	0.0	0.0$_{\pm 0.0}$	—	0.756$_{\pm 0.082}$	15.1$_{\pm 23.5}$	0.0	0.0$_{\pm 0.0}$	—	0.756$_{\pm 0.082}$	15.1$_{\pm 23.5}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	92.2$_{\pm 17.8}$	0.0	—	—	0.000$_{\pm 0.000}$	81.2$_{\pm 17.7}$	0.0	—	—	0.000$_{\pm 0.000}$	74.6$_{\pm 5.1}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	2.85$_{\pm 1.07}$	0.365$_{\pm 0.072}$	7.4$_{\pm 0.5}$	0.0	0.0$_{\pm 0.0}$	3.06$_{\pm 1.03}$	0.365$_{\pm 0.065}$	5.6$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	3.27$_{\pm 0.98}$	0.361$_{\pm 0.061}$	4.4$_{\pm 0.3}$
	PCD-SHD	5.0	0.3$_{\pm 1.4}$	0.81$_{\pm 0.55}$	0.735$_{\pm 0.061}$	3.6$_{\pm 0.1}$	5.0	0.5$_{\pm 2.0}$	0.81$_{\pm 0.53}$	0.738$_{\pm 0.060}$	3.2$_{\pm 0.6}$	5.0	0.4$_{\pm 1.7}$	0.84$_{\pm 0.50}$	0.740$_{\pm 0.058}$	2.8$_{\pm 0.8}$
	Final Proj.	85.0	76.3$_{\pm 38.4}$	0.11$_{\pm 0.24}$	0.949$_{\pm 0.027}$	5.7$_{\pm 0.7}$	85.0	78.4$_{\pm 38.7}$	0.06$_{\pm 0.21}$	0.957$_{\pm 0.023}$	3.1$_{\pm 0.7}$	85.0	79.0$_{\pm 38.6}$	0.05$_{\pm 0.20}$	0.962$_{\pm 0.020}$	5.8$_{\pm 0.7}$
	DiRecT	**100.0**	99.0$_{\pm 4.4}$	0.01$_{\pm 0.04}$	0.903$_{\pm 0.050}$	46.6$_{\pm 13.9}$	95.0	93.9$_{\pm 22.0}$	0.01$_{\pm 0.04}$	0.902$_{\pm 0.051}$	47.1$_{\pm 14.1}$	95.0	94.0$_{\pm 21.9}$	0.01$_{\pm 0.04}$	0.907$_{\pm 0.046}$	42.6$_{\pm 12.5}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.969**$_{\pm 0.100}$	4.4$_{\pm 8.8}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.969**$_{\pm 0.099}$	4.3$_{\pm 1.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.972**$_{\pm 0.055}$	0.3$_{\pm 0.1}$
12	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	13.12$_{\pm 3.83}$	0.999$_{\pm 0.001}$	0.3	0.0	0.0$_{\pm 0.0}$	13.12$_{\pm 3.83}$	0.999$_{\pm 0.001}$	0.3	0.0	0.0$_{\pm 0.0}$	13.12$_{\pm 3.83}$	0.999$_{\pm 0.001}$	0.3
	CD†	0.0	0.0$_{\pm 0.0}$	7.58$_{\pm 1.54}$	0.638$_{\pm 0.033}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	7.58$_{\pm 1.54}$	0.638$_{\pm 0.033}$	0.5$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	7.58$_{\pm 1.54}$	0.638$_{\pm 0.033}$	1.2$_{\pm 0.0}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.797$_{\pm 0.027}$	9.5$_{\pm 18.8}$	0.0	0.0$_{\pm 0.0}$	—	0.797$_{\pm 0.027}$	9.5$_{\pm 18.8}$	0.0	0.0$_{\pm 0.0}$	—	0.797$_{\pm 0.027}$	9.5$_{\pm 18.8}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	104.7$_{\pm 8.5}$	0.0	—	—	0.000$_{\pm 0.000}$	113.0$_{\pm 6.5}$	0.0	—	—	0.000$_{\pm 0.000}$	109.8$_{\pm 6.6}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	7.89$_{\pm 1.38}$	0.340$_{\pm 0.044}$	8.6$_{\pm 0.8}$	0.0	0.0$_{\pm 0.0}$	8.44$_{\pm 1.23}$	0.336$_{\pm 0.044}$	3.1$_{\pm 0.3}$	0.0	0.0$_{\pm 0.0}$	8.87$_{\pm 1.18}$	0.334$_{\pm 0.043}$	4.9$_{\pm 0.4}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	1.98$_{\pm 0.70}$	0.713$_{\pm 0.052}$	3.9$_{\pm 0.2}$	0.0	0.0$_{\pm 0.0}$	1.93$_{\pm 0.69}$	0.718$_{\pm 0.051}$	3.7$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	1.97$_{\pm 0.65}$	0.721$_{\pm 0.049}$	3.8$_{\pm 0.1}$
	Final Proj.	60.0	38.0$_{\pm 42.5}$	0.72$_{\pm 1.00}$	0.960$_{\pm 0.021}$	6.6$_{\pm 0.7}$	60.0	40.6$_{\pm 42.5}$	0.36$_{\pm 0.53}$	0.966$_{\pm 0.018}$	4.6$_{\pm 1.5}$	60.0	43.6$_{\pm 43.7}$	0.28$_{\pm 0.38}$	**0.971**$_{\pm 0.015}$	6.5$_{\pm 0.7}$
	DiRecT	**100.0**	99.2$_{\pm 2.1}$	0.00$_{\pm 0.01}$	0.905$_{\pm 0.047}$	41.5$_{\pm 0.9}$	95.0	82.7$_{\pm 35.4}$	0.00$_{\pm 0.01}$	0.906$_{\pm 0.046}$	41.4$_{\pm 0.9}$	85.0	79.8$_{\pm 39.7}$	0.04$_{\pm 0.15}$	0.915$_{\pm 0.042}$	42.4$_{\pm 1.4}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.983**$_{\pm 0.084}$	5.2$_{\pm 13.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.984**$_{\pm 0.078}$	4.7$_{\pm 1.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.968$_{\pm 0.065}$	0.4$_{\pm 0.2}$
15	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	22.25$_{\pm 4.65}$	0.999$_{\pm 0.001}$	0.3	0.0	0.0$_{\pm 0.0}$	22.25$_{\pm 4.65}$	0.999$_{\pm 0.001}$	0.3	0.0	0.0$_{\pm 0.0}$	22.25$_{\pm 4.65}$	0.999$_{\pm 0.001}$	0.4
	CD†	0.0	0.0$_{\pm 0.0}$	13.94$_{\pm 2.28}$	0.625$_{\pm 0.038}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	13.94$_{\pm 2.28}$	0.625$_{\pm 0.038}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	13.94$_{\pm 2.28}$	0.625$_{\pm 0.038}$	1.2$_{\pm 0.0}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	—	6.6$_{\pm 3.5}$	0.0	0.0$_{\pm 0.0}$	—	—	6.6$_{\pm 3.5}$	0.0	0.0$_{\pm 0.0}$	—	—	6.6$_{\pm 3.5}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	150.5$_{\pm 11.2}$	0.0	—	—	0.000$_{\pm 0.000}$	165.2$_{\pm 13.5}$	0.0	—	—	0.000$_{\pm 0.000}$	160.4$_{\pm 31.5}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	14.53$_{\pm 2.15}$	0.324$_{\pm 0.037}$	9.2$_{\pm 1.2}$	0.0	0.0$_{\pm 0.0}$	15.91$_{\pm 2.00}$	0.320$_{\pm 0.032}$	6.8$_{\pm 0.6}$	0.0	0.0$_{\pm 0.0}$	17.16$_{\pm 1.99}$	0.321$_{\pm 0.026}$	5.3$_{\pm 0.4}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	3.93$_{\pm 1.17}$	0.674$_{\pm 0.039}$	4.3$_{\pm 0.2}$	0.0	0.0$_{\pm 0.0}$	3.89$_{\pm 1.14}$	0.679$_{\pm 0.038}$	4.1$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	3.84$_{\pm 1.12}$	0.682$_{\pm 0.037}$	4.2$_{\pm 0.2}$
	Final Proj.	75.0	44.5$_{\pm 40.5}$	1.35$_{\pm 1.67}$	0.946$_{\pm 0.027}$	5.2$_{\pm 0.5}$	85.0	54.4$_{\pm 41.8}$	0.79$_{\pm 1.19}$	0.954$_{\pm 0.023}$	9.6$_{\pm 0.6}$	90.0	60.1$_{\pm 41.6}$	0.50$_{\pm 0.81}$	**0.960**$_{\pm 0.020}$	6.0$_{\pm 2.1}$
	DiRecT	**100.0**	97.7$_{\pm 4.9}$	0.03$_{\pm 0.05}$	0.887$_{\pm 0.041}$	64.3$_{\pm 2.3}$	**100.0**	92.0$_{\pm 20.0}$	0.02$_{\pm 0.05}$	0.890$_{\pm 0.039}$	71.4$_{\pm 11.9}$	95.0	91.3$_{\pm 22.3}$	0.05$_{\pm 0.12}$	0.895$_{\pm 0.038}$	72.6$_{\pm 13.9}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.001}$	5.7$_{\pm 29.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.000}$	5.3$_{\pm 11.8}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.959$_{\pm 0.082}$	0.9$_{\pm 0.3}$
18	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	28.87$_{\pm 4.75}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	28.87$_{\pm 4.75}$	0.999$_{\pm 0.001}$	0.5	0.0	0.0$_{\pm 0.0}$	28.87$_{\pm 4.75}$	0.999$_{\pm 0.001}$	0.4
	CD†	0.0	0.0$_{\pm 0.0}$	21.88$_{\pm 2.51}$	0.642$_{\pm 0.035}$	0.6$_{\pm 0.2}$	0.0	0.0$_{\pm 0.0}$	21.88$_{\pm 2.51}$	0.642$_{\pm 0.035}$	1.3$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	21.88$_{\pm 2.51}$	0.642$_{\pm 0.035}$	1.3$_{\pm 0.0}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.829$_{\pm 0.000}$	9.2$_{\pm 15.5}$	0.0	0.0$_{\pm 0.0}$	—	0.829$_{\pm 0.000}$	9.2$_{\pm 15.5}$	0.0	0.0$_{\pm 0.0}$	—	0.829$_{\pm 0.000}$	9.2$_{\pm 15.5}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	149.1$_{\pm 10.6}$	0.0	—	—	0.000$_{\pm 0.000}$	143.3$_{\pm 7.8}$	0.0	—	—	0.000$_{\pm 0.000}$	142.9$_{\pm 7.8}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	22.51$_{\pm 2.98}$	0.333$_{\pm 0.034}$	9.3$_{\pm 0.9}$	0.0	0.0$_{\pm 0.0}$	24.71$_{\pm 2.90}$	0.337$_{\pm 0.031}$	3.6$_{\pm 0.3}$	0.0	0.0$_{\pm 0.0}$	27.06$_{\pm 2.84}$	0.342$_{\pm 0.032}$	5.5$_{\pm 0.4}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	6.51$_{\pm 1.58}$	0.683$_{\pm 0.043}$	3.9$_{\pm 0.9}$	0.0	0.0$_{\pm 0.0}$	6.28$_{\pm 1.53}$	0.688$_{\pm 0.043}$	4.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	6.19$_{\pm 1.47}$	0.691$_{\pm 0.043}$	4.2$_{\pm 0.1}$
	Final Proj.	65.0	19.4$_{\pm 30.9}$	2.66$_{\pm 2.15}$	0.941$_{\pm 0.029}$	12.1$_{\pm 0.7}$	65.0	22.1$_{\pm 32.5}$	1.84$_{\pm 1.67}$	0.949$_{\pm 0.025}$	12.0$_{\pm 0.6}$	65.0	24.5$_{\pm 32.3}$	1.48$_{\pm 1.35}$	**0.956**$_{\pm 0.021}$	7.3$_{\pm 2.0}$
	DiRecT	**100.0**	92.6$_{\pm 11.6}$	0.06$_{\pm 0.11}$	0.881$_{\pm 0.045}$	93.6$_{\pm 18.4}$	90.0	78.4$_{\pm 36.7}$	0.21$_{\pm 0.39}$	0.888$_{\pm 0.044}$	92.6$_{\pm 16.9}$	90.0	77.4$_{\pm 37.0}$	0.31$_{\pm 0.60}$	0.893$_{\pm 0.042}$	84.1$_{\pm 7.1}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.000}$	6.4$_{\pm 22.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.000}$	5.3$_{\pm 4.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.953$_{\pm 0.079}$	1.0$_{\pm 0.5}$
21	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	42.00$_{\pm 5.66}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	42.00$_{\pm 5.66}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	42.00$_{\pm 5.66}$	0.999$_{\pm 0.001}$	0.8
	CD†	0.0	0.0$_{\pm 0.0}$	31.60$_{\pm 2.41}$	0.636$_{\pm 0.033}$	1.3$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	31.60$_{\pm 2.41}$	0.636$_{\pm 0.033}$	1.3$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	31.60$_{\pm 2.41}$	0.636$_{\pm 0.033}$	1.3$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	—	4.4$_{\pm 2.8}$	0.0	0.0$_{\pm 0.0}$	—	—	4.4$_{\pm 2.8}$	0.0	0.0$_{\pm 0.0}$	—	—	4.4$_{\pm 2.8}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	165.5$_{\pm 10.3}$	0.0	—	—	0.000$_{\pm 0.000}$	162.1$_{\pm 8.8}$	0.0	—	—	0.000$_{\pm 0.000}$	162.7$_{\pm 8.7}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	32.93$_{\pm 3.29}$	0.337$_{\pm 0.031}$	11.2$_{\pm 1.0}$	0.0	0.0$_{\pm 0.0}$	36.75$_{\pm 3.00}$	0.335$_{\pm 0.026}$	8.2$_{\pm 0.6}$	0.0	0.0$_{\pm 0.0}$	40.37$_{\pm 3.01}$	0.338$_{\pm 0.025}$	6.4$_{\pm 0.3}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	8.70$_{\pm 1.44}$	0.662$_{\pm 0.030}$	5.0$_{\pm 0.3}$	0.0	0.0$_{\pm 0.0}$	8.63$_{\pm 1.36}$	0.668$_{\pm 0.028}$	4.6$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	8.37$_{\pm 1.38}$	0.672$_{\pm 0.026}$	4.6$_{\pm 0.1}$
	Final Proj.	25.0	7.7$_{\pm 20.9}$	4.65$_{\pm 4.26}$	0.946$_{\pm 0.026}$	15.1$_{\pm 0.6}$	40.0	8.7$_{\pm 21.3}$	3.21$_{\pm 2.92}$	0.954$_{\pm 0.023}$	11.8$_{\pm 3.7}$	35.0	9.5$_{\pm 21.0}$	2.42$_{\pm 2.03}$	**0.960**$_{\pm 0.020}$	15.1$_{\pm 0.7}$
	DiRecT	**100.0**	93.9$_{\pm 8.7}$	**0.06**$_{\pm 0.10}$	0.875$_{\pm 0.039}$	109.4$_{\pm 9.7}$	95.0	60.2$_{\pm 38.2}$	0.11$_{\pm 0.13}$	0.880$_{\pm 0.038}$	101.1$_{\pm 21.0}$	85.0	54.4$_{\pm 43.5}$	0.64$_{\pm 0.70}$	0.894$_{\pm 0.040}$	78.0$_{\pm 0.8}$
	GlobMP (Ours)	95.0	**95.0**$_{\pm 21.8}$	2.35$_{\pm 10.24}$	**1.000**$_{\pm 0.003}$	9.7$_{\pm 47.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.002}$	6.3$_{\pm 3.8}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.945$_{\pm 0.098}$	1.5$_{\pm 2.5}$

Table 6: Multi-robot planning performance on `Dense` for $N_a \in \{3, 6, \ldots, 21\}$ under the Low, Medium and High velocity caps. *SR* and *CS* measure trajectory feasibility, *Coll* reports inter-robot collisions, *DA* adherence to the environment-specific motion pattern, and *Time* the planning time per configuration: for *GlobMP (Ours)* the median over the configurations with the standard deviation as subscript, for the baselines the mean, so the two are not directly comparable. † marks a method that applies no velocity limit, so its three blocks report the same run; MMD-CBS also returns one plan per configuration rather than 64 candidates. A dash under *Coll* marks a count that is not commensurable with the others, as MDOC and MMD-CBS count colliding time steps where every other row counts agent pairs with any collision; MDOC's *CS* is omitted because its plans that do not cover every agent were counted as feasible. MMD-CBS plans under 60 s per configuration and *GlobMP (Ours)* under 300 s per team.

		Low ($v_{\max} = 0.565$)					Medium ($v_{\max} = 0.617$)					High ($v_{\max} = 0.675$)				
N_a	Method	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓
3	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	0.56$_{\pm 0.63}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	0.56$_{\pm 0.63}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	0.56$_{\pm 0.63}$	0.999$_{\pm 0.001}$	0.5
	CD†	5.0	0.0$_{\pm 0.0}$	0.14$_{\pm 0.20}$	0.674$_{\pm 0.096}$	1.1$_{\pm 0.1}$	5.0	0.0$_{\pm 0.0}$	0.14$_{\pm 0.20}$	0.674$_{\pm 0.096}$	1.1$_{\pm 0.0}$	5.0	0.0$_{\pm 0.0}$	0.14$_{\pm 0.20}$	0.674$_{\pm 0.096}$	1.1$_{\pm 0.0}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	—	0.8$_{\pm 0.5}$	0.0	0.0$_{\pm 0.0}$	—	—	0.8$_{\pm 0.5}$	0.0	0.0$_{\pm 0.0}$	—	—	0.8$_{\pm 0.5}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	34.0$_{\pm 5.3}$	0.0	—	—	0.000$_{\pm 0.000}$	45.5$_{\pm 8.4}$	0.0	—	—	0.000$_{\pm 0.000}$	50.9$_{\pm 17.6}$
	PCD-LB	10.0	5.2$_{\pm 19.5}$	0.01$_{\pm 0.02}$	0.386$_{\pm 0.134}$	5.1$_{\pm 1.5}$	10.0	4.8$_{\pm 20.1}$	0.00$_{\pm 0.01}$	0.381$_{\pm 0.133}$	2.1$_{\pm 0.6}$	10.0	4.7$_{\pm 19.7}$	0.00$_{\pm 0.01}$	0.376$_{\pm 0.132}$	4.0$_{\pm 0.8}$
	PCD-SHD	5.0	0.1$_{\pm 0.3}$	0.11$_{\pm 0.20}$	0.798$_{\pm 0.100}$	3.1$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	0.11$_{\pm 0.20}$	0.796$_{\pm 0.099}$	3.1$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	0.12$_{\pm 0.20}$	0.794$_{\pm 0.099}$	3.0$_{\pm 0.1}$
	Final Proj.	45.0	19.2$_{\pm 35.9}$	0.11$_{\pm 0.27}$	**0.983**$_{\pm 0.021}$	5.1$_{\pm 0.7}$	45.0	19.2$_{\pm 35.9}$	0.10$_{\pm 0.27}$	**0.984**$_{\pm 0.019}$	5.1$_{\pm 0.8}$	45.0	19.2$_{\pm 35.9}$	0.09$_{\pm 0.24}$	0.986$_{\pm 0.017}$	5.1$_{\pm 0.8}$
	DiRecT	50.0	22.2$_{\pm 36.6}$	0.04$_{\pm 0.11}$	0.928$_{\pm 0.073}$	36.2$_{\pm 9.4}$	50.0	22.0$_{\pm 36.8}$	0.04$_{\pm 0.10}$	0.928$_{\pm 0.073}$	37.1$_{\pm 11.0}$	50.0	21.9$_{\pm 36.7}$	0.02$_{\pm 0.09}$	0.928$_{\pm 0.073}$	37.9$_{\pm 10.4}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.980$_{\pm 0.087}$	4.0$_{\pm 8.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.864$_{\pm 0.153}$	0.2$_{\pm 5.7}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.989**$_{\pm 0.024}$	0.2$_{\pm 0.1}$
6	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	1.42$_{\pm 0.99}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	1.42$_{\pm 0.99}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	1.42$_{\pm 0.99}$	0.999$_{\pm 0.001}$	0.4
	CD†	0.0	0.0$_{\pm 0.0}$	0.80$_{\pm 0.48}$	0.700$_{\pm 0.074}$	1.1$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	0.80$_{\pm 0.48}$	0.700$_{\pm 0.074}$	0.5$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	0.80$_{\pm 0.48}$	0.700$_{\pm 0.074}$	1.1$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	—	1.1$_{\pm 0.8}$	0.0	0.0$_{\pm 0.0}$	—	—	1.1$_{\pm 0.8}$	0.0	0.0$_{\pm 0.0}$	—	—	1.1$_{\pm 0.8}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	66.0$_{\pm 9.4}$	0.0	—	—	0.000$_{\pm 0.000}$	76.7$_{\pm 6.1}$	0.0	—	—	0.000$_{\pm 0.000}$	63.7$_{\pm 15.4}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	0.53$_{\pm 0.39}$	0.378$_{\pm 0.084}$	5.3$_{\pm 1.1}$	0.0	0.0$_{\pm 0.0}$	0.57$_{\pm 0.41}$	0.373$_{\pm 0.082}$	4.6$_{\pm 0.8}$	0.0	0.0$_{\pm 0.0}$	0.60$_{\pm 0.41}$	0.372$_{\pm 0.086}$	4.0$_{\pm 0.6}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	1.04$_{\pm 0.65}$	0.788$_{\pm 0.062}$	3.3$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	1.07$_{\pm 0.64}$	0.786$_{\pm 0.061}$	3.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	1.06$_{\pm 0.63}$	0.785$_{\pm 0.061}$	3.2$_{\pm 0.1}$
	Final Proj.	30.0	8.3$_{\pm 20.9}$	0.16$_{\pm 0.24}$	0.980$_{\pm 0.013}$	4.2$_{\pm 1.4}$	30.0	9.0$_{\pm 22.4}$	0.10$_{\pm 0.14}$	**0.981**$_{\pm 0.012}$	5.6$_{\pm 0.8}$	30.0	9.1$_{\pm 22.6}$	0.08$_{\pm 0.11}$	0.983$_{\pm 0.011}$	5.6$_{\pm 0.5}$
	DiRecT	35.0	13.0$_{\pm 26.8}$	0.04$_{\pm 0.08}$	0.922$_{\pm 0.046}$	44.0$_{\pm 12.8}$	35.0	13.2$_{\pm 26.9}$	0.03$_{\pm 0.07}$	0.923$_{\pm 0.046}$	47.7$_{\pm 14.0}$	35.0	13.4$_{\pm 27.2}$	0.02$_{\pm 0.07}$	0.923$_{\pm 0.045}$	47.4$_{\pm 14.6}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.000}$	4.2$_{\pm 5.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.915$_{\pm 0.143}$	4.1$_{\pm 2.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.985**$_{\pm 0.051}$	0.2$_{\pm 0.1}$
9	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	4.18$_{\pm 1.43}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	4.18$_{\pm 1.43}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	4.18$_{\pm 1.43}$	0.999$_{\pm 0.001}$	0.4
	CD†	0.0	0.0$_{\pm 0.0}$	3.39$_{\pm 0.77}$	0.712$_{\pm 0.056}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	3.39$_{\pm 0.77}$	0.712$_{\pm 0.056}$	1.2$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	3.39$_{\pm 0.77}$	0.712$_{\pm 0.056}$	1.2$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	—	1.0$_{\pm 0.5}$	0.0	0.0$_{\pm 0.0}$	—	—	1.0$_{\pm 0.5}$	0.0	0.0$_{\pm 0.0}$	—	—	1.0$_{\pm 0.5}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	82.2$_{\pm 12.5}$	0.0	—	—	0.000$_{\pm 0.000}$	73.9$_{\pm 5.1}$	0.0	—	—	0.000$_{\pm 0.000}$	77.0$_{\pm 6.6}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	2.95$_{\pm 1.17}$	0.403$_{\pm 0.093}$	5.2$_{\pm 1.7}$	0.0	0.0$_{\pm 0.0}$	3.12$_{\pm 1.15}$	0.400$_{\pm 0.089}$	5.5$_{\pm 1.0}$	0.0	0.0$_{\pm 0.0}$	3.30$_{\pm 1.13}$	0.399$_{\pm 0.084}$	4.8$_{\pm 0.7}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	2.08$_{\pm 0.66}$	0.799$_{\pm 0.047}$	3.6$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	2.12$_{\pm 0.62}$	0.798$_{\pm 0.046}$	1.7$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	2.17$_{\pm 0.62}$	0.796$_{\pm 0.045}$	3.5$_{\pm 0.1}$
	Final Proj.	15.0	8.8$_{\pm 25.7}$	0.46$_{\pm 0.63}$	0.984$_{\pm 0.010}$	8.2$_{\pm 0.7}$	15.0	8.8$_{\pm 25.7}$	0.36$_{\pm 0.52}$	**0.985**$_{\pm 0.009}$	5.6$_{\pm 2.1}$	15.0	8.8$_{\pm 25.7}$	0.34$_{\pm 0.50}$	**0.987**$_{\pm 0.009}$	8.4$_{\pm 0.7}$
	DiRecT	25.0	11.1$_{\pm 26.4}$	0.15$_{\pm 0.18}$	0.939$_{\pm 0.034}$	74.7$_{\pm 22.0}$	25.0	11.1$_{\pm 26.7}$	0.12$_{\pm 0.14}$	0.940$_{\pm 0.033}$	58.3$_{\pm 13.7}$	20.0	11.2$_{\pm 26.9}$	0.09$_{\pm 0.13}$	0.940$_{\pm 0.033}$	53.8$_{\pm 0.9}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.000}$	4.3$_{\pm 15.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.938$_{\pm 0.129}$	4.3$_{\pm 12.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.984$_{\pm 0.037}$	0.3$_{\pm 0.1}$
12	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	7.86$_{\pm 2.57}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	7.86$_{\pm 2.57}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	7.86$_{\pm 2.57}$	0.999$_{\pm 0.001}$	0.4
	CD†	0.0	0.0$_{\pm 0.0}$	6.72$_{\pm 1.77}$	0.745$_{\pm 0.049}$	0.6$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	6.72$_{\pm 1.77}$	0.745$_{\pm 0.049}$	1.2$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	6.72$_{\pm 1.77}$	0.745$_{\pm 0.049}$	1.2$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	—	0.9$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	—	—	0.9$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	—	—	0.9$_{\pm 0.4}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	111.9$_{\pm 6.7}$	0.0	—	—	0.000$_{\pm 0.000}$	108.8$_{\pm 6.3}$	0.0	—	—	0.000$_{\pm 0.000}$	114.3$_{\pm 6.4}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	6.12$_{\pm 2.04}$	0.405$_{\pm 0.071}$	7.2$_{\pm 1.0}$	0.0	0.0$_{\pm 0.0}$	6.57$_{\pm 2.01}$	0.404$_{\pm 0.073}$	5.9$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	7.05$_{\pm 2.03}$	0.405$_{\pm 0.074}$	5.1$_{\pm 0.6}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	4.12$_{\pm 1.60}$	0.815$_{\pm 0.039}$	3.8$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	4.12$_{\pm 1.62}$	0.814$_{\pm 0.037}$	3.7$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	4.26$_{\pm 1.62}$	0.814$_{\pm 0.037}$	3.7$_{\pm 0.1}$
	Final Proj.	0.0	0.0$_{\pm 0.0}$	1.10$_{\pm 1.28}$	0.985$_{\pm 0.009}$	10.1$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	1.00$_{\pm 1.19}$	0.986$_{\pm 0.008}$	6.3$_{\pm 1.8}$	0.0	0.0$_{\pm 0.0}$	0.92$_{\pm 1.18}$	**0.987**$_{\pm 0.007}$	10.1$_{\pm 0.7}$
	DiRecT	0.0	0.0$_{\pm 0.0}$	0.49$_{\pm 0.41}$	0.940$_{\pm 0.029}$	68.9$_{\pm 1.6}$	0.0	0.0$_{\pm 0.0}$	0.40$_{\pm 0.39}$	0.941$_{\pm 0.029}$	69.3$_{\pm 5.5}$	0.0	0.0$_{\pm 0.0}$	0.33$_{\pm 0.39}$	0.942$_{\pm 0.029}$	74.8$_{\pm 12.7}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.000}$	4.3$_{\pm 18.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.992**$_{\pm 0.054}$	4.3$_{\pm 11.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.972$_{\pm 0.073}$	0.3$_{\pm 0.1}$
15	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	12.70$_{\pm 3.56}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	12.70$_{\pm 3.56}$	0.999$_{\pm 0.001}$	0.4	0.0	0.0$_{\pm 0.0}$	12.70$_{\pm 3.56}$	0.999$_{\pm 0.001}$	0.4
	CD†	0.0	0.0$_{\pm 0.0}$	13.31$_{\pm 3.25}$	0.712$_{\pm 0.044}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	13.31$_{\pm 3.25}$	0.712$_{\pm 0.044}$	1.2$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	13.31$_{\pm 3.25}$	0.712$_{\pm 0.044}$	1.2$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	—	0.9$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	—	—	0.9$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	—	—	0.9$_{\pm 0.4}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	155.2$_{\pm 7.7}$	0.0	—	—	0.000$_{\pm 0.000}$	171.4$_{\pm 25.9}$	0.0	—	—	0.000$_{\pm 0.000}$	133.6$_{\pm 10.9}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	13.05$_{\pm 2.30}$	0.368$_{\pm 0.041}$	8.1$_{\pm 1.1}$	0.0	0.0$_{\pm 0.0}$	14.03$_{\pm 2.31}$	0.367$_{\pm 0.039}$	6.4$_{\pm 1.2}$	0.0	0.0$_{\pm 0.0}$	15.16$_{\pm 2.36}$	0.366$_{\pm 0.039}$	5.6$_{\pm 0.7}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	7.59$_{\pm 2.13}$	0.799$_{\pm 0.032}$	4.1$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	7.65$_{\pm 2.17}$	0.798$_{\pm 0.032}$	4.0$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	7.81$_{\pm 2.24}$	0.797$_{\pm 0.033}$	4.0$_{\pm 0.1}$
	Final Proj.	0.0	0.0$_{\pm 0.0}$	1.55$_{\pm 1.38}$	0.979$_{\pm 0.007}$	12.8$_{\pm 3.3}$	0.0	0.0$_{\pm 0.0}$	1.23$_{\pm 1.21}$	0.980$_{\pm 0.007}$	14.7$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	1.06$_{\pm 1.02}$	**0.982**$_{\pm 0.006}$	14.6$_{\pm 0.7}$
	DiRecT	0.0	0.0$_{\pm 0.0}$	0.75$_{\pm 0.68}$	0.922$_{\pm 0.023}$	113.1$_{\pm 16.3}$	0.0	0.0$_{\pm 0.0}$	0.53$_{\pm 0.48}$	0.923$_{\pm 0.023}$	114.6$_{\pm 19.5}$	0.0	0.0$_{\pm 0.0}$	0.36$_{\pm 0.36}$	0.923$_{\pm 0.023}$	102.4$_{\pm 7.6}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.000}$	4.6$_{\pm 33.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.982**$_{\pm 0.069}$	4.5$_{\pm 15.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.974$_{\pm 0.071}$	0.4$_{\pm 0.2}$
18	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	16.75$_{\pm 4.10}$	0.999$_{\pm 0.000}$	0.4	0.0	0.0$_{\pm 0.0}$	16.75$_{\pm 4.10}$	0.999$_{\pm 0.000}$	0.4	0.0	0.0$_{\pm 0.0}$	16.75$_{\pm 4.10}$	0.999$_{\pm 0.000}$	0.3
	CD†	0.0	0.0$_{\pm 0.0}$	19.75$_{\pm 3.14}$	0.702$_{\pm 0.038}$	1.3$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	19.75$_{\pm 3.14}$	0.702$_{\pm 0.038}$	0.6$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	19.75$_{\pm 3.14}$	0.702$_{\pm 0.038}$	1.3$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	—	0.8$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	—	—	0.8$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	—	—	0.8$_{\pm 0.4}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	147.2$_{\pm 8.4}$	0.0	—	—	0.000$_{\pm 0.000}$	143.6$_{\pm 7.6}$	0.0	—	—	0.000$_{\pm 0.000}$	143.0$_{\pm 8.6}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	20.86$_{\pm 4.27}$	0.353$_{\pm 0.037}$	8.0$_{\pm 1.0}$	0.0	0.0$_{\pm 0.0}$	22.51$_{\pm 4.22}$	0.356$_{\pm 0.035}$	6.7$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	24.22$_{\pm 4.20}$	0.359$_{\pm 0.034}$	5.7$_{\pm 0.5}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	10.38$_{\pm 2.46}$	0.792$_{\pm 0.031}$	4.3$_{\pm 0.2}$	0.0	0.0$_{\pm 0.0}$	10.52$_{\pm 2.40}$	0.790$_{\pm 0.031}$	2.0$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	10.74$_{\pm 2.40}$	0.789$_{\pm 0.030}$	4.1$_{\pm 0.1}$
	Final Proj.	0.0	0.0$_{\pm 0.0}$	2.43$_{\pm 1.79}$	0.984$_{\pm 0.007}$	14.1$_{\pm 3.8}$	0.0	0.0$_{\pm 0.0}$	2.01$_{\pm 1.52}$	**0.985**$_{\pm 0.006}$	17.3$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	1.75$_{\pm 1.40}$	**0.986**$_{\pm 0.006}$	9.2$_{\pm 0.7}$
	DiRecT	5.0	0.1$_{\pm 0.3}$	0.79$_{\pm 0.60}$	0.936$_{\pm 0.022}$	124.6$_{\pm 10.6}$	5.0	0.1$_{\pm 0.3}$	0.65$_{\pm 0.52}$	0.937$_{\pm 0.022}$	107.4$_{\pm 20.5}$	0.0	0.0$_{\pm 0.0}$	0.54$_{\pm 0.52}$	0.937$_{\pm 0.022}$	89.8$_{\pm 1.3}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.000}$	5.0$_{\pm 20.7}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.972$_{\pm 0.090}$	4.8$_{\pm 10.4}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.972$_{\pm 0.069}$	0.5$_{\pm 0.2}$
21	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	24.25$_{\pm 6.19}$	0.999$_{\pm 0.000}$	0.4	0.0	0.0$_{\pm 0.0}$	24.25$_{\pm 6.19}$	0.999$_{\pm 0.000}$	0.3	0.0	0.0$_{\pm 0.0}$	24.25$_{\pm 6.19}$	0.999$_{\pm 0.000}$	0.4
	CD†	0.0	0.0$_{\pm 0.0}$	29.53$_{\pm 5.13}$	0.706$_{\pm 0.022}$	1.3$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	29.53$_{\pm 5.13}$	0.706$_{\pm 0.022}$	1.3$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	29.53$_{\pm 5.13}$	0.706$_{\pm 0.022}$	1.3$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	—	0.9$_{\pm 0.3}$	0.0	0.0$_{\pm 0.0}$	—	—	0.9$_{\pm 0.3}$	0.0	0.0$_{\pm 0.0}$	—	—	0.9$_{\pm 0.3}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	165.3$_{\pm 8.7}$	0.0	—	—	0.000$_{\pm 0.000}$	164.1$_{\pm 8.1}$	0.0	—	—	0.000$_{\pm 0.000}$	160.9$_{\pm 8.6}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	30.67$_{\pm 3.84}$	0.356$_{\pm 0.029}$	9.6$_{\pm 1.4}$	0.0	0.0$_{\pm 0.0}$	33.64$_{\pm 3.88}$	0.357$_{\pm 0.028}$	7.9$_{\pm 1.1}$	0.0	0.0$_{\pm 0.0}$	36.36$_{\pm 3.94}$	0.359$_{\pm 0.027}$	3.5$_{\pm 0.8}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	14.48$_{\pm 2.36}$	0.796$_{\pm 0.024}$	4.9$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	14.69$_{\pm 2.44}$	0.795$_{\pm 0.024}$	4.7$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	14.90$_{\pm 2.43}$	0.794$_{\pm 0.023}$	4.6$_{\pm 0.1}$
	Final Proj.	0.0	0.0$_{\pm 0.0}$	4.31$_{\pm 3.10}$	0.982$_{\pm 0.007}$	23.1$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	3.68$_{\pm 2.74}$	**0.983**$_{\pm 0.007}$	23.1$_{\pm 0.7}$	0.0	0.0$_{\pm 0.0}$	3.20$_{\pm 2.45}$	**0.984**$_{\pm 0.006}$	16.5$_{\pm 5.1}$
	DiRecT	0.0	0.0$_{\pm 0.0}$	1.36$_{\pm 0.97}$	0.931$_{\pm 0.019}$	120.1$_{\pm 1.0}$	0.0	0.0$_{\pm 0.0}$	1.06$_{\pm 0.78}$	0.932$_{\pm 0.019}$	120.0$_{\pm 1.1}$	0.0	0.0$_{\pm 0.0}$	0.62$_{\pm 0.57}$	0.933$_{\pm 0.019}$	120.2$_{\pm 0.7}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**1.000**$_{\pm 0.000}$	5.1$_{\pm 43.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.978$_{\pm 0.081}$	5.0$_{\pm 26.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.970$_{\pm 0.071}$	0.9$_{\pm 0.3}$

Table 7: Multi-robot planning performance on `Conveyor` for $N_a \in \{3, 6, \ldots, 21\}$ under the Low, Medium and High velocity caps. *SR* and *CS* measure trajectory feasibility, *Coll* reports inter-robot collisions, *DA* adherence to the environment-specific motion pattern, and *Time* the planning time per configuration: for *GlobMP (Ours)* the median over the configurations with the standard deviation as subscript, for the baselines the mean, so the two are not directly comparable. † marks a method that applies no velocity limit, so its three blocks report the same run; MMD-CBS also returns one plan per configuration rather than 64 candidates. A dash under *Coll* marks a count that is not commensurable with the others, as MDOC and MMD-CBS count colliding time steps where every other row counts agent pairs with any collision; MDOC's *CS* is omitted because its plans that do not cover every agent were counted as feasible. MMD-CBS plans under 60 s per configuration and *GlobMP (Ours)* under 300 s per team. *DA* on this map scores an end-to-end traversal of one conveyor lane, which the uniformly drawn start and goal pairs almost never require; it is therefore below 0.07 for every method and does not separate them.

		Low ($v_{\max} = 1.21$)					Medium ($v_{\max} = 1.46$)					High ($v_{\max} = 1.76$)				
N_a	Method	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓
3	Diff. Policy†	5.0	$4.9_{\pm 21.5}$	$0.37_{\pm 0.53}$	$0.016_{\pm 0.070}$	0.6	5.0	$4.9_{\pm 21.5}$	$0.37_{\pm 0.53}$	$0.016_{\pm 0.070}$	0.6	5.0	$4.9_{\pm 21.5}$	$0.37_{\pm 0.53}$	$0.016_{\pm 0.070}$	0.6
	CD†	45.0	$17.4_{\pm 35.0}$	$0.02_{\pm 0.04}$	$0.001_{\pm 0.003}$	$0.6_{\pm 0.1}$	45.0	$18.3_{\pm 36.2}$	$0.02_{\pm 0.04}$	$0.001_{\pm 0.003}$	$1.2_{\pm 0.1}$	45.0	$19.2_{\pm 37.3}$	$0.02_{\pm 0.04}$	$0.001_{\pm 0.003}$	$1.1_{\pm 0.1}$
	MMD-CBS†	20.0	$20.0_{\pm 40.0}$	—	$0.000_{\pm 0.000}$	$1.3_{\pm 0.7}$	20.0	$20.0_{\pm 40.0}$	—	$0.000_{\pm 0.000}$	$1.3_{\pm 0.7}$	20.0	$20.0_{\pm 40.0}$	—	$0.000_{\pm 0.000}$	$1.3_{\pm 0.7}$
	MDOC†	5.0	—	—	$0.000_{\pm 0.000}$	$40.1_{\pm 7.0}$	5.0	—	—	$0.000_{\pm 0.000}$	$36.4_{\pm 2.6}$	5.0	—	—	$0.000_{\pm 0.000}$	$36.3_{\pm 2.8}$
	PCD-LB	70.0	$36.2_{\pm 42.5}$	$0.01_{\pm 0.03}$	$0.004_{\pm 0.016}$	$3.0_{\pm 0.1}$	65.0	$37.1_{\pm 43.6}$	$0.01_{\pm 0.02}$	$0.004_{\pm 0.016}$	$3.1_{\pm 0.1}$	65.0	$36.8_{\pm 43.5}$	$0.01_{\pm 0.02}$	$0.004_{\pm 0.017}$	$3.1_{\pm 0.1}$
	PCD-SHD	10.0	$9.5_{\pm 28.4}$	$0.01_{\pm 0.03}$	$0.003_{\pm 0.011}$	$2.1_{\pm 0.7}$	10.0	$9.5_{\pm 28.4}$	$0.01_{\pm 0.02}$	$0.002_{\pm 0.010}$	$3.0_{\pm 0.1}$	10.0	$9.6_{\pm 28.9}$	$0.01_{\pm 0.02}$	$0.002_{\pm 0.009}$	$3.0_{\pm 0.1}$
	Final Proj.	10.0	$10.0_{\pm 30.0}$	$0.00_{\pm 0.01}$	$0.000_{\pm 0.000}$	$3.6_{\pm 0.5}$	10.0	$10.0_{\pm 30.0}$	$0.00_{\pm 0.00}$	$0.000_{\pm 0.000}$	$3.7_{\pm 0.7}$	10.0	$10.0_{\pm 30.0}$	**0.00**$_{\pm 0.00}$	$0.000_{\pm 0.000}$	$3.7_{\pm 0.6}$
	DiRecT	10.0	$9.9_{\pm 29.8}$	**0.00**$_{\pm 0.00}$	$0.000_{\pm 0.000}$	$24.3_{\pm 5.8}$	10.0	$10.0_{\pm 30.0}$	**0.00**$_{\pm 0.00}$	$0.000_{\pm 0.000}$	$24.5_{\pm 5.7}$	10.0	$10.0_{\pm 30.0}$	**0.00**$_{\pm 0.00}$	$0.000_{\pm 0.000}$	$26.4_{\pm 6.7}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.017**$_{\pm 0.128}$	$0.4_{\pm 0.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.033**$_{\pm 0.180}$	$0.5_{\pm 0.4}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.067**$_{\pm 0.249}$	$0.7_{\pm 0.6}$
6	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$1.22_{\pm 1.04}$	$0.000_{\pm 0.000}$	0.6	0.0	$0.0_{\pm 0.0}$	$1.22_{\pm 1.04}$	$0.000_{\pm 0.000}$	0.6	0.0	$0.0_{\pm 0.0}$	$1.22_{\pm 1.04}$	$0.000_{\pm 0.000}$	0.6
	CD†	5.0	$0.1_{\pm 0.3}$	$0.25_{\pm 0.29}$	$0.008_{\pm 0.032}$	$1.2_{\pm 0.1}$	5.0	$0.1_{\pm 0.3}$	$0.25_{\pm 0.29}$	$0.008_{\pm 0.032}$	$1.2_{\pm 0.1}$	5.0	$0.2_{\pm 0.7}$	$0.25_{\pm 0.29}$	$0.008_{\pm 0.032}$	$1.1_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.5_{\pm 1.1}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.5_{\pm 1.1}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.5_{\pm 1.1}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$62.5_{\pm 3.3}$	0.0	—	—	$0.000_{\pm 0.000}$	$82.6_{\pm 23.1}$	0.0	—	—	$0.000_{\pm 0.000}$	$85.2_{\pm 20.8}$
	PCD-LB	25.0	$6.1_{\pm 14.7}$	$0.85_{\pm 0.74}$	$0.006_{\pm 0.013}$	$2.8_{\pm 0.7}$	40.0	$7.8_{\pm 16.5}$	$0.72_{\pm 0.60}$	$0.007_{\pm 0.013}$	$3.1_{\pm 0.4}$	45.0	$13.5_{\pm 20.5}$	$0.44_{\pm 0.34}$	$0.007_{\pm 0.014}$	$3.3_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$0.10_{\pm 0.18}$	$0.001_{\pm 0.006}$	$3.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.06_{\pm 0.09}$	$0.001_{\pm 0.006}$	$3.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.03_{\pm 0.05}$	$0.001_{\pm 0.005}$	$3.2_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.05_{\pm 0.14}$	$0.000_{\pm 0.000}$	$1.9_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	$0.03_{\pm 0.10}$	$0.000_{\pm 0.000}$	$3.4_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	$0.01_{\pm 0.03}$	$0.000_{\pm 0.000}$	$3.4_{\pm 0.6}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	$0.00_{\pm 0.01}$	$0.001_{\pm 0.005}$	$23.4_{\pm 6.1}$	0.0	$0.0_{\pm 0.0}$	$0.01_{\pm 0.02}$	$0.001_{\pm 0.002}$	$24.0_{\pm 6.6}$	0.0	$0.0_{\pm 0.0}$	$0.01_{\pm 0.02}$	$0.000_{\pm 0.001}$	$25.0_{\pm 7.0}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.008**$_{\pm 0.091}$	$0.8_{\pm 0.8}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.008**$_{\pm 0.091}$	$1.2_{\pm 2.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.017**$_{\pm 0.128}$	$1.8_{\pm 10.5}$
9	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$3.16_{\pm 1.58}$	$0.006_{\pm 0.024}$	0.6	0.0	$0.0_{\pm 0.0}$	$3.16_{\pm 1.58}$	$0.006_{\pm 0.024}$	0.3	0.0	$0.0_{\pm 0.0}$	$3.16_{\pm 1.58}$	$0.006_{\pm 0.024}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$1.47_{\pm 0.91}$	$0.000_{\pm 0.002}$	$0.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$1.47_{\pm 0.91}$	$0.000_{\pm 0.002}$	$1.1_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$1.47_{\pm 0.91}$	$0.000_{\pm 0.002}$	$1.1_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.8_{\pm 1.1}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.8_{\pm 1.1}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.8_{\pm 1.1}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$113.3_{\pm 17.4}$	0.0	—	—	$0.000_{\pm 0.000}$	$97.2_{\pm 18.9}$	0.0	—	—	$0.000_{\pm 0.000}$	$74.7_{\pm 4.0}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$4.20_{\pm 1.42}$	$0.004_{\pm 0.008}$	$3.5_{\pm 0.1}$	10.0	$0.4_{\pm 1.4}$	$3.09_{\pm 1.02}$	$0.004_{\pm 0.010}$	$3.5_{\pm 0.1}$	25.0	$2.4_{\pm 5.4}$	$1.54_{\pm 0.57}$	$0.005_{\pm 0.011}$	$3.5_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$0.13_{\pm 0.16}$	$0.003_{\pm 0.014}$	$3.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.08_{\pm 0.10}$	$0.003_{\pm 0.014}$	$3.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.05_{\pm 0.06}$	$0.003_{\pm 0.014}$	$3.6_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.05_{\pm 0.10}$	$0.005_{\pm 0.024}$	$4.3_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	$0.02_{\pm 0.04}$	$0.005_{\pm 0.024}$	$4.3_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	**0.02**$_{\pm 0.04}$	$0.005_{\pm 0.024}$	$4.3_{\pm 0.6}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	$0.04_{\pm 0.06}$	$0.007_{\pm 0.025}$	$34.5_{\pm 9.7}$	0.0	$0.0_{\pm 0.0}$	$0.01_{\pm 0.02}$	$0.006_{\pm 0.024}$	$35.6_{\pm 10.5}$	0.0	$0.0_{\pm 0.0}$	$0.02_{\pm 0.03}$	$0.006_{\pm 0.024}$	$25.9_{\pm 1.3}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.028**$_{\pm 0.164}$	$2.0_{\pm 8.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.028**$_{\pm 0.164}$	$2.6_{\pm 2.7}$	**95.0**	**95.0**$_{\pm 21.8}$	$0.10_{\pm 0.44}$	**0.028**$_{\pm 0.164}$	$5.5_{\pm 64.2}$
12	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$6.33_{\pm 2.50}$	$0.012_{\pm 0.029}$	0.6	0.0	$0.0_{\pm 0.0}$	$6.33_{\pm 2.50}$	$0.012_{\pm 0.029}$	0.6	0.0	$0.0_{\pm 0.0}$	$6.33_{\pm 2.50}$	$0.012_{\pm 0.029}$	0.6
	CD†	0.0	$0.0_{\pm 0.0}$	$3.96_{\pm 1.28}$	$0.018_{\pm 0.026}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$3.96_{\pm 1.28}$	$0.018_{\pm 0.026}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$3.96_{\pm 1.28}$	$0.018_{\pm 0.026}$	$0.6_{\pm 0.0}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.9_{\pm 1.6}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.9_{\pm 1.6}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.9_{\pm 1.6}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$107.9_{\pm 7.1}$	0.0	—	—	$0.000_{\pm 0.000}$	$120.6_{\pm 5.3}$	0.0	—	—	$0.000_{\pm 0.000}$	$127.3_{\pm 7.3}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$9.25_{\pm 1.68}$	$0.005_{\pm 0.007}$	$3.7_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$7.14_{\pm 1.37}$	$0.005_{\pm 0.007}$	$1.9_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$3.76_{\pm 0.86}$	$0.005_{\pm 0.006}$	$3.7_{\pm 0.5}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$0.62_{\pm 0.51}$	$0.013_{\pm 0.022}$	$3.9_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.37_{\pm 0.32}$	$0.012_{\pm 0.022}$	$3.9_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.19_{\pm 0.18}$	$0.012_{\pm 0.022}$	$1.8_{\pm 0.2}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.20_{\pm 0.29}$	$0.010_{\pm 0.025}$	$5.4_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	**0.05**$_{\pm 0.08}$	$0.010_{\pm 0.025}$	$5.4_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	**0.06**$_{\pm 0.11}$	$0.010_{\pm 0.025}$	$5.4_{\pm 0.7}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	**0.06**$_{\pm 0.08}$	$0.012_{\pm 0.026}$	$33.6_{\pm 1.4}$	0.0	$0.0_{\pm 0.0}$	$0.07_{\pm 0.09}$	$0.013_{\pm 0.028}$	$34.0_{\pm 1.9}$	0.0	$0.0_{\pm 0.0}$	$0.09_{\pm 0.16}$	$0.014_{\pm 0.029}$	$34.2_{\pm 1.9}$
	GlobMP (Ours)	**90.0**	**90.0**$_{\pm 30.0}$	$0.85_{\pm 2.55}$	**0.042**$_{\pm 0.200}$	$3.3_{\pm 91.3}$	**95.0**	**95.0**$_{\pm 21.8}$	$0.45_{\pm 1.96}$	**0.037**$_{\pm 0.190}$	$5.7_{\pm 72.9}$	**90.0**	**90.0**$_{\pm 30.0}$	$0.60_{\pm 1.80}$	**0.033**$_{\pm 0.180}$	$13.3_{\pm 79.3}$
15	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$9.77_{\pm 2.61}$	$0.003_{\pm 0.013}$	0.3	0.0	$0.0_{\pm 0.0}$	$9.77_{\pm 2.61}$	$0.003_{\pm 0.013}$	0.3	0.0	$0.0_{\pm 0.0}$	$9.77_{\pm 2.61}$	$0.003_{\pm 0.013}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$7.35_{\pm 1.96}$	$0.003_{\pm 0.009}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$7.35_{\pm 1.96}$	$0.003_{\pm 0.009}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$7.35_{\pm 1.96}$	$0.003_{\pm 0.009}$	$1.2_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.9_{\pm 1.4}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.9_{\pm 1.4}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.9_{\pm 1.4}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$155.4_{\pm 11.8}$	0.0	—	—	$0.000_{\pm 0.000}$	$172.5_{\pm 14.6}$	0.0	—	—	$0.000_{\pm 0.000}$	$174.3_{\pm 27.4}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$20.74_{\pm 3.01}$	$0.005_{\pm 0.006}$	$4.1_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$17.02_{\pm 2.64}$	$0.004_{\pm 0.006}$	$4.1_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$11.84_{\pm 1.88}$	$0.005_{\pm 0.006}$	$4.1_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$0.75_{\pm 0.40}$	$0.003_{\pm 0.010}$	$4.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.48_{\pm 0.28}$	$0.003_{\pm 0.010}$	$4.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.29_{\pm 0.19}$	$0.003_{\pm 0.010}$	$4.3_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.49_{\pm 0.42}$	$0.003_{\pm 0.015}$	$3.7_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	$0.23_{\pm 0.25}$	$0.003_{\pm 0.015}$	$6.9_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	**0.18**$_{\pm 0.24}$	$0.003_{\pm 0.015}$	$6.0_{\pm 1.6}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	**0.14**$_{\pm 0.15}$	$0.003_{\pm 0.013}$	$44.0_{\pm 1.3}$	0.0	$0.0_{\pm 0.0}$	**0.16**$_{\pm 0.12}$	$0.003_{\pm 0.013}$	$44.9_{\pm 2.2}$	0.0	$0.0_{\pm 0.0}$	**0.18**$_{\pm 0.15}$	$0.003_{\pm 0.013}$	$47.2_{\pm 5.5}$
	GlobMP (Ours)	**90.0**	**90.0**$_{\pm 30.0}$	$1.00_{\pm 3.15}$	**0.027**$_{\pm 0.161}$	$5.5_{\pm 68.3}$	**85.0**	**85.0**$_{\pm 35.7}$	$1.45_{\pm 3.58}$	**0.023**$_{\pm 0.151}$	$11.2_{\pm 74.1}$	**90.0**	**90.0**$_{\pm 30.0}$	$0.90_{\pm 2.70}$	**0.023**$_{\pm 0.151}$	$22.8_{\pm 91.2}$
18	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$15.01_{\pm 3.59}$	$0.000_{\pm 0.000}$	0.3	0.0	$0.0_{\pm 0.0}$	$15.01_{\pm 3.59}$	$0.000_{\pm 0.000}$	0.3	0.0	$0.0_{\pm 0.0}$	$15.01_{\pm 3.59}$	$0.000_{\pm 0.000}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$12.92_{\pm 1.90}$	$0.003_{\pm 0.006}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$12.92_{\pm 1.90}$	$0.003_{\pm 0.006}$	$1.2_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$12.92_{\pm 1.90}$	$0.003_{\pm 0.006}$	$1.2_{\pm 0.0}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.4_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.4_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.4_{\pm 0.9}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$162.1_{\pm 8.6}$	0.0	—	—	$0.000_{\pm 0.000}$	$156.7_{\pm 7.6}$	0.0	—	—	$0.000_{\pm 0.000}$	$157.5_{\pm 7.0}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$35.56_{\pm 3.24}$	$0.002_{\pm 0.003}$	$4.2_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$28.42_{\pm 2.79}$	$0.002_{\pm 0.003}$	$4.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$19.50_{\pm 1.96}$	$0.003_{\pm 0.004}$	$2.8_{\pm 0.9}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$1.18_{\pm 0.69}$	$0.003_{\pm 0.004}$	$4.4_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.80_{\pm 0.52}$	$0.003_{\pm 0.004}$	$4.4_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.55_{\pm 0.37}$	$0.002_{\pm 0.004}$	$4.4_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.84_{\pm 0.83}$	$0.000_{\pm 0.000}$	$8.7_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	$0.33_{\pm 0.33}$	$0.000_{\pm 0.000}$	$7.3_{\pm 2.1}$	0.0	$0.0_{\pm 0.0}$	**0.23**$_{\pm 0.23}$	$0.000_{\pm 0.000}$	$8.7_{\pm 0.7}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	**0.20**$_{\pm 0.15}$	$0.003_{\pm 0.009}$	$65.8_{\pm 10.7}$	0.0	$0.0_{\pm 0.0}$	**0.22**$_{\pm 0.18}$	$0.003_{\pm 0.008}$	$64.5_{\pm 10.8}$	0.0	$0.0_{\pm 0.0}$	$0.26_{\pm 0.22}$	$0.003_{\pm 0.008}$	$64.4_{\pm 9.0}$
	GlobMP (Ours)	**90.0**	**90.0**$_{\pm 30.0}$	$1.85_{\pm 5.55}$	**0.047**$_{\pm 0.212}$	$12.4_{\pm 89.4}$	**85.0**	**85.0**$_{\pm 35.7}$	$2.75_{\pm 6.77}$	**0.044**$_{\pm 0.206}$	$14.2_{\pm 102.5}$	**75.0**	**75.0**$_{\pm 43.3}$	$3.60_{\pm 6.61}$	**0.039**$_{\pm 0.193}$	$36.0_{\pm 118.0}$
21	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$20.41_{\pm 2.03}$	$0.002_{\pm 0.010}$	0.3	0.0	$0.0_{\pm 0.0}$	$20.41_{\pm 2.03}$	$0.002_{\pm 0.010}$	0.3	0.0	$0.0_{\pm 0.0}$	$20.41_{\pm 2.03}$	$0.002_{\pm 0.010}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$20.28_{\pm 2.30}$	$0.011_{\pm 0.016}$	$1.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$20.28_{\pm 2.30}$	$0.011_{\pm 0.016}$	$1.2_{\pm 0.0}$	0.0	$0.0_{\pm 0.0}$	$20.28_{\pm 2.30}$	$0.011_{\pm 0.016}$	$1.3_{\pm 0.0}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.7_{\pm 1.4}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.7_{\pm 1.4}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.7_{\pm 1.4}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$182.3_{\pm 8.9}$	0.0	—	—	$0.000_{\pm 0.000}$	$181.9_{\pm 9.0}$	0.0	—	—	$0.000_{\pm 0.000}$	$163.8_{\pm 8.1}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$53.35_{\pm 2.98}$	$0.002_{\pm 0.003}$	$4.7_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$40.89_{\pm 3.07}$	$0.002_{\pm 0.003}$	$4.8_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$26.92_{\pm 2.55}$	$0.002_{\pm 0.003}$	$4.9_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$1.60_{\pm 0.65}$	$0.006_{\pm 0.011}$	$5.0_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$1.09_{\pm 0.49}$	$0.006_{\pm 0.011}$	$3.0_{\pm 1.0}$	0.0	$0.0_{\pm 0.0}$	$0.68_{\pm 0.32}$	$0.006_{\pm 0.011}$	$5.0_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.98_{\pm 0.65}$	$0.003_{\pm 0.010}$	$11.5_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	$0.43_{\pm 0.32}$	$0.003_{\pm 0.010}$	$10.8_{\pm 1.3}$	0.0	$0.0_{\pm 0.0}$	$0.28_{\pm 0.19}$	$0.003_{\pm 0.010}$	$11.4_{\pm 0.7}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	**0.20**$_{\pm 0.14}$	$0.006_{\pm 0.013}$	$89.0_{\pm 12.7}$	0.0	$0.0_{\pm 0.0}$	**0.21**$_{\pm 0.11}$	$0.007_{\pm 0.015}$	$84.7_{\pm 5.9}$	0.0	$0.0_{\pm 0.0}$	**0.22**$_{\pm 0.13}$	$0.007_{\pm 0.015}$	$89.1_{\pm 11.6}$
	GlobMP (Ours)	**80.0**	**80.0**$_{\pm 40.0}$	$5.00_{\pm 10.27}$	**0.038**$_{\pm 0.191}$	$16.4_{\pm 112.4}$	**70.0**	**70.0**$_{\pm 45.8}$	$6.30_{\pm 9.77}$	**0.026**$_{\pm 0.160}$	$41.4_{\pm 119.1}$	**65.0**	**65.0**$_{\pm 47.7}$	$5.95_{\pm 9.13}$	**0.026**$_{\pm 0.160}$	$108.4_{\pm 116.6}$

Table 8: Multi-robot planning performance on `Highways` for $N_a \in \{3, 6, \ldots, 21\}$ under the Low, Medium and High velocity caps. *SR* and *CS* measure trajectory feasibility, *Coll* reports inter-robot collisions, *DA* adherence to the environment-specific motion pattern, and *Time* the planning time per configuration: for *GlobMP (Ours)* the median over the configurations with the standard deviation as subscript, for the baselines the mean, so the two are not directly comparable. † marks a method that applies no velocity limit, so its three blocks report the same run; MMD-CBS also returns one plan per configuration rather than 64 candidates. A dash under *Coll* marks a count that is not commensurable with the others, as MDOC and MMD-CBS count colliding time steps where every other row counts agent pairs with any collision; MDOC's *CS* is omitted because its plans that do not cover every agent were counted as feasible. MMD-CBS plans under 60 s per configuration and *GlobMP (Ours)* under 300 s per team.

		Low ($v_{\max} = 0.647$)					Medium ($v_{\max} = 0.783$)					High ($v_{\max} = 0.878$)				
N_a	Method	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓
3	Diff. Policy†	10.0	10.0$_{\pm 30.0}$	0.15$_{\pm 0.35}$	0.567$_{\pm 0.298}$	0.3	10.0	10.0$_{\pm 30.0}$	0.15$_{\pm 0.35}$	0.567$_{\pm 0.298}$	0.3	10.0	10.0$_{\pm 30.0}$	0.15$_{\pm 0.35}$	0.567$_{\pm 0.298}$	0.3
	CD†	100.0	0.0$_{\pm 0.0}$	0.25$_{\pm 0.32}$	0.564$_{\pm 0.289}$	1.1$_{\pm 0.1}$	100.0	0.0$_{\pm 0.0}$	0.25$_{\pm 0.32}$	0.564$_{\pm 0.289}$	1.2$_{\pm 0.1}$	100.0	0.0$_{\pm 0.0}$	0.25$_{\pm 0.32}$	0.564$_{\pm 0.289}$	1.2$_{\pm 0.1}$
	MMD-CBS†	90.0	45.0$_{\pm 49.7}$	—	0.567$_{\pm 0.318}$	3.8$_{\pm 3.2}$	90.0	55.0$_{\pm 49.7}$	—	0.567$_{\pm 0.318}$	3.8$_{\pm 3.2}$	90.0	55.0$_{\pm 49.7}$	—	0.567$_{\pm 0.318}$	3.8$_{\pm 3.2}$
	MDOC†	15.0	—	—	0.083$_{\pm 0.233}$	41.3$_{\pm 5.6}$	15.0	—	—	0.083$_{\pm 0.233}$	37.2$_{\pm 3.4}$	15.0	—	—	0.083$_{\pm 0.233}$	36.8$_{\pm 3.3}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	0.02$_{\pm 0.02}$	0.565$_{\pm 0.293}$	3.5$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	0.05$_{\pm 0.11}$	0.564$_{\pm 0.292}$	3.0$_{\pm 0.2}$	0.0	0.0$_{\pm 0.0}$	0.05$_{\pm 0.13}$	0.564$_{\pm 0.292}$	3.0$_{\pm 0.1}$
	PCD-SHD	**100.0**	89.8$_{\pm 15.8}$	0.09$_{\pm 0.17}$	0.561$_{\pm 0.281}$	2.8$_{\pm 0.1}$	**100.0**	90.9$_{\pm 13.8}$	0.08$_{\pm 0.14}$	0.562$_{\pm 0.279}$	2.8$_{\pm 0.1}$	**100.0**	90.4$_{\pm 13.7}$	0.08$_{\pm 0.14}$	0.562$_{\pm 0.279}$	2.8$_{\pm 0.1}$
	Final Proj.	95.0	85.9$_{\pm 33.0}$	0.18$_{\pm 0.47}$	0.567$_{\pm 0.298}$	3.8$_{\pm 0.6}$	**100.0**	86.1$_{\pm 32.4}$	0.18$_{\pm 0.46}$	0.567$_{\pm 0.298}$	3.9$_{\pm 0.7}$	**100.0**	84.6$_{\pm 32.6}$	0.20$_{\pm 0.46}$	0.567$_{\pm 0.298}$	3.9$_{\pm 0.6}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.568$_{\pm 0.294}$	26.4$_{\pm 6.4}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.568**$_{\pm 0.294}$	27.6$_{\pm 7.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.568**$_{\pm 0.294}$	29.3$_{\pm 7.7}$
	GlobMP (Ours)	95.0	95.0$_{\pm 21.8}$	**0.00**$_{\pm 0.00}$	**0.583**$_{\pm 0.493}$	0.2$_{\pm 15.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.567$_{\pm 0.496}$	0.3$_{\pm 0.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.550$_{\pm 0.497}$	0.2$_{\pm 0.2}$
6	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	1.39$_{\pm 0.78}$	0.582$_{\pm 0.187}$	0.3	0.0	0.0$_{\pm 0.0}$	1.39$_{\pm 0.78}$	0.582$_{\pm 0.187}$	0.3	0.0	0.0$_{\pm 0.0}$	1.39$_{\pm 0.78}$	0.582$_{\pm 0.187}$	0.3
	CD†	30.0	0.0$_{\pm 0.0}$	2.61$_{\pm 0.80}$	0.555$_{\pm 0.192}$	1.2$_{\pm 0.1}$	30.0	0.0$_{\pm 0.0}$	2.61$_{\pm 0.80}$	0.555$_{\pm 0.192}$	1.2$_{\pm 0.1}$	30.0	0.0$_{\pm 0.0}$	2.61$_{\pm 0.80}$	0.555$_{\pm 0.192}$	0.5$_{\pm 0.0}$
	MMD-CBS†	75.0	0.0$_{\pm 0.0}$	—	0.558$_{\pm 0.219}$	32.0$_{\pm 16.7}$	75.0	0.0$_{\pm 0.0}$	—	0.558$_{\pm 0.219}$	32.0$_{\pm 16.7}$	75.0	0.0$_{\pm 0.0}$	—	0.558$_{\pm 0.219}$	32.0$_{\pm 16.7}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	63.5$_{\pm 3.2}$	0.0	—	—	0.000$_{\pm 0.000}$	121.4$_{\pm 150.4}$	0.0	—	—	0.000$_{\pm 0.000}$	72.0$_{\pm 6.6}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	0.26$_{\pm 0.26}$	0.539$_{\pm 0.205}$	3.6$_{\pm 0.5}$	0.0	0.0$_{\pm 0.0}$	0.19$_{\pm 0.16}$	0.540$_{\pm 0.204}$	1.6$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	0.15$_{\pm 0.15}$	0.540$_{\pm 0.203}$	3.1$_{\pm 0.1}$
	PCD-SHD	**100.0**	52.6$_{\pm 20.0}$	0.59$_{\pm 0.34}$	0.561$_{\pm 0.180}$	3.0$_{\pm 0.1}$	**100.0**	52.8$_{\pm 21.6}$	0.59$_{\pm 0.35}$	0.562$_{\pm 0.182}$	2.6$_{\pm 0.7}$	**100.0**	53.3$_{\pm 21.7}$	0.57$_{\pm 0.36}$	0.563$_{\pm 0.183}$	2.9$_{\pm 0.5}$
	Final Proj.	**100.0**	69.9$_{\pm 39.6}$	0.53$_{\pm 0.90}$	**0.582**$_{\pm 0.187}$	2.5$_{\pm 1.2}$	**100.0**	75.9$_{\pm 33.4}$	0.33$_{\pm 0.56}$	**0.582**$_{\pm 0.187}$	4.1$_{\pm 0.6}$	**100.0**	79.1$_{\pm 31.5}$	0.29$_{\pm 0.51}$	**0.582**$_{\pm 0.187}$	4.1$_{\pm 0.7}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.570$_{\pm 0.187}$	32.0$_{\pm 8.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.570$_{\pm 0.187}$	32.8$_{\pm 9.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.570$_{\pm 0.187}$	34.9$_{\pm 9.7}$
	GlobMP (Ours)	80.0	80.0$_{\pm 40.0}$	0.10$_{\pm 0.44}$	0.575$_{\pm 0.494}$	0.7$_{\pm 68.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.567$_{\pm 0.496}$	0.6$_{\pm 10.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.567$_{\pm 0.496}$	0.7$_{\pm 0.7}$
9	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	4.58$_{\pm 1.91}$	0.450$_{\pm 0.172}$	0.3	0.0	0.0$_{\pm 0.0}$	4.58$_{\pm 1.91}$	0.450$_{\pm 0.172}$	0.3	0.0	0.0$_{\pm 0.0}$	4.58$_{\pm 1.91}$	0.450$_{\pm 0.172}$	0.3
	CD†	0.0	0.0$_{\pm 0.0}$	7.04$_{\pm 1.79}$	0.463$_{\pm 0.170}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	7.04$_{\pm 1.79}$	0.463$_{\pm 0.170}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	7.04$_{\pm 1.79}$	0.463$_{\pm 0.170}$	1.2$_{\pm 0.1}$
	MMD-CBS†	5.0	0.0$_{\pm 0.0}$	—	0.478$_{\pm 0.186}$	59.0$_{\pm 9.8}$	5.0	0.0$_{\pm 0.0}$	—	0.478$_{\pm 0.186}$	59.0$_{\pm 9.8}$	5.0	0.0$_{\pm 0.0}$	—	0.478$_{\pm 0.186}$	59.0$_{\pm 9.8}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	121.0$_{\pm 15.7}$	0.0	—	—	0.000$_{\pm 0.000}$	91.0$_{\pm 13.2}$	0.0	—	—	0.000$_{\pm 0.000}$	82.7$_{\pm 3.8}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	1.29$_{\pm 0.58}$	**0.469**$_{\pm 0.162}$	4.4$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	1.26$_{\pm 0.55}$	**0.469**$_{\pm 0.161}$	3.4$_{\pm 0.5}$	0.0	0.0$_{\pm 0.0}$	1.28$_{\pm 0.56}$	0.469$_{\pm 0.160}$	3.5$_{\pm 0.1}$
	PCD-SHD	80.0	9.9$_{\pm 14.2}$	2.60$_{\pm 0.95}$	0.460$_{\pm 0.166}$	3.3$_{\pm 0.1}$	90.0	10.8$_{\pm 15.0}$	2.38$_{\pm 0.89}$	0.461$_{\pm 0.166}$	3.4$_{\pm 0.1}$	90.0	10.9$_{\pm 13.2}$	2.33$_{\pm 0.93}$	0.462$_{\pm 0.166}$	3.4$_{\pm 0.1}$
	Final Proj.	70.0	41.5$_{\pm 40.8}$	1.60$_{\pm 1.72}$	0.452$_{\pm 0.174}$	5.5$_{\pm 0.9}$	80.0	44.0$_{\pm 39.7}$	1.23$_{\pm 1.35}$	0.452$_{\pm 0.174}$	5.6$_{\pm 0.7}$	80.0	41.5$_{\pm 37.4}$	1.12$_{\pm 1.13}$	0.451$_{\pm 0.174}$	5.6$_{\pm 0.7}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.454$_{\pm 0.176}$	49.0$_{\pm 13.7}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.454$_{\pm 0.175}$	36.4$_{\pm 7.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.454$_{\pm 0.176}$	35.2$_{\pm 1.3}$
	GlobMP (Ours)	65.0	65.0$_{\pm 47.7}$	1.00$_{\pm 1.76}$	0.450$_{\pm 0.497}$	2.5$_{\pm 127.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.467$_{\pm 0.499}$	1.1$_{\pm 5.4}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.483**$_{\pm 0.500}$	1.2$_{\pm 13.1}$
12	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	8.75$_{\pm 3.11}$	0.464$_{\pm 0.154}$	0.3	0.0	0.0$_{\pm 0.0}$	8.75$_{\pm 3.11}$	0.464$_{\pm 0.154}$	0.3	0.0	0.0$_{\pm 0.0}$	8.75$_{\pm 3.11}$	0.464$_{\pm 0.154}$	0.3
	CD†	0.0	0.0$_{\pm 0.0}$	15.44$_{\pm 3.23}$	0.466$_{\pm 0.145}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	15.44$_{\pm 3.23}$	0.466$_{\pm 0.145}$	1.2$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	15.44$_{\pm 3.23}$	0.466$_{\pm 0.145}$	0.6$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.450$_{\pm 0.150}$	61.2$_{\pm 0.9}$	0.0	0.0$_{\pm 0.0}$	—	0.450$_{\pm 0.150}$	61.2$_{\pm 0.9}$	0.0	0.0$_{\pm 0.0}$	—	0.450$_{\pm 0.150}$	61.2$_{\pm 0.9}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	100.8$_{\pm 7.3}$	0.0	—	—	0.000$_{\pm 0.000}$	124.5$_{\pm 6.1}$	0.0	—	—	0.000$_{\pm 0.000}$	122.5$_{\pm 5.6}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	5.02$_{\pm 0.76}$	0.456$_{\pm 0.140}$	4.4$_{\pm 0.4}$	0.0	0.0$_{\pm 0.0}$	6.04$_{\pm 0.83}$	0.457$_{\pm 0.140}$	3.7$_{\pm 0.2}$	0.0	0.0$_{\pm 0.0}$	6.68$_{\pm 0.77}$	0.459$_{\pm 0.140}$	3.7$_{\pm 0.1}$
	PCD-SHD	20.0	0.6$_{\pm 1.3}$	5.75$_{\pm 1.94}$	**0.468**$_{\pm 0.142}$	3.6$_{\pm 0.1}$	20.0	0.5$_{\pm 1.1}$	5.46$_{\pm 1.97}$	0.469$_{\pm 0.142}$	3.6$_{\pm 0.1}$	20.0	0.9$_{\pm 2.0}$	5.23$_{\pm 1.77}$	**0.469**$_{\pm 0.142}$	3.7$_{\pm 0.1}$
	Final Proj.	50.0	29.1$_{\pm 39.5}$	3.49$_{\pm 3.31}$	0.463$_{\pm 0.154}$	6.3$_{\pm 1.4}$	55.0	29.2$_{\pm 40.0}$	2.76$_{\pm 2.80}$	0.463$_{\pm 0.154}$	6.8$_{\pm 0.7}$	65.0	30.2$_{\pm 39.5}$	2.23$_{\pm 2.24}$	0.463$_{\pm 0.154}$	6.8$_{\pm 0.7}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.467$_{\pm 0.148}$	44.1$_{\pm 2.2}$	**100.0**	99.9$_{\pm 0.3}$	0.00$_{\pm 0.01}$	0.466$_{\pm 0.148}$	43.4$_{\pm 1.8}$	**100.0**	99.9$_{\pm 0.3}$	0.00$_{\pm 0.00}$	0.466$_{\pm 0.149}$	43.7$_{\pm 1.7}$
	GlobMP (Ours)	60.0	60.0$_{\pm 49.0}$	3.45$_{\pm 4.99}$	0.467$_{\pm 0.499}$	25.2$_{\pm 102.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.492**$_{\pm 0.500}$	2.1$_{\pm 24.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.458$_{\pm 0.498}$	2.2$_{\pm 12.3}$
15	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	14.06$_{\pm 4.00}$	0.484$_{\pm 0.109}$	0.3	0.0	0.0$_{\pm 0.0}$	14.06$_{\pm 4.00}$	0.484$_{\pm 0.109}$	0.3	0.0	0.0$_{\pm 0.0}$	14.06$_{\pm 4.00}$	0.484$_{\pm 0.109}$	0.4
	CD†	0.0	0.0$_{\pm 0.0}$	26.51$_{\pm 3.99}$	0.481$_{\pm 0.107}$	1.2$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	26.51$_{\pm 3.99}$	0.481$_{\pm 0.107}$	1.3$_{\pm 0.0}$	0.0	0.0$_{\pm 0.0}$	26.51$_{\pm 3.99}$	0.481$_{\pm 0.107}$	1.2$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.483$_{\pm 0.111}$	63.0$_{\pm 2.1}$	0.0	0.0$_{\pm 0.0}$	—	0.483$_{\pm 0.111}$	63.0$_{\pm 2.1}$	0.0	0.0$_{\pm 0.0}$	—	0.483$_{\pm 0.111}$	63.0$_{\pm 2.1}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	141.9$_{\pm 12.5}$	0.0	—	—	0.000$_{\pm 0.000}$	168.6$_{\pm 16.0}$	0.0	—	—	0.000$_{\pm 0.000}$	177.0$_{\pm 31.2}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	11.99$_{\pm 1.72}$	0.485$_{\pm 0.110}$	5.1$_{\pm 0.6}$	0.0	0.0$_{\pm 0.0}$	14.72$_{\pm 1.73}$	**0.485**$_{\pm 0.109}$	4.2$_{\pm 0.3}$	0.0	0.0$_{\pm 0.0}$	16.38$_{\pm 1.74}$	**0.486**$_{\pm 0.109}$	2.8$_{\pm 1.0}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	10.01$_{\pm 2.01}$	0.480$_{\pm 0.106}$	3.3$_{\pm 0.9}$	0.0	0.0$_{\pm 0.0}$	9.49$_{\pm 1.89}$	0.481$_{\pm 0.106}$	4.0$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	9.22$_{\pm 1.82}$	0.481$_{\pm 0.105}$	4.1$_{\pm 0.1}$
	Final Proj.	30.0	11.0$_{\pm 26.4}$	5.93$_{\pm 4.23}$	0.485$_{\pm 0.109}$	6.7$_{\pm 2.3}$	35.0	11.1$_{\pm 23.1}$	4.71$_{\pm 3.34}$	0.485$_{\pm 0.109}$	9.1$_{\pm 0.8}$	35.0	10.9$_{\pm 23.2}$	4.14$_{\pm 2.88}$	0.485$_{\pm 0.109}$	9.1$_{\pm 0.7}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.482$_{\pm 0.109}$	60.0$_{\pm 2.7}$	**100.0**	99.9$_{\pm 0.3}$	0.00$_{\pm 0.01}$	0.482$_{\pm 0.109}$	69.0$_{\pm 10.6}$	**100.0**	99.7$_{\pm 1.4}$	0.01$_{\pm 0.03}$	0.482$_{\pm 0.109}$	70.2$_{\pm 10.7}$
	GlobMP (Ours)	30.0	30.0$_{\pm 45.8}$	9.00$_{\pm 7.40}$	**0.487**$_{\pm 0.500}$	300.1$_{\pm 136.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.467$_{\pm 0.499}$	2.1$_{\pm 74.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.467$_{\pm 0.499}$	2.6$_{\pm 61.0}$
18	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	19.84$_{\pm 5.64}$	0.504$_{\pm 0.119}$	0.3	0.0	0.0$_{\pm 0.0}$	19.84$_{\pm 5.64}$	0.504$_{\pm 0.119}$	0.3	0.0	0.0$_{\pm 0.0}$	19.84$_{\pm 5.64}$	0.504$_{\pm 0.119}$	0.3
	CD†	0.0	0.0$_{\pm 0.0}$	39.00$_{\pm 3.90}$	0.508$_{\pm 0.116}$	1.3$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	39.00$_{\pm 3.90}$	0.508$_{\pm 0.116}$	1.3$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	39.00$_{\pm 3.90}$	0.508$_{\pm 0.116}$	1.3$_{\pm 0.1}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.503$_{\pm 0.121}$	64.8$_{\pm 2.5}$	0.0	0.0$_{\pm 0.0}$	—	0.503$_{\pm 0.121}$	64.8$_{\pm 2.5}$	0.0	0.0$_{\pm 0.0}$	—	0.503$_{\pm 0.121}$	64.8$_{\pm 2.5}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	163.3$_{\pm 8.1}$	0.0	—	—	0.000$_{\pm 0.000}$	159.7$_{\pm 7.1}$	0.0	—	—	0.000$_{\pm 0.000}$	160.6$_{\pm 7.0}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	20.39$_{\pm 2.78}$	**0.511**$_{\pm 0.110}$	5.0$_{\pm 0.5}$	0.0	0.0$_{\pm 0.0}$	25.36$_{\pm 2.89}$	0.511$_{\pm 0.109}$	4.2$_{\pm 0.2}$	0.0	0.0$_{\pm 0.0}$	28.30$_{\pm 2.87}$	0.511$_{\pm 0.108}$	4.1$_{\pm 0.1}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	15.17$_{\pm 2.97}$	0.508$_{\pm 0.114}$	4.1$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	14.31$_{\pm 2.75}$	0.509$_{\pm 0.114}$	4.1$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	14.20$_{\pm 2.61}$	0.509$_{\pm 0.113}$	4.2$_{\pm 0.1}$
	Final Proj.	15.0	3.1$_{\pm 12.2}$	10.49$_{\pm 5.78}$	0.504$_{\pm 0.118}$	7.5$_{\pm 2.3}$	25.0	4.7$_{\pm 17.3}$	8.44$_{\pm 5.26}$	0.504$_{\pm 0.118}$	11.5$_{\pm 0.7}$	35.0	4.8$_{\pm 14.9}$	7.71$_{\pm 4.86}$	0.504$_{\pm 0.118}$	10.0$_{\pm 2.6}$
	DiRecT	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.507$_{\pm 0.118}$	87.5$_{\pm 9.0}$	**100.0**	**99.8**$_{\pm 0.7}$	**0.01**$_{\pm 0.02}$	0.507$_{\pm 0.118}$	88.4$_{\pm 12.7}$	**100.0**	**99.9**$_{\pm 0.3}$	**0.00**$_{\pm 0.01}$	0.507$_{\pm 0.118}$	77.5$_{\pm 8.1}$
	GlobMP (Ours)	0.0	0.0$_{\pm 0.0}$	19.10$_{\pm 5.20}$	0.503$_{\pm 0.500}$	300.1$_{\pm 0.3}$	90.0	90.0$_{\pm 30.0}$	2.85$_{\pm 8.79}$	**0.525**$_{\pm 0.499}$	9.2$_{\pm 101.2}$	90.0	90.0$_{\pm 30.0}$	3.55$_{\pm 10.66}$	**0.519**$_{\pm 0.500}$	4.3$_{\pm 101.1}$
21	Diff. Policy†	0.0	0.0$_{\pm 0.0}$	23.25$_{\pm 7.73}$	0.500$_{\pm 0.121}$	0.3	0.0	0.0$_{\pm 0.0}$	23.25$_{\pm 7.73}$	0.500$_{\pm 0.121}$	0.3	0.0	0.0$_{\pm 0.0}$	23.25$_{\pm 7.73}$	0.500$_{\pm 0.121}$	0.3
	CD†	0.0	0.0$_{\pm 0.0}$	52.34$_{\pm 7.72}$	0.500$_{\pm 0.118}$	0.7$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	52.34$_{\pm 7.72}$	0.500$_{\pm 0.118}$	1.3$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	52.34$_{\pm 7.72}$	0.500$_{\pm 0.118}$	1.3$_{\pm 0.0}$
	MMD-CBS†	0.0	0.0$_{\pm 0.0}$	—	0.502$_{\pm 0.122}$	66.3$_{\pm 4.0}$	0.0	0.0$_{\pm 0.0}$	—	0.502$_{\pm 0.122}$	66.3$_{\pm 4.0}$	0.0	0.0$_{\pm 0.0}$	—	0.502$_{\pm 0.122}$	66.3$_{\pm 4.0}$
	MDOC†	0.0	—	—	0.000$_{\pm 0.000}$	165.8$_{\pm 8.8}$	0.0	—	—	0.000$_{\pm 0.000}$	183.6$_{\pm 8.1}$	0.0	—	—	0.000$_{\pm 0.000}$	180.1$_{\pm 8.2}$
	PCD-LB	0.0	0.0$_{\pm 0.0}$	33.40$_{\pm 2.61}$	0.502$_{\pm 0.113}$	5.5$_{\pm 0.8}$	0.0	0.0$_{\pm 0.0}$	40.60$_{\pm 3.30}$	0.502$_{\pm 0.113}$	4.7$_{\pm 0.3}$	0.0	0.0$_{\pm 0.0}$	44.22$_{\pm 3.72}$	0.502$_{\pm 0.113}$	4.7$_{\pm 0.1}$
	PCD-SHD	0.0	0.0$_{\pm 0.0}$	19.55$_{\pm 3.93}$	0.500$_{\pm 0.115}$	4.5$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	18.93$_{\pm 3.44}$	0.500$_{\pm 0.115}$	4.6$_{\pm 0.1}$	0.0	0.0$_{\pm 0.0}$	18.51$_{\pm 3.52}$	0.500$_{\pm 0.115}$	4.7$_{\pm 0.1}$
	Final Proj.	25.0	3.2$_{\pm 8.2}$	7.22$_{\pm 6.39}$	0.501$_{\pm 0.122}$	14.9$_{\pm 0.7}$	40.0	5.4$_{\pm 11.2}$	5.81$_{\pm 5.73}$	0.501$_{\pm 0.122}$	14.9$_{\pm 0.8}$	45.0	6.6$_{\pm 14.1}$	5.29$_{\pm 5.39}$	0.501$_{\pm 0.122}$	10.2$_{\pm 3.4}$
	DiRecT	**100.0**	**99.9**$_{\pm 0.3}$	**0.00**$_{\pm 0.00}$	0.501$_{\pm 0.123}$	99.2$_{\pm 1.5}$	**100.0**	**99.8**$_{\pm 0.5}$	**0.00**$_{\pm 0.01}$	0.501$_{\pm 0.123}$	97.0$_{\pm 8.7}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	0.501$_{\pm 0.123}$	77.5$_{\pm 0.7}$
	GlobMP (Ours)	15.0	15.0$_{\pm 35.7}$	20.90$_{\pm 11.92}$	**0.502**$_{\pm 0.500}$	300.1$_{\pm 68.7}$	55.0	55.0$_{\pm 49.7}$	11.35$_{\pm 14.57}$	**0.519**$_{\pm 0.500}$	35.2$_{\pm 143.5}$	65.0	65.0$_{\pm 47.7}$	10.75$_{\pm 15.56}$	**0.507**$_{\pm 0.500}$	19.1$_{\pm 136.7}$

Table 9: Multi-robot planning performance on `Drop-Region` for $N_a \in \{3, 6, \ldots, 21\}$ under the Low, Medium and High velocity caps. *SR* and *CS* measure trajectory feasibility, *Coll* reports inter-robot collisions, *DA* adherence to the environment-specific motion pattern, and *Time* the planning time per configuration: for *GlobMP (Ours)* the median over the configurations with the standard deviation as subscript, for the baselines the mean, so the two are not directly comparable. † marks a method that applies no velocity limit, so its three blocks report the same run; MMD-CBS also returns one plan per configuration rather than 64 candidates. A dash under *Coll* marks a count that is not commensurable with the others, as MDOC and MMD-CBS count colliding time steps where every other row counts agent pairs with any collision; MDOC's *CS* is omitted because its plans that do not cover every agent were counted as feasible. MMD-CBS plans under 60 s per configuration and *GlobMP (Ours)* under 300 s per team.

		Low ($v_{\max} = 0.928$)					Medium ($v_{\max} = 1.13$)					High ($v_{\max} = 1.34$)				
N_a	Method	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓
3	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$0.13_{\pm 0.34}$	$0.139_{\pm 0.175}$	0.3	0.0	$0.0_{\pm 0.0}$	$0.13_{\pm 0.34}$	$0.139_{\pm 0.175}$	0.3	0.0	$0.0_{\pm 0.0}$	$0.13_{\pm 0.34}$	$0.139_{\pm 0.175}$	0.3
	CD†	45.0	$5.2_{\pm 21.7}$	$0.11_{\pm 0.32}$	$0.000_{\pm 0.001}$	$0.5_{\pm 0.0}$	45.0	$11.9_{\pm 24.6}$	$0.11_{\pm 0.32}$	$0.000_{\pm 0.001}$	$1.0_{\pm 0.2}$	45.0	$22.2_{\pm 34.5}$	$0.11_{\pm 0.32}$	$0.000_{\pm 0.001}$	$1.2_{\pm 0.0}$
	MMD-CBS†	30.0	$25.0_{\pm 43.3}$	—	$0.000_{\pm 0.000}$	$2.2_{\pm 1.5}$	30.0	$30.0_{\pm 45.8}$	—	$0.000_{\pm 0.000}$	$2.2_{\pm 1.5}$	30.0	$30.0_{\pm 45.8}$	—	$0.000_{\pm 0.000}$	$2.2_{\pm 1.5}$
	MDOC†	25.0	—	—	$0.000_{\pm 0.000}$	$42.7_{\pm 6.5}$	25.0	—	—	$0.000_{\pm 0.000}$	$33.2_{\pm 2.7}$	25.0	—	—	$0.000_{\pm 0.000}$	$38.6_{\pm 3.8}$
	PCD-LB	75.0	$55.9_{\pm 44.1}$	$0.00_{\pm 0.01}$	$0.000_{\pm 0.000}$	$3.1_{\pm 0.1}$	75.0	$56.8_{\pm 44.0}$	$0.01_{\pm 0.02}$	$0.000_{\pm 0.000}$	$2.7_{\pm 0.6}$	80.0	$57.0_{\pm 43.5}$	$0.01_{\pm 0.03}$	$0.000_{\pm 0.000}$	$3.0_{\pm 0.1}$
	PCD-SHD	30.0	$11.3_{\pm 28.3}$	$0.09_{\pm 0.23}$	$0.047_{\pm 0.094}$	$1.2_{\pm 0.1}$	20.0	$11.2_{\pm 28.4}$	$0.08_{\pm 0.21}$	$0.049_{\pm 0.094}$	$1.2_{\pm 0.1}$	20.0	$11.3_{\pm 28.5}$	$0.06_{\pm 0.15}$	$0.049_{\pm 0.092}$	$1.2_{\pm 0.1}$
	Final Proj.	95.0	$86.2_{\pm 26.3}$	$0.11_{\pm 0.28}$	$\mathbf{0.194}_{\pm 0.211}$	$3.4_{\pm 0.5}$	95.0	$86.2_{\pm 26.4}$	$0.11_{\pm 0.28}$	$\mathbf{0.204}_{\pm 0.215}$	$3.6_{\pm 0.6}$	**100.0**	$96.1_{\pm 5.1}$	$0.01_{\pm 0.03}$	$\mathbf{0.234}_{\pm 0.215}$	$3.6_{\pm 0.6}$
	DiRecT	**100.0**	$97.5_{\pm 8.1}$	$0.03_{\pm 0.09}$	$0.114_{\pm 0.149}$	$24.6_{\pm 6.3}$	**100.0**	$96.3_{\pm 12.5}$	$0.04_{\pm 0.13}$	$0.115_{\pm 0.150}$	$26.2_{\pm 6.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.115_{\pm 0.150}$	$26.2_{\pm 6.9}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.083_{\pm 0.276}$	$0.3_{\pm 0.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.083_{\pm 0.276}$	$0.5_{\pm 0.2}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.083_{\pm 0.276}$	$0.4_{\pm 0.2}$
6	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$1.16_{\pm 0.91}$	$0.148_{\pm 0.107}$	0.3	0.0	$0.0_{\pm 0.0}$	$1.16_{\pm 0.91}$	$0.148_{\pm 0.107}$	0.3	0.0	$0.0_{\pm 0.0}$	$1.16_{\pm 0.91}$	$0.148_{\pm 0.107}$	0.3
	CD†	15.0	$0.0_{\pm 0.0}$	$0.91_{\pm 0.78}$	$0.001_{\pm 0.002}$	$1.2_{\pm 0.0}$	15.0	$0.0_{\pm 0.0}$	$0.91_{\pm 0.78}$	$0.001_{\pm 0.002}$	$1.2_{\pm 0.1}$	15.0	$0.4_{\pm 1.0}$	$0.91_{\pm 0.78}$	$0.001_{\pm 0.002}$	$1.1_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$2.2_{\pm 1.3}$	0.0	$0.0_{\pm 0.0}$	—	—	$2.2_{\pm 1.3}$	0.0	$0.0_{\pm 0.0}$	—	—	$2.2_{\pm 1.3}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$62.9_{\pm 3.7}$	0.0	—	—	$0.000_{\pm 0.000}$	$83.6_{\pm 24.8}$	0.0	—	—	$0.000_{\pm 0.000}$	$83.0_{\pm 17.9}$
	PCD-LB	5.0	$0.1_{\pm 0.3}$	$1.32_{\pm 0.80}$	$0.000_{\pm 0.000}$	$3.2_{\pm 0.2}$	10.0	$0.4_{\pm 1.4}$	$1.36_{\pm 0.81}$	$0.000_{\pm 0.000}$	$3.1_{\pm 0.1}$	20.0	$0.9_{\pm 2.3}$	$1.31_{\pm 0.79}$	$0.000_{\pm 0.000}$	$3.1_{\pm 0.1}$
	PCD-SHD	5.0	$0.1_{\pm 0.3}$	$0.24_{\pm 0.35}$	$0.027_{\pm 0.047}$	$1.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.21_{\pm 0.30}$	$0.028_{\pm 0.049}$	$1.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.17_{\pm 0.23}$	$0.027_{\pm 0.049}$	$1.4_{\pm 0.1}$
	Final Proj.	**100.0**	$47.8_{\pm 35.8}$	$0.56_{\pm 0.55}$	$\mathbf{0.161}_{\pm 0.127}$	$3.4_{\pm 0.6}$	90.0	$45.7_{\pm 37.1}$	$0.57_{\pm 0.50}$	$\mathbf{0.168}_{\pm 0.127}$	$3.4_{\pm 0.6}$	**100.0**	$62.5_{\pm 32.4}$	$0.33_{\pm 0.40}$	$\mathbf{0.212}_{\pm 0.137}$	$3.4_{\pm 0.7}$
	DiRecT	**100.0**	$98.2_{\pm 6.1}$	$0.02_{\pm 0.06}$	$0.073_{\pm 0.106}$	$24.9_{\pm 6.8}$	**100.0**	$97.8_{\pm 6.2}$	$0.02_{\pm 0.06}$	$0.075_{\pm 0.106}$	$25.7_{\pm 6.8}$	**100.0**	$99.8_{\pm 0.7}$	$0.00_{\pm 0.01}$	$0.076_{\pm 0.107}$	$26.7_{\pm 7.4}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.117_{\pm 0.321}$	$0.6_{\pm 0.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.125_{\pm 0.331}$	$0.9_{\pm 1.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.133_{\pm 0.340}$	$1.1_{\pm 3.6}$
9	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$2.92_{\pm 1.22}$	$0.124_{\pm 0.096}$	0.3	0.0	$0.0_{\pm 0.0}$	$2.92_{\pm 1.22}$	$0.124_{\pm 0.096}$	0.3	0.0	$0.0_{\pm 0.0}$	$2.92_{\pm 1.22}$	$0.124_{\pm 0.096}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$3.40_{\pm 1.07}$	$0.001_{\pm 0.002}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$3.40_{\pm 1.07}$	$0.001_{\pm 0.002}$	$1.2_{\pm 0.0}$	0.0	$0.0_{\pm 0.0}$	$3.40_{\pm 1.07}$	$0.001_{\pm 0.002}$	$1.2_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$2.3_{\pm 1.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$2.3_{\pm 1.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$2.3_{\pm 1.9}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$108.4_{\pm 21.4}$	0.0	—	—	$0.000_{\pm 0.000}$	$95.4_{\pm 14.3}$	0.0	—	—	$0.000_{\pm 0.000}$	$83.7_{\pm 3.8}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$4.77_{\pm 1.42}$	$0.000_{\pm 0.000}$	$3.5_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$5.00_{\pm 1.45}$	$0.000_{\pm 0.000}$	$3.4_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$4.58_{\pm 1.31}$	$0.000_{\pm 0.000}$	$3.5_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$1.06_{\pm 0.75}$	$0.031_{\pm 0.043}$	$1.5_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.98_{\pm 0.67}$	$0.034_{\pm 0.044}$	$1.5_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.78_{\pm 0.56}$	$0.035_{\pm 0.043}$	$1.5_{\pm 0.1}$
	Final Proj.	50.0	$17.3_{\pm 31.9}$	$1.92_{\pm 1.82}$	$\mathbf{0.180}_{\pm 0.109}$	$4.3_{\pm 0.7}$	40.0	$13.0_{\pm 26.5}$	$2.00_{\pm 1.72}$	$\mathbf{0.209}_{\pm 0.118}$	$3.9_{\pm 1.1}$	80.0	$32.2_{\pm 32.6}$	$1.22_{\pm 1.33}$	$\mathbf{0.244}_{\pm 0.123}$	$4.4_{\pm 0.7}$
	DiRecT	90.0	$74.3_{\pm 35.6}$	$0.28_{\pm 0.40}$	$0.080_{\pm 0.084}$	$37.2_{\pm 9.7}$	95.0	$69.5_{\pm 37.5}$	$0.34_{\pm 0.44}$	$0.081_{\pm 0.085}$	$30.0_{\pm 6.7}$	**100.0**	**99.6**$_{\pm 1.4}$	**0.00**$_{\pm 0.01}$	$0.082_{\pm 0.086}$	$31.1_{\pm 7.7}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.061_{\pm 0.240}$	$1.3_{\pm 0.4}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.067_{\pm 0.249}$	$2.0_{\pm 2.7}$	95.0	$95.0_{\pm 21.8}$	$0.35_{\pm 1.53}$	$0.067_{\pm 0.249}$	$2.1_{\pm 66.9}$
12	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$6.19_{\pm 2.67}$	$0.142_{\pm 0.077}$	0.3	0.0	$0.0_{\pm 0.0}$	$6.19_{\pm 2.67}$	$0.142_{\pm 0.077}$	0.3	0.0	$0.0_{\pm 0.0}$	$6.19_{\pm 2.67}$	$0.142_{\pm 0.077}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$7.84_{\pm 1.65}$	$0.002_{\pm 0.005}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$7.84_{\pm 1.65}$	$0.002_{\pm 0.005}$	$0.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$7.84_{\pm 1.65}$	$0.002_{\pm 0.005}$	$1.2_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.7_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.7_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.7_{\pm 0.9}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$110.6_{\pm 10.0}$	0.0	—	—	$0.000_{\pm 0.000}$	$121.3_{\pm 5.9}$	0.0	—	—	$0.000_{\pm 0.000}$	$107.6_{\pm 5.6}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$11.55_{\pm 2.61}$	$0.000_{\pm 0.002}$	$1.7_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$11.87_{\pm 2.46}$	$0.000_{\pm 0.001}$	$3.7_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$11.07_{\pm 2.25}$	$0.000_{\pm 0.002}$	$3.8_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$2.48_{\pm 1.14}$	$0.021_{\pm 0.022}$	$1.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$2.31_{\pm 1.05}$	$0.022_{\pm 0.023}$	$1.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$1.88_{\pm 0.92}$	$0.024_{\pm 0.023}$	$1.6_{\pm 0.1}$
	Final Proj.	10.0	$0.8_{\pm 3.1}$	$4.75_{\pm 2.81}$	$\mathbf{0.175}_{\pm 0.078}$	$5.4_{\pm 0.7}$	5.0	$0.7_{\pm 3.1}$	$4.67_{\pm 2.64}$	$\mathbf{0.206}_{\pm 0.089}$	$5.4_{\pm 0.7}$	30.0	$10.5_{\pm 24.4}$	$2.49_{\pm 1.41}$	$\mathbf{0.255}_{\pm 0.096}$	$5.4_{\pm 0.7}$
	DiRecT	55.0	$32.3_{\pm 37.5}$	$1.09_{\pm 0.78}$	$0.054_{\pm 0.048}$	$41.0_{\pm 10.3}$	55.0	$30.4_{\pm 37.4}$	$1.21_{\pm 0.85}$	$0.056_{\pm 0.050}$	$45.0_{\pm 12.0}$	**100.0**	**98.4**$_{\pm 2.7}$	**0.02**$_{\pm 0.03}$	$0.057_{\pm 0.051}$	$47.4_{\pm 12.7}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.067_{\pm 0.249}$	$2.0_{\pm 8.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.071_{\pm 0.257}$	$3.2_{\pm 1.6}$	95.0	$95.0_{\pm 21.8}$	$0.50_{\pm 2.18}$	$0.071_{\pm 0.257}$	$4.0_{\pm 78.9}$
15	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$8.53_{\pm 2.25}$	$0.183_{\pm 0.062}$	0.3	0.0	$0.0_{\pm 0.0}$	$8.53_{\pm 2.25}$	$0.183_{\pm 0.062}$	0.3	0.0	$0.0_{\pm 0.0}$	$8.53_{\pm 2.25}$	$0.183_{\pm 0.062}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$13.91_{\pm 2.49}$	$0.001_{\pm 0.002}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$13.91_{\pm 2.49}$	$0.001_{\pm 0.002}$	$1.2_{\pm 0.0}$	0.0	$0.0_{\pm 0.0}$	$13.91_{\pm 2.49}$	$0.001_{\pm 0.002}$	$1.2_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.2_{\pm 0.8}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.2_{\pm 0.8}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.2_{\pm 0.8}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$158.3_{\pm 11.1}$	0.0	—	—	$0.000_{\pm 0.000}$	$175.3_{\pm 13.5}$	0.0	—	—	$0.000_{\pm 0.000}$	$167.0_{\pm 72.6}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$22.65_{\pm 2.75}$	$0.000_{\pm 0.000}$	$4.1_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$23.62_{\pm 2.63}$	$0.000_{\pm 0.000}$	$4.1_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$22.71_{\pm 2.48}$	$0.000_{\pm 0.000}$	$4.1_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$2.78_{\pm 1.30}$	$0.026_{\pm 0.028}$	$1.7_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$2.60_{\pm 1.24}$	$0.025_{\pm 0.027}$	$1.8_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$2.11_{\pm 1.00}$	$0.025_{\pm 0.027}$	$1.8_{\pm 0.1}$
	Final Proj.	15.0	$3.2_{\pm 12.9}$	$5.40_{\pm 2.82}$	$\mathbf{0.211}_{\pm 0.071}$	$7.0_{\pm 0.7}$	15.0	$2.0_{\pm 7.8}$	$5.43_{\pm 2.64}$	$\mathbf{0.254}_{\pm 0.074}$	$4.7_{\pm 1.8}$	50.0	$7.0_{\pm 18.1}$	$2.96_{\pm 1.97}$	$\mathbf{0.285}_{\pm 0.076}$	$7.0_{\pm 0.7}$
	DiRecT	70.0	$41.2_{\pm 40.4}$	$1.16_{\pm 1.06}$	$0.083_{\pm 0.037}$	$57.0_{\pm 14.4}$	65.0	$38.3_{\pm 40.2}$	$1.28_{\pm 1.12}$	$0.085_{\pm 0.036}$	$51.8_{\pm 7.9}$	**100.0**	$97.5_{\pm 7.6}$	$0.03_{\pm 0.08}$	$0.087_{\pm 0.036}$	$52.8_{\pm 7.9}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.047_{\pm 0.211}$	$2.3_{\pm 2.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.063_{\pm 0.244}$	$4.3_{\pm 3.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.067_{\pm 0.249}$	$11.8_{\pm 18.1}$
18	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$14.42_{\pm 4.49}$	$0.124_{\pm 0.051}$	0.3	0.0	$0.0_{\pm 0.0}$	$14.42_{\pm 4.49}$	$0.124_{\pm 0.051}$	0.3	0.0	$0.0_{\pm 0.0}$	$14.42_{\pm 4.49}$	$0.124_{\pm 0.051}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$23.43_{\pm 2.96}$	$0.002_{\pm 0.004}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$23.43_{\pm 2.96}$	$0.002_{\pm 0.004}$	$1.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$23.43_{\pm 2.96}$	$0.002_{\pm 0.004}$	$1.2_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$2.1_{\pm 1.2}$	0.0	$0.0_{\pm 0.0}$	—	—	$2.1_{\pm 1.2}$	0.0	$0.0_{\pm 0.0}$	—	—	$2.1_{\pm 1.2}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$166.5_{\pm 9.9}$	0.0	—	—	$0.000_{\pm 0.000}$	$159.0_{\pm 7.7}$	0.0	—	—	$0.000_{\pm 0.000}$	$142.2_{\pm 7.0}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$37.15_{\pm 5.84}$	$0.000_{\pm 0.000}$	$4.1_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$38.50_{\pm 5.83}$	$0.000_{\pm 0.000}$	$4.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$36.52_{\pm 5.40}$	$0.000_{\pm 0.000}$	$4.0_{\pm 0.6}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$4.46_{\pm 2.06}$	$0.020_{\pm 0.023}$	$1.8_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$4.29_{\pm 1.94}$	$0.022_{\pm 0.026}$	$1.8_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$3.56_{\pm 1.70}$	$0.023_{\pm 0.026}$	$1.8_{\pm 0.1}$
	Final Proj.	10.0	$1.5_{\pm 4.8}$	$8.88_{\pm 5.02}$	$\mathbf{0.140}_{\pm 0.049}$	$8.6_{\pm 0.8}$	10.0	$0.9_{\pm 2.6}$	$8.67_{\pm 4.89}$	$\mathbf{0.166}_{\pm 0.052}$	$7.4_{\pm 1.7}$	10.0	$0.7_{\pm 2.2}$	$5.66_{\pm 2.88}$	$\mathbf{0.199}_{\pm 0.066}$	$8.6_{\pm 0.7}$
	DiRecT	60.0	$31.9_{\pm 37.5}$	$2.05_{\pm 2.15}$	$0.059_{\pm 0.043}$	$68.0_{\pm 9.8}$	55.0	$30.5_{\pm 37.9}$	$2.19_{\pm 2.27}$	$0.060_{\pm 0.042}$	$64.9_{\pm 9.0}$	**100.0**	**91.6**$_{\pm 15.7}$	**0.11**$_{\pm 0.22}$	$0.062_{\pm 0.042}$	$64.4_{\pm 8.0}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.056_{\pm 0.229}$	$3.6_{\pm 9.4}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.056_{\pm 0.229}$	$9.5_{\pm 7.6}$	90.0	$90.0_{\pm 30.0}$	$2.15_{\pm 6.45}$	$0.067_{\pm 0.249}$	$24.2_{\pm 86.0}$
21	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$18.84_{\pm 3.94}$	$0.161_{\pm 0.078}$	0.4	0.0	$0.0_{\pm 0.0}$	$18.84_{\pm 3.94}$	$0.161_{\pm 0.078}$	0.3	0.0	$0.0_{\pm 0.0}$	$18.84_{\pm 3.94}$	$0.161_{\pm 0.078}$	0.4
	CD†	0.0	$0.0_{\pm 0.0}$	$33.21_{\pm 2.01}$	$0.002_{\pm 0.003}$	$0.8_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$33.21_{\pm 2.01}$	$0.002_{\pm 0.003}$	$1.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$33.21_{\pm 2.01}$	$0.002_{\pm 0.003}$	$1.3_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$2.6_{\pm 1.7}$	0.0	$0.0_{\pm 0.0}$	—	—	$2.6_{\pm 1.7}$	0.0	$0.0_{\pm 0.0}$	—	—	$2.6_{\pm 1.7}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$185.4_{\pm 9.2}$	0.0	—	—	$0.000_{\pm 0.000}$	$181.0_{\pm 8.7}$	0.0	—	—	$0.000_{\pm 0.000}$	$178.6_{\pm 8.8}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$53.13_{\pm 5.24}$	$0.000_{\pm 0.000}$	$4.6_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$54.89_{\pm 5.28}$	$0.000_{\pm 0.000}$	$4.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$51.21_{\pm 4.97}$	$0.000_{\pm 0.000}$	$4.6_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$6.27_{\pm 1.83}$	$0.027_{\pm 0.023}$	$2.0_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$5.99_{\pm 1.81}$	$0.029_{\pm 0.024}$	$2.0_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$4.79_{\pm 1.53}$	$0.030_{\pm 0.024}$	$2.0_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$14.01_{\pm 6.39}$	$\mathbf{0.160}_{\pm 0.079}$	$8.4_{\pm 2.8}$	0.0	$0.0_{\pm 0.0}$	$13.54_{\pm 6.03}$	$\mathbf{0.194}_{\pm 0.084}$	$11.9_{\pm 0.8}$	10.0	$0.2_{\pm 0.7}$	$8.83_{\pm 4.51}$	$\mathbf{0.232}_{\pm 0.086}$	$11.8_{\pm 0.7}$
	DiRecT	20.0	$12.2_{\pm 26.9}$	**3.57**$_{\pm 2.00}$	$0.062_{\pm 0.041}$	$89.9_{\pm 13.2}$	20.0	$9.7_{\pm 23.9}$	$3.73_{\pm 2.03}$	$0.066_{\pm 0.043}$	$78.6_{\pm 1.2}$	**100.0**	**88.0**$_{\pm 10.0}$	**0.14**$_{\pm 0.12}$	$0.070_{\pm 0.046}$	$79.9_{\pm 1.2}$
	GlobMP (Ours)	**65.0**	$65.0_{\pm 47.7}$	$5.45_{\pm 7.65}$	$0.095_{\pm 0.294}$	$15.2_{\pm 136.9}$	**95.0**	**95.0**$_{\pm 21.8}$	**1.05**$_{\pm 4.58}$	$0.086_{\pm 0.280}$	$15.1_{\pm 64.4}$	85.0	$85.0_{\pm 35.7}$	$3.95_{\pm 9.98}$	$0.093_{\pm 0.290}$	$27.1_{\pm 97.5}$

Table 10: Multi-robot planning performance on `Corridor-Bay` for $N_a \in \{3, 6, \ldots, 21\}$ under the Low, Medium and High velocity caps. *SR* and *CS* measure trajectory feasibility, *Coll* reports inter-robot collisions, *DA* adherence to the environment-specific motion pattern, and *Time* the planning time per configuration: for *GlobMP (Ours)* the median over the configurations with the standard deviation as subscript, for the baselines the mean, so the two are not directly comparable. † marks a method that applies no velocity limit, so its three blocks report the same run; MMD-CBS also returns one plan per configuration rather than 64 candidates. A dash under *Coll* marks a count that is not commensurable with the others, as MDOC and MMD-CBS count colliding time steps where every other row counts agent pairs with any collision; MDOC's *CS* is omitted because its plans that do not cover every agent were counted as feasible. MMD-CBS plans under 60 s per configuration and *GlobMP (Ours)* under 300 s per team.

		Low ($v_{\max} = 1.21$)					Medium ($v_{\max} = 1.46$)					High ($v_{\max} = 1.76$)				
N_a	Method	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓
3	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$0.31_{\pm 0.53}$	$0.997_{\pm 0.002}$	0.3	0.0	$0.0_{\pm 0.0}$	$0.31_{\pm 0.53}$	$0.997_{\pm 0.002}$	0.3	0.0	$0.0_{\pm 0.0}$	$0.31_{\pm 0.53}$	$0.997_{\pm 0.002}$	0.3
	CD†	85.0	$12.5_{\pm 20.0}$	$0.12_{\pm 0.26}$	$0.683_{\pm 0.079}$	$1.2_{\pm 0.0}$	85.0	$16.3_{\pm 21.3}$	$0.12_{\pm 0.26}$	$0.683_{\pm 0.079}$	$1.1_{\pm 0.0}$	85.0	$18.0_{\pm 21.5}$	$0.12_{\pm 0.26}$	$0.683_{\pm 0.079}$	$1.2_{\pm 0.0}$
	MMD-CBS†	45.0	$45.0_{\pm 49.7}$	—	$0.890_{\pm 0.068}$	$5.7_{\pm 6.5}$	45.0	$45.0_{\pm 49.7}$	—	$0.890_{\pm 0.068}$	$5.7_{\pm 6.5}$	45.0	$45.0_{\pm 49.7}$	—	$0.890_{\pm 0.068}$	$5.7_{\pm 6.5}$
	MDOC†	10.0	—	—	$0.072_{\pm 0.215}$	$36.1_{\pm 6.6}$	10.0	—	—	$0.072_{\pm 0.215}$	$36.4_{\pm 2.9}$	10.0	—	—	$0.072_{\pm 0.215}$	$37.0_{\pm 3.6}$
	PCD-LB	10.0	$0.5_{\pm 1.7}$	$0.02_{\pm 0.06}$	—	$3.0_{\pm 0.0}$	10.0	$0.5_{\pm 2.1}$	$0.01_{\pm 0.05}$	—	$3.1_{\pm 0.1}$	20.0	$1.5_{\pm 3.8}$	$0.01_{\pm 0.03}$	—	$3.1_{\pm 0.1}$
	PCD-SHD	45.0	$25.9_{\pm 36.9}$	$0.06_{\pm 0.09}$	—	$1.9_{\pm 0.8}$	50.0	$26.9_{\pm 38.0}$	$0.07_{\pm 0.12}$	—	$3.1_{\pm 0.1}$	40.0	$26.9_{\pm 38.5}$	$0.06_{\pm 0.09}$	—	$3.1_{\pm 0.1}$
	Final Proj.	80.0	$53.4_{\pm 42.9}$	$0.00_{\pm 0.00}$	$0.912_{\pm 0.039}$	$2.3_{\pm 0.9}$	85.0	$51.9_{\pm 43.4}$	$0.02_{\pm 0.08}$	$0.911_{\pm 0.040}$	$3.7_{\pm 0.6}$	85.0	$51.9_{\pm 43.4}$	**0.00**$_{\pm 0.00}$	$0.914_{\pm 0.039}$	$3.7_{\pm 0.6}$
	DiRecT	**95.0**	$87.2_{\pm 24.1}$	**0.00**$_{\pm 0.00}$	**0.920**$_{\pm 0.056}$	$18.0_{\pm 0.6}$	**95.0**	$82.0_{\pm 25.8}$	**0.00**$_{\pm 0.00}$	**0.920**$_{\pm 0.056}$	$18.2_{\pm 0.7}$	**95.0**	$80.0_{\pm 27.4}$	**0.00**$_{\pm 0.00}$	**0.921**$_{\pm 0.054}$	$18.2_{\pm 0.7}$
	GlobMP (Ours)	**95.0**	**95.0**$_{\pm 21.8}$	$0.05_{\pm 0.22}$	$0.914_{\pm 0.070}$	$0.7_{\pm 65.2}$	**95.0**	**95.0**$_{\pm 21.8}$	$0.05_{\pm 0.22}$	$0.918_{\pm 0.069}$	$0.9_{\pm 65.2}$	**95.0**	**95.0**$_{\pm 21.8}$	$0.05_{\pm 0.22}$	$0.920_{\pm 0.068}$	$0.7_{\pm 65.2}$
6	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$1.27_{\pm 1.05}$	$0.996_{\pm 0.002}$	0.3	0.0	$0.0_{\pm 0.0}$	$1.27_{\pm 1.05}$	$0.996_{\pm 0.002}$	0.3	0.0	$0.0_{\pm 0.0}$	$1.27_{\pm 1.05}$	$0.996_{\pm 0.002}$	0.3
	CD†	40.0	$3.8_{\pm 12.3}$	$0.70_{\pm 0.78}$	$0.669_{\pm 0.032}$	$1.2_{\pm 0.1}$	40.0	$4.8_{\pm 12.6}$	$0.70_{\pm 0.78}$	$0.669_{\pm 0.032}$	$0.5_{\pm 0.0}$	40.0	$5.8_{\pm 13.6}$	$0.70_{\pm 0.78}$	$0.669_{\pm 0.032}$	$1.2_{\pm 0.1}$
	MMD-CBS†	50.0	$35.0_{\pm 47.7}$	—	$0.880_{\pm 0.034}$	$13.5_{\pm 13.0}$	50.0	$50.0_{\pm 50.0}$	—	$0.880_{\pm 0.034}$	$13.5_{\pm 13.0}$	50.0	$50.0_{\pm 50.0}$	—	$0.880_{\pm 0.034}$	$13.5_{\pm 13.0}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$55.8_{\pm 3.5}$	0.0	—	—	$0.000_{\pm 0.000}$	$74.4_{\pm 18.6}$	0.0	—	—	$0.000_{\pm 0.000}$	$92.2_{\pm 23.6}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$0.84_{\pm 0.63}$	—	$1.4_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$0.66_{\pm 0.47}$	—	$3.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.39_{\pm 0.30}$	—	$3.4_{\pm 0.1}$
	PCD-SHD	40.0	$18.4_{\pm 34.0}$	$0.44_{\pm 0.47}$	—	$3.2_{\pm 0.1}$	40.0	$19.1_{\pm 33.2}$	$0.39_{\pm 0.39}$	—	$3.2_{\pm 0.1}$	50.0	$19.2_{\pm 33.2}$	$0.32_{\pm 0.28}$	—	$3.2_{\pm 0.1}$
	Final Proj.	60.0	$24.5_{\pm 37.2}$	$0.02_{\pm 0.07}$	$0.897_{\pm 0.042}$	$3.8_{\pm 0.6}$	60.0	$23.2_{\pm 36.9}$	$0.01_{\pm 0.03}$	$0.899_{\pm 0.039}$	$2.0_{\pm 0.6}$	60.0	$23.1_{\pm 37.0}$	$0.01_{\pm 0.03}$	$0.903_{\pm 0.037}$	$3.8_{\pm 0.7}$
	DiRecT	**100.0**	$65.8_{\pm 36.9}$	$0.00_{\pm 0.01}$	**0.915**$_{\pm 0.037}$	$18.3_{\pm 0.7}$	**100.0**	$56.8_{\pm 38.4}$	$0.00_{\pm 0.00}$	$0.915_{\pm 0.036}$	$18.3_{\pm 0.7}$	**95.0**	$51.8_{\pm 37.4}$	**0.00**$_{\pm 0.00}$	**0.917**$_{\pm 0.035}$	$18.3_{\pm 0.7}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.915_{\pm 0.071}$	$0.4_{\pm 2.6}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	**0.917**$_{\pm 0.069}$	$0.6_{\pm 1.5}$	85.0	**85.0**$_{\pm 35.7}$	$0.20_{\pm 0.51}$	$0.917_{\pm 0.071}$	$1.7_{\pm 106.5}$
9	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$3.48_{\pm 1.58}$	$0.997_{\pm 0.001}$	0.3	0.0	$0.0_{\pm 0.0}$	$3.48_{\pm 1.58}$	$0.997_{\pm 0.001}$	0.6	0.0	$0.0_{\pm 0.0}$	$3.48_{\pm 1.58}$	$0.997_{\pm 0.001}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$2.86_{\pm 1.16}$	$0.698_{\pm 0.034}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$2.86_{\pm 1.16}$	$0.698_{\pm 0.034}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$2.86_{\pm 1.16}$	$0.698_{\pm 0.034}$	$1.2_{\pm 0.1}$
	MMD-CBS†	5.0	$0.0_{\pm 0.0}$	—	$0.901_{\pm 0.039}$	$22.1_{\pm 32.2}$	5.0	$0.0_{\pm 0.0}$	—	$0.901_{\pm 0.039}$	$22.1_{\pm 32.2}$	5.0	$5.0_{\pm 21.8}$	—	$0.901_{\pm 0.039}$	$22.1_{\pm 32.2}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$117.4_{\pm 19.9}$	0.0	—	—	$0.000_{\pm 0.000}$	$98.4_{\pm 14.9}$	0.0	—	—	$0.000_{\pm 0.000}$	$82.5_{\pm 4.3}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$2.66_{\pm 1.28}$	—	$3.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$2.05_{\pm 0.97}$	—	$3.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$1.13_{\pm 0.50}$	—	$3.7_{\pm 0.1}$
	PCD-SHD	5.0	$0.5_{\pm 2.4}$	$1.54_{\pm 0.58}$	—	$3.6_{\pm 0.1}$	10.0	$0.9_{\pm 3.4}$	$1.19_{\pm 0.42}$	—	$3.6_{\pm 0.1}$	10.0	$1.0_{\pm 3.8}$	$1.04_{\pm 0.34}$	—	$3.7_{\pm 0.1}$
	Final Proj.	75.0	$22.0_{\pm 28.3}$	$0.03_{\pm 0.03}$	$0.886_{\pm 0.024}$	$4.9_{\pm 0.6}$	70.0	$19.5_{\pm 27.8}$	$0.03_{\pm 0.09}$	$0.889_{\pm 0.025}$	$4.9_{\pm 0.7}$	70.0	$19.5_{\pm 27.8}$	$0.00_{\pm 0.01}$	$0.894_{\pm 0.024}$	$2.6_{\pm 0.6}$
	DiRecT	**95.0**	$76.7_{\pm 36.3}$	**0.00**$_{\pm 0.00}$	**0.919**$_{\pm 0.024}$	$25.0_{\pm 0.8}$	**90.0**	$67.1_{\pm 34.6}$	**0.00**$_{\pm 0.00}$	**0.920**$_{\pm 0.024}$	$25.0_{\pm 0.7}$	**90.0**	$60.2_{\pm 34.5}$	**0.00**$_{\pm 0.00}$	**0.922**$_{\pm 0.023}$	$25.0_{\pm 0.9}$
	GlobMP (Ours)	**95.0**	**95.0**$_{\pm 21.8}$	$0.35_{\pm 1.53}$	$0.911_{\pm 0.064}$	$1.8_{\pm 69.7}$	75.0	**75.0**$_{\pm 43.3}$	$1.10_{\pm 2.07}$	$0.909_{\pm 0.066}$	$2.4_{\pm 128.5}$	75.0	**75.0**$_{\pm 43.3}$	$0.85_{\pm 1.53}$	$0.912_{\pm 0.064}$	$12.5_{\pm 126.4}$
12	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$5.98_{\pm 1.89}$	$0.997_{\pm 0.001}$	0.6	0.0	$0.0_{\pm 0.0}$	$5.98_{\pm 1.89}$	$0.997_{\pm 0.001}$	0.3	0.0	$0.0_{\pm 0.0}$	$5.98_{\pm 1.89}$	$0.997_{\pm 0.001}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$5.62_{\pm 1.27}$	$0.694_{\pm 0.024}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$5.62_{\pm 1.27}$	$0.694_{\pm 0.024}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$5.62_{\pm 1.27}$	$0.694_{\pm 0.024}$	$0.6_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	$0.921_{\pm 0.000}$	$5.1_{\pm 12.7}$	0.0	$0.0_{\pm 0.0}$	—	$0.921_{\pm 0.000}$	$5.1_{\pm 12.7}$	0.0	$0.0_{\pm 0.0}$	—	$0.921_{\pm 0.000}$	$5.1_{\pm 12.7}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$108.7_{\pm 7.5}$	0.0	—	—	$0.000_{\pm 0.000}$	$124.3_{\pm 6.0}$	0.0	—	—	$0.000_{\pm 0.000}$	$128.6_{\pm 6.3}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$7.83_{\pm 1.80}$	—	$3.8_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$5.87_{\pm 1.42}$	—	$4.0_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$3.64_{\pm 0.99}$	—	$1.9_{\pm 0.2}$
	PCD-SHD	5.0	$0.1_{\pm 0.3}$	$2.98_{\pm 1.29}$	—	$3.9_{\pm 0.1}$	5.0	$0.1_{\pm 0.3}$	$2.48_{\pm 1.12}$	—	$2.0_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$2.15_{\pm 0.97}$	—	$4.0_{\pm 0.1}$
	Final Proj.	45.0	$16.4_{\pm 29.4}$	$0.09_{\pm 0.10}$	$0.878_{\pm 0.036}$	$6.1_{\pm 0.7}$	40.0	$15.5_{\pm 30.9}$	$0.04_{\pm 0.08}$	$0.882_{\pm 0.035}$	$6.1_{\pm 0.6}$	40.0	$15.4_{\pm 31.0}$	$0.00_{\pm 0.00}$	$0.887_{\pm 0.033}$	$6.1_{\pm 0.7}$
	DiRecT	**95.0**	$63.2_{\pm 37.9}$	**0.00**$_{\pm 0.01}$	**0.921**$_{\pm 0.027}$	$31.1_{\pm 0.7}$	**95.0**	$54.1_{\pm 38.4}$	**0.00**$_{\pm 0.00}$	**0.923**$_{\pm 0.026}$	$31.1_{\pm 0.7}$	**95.0**	$49.1_{\pm 39.8}$	**0.00**$_{\pm 0.00}$	**0.925**$_{\pm 0.024}$	$31.1_{\pm 0.7}$
	GlobMP (Ours)	85.0	**85.0**$_{\pm 35.7}$	$1.15_{\pm 2.74}$	$0.903_{\pm 0.067}$	$4.5_{\pm 105.6}$	75.0	**75.0**$_{\pm 43.3}$	$1.75_{\pm 3.05}$	$0.905_{\pm 0.068}$	$8.4_{\pm 127.3}$	70.0	**70.0**$_{\pm 45.8}$	$1.75_{\pm 2.77}$	$0.908_{\pm 0.066}$	$17.1_{\pm 130.8}$
15	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$10.78_{\pm 2.90}$	$0.996_{\pm 0.001}$	0.3	0.0	$0.0_{\pm 0.0}$	$10.78_{\pm 2.90}$	$0.996_{\pm 0.001}$	0.3	0.0	$0.0_{\pm 0.0}$	$10.78_{\pm 2.90}$	$0.996_{\pm 0.001}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$10.72_{\pm 2.10}$	$0.686_{\pm 0.021}$	$1.2_{\pm 0.0}$	0.0	$0.0_{\pm 0.0}$	$10.72_{\pm 2.10}$	$0.686_{\pm 0.021}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$10.72_{\pm 2.10}$	$0.686_{\pm 0.021}$	$1.2_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	$0.844_{\pm 0.000}$	$8.1_{\pm 17.6}$	0.0	$0.0_{\pm 0.0}$	—	$0.844_{\pm 0.000}$	$8.1_{\pm 17.6}$	0.0	$0.0_{\pm 0.0}$	—	$0.844_{\pm 0.000}$	$8.1_{\pm 17.6}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$164.4_{\pm 13.3}$	0.0	—	—	$0.000_{\pm 0.000}$	$175.2_{\pm 19.9}$	0.0	—	—	$0.000_{\pm 0.000}$	$171.8_{\pm 26.1}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$18.53_{\pm 1.79}$	—	$4.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$15.04_{\pm 1.50}$	—	$4.4_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$10.75_{\pm 1.19}$	—	$4.4_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$5.32_{\pm 1.80}$	—	$4.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$4.44_{\pm 1.50}$	—	$4.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$3.73_{\pm 1.21}$	—	$4.4_{\pm 0.1}$
	Final Proj.	40.0	$5.2_{\pm 13.1}$	$0.22_{\pm 0.24}$	$0.851_{\pm 0.030}$	$4.3_{\pm 1.0}$	30.0	$6.7_{\pm 17.7}$	$0.08_{\pm 0.13}$	$0.858_{\pm 0.030}$	$7.8_{\pm 0.7}$	30.0	$7.0_{\pm 18.3}$	$0.02_{\pm 0.04}$	$0.863_{\pm 0.028}$	$7.8_{\pm 0.7}$
	DiRecT	**100.0**	$60.2_{\pm 30.4}$	**0.00**$_{\pm 0.01}$	**0.904**$_{\pm 0.018}$	$40.4_{\pm 0.7}$	**95.0**	$39.7_{\pm 33.0}$	**0.00**$_{\pm 0.00}$	**0.906**$_{\pm 0.017}$	$40.4_{\pm 0.7}$	**90.0**	$33.6_{\pm 32.8}$	**0.00**$_{\pm 0.00}$	**0.908**$_{\pm 0.017}$	$40.5_{\pm 0.9}$
	GlobMP (Ours)	75.0	**75.0**$_{\pm 43.3}$	$2.85_{\pm 4.99}$	$0.892_{\pm 0.074}$	$3.9_{\pm 127.2}$	65.0	**65.0**$_{\pm 47.7}$	$3.70_{\pm 5.45}$	$0.895_{\pm 0.072}$	$36.9_{\pm 133.7}$	65.0	**65.0**$_{\pm 47.7}$	$3.15_{\pm 4.63}$	$0.899_{\pm 0.072}$	$93.9_{\pm 129.5}$
18	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$15.87_{\pm 4.43}$	$0.996_{\pm 0.001}$	0.3	0.0	$0.0_{\pm 0.0}$	$15.87_{\pm 4.43}$	$0.996_{\pm 0.001}$	0.3	0.0	$0.0_{\pm 0.0}$	$15.87_{\pm 4.43}$	$0.996_{\pm 0.001}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$16.96_{\pm 3.44}$	$0.684_{\pm 0.022}$	$1.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$16.96_{\pm 3.44}$	$0.684_{\pm 0.022}$	$1.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$16.96_{\pm 3.44}$	$0.684_{\pm 0.022}$	$1.3_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	$0.898_{\pm 0.016}$	$8.5_{\pm 18.7}$	0.0	$0.0_{\pm 0.0}$	—	$0.898_{\pm 0.016}$	$8.5_{\pm 18.7}$	0.0	$0.0_{\pm 0.0}$	—	$0.898_{\pm 0.016}$	$8.5_{\pm 18.7}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$147.2_{\pm 7.4}$	0.0	—	—	$0.000_{\pm 0.000}$	$157.6_{\pm 7.0}$	0.0	—	—	$0.000_{\pm 0.000}$	$159.9_{\pm 7.2}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$30.77_{\pm 4.62}$	—	$4.4_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$24.93_{\pm 3.85}$	—	$4.5_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$18.60_{\pm 2.71}$	—	$4.6_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$8.11_{\pm 2.72}$	—	$4.4_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$6.79_{\pm 2.30}$	—	$4.4_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$5.77_{\pm 1.97}$	—	$4.5_{\pm 0.1}$
	Final Proj.	20.0	$6.6_{\pm 18.2}$	$0.38_{\pm 0.36}$	$0.837_{\pm 0.040}$	$10.2_{\pm 0.7}$	20.0	$6.6_{\pm 15.8}$	$0.09_{\pm 0.09}$	$0.846_{\pm 0.038}$	$8.8_{\pm 2.0}$	20.0	$5.5_{\pm 13.1}$	$0.02_{\pm 0.03}$	$0.853_{\pm 0.038}$	$10.2_{\pm 0.8}$
	DiRecT	**90.0**	$43.4_{\pm 34.4}$	**0.03**$_{\pm 0.04}$	**0.902**$_{\pm 0.026}$	$53.3_{\pm 0.9}$	**90.0**	$30.7_{\pm 30.5}$	**0.00**$_{\pm 0.00}$	**0.905**$_{\pm 0.025}$	$53.3_{\pm 0.9}$	**85.0**	$23.4_{\pm 28.2}$	**0.00**$_{\pm 0.00}$	**0.908**$_{\pm 0.025}$	$53.1_{\pm 0.7}$
	GlobMP (Ours)	65.0	**65.0**$_{\pm 47.7}$	$6.05_{\pm 8.70}$	$0.883_{\pm 0.071}$	$28.6_{\pm 135.8}$	60.0	**60.0**$_{\pm 49.0}$	$7.85_{\pm 10.09}$	$0.892_{\pm 0.070}$	$11.2_{\pm 143.1}$	40.0	**40.0**$_{\pm 49.0}$	$9.95_{\pm 8.68}$	$0.894_{\pm 0.068}$	$300.0_{\pm 137.3}$
21	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$20.72_{\pm 5.04}$	$0.997_{\pm 0.001}$	0.3	0.0	$0.0_{\pm 0.0}$	$20.72_{\pm 5.04}$	$0.997_{\pm 0.001}$	0.3	0.0	$0.0_{\pm 0.0}$	$20.72_{\pm 5.04}$	$0.997_{\pm 0.001}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$24.06_{\pm 3.00}$	$0.692_{\pm 0.016}$	$1.3_{\pm 0.0}$	0.0	$0.0_{\pm 0.0}$	$24.06_{\pm 3.00}$	$0.692_{\pm 0.016}$	$0.8_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$24.06_{\pm 3.00}$	$0.692_{\pm 0.016}$	$1.3_{\pm 0.0}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	$0.954_{\pm 0.015}$	$19.8_{\pm 41.4}$	0.0	$0.0_{\pm 0.0}$	—	$0.954_{\pm 0.015}$	$19.8_{\pm 41.4}$	0.0	$0.0_{\pm 0.0}$	—	$0.954_{\pm 0.015}$	$19.8_{\pm 41.4}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$183.7_{\pm 8.7}$	0.0	—	—	$0.000_{\pm 0.000}$	$180.8_{\pm 8.7}$	0.0	—	—	$0.000_{\pm 0.000}$	$179.5_{\pm 8.6}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$42.63_{\pm 4.11}$	—	$5.0_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$34.30_{\pm 3.31}$	—	$5.1_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$26.27_{\pm 2.54}$	—	$4.4_{\pm 1.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$10.66_{\pm 3.55}$	—	$4.8_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$9.02_{\pm 2.99}$	—	$4.9_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$7.59_{\pm 2.61}$	—	$2.5_{\pm 0.2}$
	Final Proj.	25.0	$8.2_{\pm 21.8}$	$0.70_{\pm 0.81}$	$0.832_{\pm 0.033}$	$13.0_{\pm 0.8}$	30.0	$6.7_{\pm 19.6}$	$0.23_{\pm 0.30}$	$0.840_{\pm 0.032}$	$9.6_{\pm 2.9}$	35.0	$6.7_{\pm 19.5}$	$0.07_{\pm 0.12}$	$0.846_{\pm 0.032}$	$13.1_{\pm 0.8}$
	DiRecT	**100.0**	**52.2**$_{\pm 29.1}$	**0.04**$_{\pm 0.06}$	**0.906**$_{\pm 0.024}$	$68.5_{\pm 0.7}$	**100.0**	$35.5_{\pm 29.5}$	**0.00**$_{\pm 0.00}$	**0.910**$_{\pm 0.022}$	$68.5_{\pm 0.7}$	**85.0**	$26.8_{\pm 29.7}$	**0.00**$_{\pm 0.00}$	**0.913**$_{\pm 0.021}$	$68.5_{\pm 0.7}$
	GlobMP (Ours)	45.0	$45.0_{\pm 49.7}$	$12.90_{\pm 12.22}$	$0.886_{\pm 0.077}$	$300.0_{\pm 141.1}$	60.0	**60.0**$_{\pm 49.0}$	$8.45_{\pm 10.77}$	$0.900_{\pm 0.068}$	$33.5_{\pm 137.9}$	70.0	**70.0**$_{\pm 45.8}$	$5.60_{\pm 8.79}$	$0.899_{\pm 0.071}$	$83.2_{\pm 116.2}$

Table 11: Multi-robot planning performance on `Cross-Room` for $N_a \in \{3, 6, \ldots, 21\}$ under the Low, Medium and High velocity caps. *SR* and *CS* measure trajectory feasibility, *Coll* reports inter-robot collisions, *DA* adherence to the environment-specific motion pattern, and *Time* the planning time per configuration: for *GlobMP (Ours)* the median over the configurations with the standard deviation as subscript, for the baselines the mean, so the two are not directly comparable. † marks a method that applies no velocity limit, so its three blocks report the same run; MMD-CBS also returns one plan per configuration rather than 64 candidates. A dash under *Coll* marks a count that is not commensurable with the others, as MDOC and MMD-CBS count colliding time steps where every other row counts agent pairs with any collision; MDOC's *CS* is omitted because its plans that do not cover every agent were counted as feasible. MMD-CBS plans under 60 s per configuration and *GlobMP (Ours)* under 300 s per team.

		Low ($v_{\max} = 0.928$)					Medium ($v_{\max} = 1.13$)					High ($v_{\max} = 1.34$)				
N_a	Method	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓	SR↑	CS↑	Coll↓	DA↑	Time↓
3	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$0.30_{\pm 0.55}$	$0.985_{\pm 0.007}$	0.6	0.0	$0.0_{\pm 0.0}$	$0.30_{\pm 0.55}$	$0.985_{\pm 0.007}$	0.5	0.0	$0.0_{\pm 0.0}$	$0.30_{\pm 0.55}$	$0.985_{\pm 0.007}$	0.5
	CD†	20.0	$5.6_{\pm 15.6}$	$0.02_{\pm 0.05}$	$0.662_{\pm 0.070}$	$1.1_{\pm 0.0}$	20.0	$9.3_{\pm 25.5}$	$0.02_{\pm 0.05}$	$0.662_{\pm 0.070}$	$1.1_{\pm 0.1}$	20.0	$10.1_{\pm 27.3}$	$0.02_{\pm 0.05}$	$0.662_{\pm 0.070}$	$1.1_{\pm 0.0}$
	MMD-CBS†	10.0	$10.0_{\pm 30.0}$	—	$0.887_{\pm 0.049}$	$0.7_{\pm 0.4}$	10.0	$10.0_{\pm 30.0}$	—	$0.887_{\pm 0.049}$	$0.7_{\pm 0.4}$	10.0	$10.0_{\pm 30.0}$	—	$0.887_{\pm 0.049}$	$0.7_{\pm 0.4}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$37.9_{\pm 2.4}$	0.0	—	—	$0.000_{\pm 0.000}$	$36.2_{\pm 2.5}$	0.0	—	—	$0.000_{\pm 0.000}$	$35.4_{\pm 2.6}$
	PCD-LB	55.0	$16.6_{\pm 30.6}$	$0.05_{\pm 0.17}$	—	$2.7_{\pm 0.2}$	45.0	$15.6_{\pm 30.8}$	$0.05_{\pm 0.15}$	—	$1.5_{\pm 0.1}$	45.0	$16.3_{\pm 30.6}$	$0.04_{\pm 0.14}$	—	$1.5_{\pm 0.1}$
	PCD-SHD	10.0	$4.0_{\pm 16.3}$	$0.00_{\pm 0.01}$	—	$2.0_{\pm 0.6}$	10.0	$3.8_{\pm 15.3}$	$0.00_{\pm 0.01}$	—	$1.6_{\pm 0.4}$	10.0	$4.0_{\pm 16.3}$	$0.00_{\pm 0.01}$	—	$1.5_{\pm 0.1}$
	Final Proj.	5.0	$2.7_{\pm 11.9}$	$0.01_{\pm 0.02}$	**0.910**$_{\pm 0.035}$	$4.4_{\pm 0.9}$	5.0	$2.7_{\pm 11.9}$	$0.01_{\pm 0.01}$	**0.907**$_{\pm 0.035}$	$4.4_{\pm 0.9}$	5.0	$2.7_{\pm 11.9}$	$0.00_{\pm 0.01}$	**0.909**$_{\pm 0.035}$	$2.4_{\pm 1.0}$
	DiRecT	10.0	$5.0_{\pm 21.1}$	$0.03_{\pm 0.08}$	$0.899_{\pm 0.038}$	$33.1_{\pm 6.8}$	10.0	$5.0_{\pm 21.1}$	$0.00_{\pm 0.01}$	$0.902_{\pm 0.038}$	$30.4_{\pm 8.1}$	10.0	$4.8_{\pm 20.8}$	$0.00_{\pm 0.00}$	$0.904_{\pm 0.038}$	$29.1_{\pm 8.4}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.804_{\pm 0.142}$	$0.4_{\pm 0.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.802_{\pm 0.133}$	$0.4_{\pm 0.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.809_{\pm 0.143}$	$0.6_{\pm 0.7}$
6	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$1.59_{\pm 0.98}$	$0.985_{\pm 0.004}$	0.6	0.0	$0.0_{\pm 0.0}$	$1.59_{\pm 0.98}$	$0.985_{\pm 0.004}$	0.6	0.0	$0.0_{\pm 0.0}$	$1.59_{\pm 0.98}$	$0.985_{\pm 0.004}$	0.6
	CD†	10.0	$1.2_{\pm 5.4}$	$0.40_{\pm 0.41}$	$0.671_{\pm 0.041}$	$1.1_{\pm 0.1}$	10.0	$3.2_{\pm 14.0}$	$0.40_{\pm 0.41}$	$0.671_{\pm 0.041}$	$0.5_{\pm 0.1}$	10.0	$3.7_{\pm 16.0}$	$0.40_{\pm 0.41}$	$0.671_{\pm 0.041}$	$1.1_{\pm 0.0}$
	MMD-CBS†	5.0	$0.0_{\pm 0.0}$	—	$0.849_{\pm 0.000}$	$1.0_{\pm 1.1}$	5.0	$0.0_{\pm 0.0}$	—	$0.849_{\pm 0.000}$	$1.0_{\pm 1.1}$	5.0	$0.0_{\pm 0.0}$	—	$0.849_{\pm 0.000}$	$1.0_{\pm 1.1}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$66.3_{\pm 3.8}$	0.0	—	—	$0.000_{\pm 0.000}$	$64.4_{\pm 6.6}$	0.0	—	—	$0.000_{\pm 0.000}$	$61.7_{\pm 6.2}$
	PCD-LB	25.0	$8.6_{\pm 22.8}$	$1.01_{\pm 0.48}$	—	$2.7_{\pm 0.7}$	20.0	$7.7_{\pm 22.1}$	$1.07_{\pm 0.46}$	—	$2.5_{\pm 0.5}$	40.0	$7.7_{\pm 21.9}$	$1.01_{\pm 0.46}$	—	$1.9_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$0.17_{\pm 0.26}$	—	$2.1_{\pm 0.6}$	0.0	$0.0_{\pm 0.0}$	$0.14_{\pm 0.20}$	—	$2.1_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	$0.12_{\pm 0.16}$	—	$1.8_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.08_{\pm 0.12}$	$0.906_{\pm 0.029}$	$3.9_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	$0.07_{\pm 0.11}$	$0.902_{\pm 0.030}$	$3.9_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	$0.06_{\pm 0.09}$	$0.903_{\pm 0.031}$	$3.9_{\pm 0.8}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	$0.07_{\pm 0.14}$	**0.908**$_{\pm 0.031}$	$24.5_{\pm 7.3}$	0.0	$0.0_{\pm 0.0}$	$0.03_{\pm 0.11}$	**0.908**$_{\pm 0.033}$	$24.4_{\pm 7.3}$	0.0	$0.0_{\pm 0.0}$	$0.01_{\pm 0.02}$	**0.909**$_{\pm 0.034}$	$27.1_{\pm 7.2}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.783_{\pm 0.133}$	$1.2_{\pm 52.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.786_{\pm 0.136}$	$1.5_{\pm 1.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.787_{\pm 0.133}$	$2.7_{\pm 2.3}$
9	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$3.23_{\pm 1.34}$	$0.985_{\pm 0.003}$	0.6	0.0	$0.0_{\pm 0.0}$	$3.23_{\pm 1.34}$	$0.985_{\pm 0.003}$	0.6	0.0	$0.0_{\pm 0.0}$	$3.23_{\pm 1.34}$	$0.985_{\pm 0.003}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$1.42_{\pm 0.71}$	$0.659_{\pm 0.030}$	$1.1_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$1.42_{\pm 0.71}$	$0.659_{\pm 0.030}$	$1.1_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$1.42_{\pm 0.71}$	$0.659_{\pm 0.030}$	$1.1_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.0_{\pm 0.8}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.0_{\pm 0.8}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.0_{\pm 0.8}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$94.2_{\pm 4.1}$	0.0	—	—	$0.000_{\pm 0.000}$	$92.9_{\pm 3.9}$	0.0	—	—	$0.000_{\pm 0.000}$	$93.0_{\pm 6.6}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$3.96_{\pm 1.41}$	—	$2.1_{\pm 0.4}$	0.0	$0.0_{\pm 0.0}$	$4.11_{\pm 1.38}$	—	$3.1_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	$3.68_{\pm 1.18}$	—	$2.2_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$0.28_{\pm 0.27}$	—	$2.0_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.26_{\pm 0.22}$	—	$2.1_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.22_{\pm 0.16}$	—	$2.0_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.14_{\pm 0.17}$	$0.895_{\pm 0.032}$	$2.8_{\pm 0.8}$	0.0	$0.0_{\pm 0.0}$	$0.11_{\pm 0.13}$	$0.891_{\pm 0.034}$	$5.1_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	$0.08_{\pm 0.10}$	$0.893_{\pm 0.034}$	$5.1_{\pm 1.0}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	$0.20_{\pm 0.26}$	**0.902**$_{\pm 0.027}$	$35.4_{\pm 9.8}$	0.0	$0.0_{\pm 0.0}$	$0.06_{\pm 0.08}$	**0.906**$_{\pm 0.028}$	$36.8_{\pm 9.8}$	0.0	$0.0_{\pm 0.0}$	$0.02_{\pm 0.03}$	**0.908**$_{\pm 0.027}$	$38.5_{\pm 9.6}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.782_{\pm 0.133}$	$2.3_{\pm 1.1}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.783_{\pm 0.124}$	$3.3_{\pm 3.0}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.782_{\pm 0.132}$	$7.3_{\pm 20.9}$
12	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$7.06_{\pm 2.70}$	$0.985_{\pm 0.003}$	0.3	0.0	$0.0_{\pm 0.0}$	$7.06_{\pm 2.70}$	$0.985_{\pm 0.003}$	0.6	0.0	$0.0_{\pm 0.0}$	$7.06_{\pm 2.70}$	$0.985_{\pm 0.003}$	0.6
	CD†	0.0	$0.0_{\pm 0.0}$	$4.08_{\pm 1.26}$	$0.663_{\pm 0.020}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$4.08_{\pm 1.26}$	$0.663_{\pm 0.020}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$4.08_{\pm 1.26}$	$0.663_{\pm 0.020}$	$0.6_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.1_{\pm 1.2}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.1_{\pm 1.2}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.1_{\pm 1.2}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$122.0_{\pm 5.3}$	0.0	—	—	$0.000_{\pm 0.000}$	$110.7_{\pm 5.9}$	0.0	—	—	$0.000_{\pm 0.000}$	$108.8_{\pm 4.8}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$9.93_{\pm 1.54}$	—	$3.4_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	$10.31_{\pm 1.59}$	—	$2.4_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$9.16_{\pm 1.46}$	—	$4.5_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$0.60_{\pm 0.36}$	—	$4.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.53_{\pm 0.26}$	—	$2.4_{\pm 0.3}$	0.0	$0.0_{\pm 0.0}$	$0.46_{\pm 0.26}$	—	$4.1_{\pm 0.8}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.41_{\pm 0.42}$	$0.881_{\pm 0.026}$	$5.4_{\pm 2.1}$	0.0	$0.0_{\pm 0.0}$	$0.33_{\pm 0.32}$	$0.876_{\pm 0.028}$	$7.5_{\pm 1.0}$	0.0	$0.0_{\pm 0.0}$	$0.33_{\pm 0.28}$	$0.877_{\pm 0.029}$	$7.5_{\pm 0.9}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	$0.44_{\pm 0.48}$	**0.898**$_{\pm 0.020}$	$49.1_{\pm 12.6}$	0.0	$0.0_{\pm 0.0}$	$0.10_{\pm 0.14}$	**0.900**$_{\pm 0.020}$	$49.2_{\pm 12.6}$	0.0	$0.0_{\pm 0.0}$	$0.04_{\pm 0.05}$	**0.902**$_{\pm 0.020}$	$49.8_{\pm 12.5}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.773_{\pm 0.140}$	$3.1_{\pm 57.3}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.778_{\pm 0.135}$	$5.8_{\pm 11.5}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.778_{\pm 0.140}$	$9.8_{\pm 42.3}$
15	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$9.58_{\pm 2.09}$	$0.985_{\pm 0.003}$	0.6	0.0	$0.0_{\pm 0.0}$	$9.58_{\pm 2.09}$	$0.985_{\pm 0.003}$	0.6	0.0	$0.0_{\pm 0.0}$	$9.58_{\pm 2.09}$	$0.985_{\pm 0.003}$	0.6
	CD†	0.0	$0.0_{\pm 0.0}$	$6.69_{\pm 1.73}$	$0.662_{\pm 0.016}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$6.69_{\pm 1.73}$	$0.662_{\pm 0.016}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$6.69_{\pm 1.73}$	$0.662_{\pm 0.016}$	$1.2_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.0_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.0_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.0_{\pm 0.9}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$140.7_{\pm 6.7}$	0.0	—	—	$0.000_{\pm 0.000}$	$134.0_{\pm 6.8}$	0.0	—	—	$0.000_{\pm 0.000}$	$134.2_{\pm 6.1}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$20.03_{\pm 2.29}$	—	$3.0_{\pm 0.6}$	0.0	$0.0_{\pm 0.0}$	$20.97_{\pm 2.25}$	—	$2.9_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$19.26_{\pm 1.97}$	—	$3.0_{\pm 0.4}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$0.84_{\pm 0.37}$	—	$3.9_{\pm 1.1}$	0.0	$0.0_{\pm 0.0}$	$0.78_{\pm 0.35}$	—	$2.4_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.68_{\pm 0.31}$	—	$2.4_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	$0.59_{\pm 0.46}$	$0.872_{\pm 0.019}$	$7.9_{\pm 2.4}$	0.0	$0.0_{\pm 0.0}$	$0.51_{\pm 0.37}$	$0.867_{\pm 0.020}$	$11.2_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	$0.44_{\pm 0.31}$	$0.868_{\pm 0.020}$	$8.8_{\pm 2.8}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	$0.64_{\pm 0.50}$	**0.892**$_{\pm 0.014}$	$68.2_{\pm 15.4}$	0.0	$0.0_{\pm 0.0}$	$0.20_{\pm 0.24}$	**0.894**$_{\pm 0.014}$	$69.0_{\pm 15.5}$	0.0	$0.0_{\pm 0.0}$	**0.07**$_{\pm 0.08}$	**0.896**$_{\pm 0.014}$	$75.8_{\pm 12.1}$
	GlobMP (Ours)	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.774_{\pm 0.142}$	$3.9_{\pm 23.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.772_{\pm 0.134}$	$9.2_{\pm 44.7}$	**95.0**	**95.0**$_{\pm 21.8}$	$0.80_{\pm 3.49}$	$0.775_{\pm 0.139}$	$21.9_{\pm 62.5}$
18	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$13.49_{\pm 2.49}$	$0.985_{\pm 0.002}$	0.3	0.0	$0.0_{\pm 0.0}$	$13.49_{\pm 2.49}$	$0.985_{\pm 0.002}$	0.6	0.0	$0.0_{\pm 0.0}$	$13.49_{\pm 2.49}$	$0.985_{\pm 0.002}$	0.3
	CD†	0.0	$0.0_{\pm 0.0}$	$11.98_{\pm 2.21}$	$0.665_{\pm 0.016}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$11.98_{\pm 2.21}$	$0.665_{\pm 0.016}$	$1.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$11.98_{\pm 2.21}$	$0.665_{\pm 0.016}$	$1.2_{\pm 0.1}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$0.9_{\pm 0.6}$	0.0	$0.0_{\pm 0.0}$	—	—	$0.9_{\pm 0.6}$	0.0	$0.0_{\pm 0.0}$	—	—	$0.9_{\pm 0.6}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$161.4_{\pm 7.5}$	0.0	—	—	$0.000_{\pm 0.000}$	$199.2_{\pm 11.4}$	0.0	—	—	$0.000_{\pm 0.000}$	$173.8_{\pm 8.0}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$32.89_{\pm 2.64}$	—	$3.1_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$34.26_{\pm 2.78}$	—	$5.7_{\pm 0.5}$	0.0	$0.0_{\pm 0.0}$	$31.50_{\pm 2.87}$	—	$3.4_{\pm 0.1}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$1.07_{\pm 0.63}$	—	$3.2_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$0.99_{\pm 0.57}$	—	$2.9_{\pm 0.2}$	0.0	$0.0_{\pm 0.0}$	$0.88_{\pm 0.43}$	—	$3.3_{\pm 0.1}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	**0.56**$_{\pm 0.38}$	$0.869_{\pm 0.016}$	$14.9_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	$0.45_{\pm 0.33}$	$0.865_{\pm 0.017}$	$11.6_{\pm 3.7}$	0.0	$0.0_{\pm 0.0}$	$0.38_{\pm 0.26}$	$0.866_{\pm 0.018}$	$14.9_{\pm 0.9}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	$1.02_{\pm 0.65}$	**0.900**$_{\pm 0.012}$	$89.6_{\pm 24.7}$	0.0	$0.0_{\pm 0.0}$	**0.27**$_{\pm 0.21}$	**0.903**$_{\pm 0.012}$	$104.4_{\pm 3.8}$	0.0	$0.0_{\pm 0.0}$	$0.09_{\pm 0.06}$	**0.905**$_{\pm 0.011}$	$58.6_{\pm 25.2}$
	GlobMP (Ours)	**95.0**	**95.0**$_{\pm 21.8}$	$1.00_{\pm 4.36}$	$0.789_{\pm 0.134}$	$8.1_{\pm 96.3}$	**95.0**	**95.0**$_{\pm 21.8}$	$1.00_{\pm 4.36}$	$0.791_{\pm 0.130}$	$11.1_{\pm 61.9}$	**100.0**	**100.0**$_{\pm 0.0}$	**0.00**$_{\pm 0.00}$	$0.779_{\pm 0.130}$	$60.5_{\pm 84.3}$
21	Diff. Policy†	0.0	$0.0_{\pm 0.0}$	$20.13_{\pm 3.78}$	$0.985_{\pm 0.002}$	0.7	0.0	$0.0_{\pm 0.0}$	$20.13_{\pm 3.78}$	$0.985_{\pm 0.002}$	0.7	0.0	$0.0_{\pm 0.0}$	$20.13_{\pm 3.78}$	$0.985_{\pm 0.002}$	0.7
	CD†	0.0	$0.0_{\pm 0.0}$	$18.02_{\pm 3.46}$	$0.659_{\pm 0.017}$	$1.2_{\pm 0.0}$	0.0	$0.0_{\pm 0.0}$	$18.02_{\pm 3.46}$	$0.659_{\pm 0.017}$	$0.6_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$18.02_{\pm 3.46}$	$0.659_{\pm 0.017}$	$1.3_{\pm 0.0}$
	MMD-CBS†	0.0	$0.0_{\pm 0.0}$	—	—	$1.1_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.1_{\pm 0.9}$	0.0	$0.0_{\pm 0.0}$	—	—	$1.1_{\pm 0.9}$
	MDOC†	0.0	—	—	$0.000_{\pm 0.000}$	$214.2_{\pm 10.8}$	0.0	—	—	$0.000_{\pm 0.000}$	$229.0_{\pm 40.8}$	0.0	—	—	$0.000_{\pm 0.000}$	$188.8_{\pm 17.4}$
	PCD-LB	0.0	$0.0_{\pm 0.0}$	$47.76_{\pm 4.94}$	—	$3.8_{\pm 0.3}$	0.0	$0.0_{\pm 0.0}$	$50.44_{\pm 4.95}$	—	$4.0_{\pm 0.4}$	0.0	$0.0_{\pm 0.0}$	$46.76_{\pm 4.28}$	—	$6.1_{\pm 1.6}$
	PCD-SHD	0.0	$0.0_{\pm 0.0}$	$2.04_{\pm 0.52}$	—	$3.9_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$1.97_{\pm 0.41}$	—	$3.3_{\pm 0.1}$	0.0	$0.0_{\pm 0.0}$	$1.75_{\pm 0.36}$	—	$3.4_{\pm 0.2}$
	Final Proj.	0.0	$0.0_{\pm 0.0}$	**1.34**$_{\pm 1.05}$	$0.842_{\pm 0.021}$	$17.5_{\pm 4.4}$	0.0	$0.0_{\pm 0.0}$	$1.08_{\pm 0.87}$	$0.837_{\pm 0.022}$	$20.4_{\pm 1.0}$	0.0	$0.0_{\pm 0.0}$	$0.90_{\pm 0.71}$	$0.837_{\pm 0.024}$	$16.0_{\pm 4.8}$
	DiRecT	0.0	$0.0_{\pm 0.0}$	$1.52_{\pm 0.87}$	**0.883**$_{\pm 0.018}$	$80.9_{\pm 35.1}$	0.0	$0.0_{\pm 0.0}$	**0.37**$_{\pm 0.35}$	**0.886**$_{\pm 0.018}$	$57.5_{\pm 0.7}$	0.0	$0.0_{\pm 0.0}$	**0.15**$_{\pm 0.12}$	**0.889**$_{\pm 0.017}$	$57.5_{\pm 0.7}$
	GlobMP (Ours)	**85.0**	**85.0**$_{\pm 35.7}$	$5.30_{\pm 13.46}$	$0.765_{\pm 0.146}$	$14.4_{\pm 116.6}$	**75.0**	**75.0**$_{\pm 43.3}$	$9.80_{\pm 17.31}$	$0.753_{\pm 0.139}$	$197.1_{\pm 116.2}$	**90.0**	**90.0**$_{\pm 30.0}$	$4.35_{\pm 13.07}$	$0.761_{\pm 0.138}$	$98.2_{\pm 103.0}$

Performance generally decreases as the number of robots increases, although the effect depends strongly on the environment. At the Medium cap, DiRecT maintains $SR = 1.00$ on `Empty` and `Highways` at every team size and at least 0.90 on `Basic`, but at most 0.50 on `Dense`. `Conveyor` and `Cross-Room` separate the methods most sharply: the best baseline is PCD-LB, at 0.65 and 0.40 on `Conveyor` and 0.45 and 0.20 on `Cross-Room` for $N_a = 3$ and 6, and from $N_a = 9$ no baseline exceeds 0.10, while *GlobMP (Ours)* stays at 1.00 through $N_a = 15$ on `Cross-Room` and ranges from 1.00 to 0.70 on `Conveyor`. `Drop-Region` and `Corridor-Bay` provide intermediate difficulty, with DiRecT attaining mean *SR* values of 0.70 and 0.95, respectively; DiRecT exceeds GlobMP on `Corridor-Bay` from $N_a = 9$ (at least 0.90 against 0.60 to 0.75) and on `Highways` at $N_a \geq 18$ (1.00 against 0.90 and 0.55), and edges it on `Empty` at $N_a = 21$ (1.00 against 0.95). Changing the velocity cap has a comparatively small effect (four of the nine rows, marked †, ignore it), except for DiRecT on `Drop-Region`, which reaches 1.00 at every team size only under the High cap, and for GlobMP on `Highways`, whose Low-cap SR falls from 0.80 at $N_a = 6$ to 0.60 at $N_a = 12$ and whose cells from $N_a = 15$ hit the planning budget (0.30, 0 and 0.15), suggesting that environment geometry and coupled collision avoidance dominate the difficulty of the harder instances.

The *GlobMP (Ours)* rows use the same shared instances across all eight maps, all velocity caps, and all 168 experimental cells, with a 300 s planning budget per team; MMD-CBS has 60 s per configuration. Planning times are measured on pinned CPU cores with four cores per job. Cells that reach the full 300 s budget were repeated on an idle machine and showed the same behaviour, so these cases reflect planning difficulty, not resource contention: on `Highways` at the Low cap for $N_a \geq 15$ and on `Corridor-Bay` at $N_a = 18$ (High) and $N_a = 21$ (Low).

The suite's success rate is a best-of-64 quantity: an instance counts as solved when any of the 64 candidates a baseline samples is collision-free, whereas our method returns one trajectory. Figure 16 therefore scores every method with per-candidate data on the success of a single returned plan, and Figure 15 shows the best-of-64 success rate the tables report.

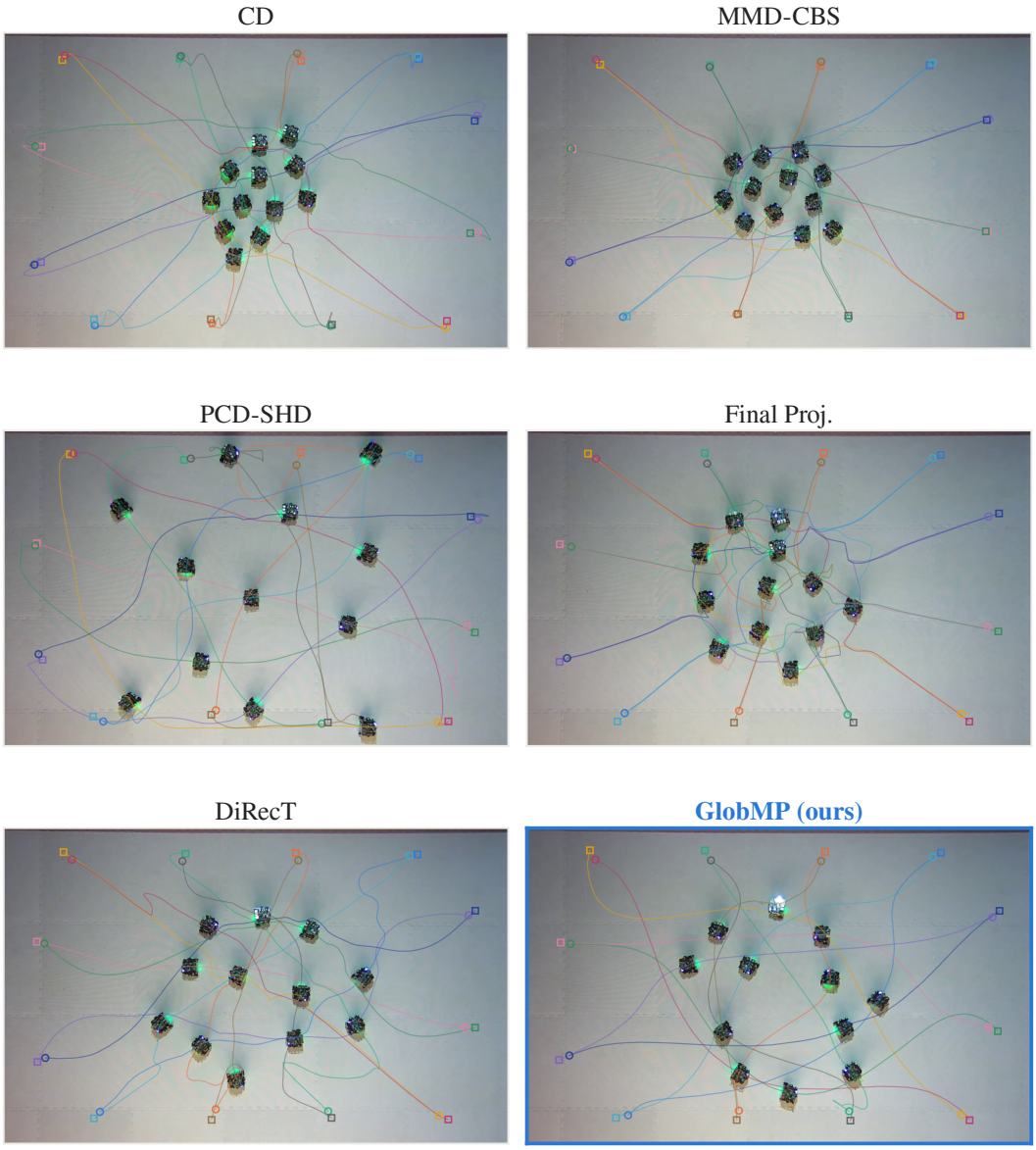


Figure 17: Hardware execution on `Empty`: all methods, twelve GRITSBots on the Georgia Tech Robotarium. The tracking controller and the virtual obstacles are those of Appendix A4.1. Each panel reports one execution of the ring instance by the method named above it: a single overhead frame, with the trajectories recovered from the video drawn over it, each marked by a circle at its start and a square at its terminal state. A robot is counted as being in collision when its footprint overlaps an obstacle region. The frame shown is the midpoint of the tracked window for an execution in which no robot overlaps a region at any time, and otherwise the frame at which the number of overlapping robots is maximal; only the midpoints are therefore proportional to the duration of an execution. `Empty` carries no obstacles, so no execution can overlap a region and the panels differ only in the routing of the exchange, with midpoints at 25 s for GlobMP, 27 s for PCD-SHD, 28 s for Final Proj., 29 s for DiRecT, 34 s for CD and 57 s for MMD-CBS.

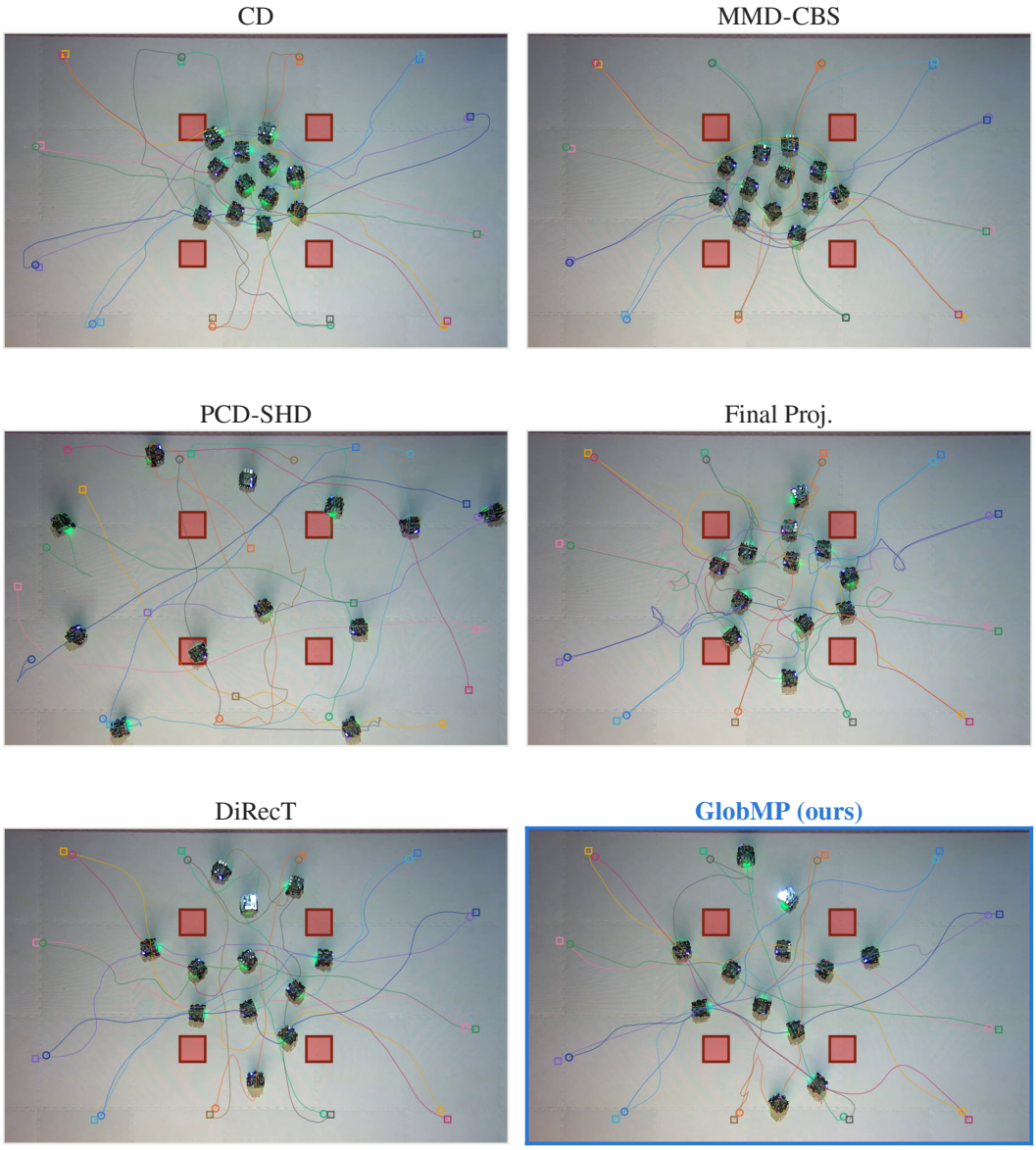


Figure 18: Hardware execution on `Drop-Region`: all methods, with the protocol of Figure 17. Three of the six methods complete the exchange with no footprint overlap: MMD-CBS (midpoint frame at 61 s), DiRecT (31 s) and GlobMP (31 s). For the remaining three the panel shows the frame of maximal overlap: two of the twelve robots overlap a keep-out region under Final Proj. (40 s), three under CD (43 s) and five under PCD-SHD (23 s). PCD-SHD is additionally the only execution on this map whose robots do not reach their goals; its terminal states lie well away from the goal ring. This map discriminates least among the four maps with obstacles, and the discrimination that remains lies in when the overlaps occur rather than in how many robots they involve.

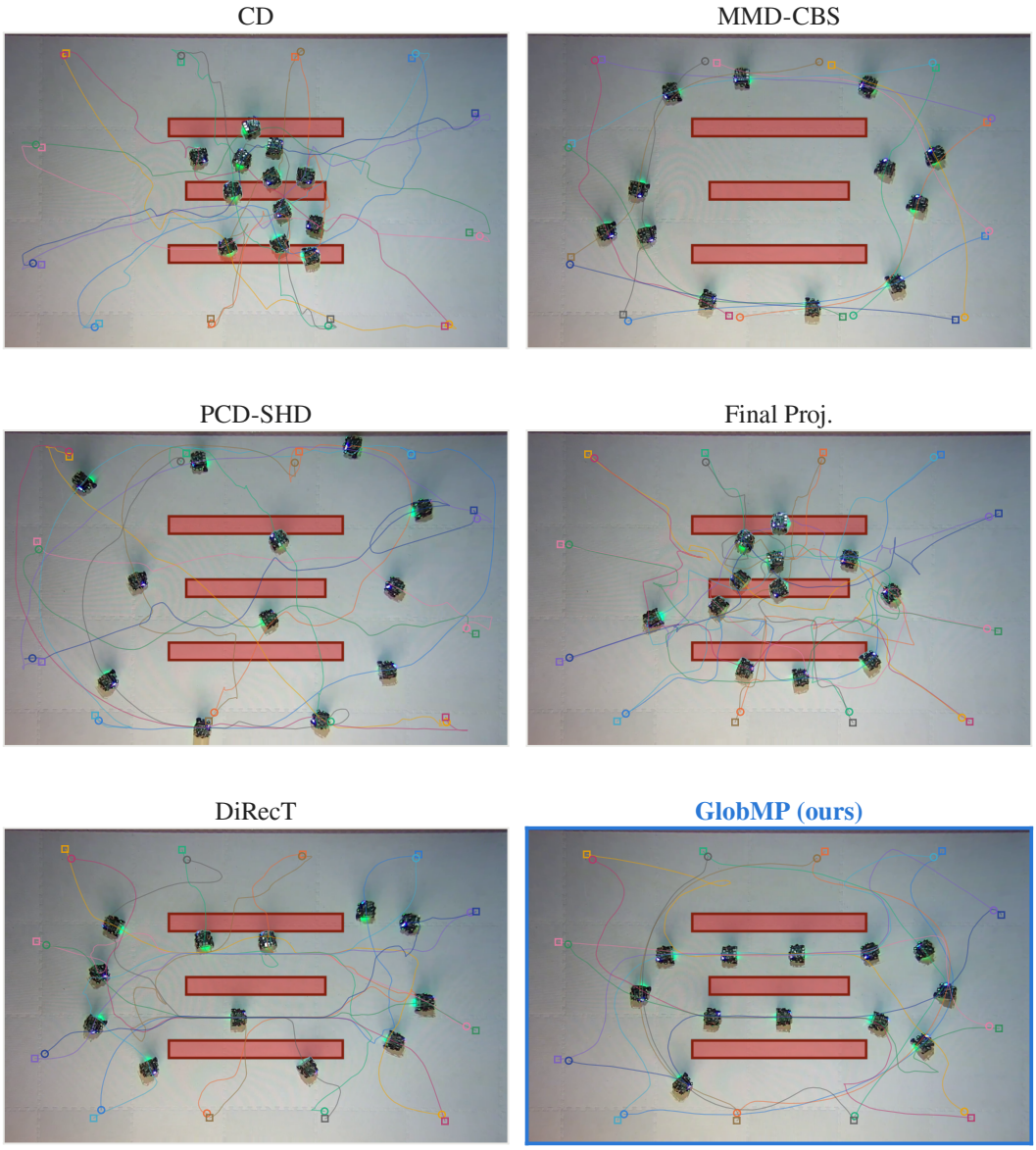


Figure 19: Hardware execution on `Conveyor`: all methods, with the protocol of Figure 17. The three belts sit in the middle of the arena and leave two gaps between them, so the exchange requires an ordering of the team as well as a route; the executions differ principally in that ordering. GlobMP completes the instance with no footprint overlap (midpoint frame at 37 s). PCD-SHD and DiRecT each leave five of the twelve robots overlapping a belt at their frames of maximal overlap, 44 s and 65 s respectively, the latter being the latest frame on this map. CD and Final Proj. leave all twelve overlapping, at 40 s and 48 s: both traverse the belts as a single block, so the entire team is inside a region at one instant. MMD-CBS returned no plan on this map; its robots therefore remain on the start ring, and its panel is marked rather than scored, since counting overlaps for a team that never moved would record a success where no trajectory was produced.

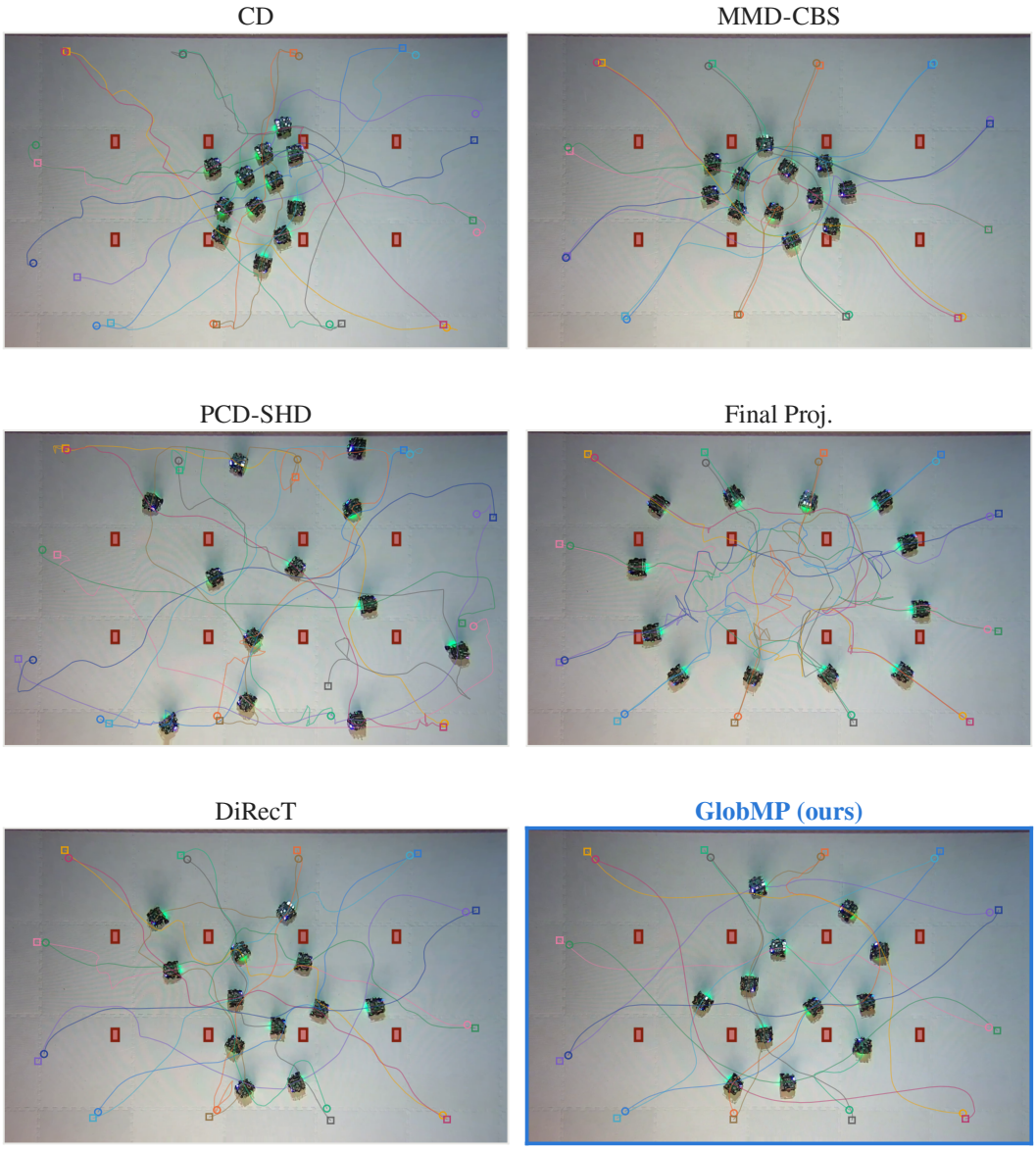


Figure 20: Hardware execution on `Corridor-Bay`: all methods, with the protocol of Figure 17. As on `Drop-Region`, three methods complete the exchange with no footprint overlap: PCD-SHD (midpoint frame at 45 s), DiRecT (32 s) and GlobMP (26 s). The remaining three overlap the keep-out regions at their frames of maximal overlap: three of the twelve robots under MMD-CBS (42 s), four under Final Proj. (78 s), the latest frame of maximal overlap in this gallery, and eight under CD (49 s). Final Proj. remains clear for the first minute and overlaps only as the team disperses towards its goals, whereas CD overlaps from early in the execution. MMD-CBS, which returns no plan on two of the other maps, produces one here and brings nine of the twelve robots through without overlap.

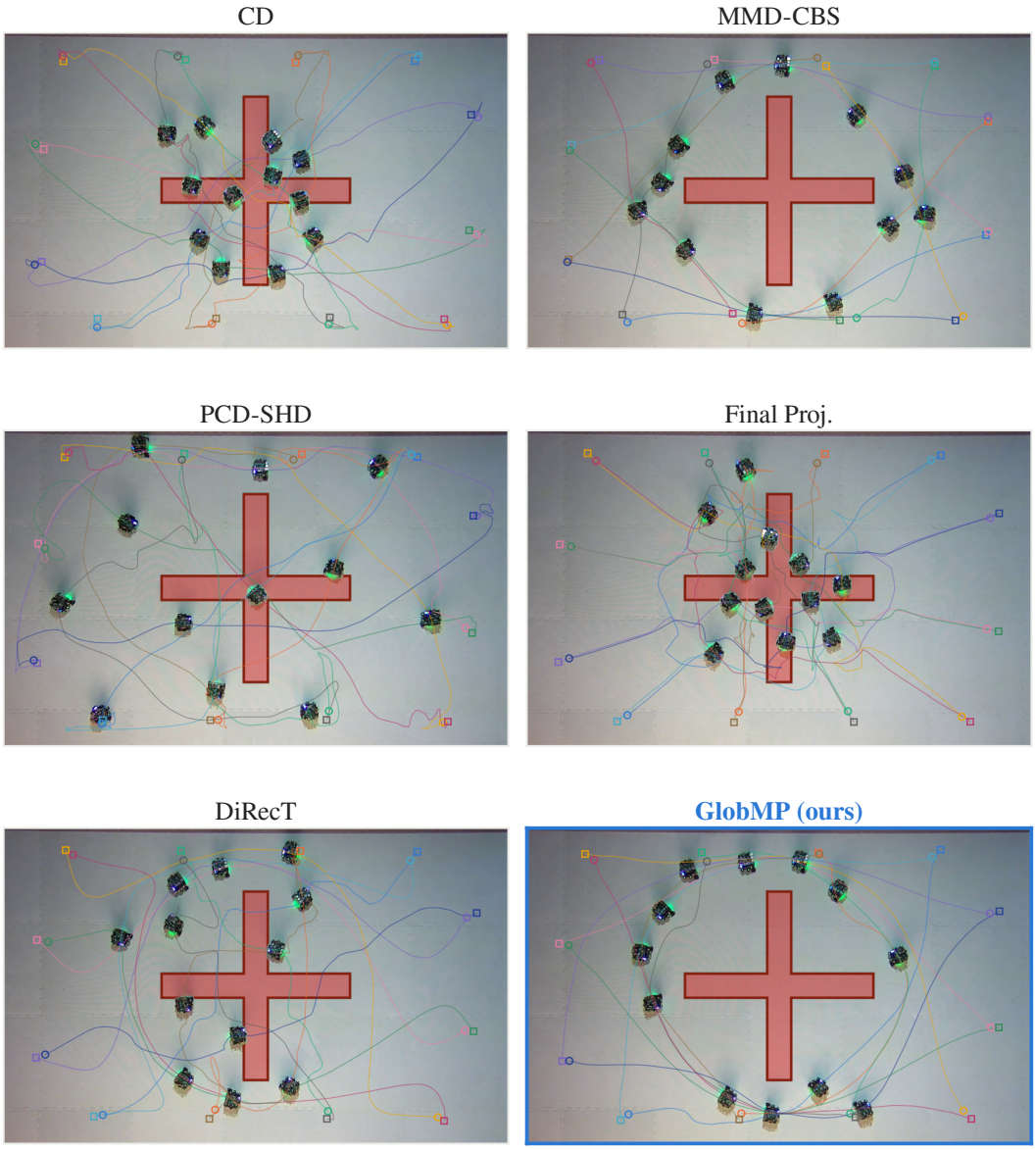


Figure 21: Hardware execution on `Cross-Room`: all methods, with the protocol of Figure 17. The cross occupies the centre of the arena and leaves four corner regions connected only along the walls, so the exchange must be routed around the arms; a team that advances directly towards its goals enters one of them. GlobMP is the only method that completes the instance with no footprint overlap (midpoint frame at 26 s). At their frames of maximal overlap, six of the twelve robots overlap the cross under DiRecT (37 s) and seven under PCD-SHD (41 s); both route part of the team around the arms and lose the remainder. CD and Final Proj. leave all twelve overlapping, at 38 s and 34 s. MMD-CBS returned no plan on this map as well, so its panel again shows the untouched start ring and is marked rather than scored. With `Conveyor`, this is the map on which the margin between GlobMP and the baselines is largest under hardware execution.

A4.6 IMPLEMENTATION DETAILS

This subsection lists how GlobMP is run in the experiments: the curve class of each experiment, the settings of Algorithm 2, the cone solve, and the checks of the implementation.

Curve class. Table 12 gives the degree, the cost order k and the boundary order ℓ of each experiment; the synthetic study uses 7 pieces (4 in the scaling study). The synthetic study uses the minimum-snap class of the baselines it is compared against ($k=\ell=4$); the multi-robot tasks prescribe only where each robot starts and stops, so they fix position and velocity ($\ell=1$) and minimize acceleration ($k=2$); Maze2D uses $k=\ell=2$. In every row $2(\ell+1)\geq k$, so Assumption B holds, and every degree is at least $\max\{2k-1,\ell\}$, so the synthetic class is that of Theorem 1, with rate N^{-1} ($\ell=k$); Maze2D joins its pieces C^{ℓ}, a subclass. Every row starts from heuristic seeds rather than the grid start, the multi-robot rows add a speed limit, and Maze2D adds corridor blocks, each box binding one piece, and keep-out blocks outside (P_∞), so the theorem certifies no row's cost as optimal.

Table 12: Curve class of each planning experiment.

Experiment	Section	degree d	k	ℓ
Synthetic study	§4.1	9	4	4
Multi-robot suite	§A4.5	9	2	1
Robotarium	§4.2	12	2	1
Maze2D	§A4.3	9	2	2

Piecewise curves. With knots $0=t_0<\cdots<t_N=1$, $\Delta_i=t_i-t_{i-1}$ and $u=(t-t_{i-1})/\Delta_i$ on piece i, the curve is $\boldsymbol{\gamma}(t)=\boldsymbol{\Gamma}^{(i)}\mathbf{b}_d(u)$ and $\boldsymbol{\gamma}^{(s)}(t)=\Delta_i^{-s}\boldsymbol{\Gamma}^{(i)}\mathbf{b}_d^{(s)}(u)$. The endpoint conditions of order $s\leq\ell$ and the C^{k-1} (or smoother) continuity conditions at the $N-1$ junctions are linear in the stacked control points, so $\boldsymbol{\Gamma}\mathbf{A}=\mathbf{B}$ gives the same orthonormal chart as for one piece, with $d_f=N(d+1)-\operatorname{rank}(\mathbf{A})$. The cost matrix is $\operatorname{blkdiag}(\Delta_i^{1-2k}\mathbf{G}^{(k)})$, and the constraint blocks are built piece by piece with the same scaling. With these junctions the curves lie in $\mathscr{H}^k$, and a zero-cost perturbation is a polynomial of degree below k, zero when $k\leq 2(\ell+1)$.

Restoration (Algorithm 2). The provable step $1/L_{\mathrm{proj}}$ (Appendix A3.1.4) is too small to be practical, so the implementation calibrates the step: it starts at 10^{-3}, caps it at $2V/\|\nabla V\|^2$ for the current merit value V, halves it until the merit (2) decreases, and then grows it by a factor 1.3 (Barzilai–Borwein steps on Maze2D and the Robotarium); blocks within their tolerance are left out of the merit and its gradient, so V and ∇V above are over the remaining blocks and the bound of Appendix A3.1.4 does not cover these steps. A curve is accepted when every block satisfies $\operatorname{dist}^2(\boldsymbol{\pi}_j,\mathscr{K}_j)\leq \mathrm{tol}_j$, with $\mathrm{tol}_j=(2r_j\delta+\delta^2)^2$ for a ball, pair or moving-ball block whose radius is inflated by δ, δ^2 for a box, and 10^{-12} otherwise.

Cone solve. Each projection onto $\mathscr{K}_j$ uses Moreau's decomposition and fully corrective nonnegative matching pursuit (Algorithm 3), with the block rescaled to unit norm. The implementation maximizes the unnormalized correlation $\langle\mathbf{v},\mathbf{b}_{m_j}(t)\rangle$ (Algorithm 3 normalizes it, and the rate of Appendix A3.1.4 is for that version), a polynomial in t, at the real roots of its derivative and the two endpoints (by Bernstein branch-and-bound above degree 24), then refits the nonnegative weights and drops atoms whose weight reaches zero; it stops at the relative threshold $\varepsilon=10^{-6}$. The number of atoms is bounded by Carathéodory's theorem, which allows at most m_j+1 atoms ($2d+1$ for a collision block of degree $2d$).

Finishing step. In the synthetic study, the minimum-snap curvature ratio (13) $\beta/\alpha\approx 4\times10^7$ ($N=7$; 4×10^5 at $N=4$) makes first-order descent slow along flat directions, so each main-comparison run is finished by Newton's method on the first-order conditions and the active contact equations. An interior contact at a time t_* contributes $g_j^{\boldsymbol{\Gamma}}(t_*)=0$ and $\partial_t g_j^{\boldsymbol{\Gamma}}(t_*)=0$; the unknowns are the free control points, the contact times and nonnegative multipliers. This step lowered the cost by a further

factor of 2 to 3; the reported curves pass the dense recheck, but the step is numerical and is not covered by the theory. As in Maze2D (§A4.3), the step follows an SQP polish of the output of Algorithm 1, also outside the theory.

Several robots. The free control points of all robots are stacked; the robots share d, k and ℓ, and hence $\mathbf{A}$, $\mathbf{N}$ and the cost matrix, and each has its own boundary data. There is one collision block per robot and obstacle and one per pair of robots, the latter the obstacle block applied to the difference curve $\boldsymbol{\Gamma}_a - \boldsymbol{\Gamma}_b$ at the separation radius, so Algorithms 1 and 2 are unchanged.

Workspace bounds. A bound $\underline{x}_i \le \gamma_i(t) \le \bar{x}_i$ is two nonnegativity conditions with Bernstein coefficients $\boldsymbol{\Gamma}_{i,:}^\top - \underline{x}_i \mathbf{u}$ and $\bar{x}_i \mathbf{u} - \boldsymbol{\Gamma}_{i,:}^\top$. Since $\mathbf{u}^\top \mathbf{b}_d \equiv 1$, a degree-$d$ coefficient vector $\boldsymbol{\zeta}$ is raised to degree $2d$ by $S_d\big(\frac{1}{2}(\boldsymbol{\zeta}\mathbf{u}^\top + \mathbf{u}\boldsymbol{\zeta}^\top)\big)$, so these blocks share the collision blocks' cone and are affine in $\boldsymbol{\Gamma}$.

Checks of the implementation. Each of the following holds to machine precision on the benchmark instances: S_d reproduces the Bernstein coefficients of $\mathbf{b}_d^\top \mathbf{X} \mathbf{b}_d$; the chart satisfies $\boldsymbol{\Gamma}\mathbf{A} = \mathbf{B}$ and $\mathbf{N}^\top \mathbf{A} = \mathbf{0}$; $\boldsymbol{\pi}_j$ reproduces $\|\boldsymbol{\gamma}(t) - \mathbf{c}_j\|^2 - r_j^2$; the cost matches $\int_0^1 \|\boldsymbol{\gamma}^{(k)}\|^2 dt$; and the cone test is exact in both directions, returning zero for collision-free curves and a positive value, increasing with the penetration depth, for colliding ones.